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ORIENTED SUPERSINGULAR ELLIPTIC CURVES & CLASS GROUP ACTIONS

LEONARDO COLÒ

Institut de Mathématiques de Marseille

PhD Thesis Defense

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- Introduction.
- Orientations and class group actions.
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INTRODUCTION

Definition

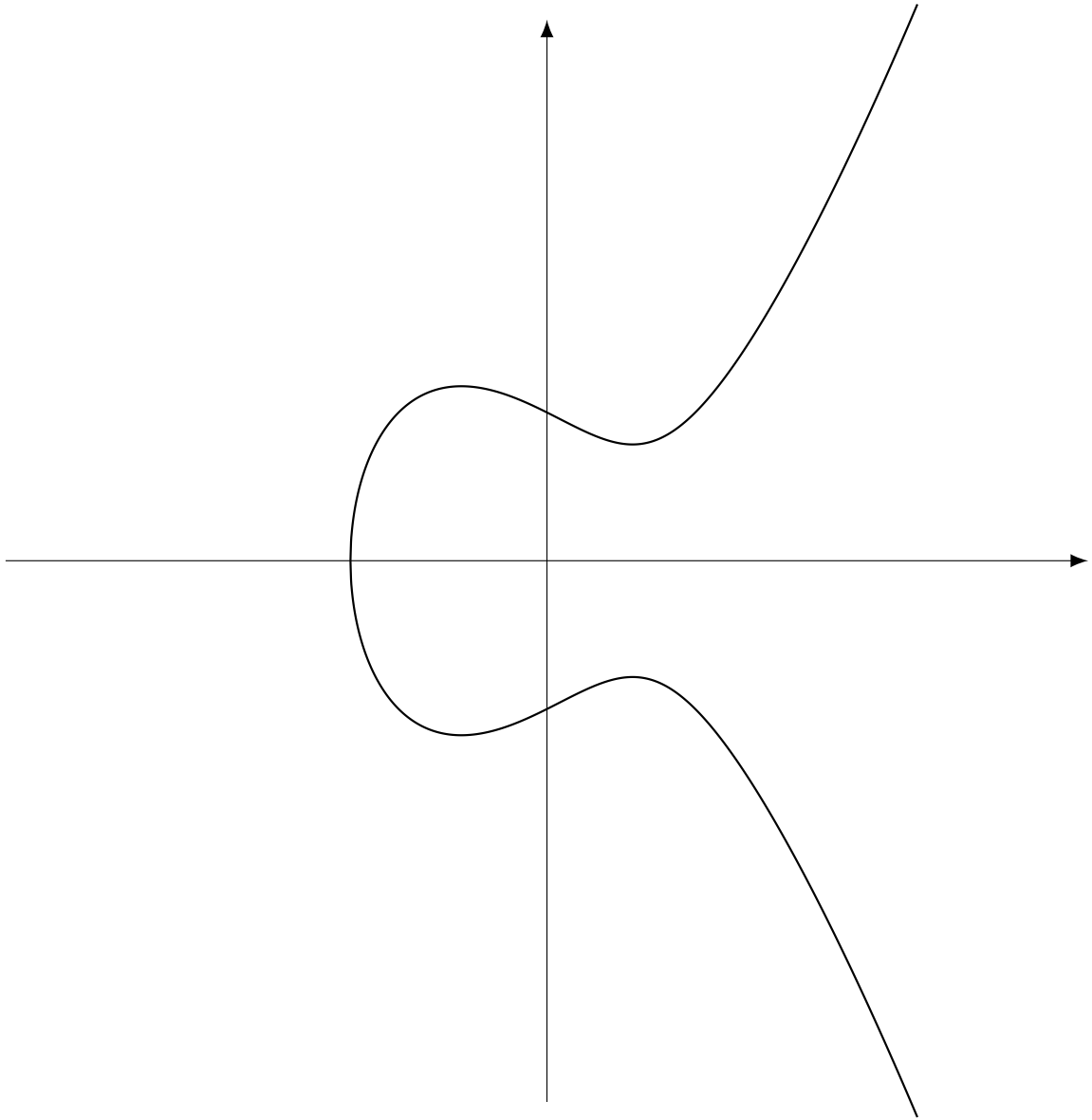
Let k be a field of characteristic $\neq 2, 3$. An elliptic curve E defined over k is a smooth projective curve of genus 1 defined by a Weierstrass equation

$$E : Y^2Z = X^3 + aXZ^2 + bZ^3$$

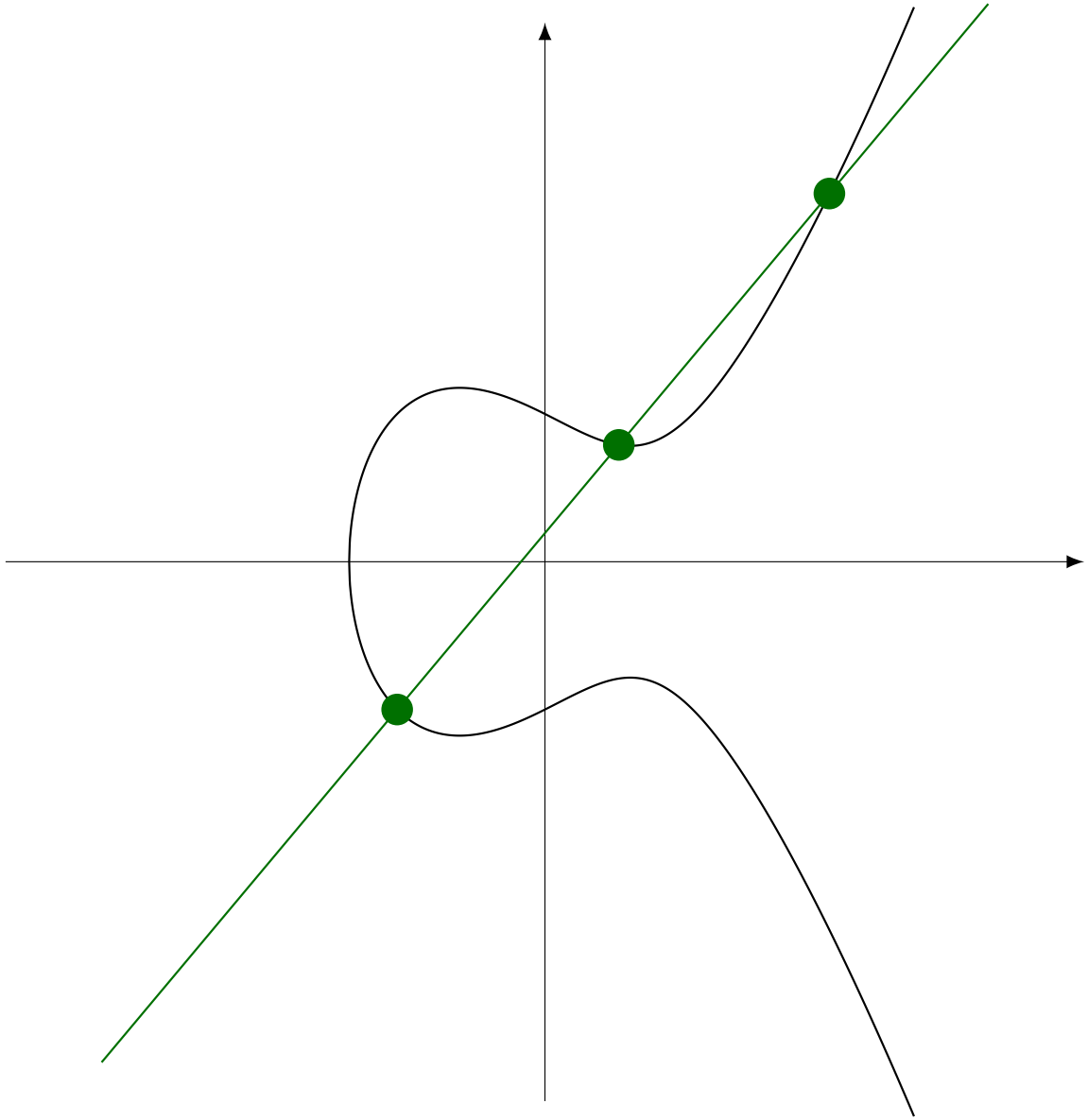
where $a, b \in k$ are such that $4a^3 + 27b^2 \neq 0$.

In general we work with the affine equation of E, i.e., $E : y^2 = x^3 + ax + b$ together with the distinguished point $O = (0 : 1 : 0)$ (said to be the *point at infinity*).

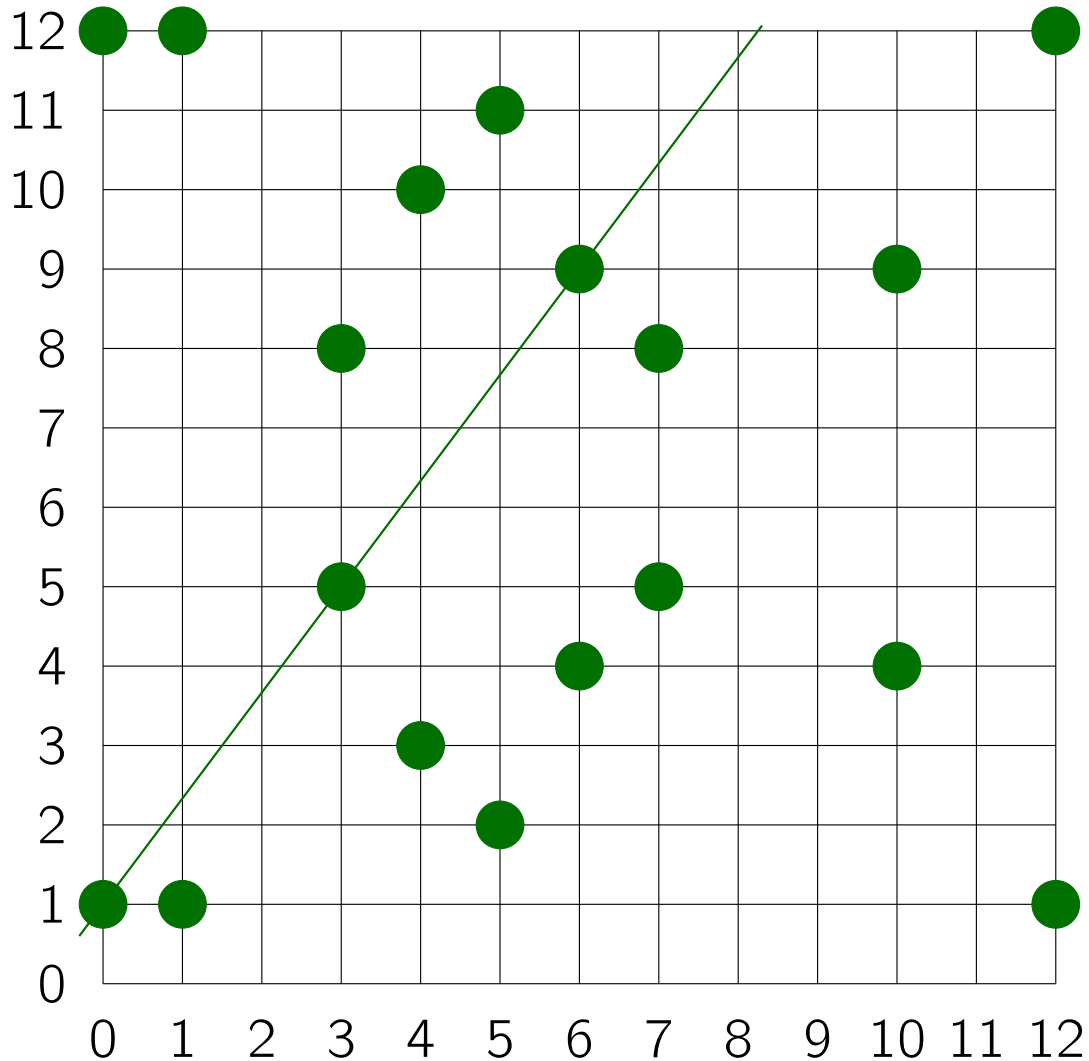
ELLIPTIC CURVES - EXAMPLES



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Definition

Given an elliptic curve E over k, and a finite set of primes S, we can associate an isogeny graph $G_S(E)$

- whose vertices are elliptic curves isogenous to E over $\bar{k}$, and
- whose edges are isogenies of degree $\ell \in S$.

If $S = \{\ell\}$, then we write $G_\ell(E)$, the ℓ-isogeny graph.

The vertices are defined up to $\bar{k}$-isomorphism and the edges from a given vertex are defined up to a $\bar{k}$-isomorphism of the codomain.

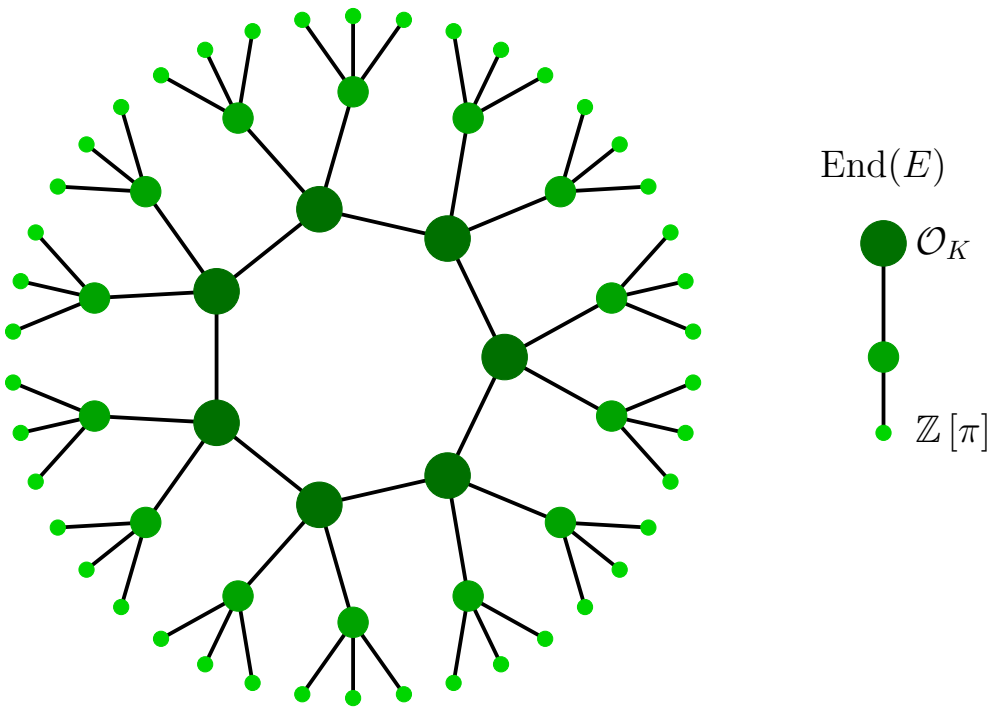
The ℓ-isogeny graph of E is $(\ell + 1)$-regular (as a directed multigraph).

ORDINARY ISOGENY GRAPHS: VOLCANOES

Let $\mathbf{End}(E) = \mathcal{O} \subseteq K$, an imaginary quadratic field. The class group $\mathrm{Cl}(\mathcal{O})$ acts faithfully and transitively on the set of elliptic curves with endomorphism ring $\mathcal{O}$:

$$E \longrightarrow E/E[\mathfrak{a}] \qquad E[\mathfrak{a}] = \{P \in E \mid \alpha(P) = 0 \ \forall \alpha \in \mathfrak{a}\}$$

Thus, the CM isogeny graphs can be modelled by an equivalent category of fractional ideals of K.



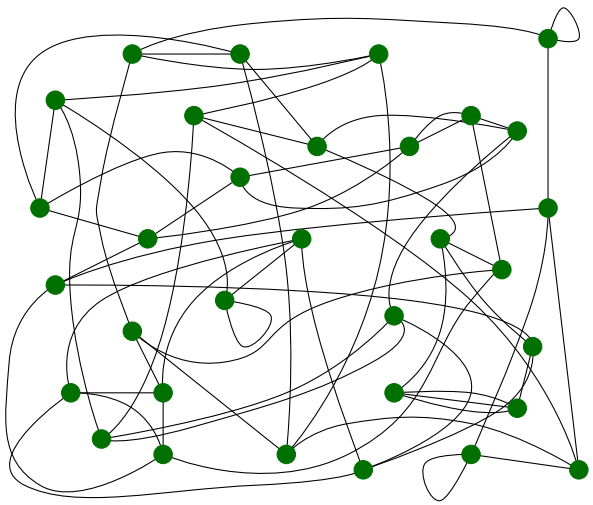
SUPERSINGULAR ISOGENY GRAPHS

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The supersingular isogeny graphs are remarkable because the vertex sets are finite : there are $(p+1)/12 + \epsilon_p$ curves. Moreover

- every supersingular elliptic curve can be defined over $\mathbb{F}_{p^2}$;
- all ℓ-isogenies are defined over $\mathbb{F}_{p^2}$;
- every endomorphism of E is defined over $\mathbb{F}_{p^2}$.

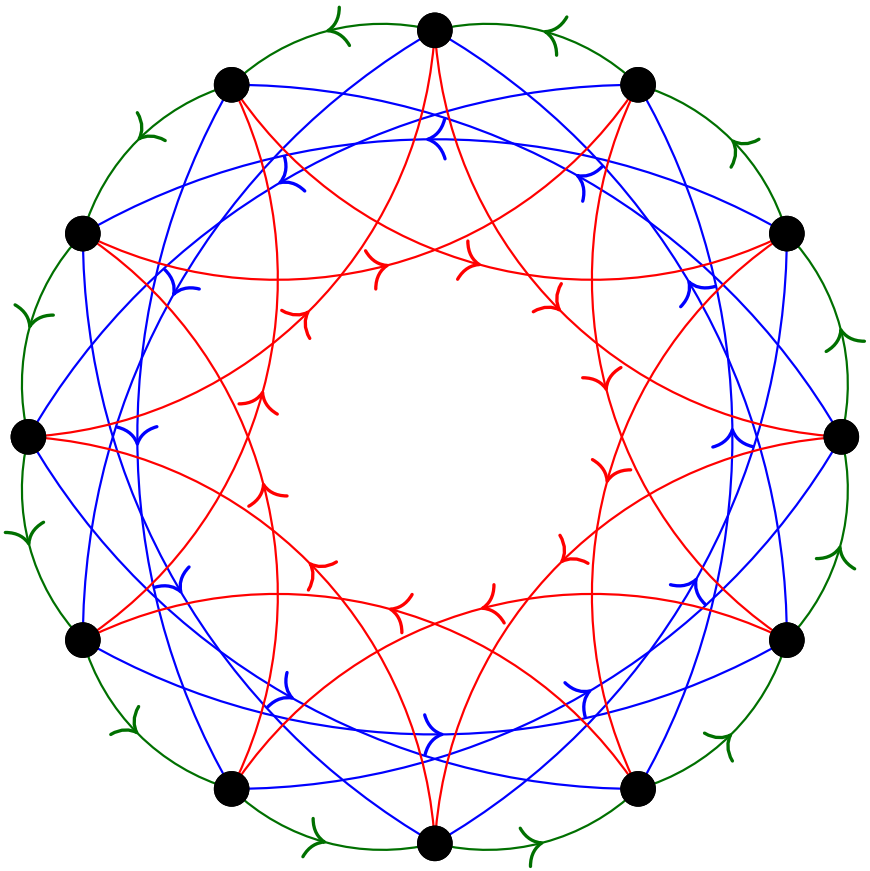
The lack of a commutative group acting on the set of supersingular elliptic curves/$\mathbb{F}_{p^2}$ makes the isogeny graph more complicated.



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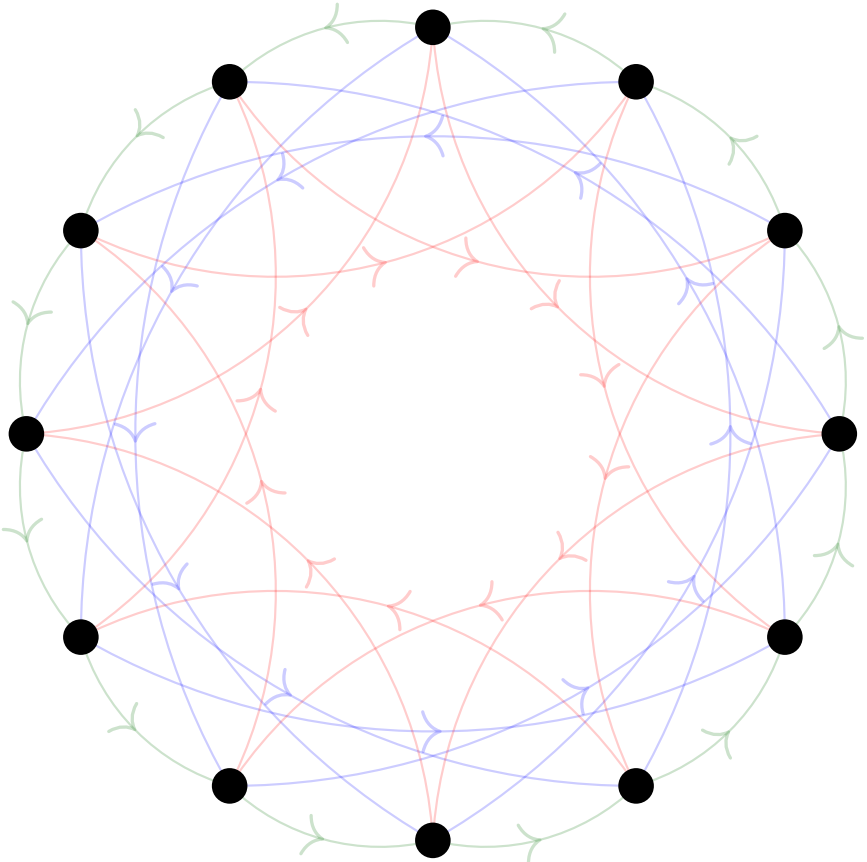
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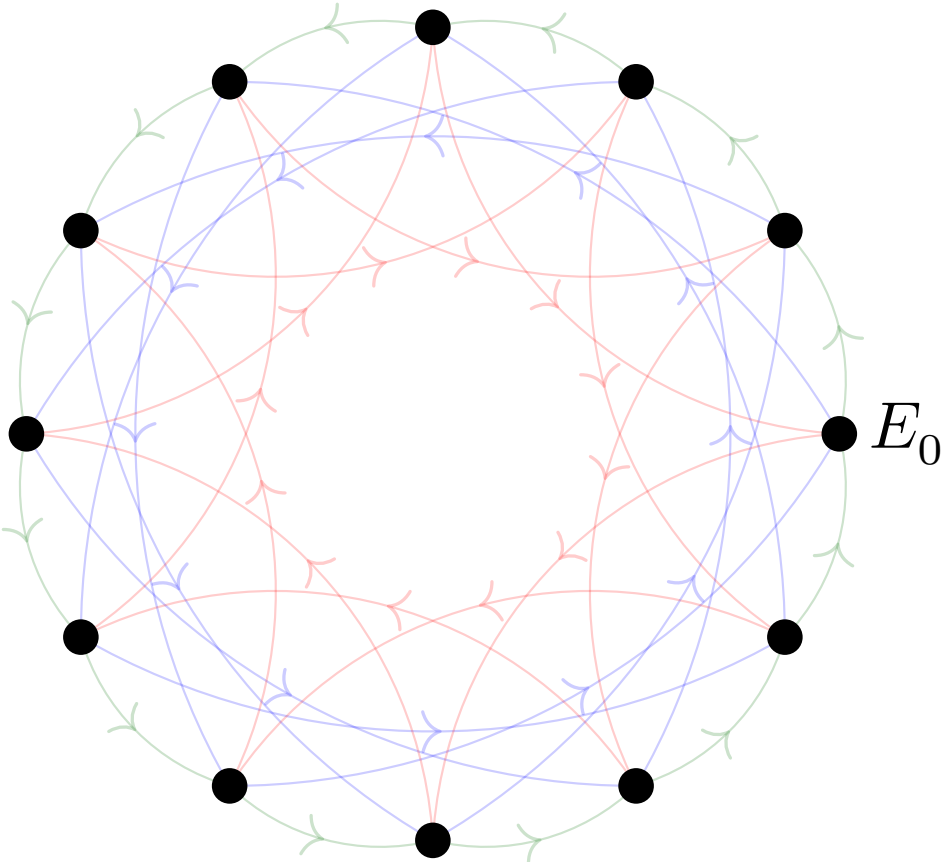
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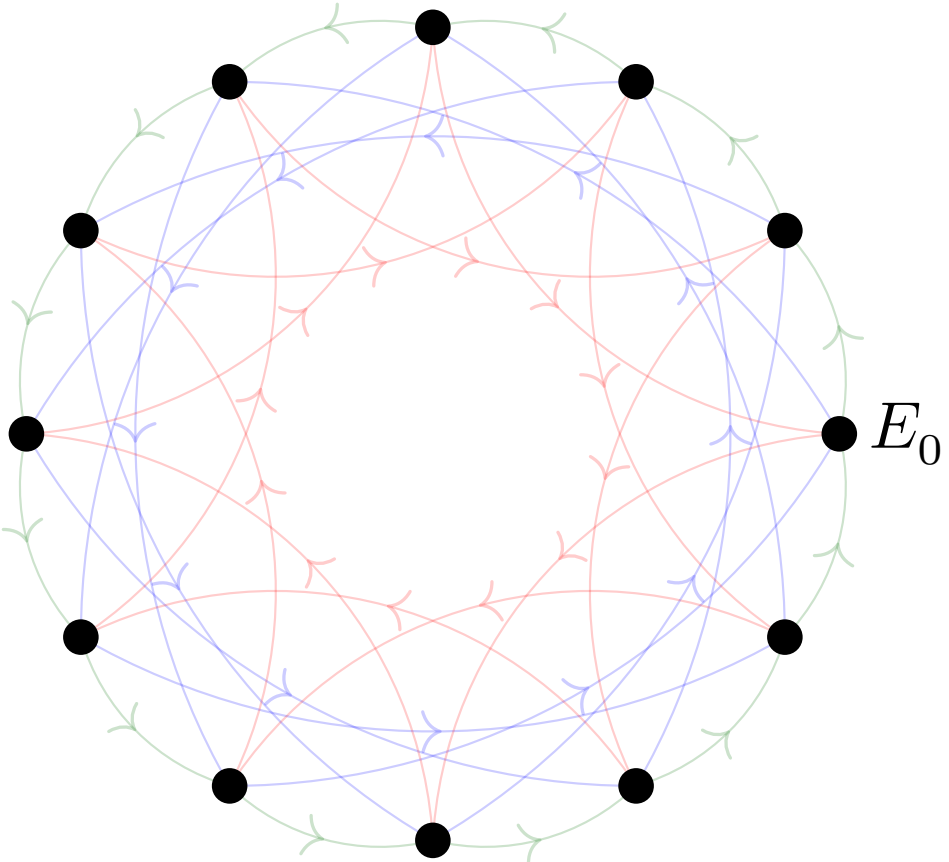
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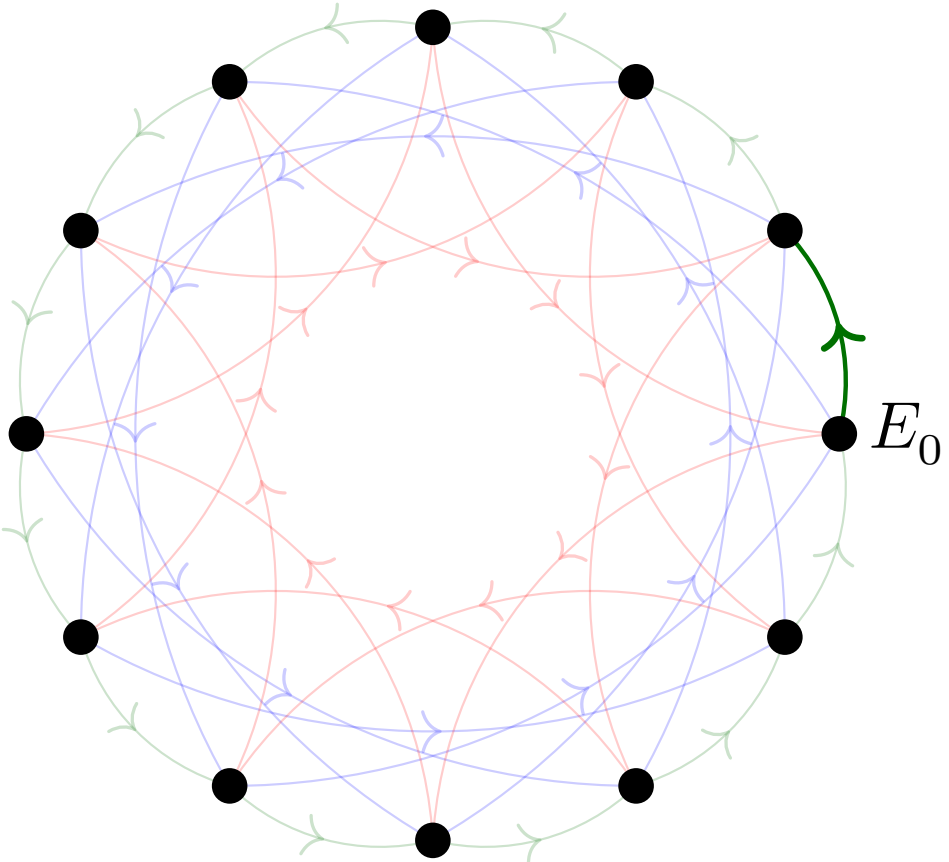
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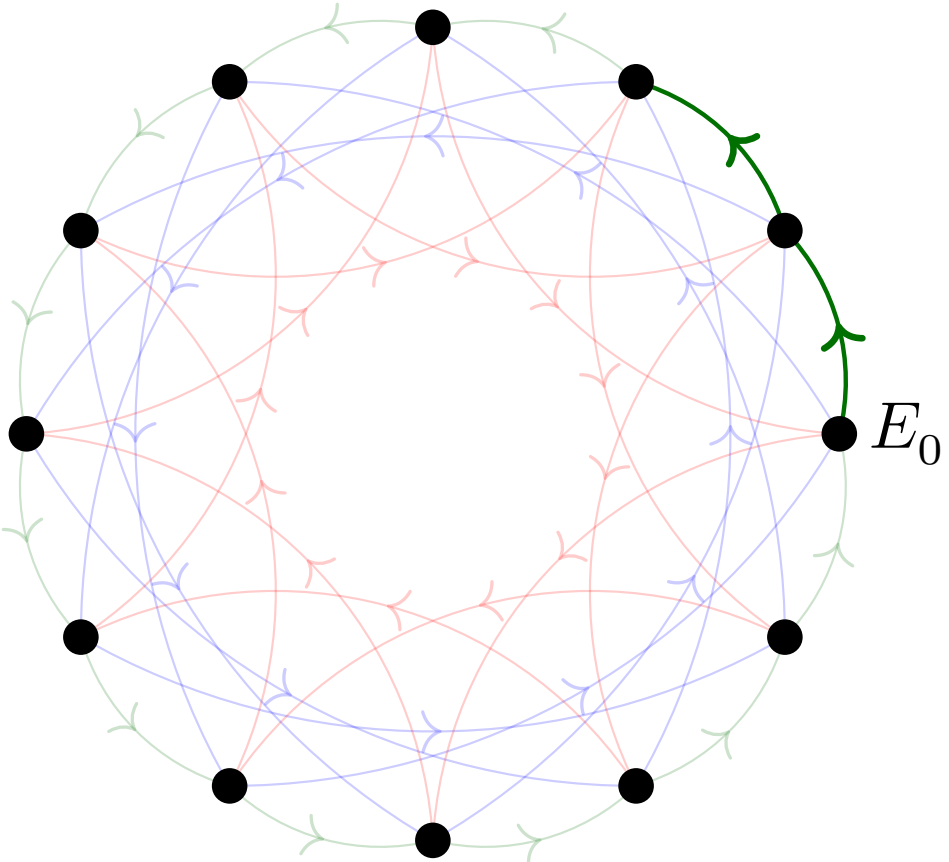
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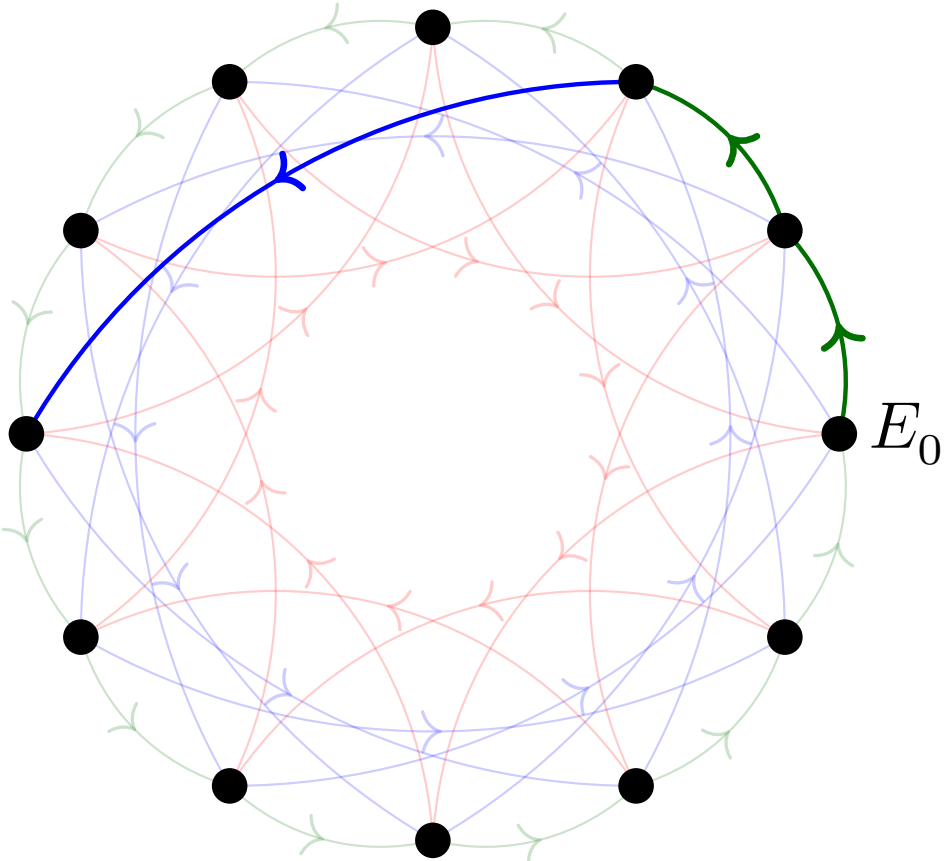
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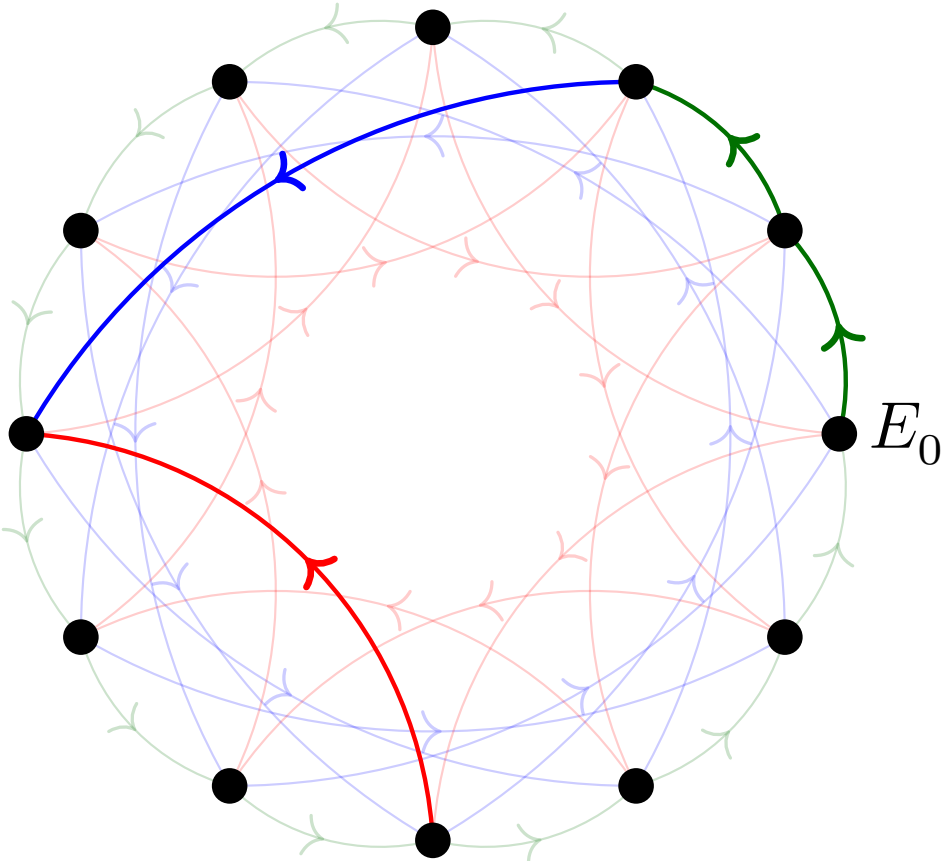
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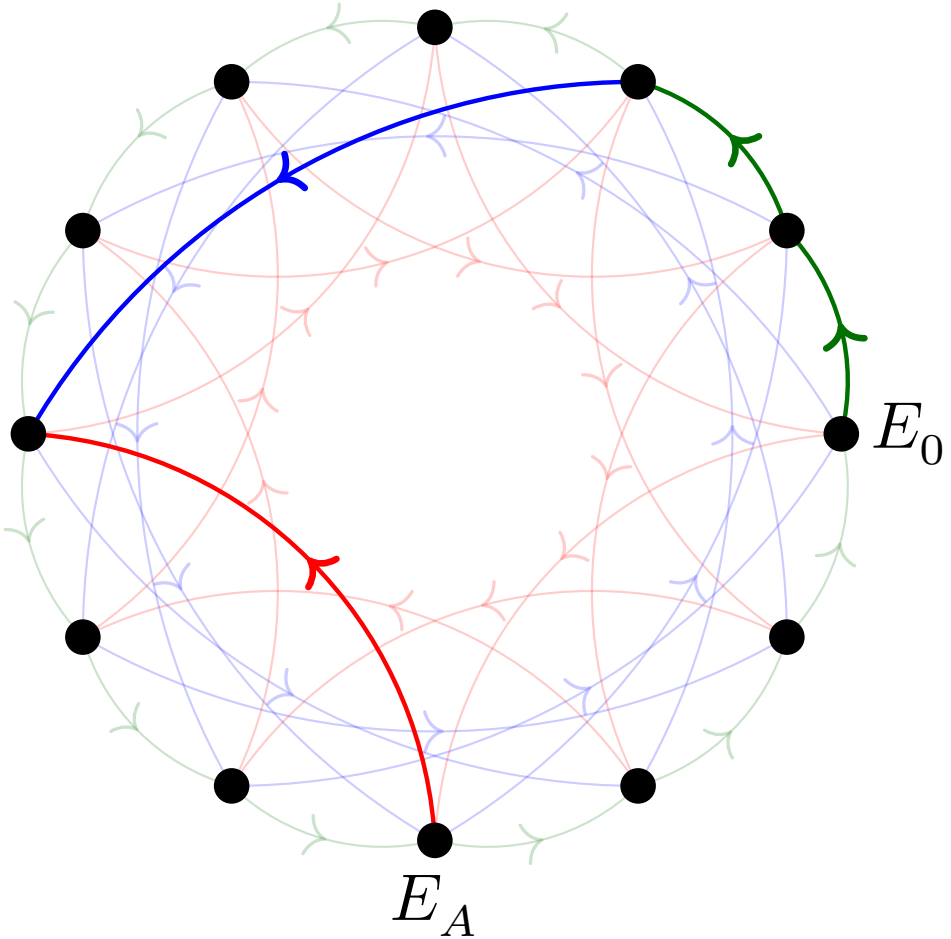
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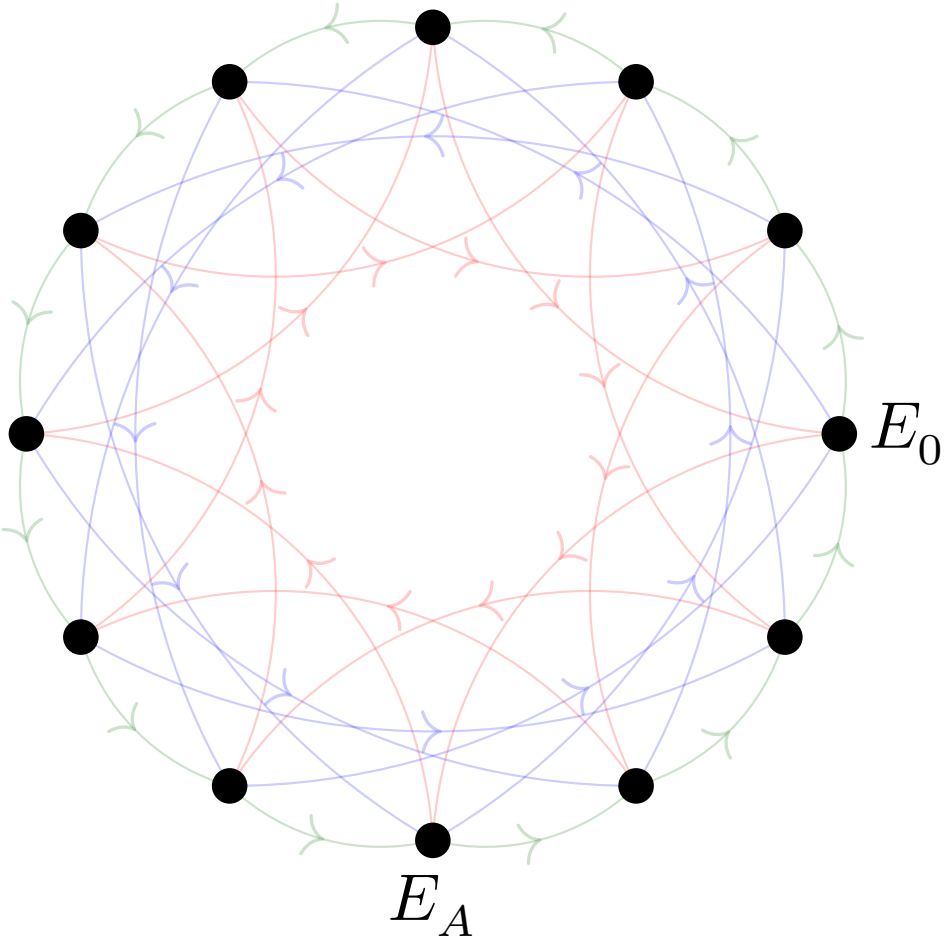
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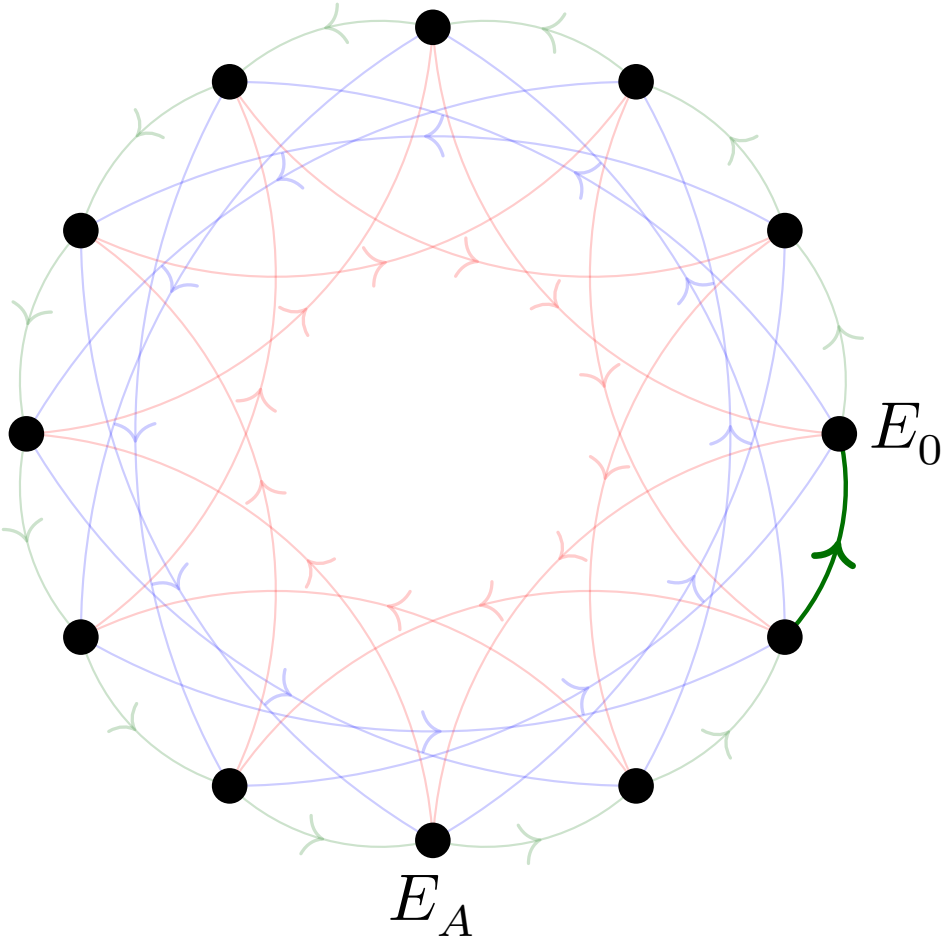
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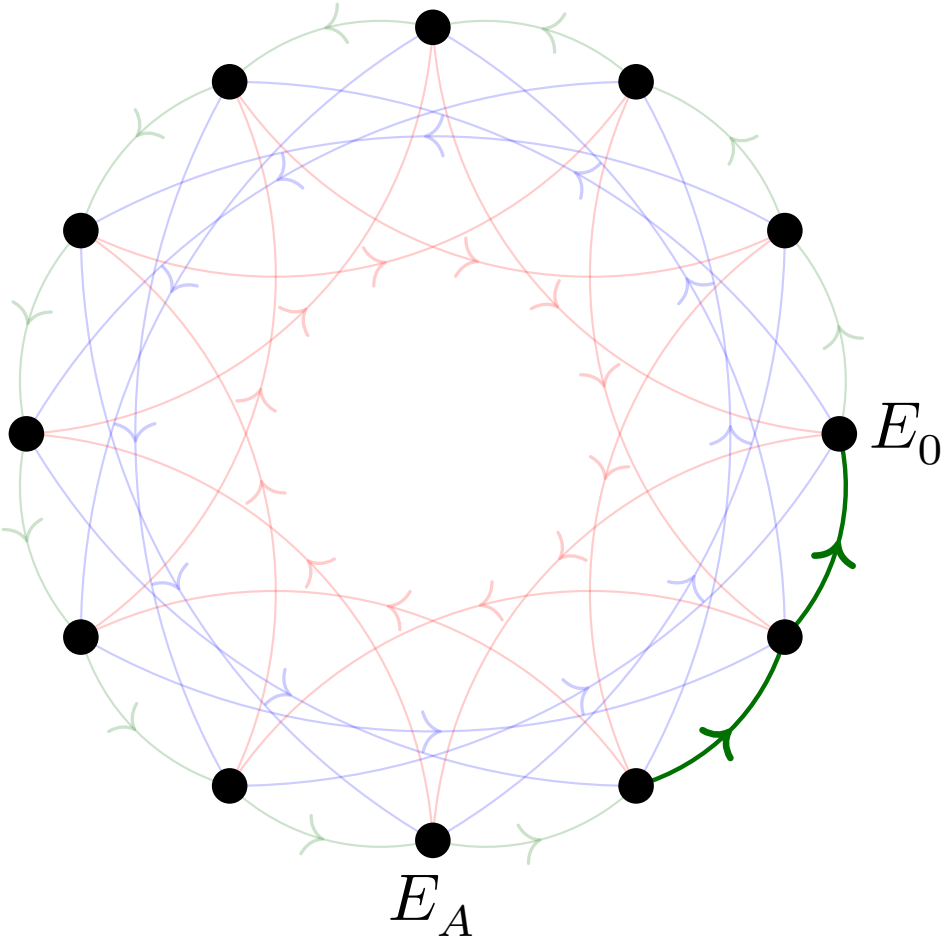
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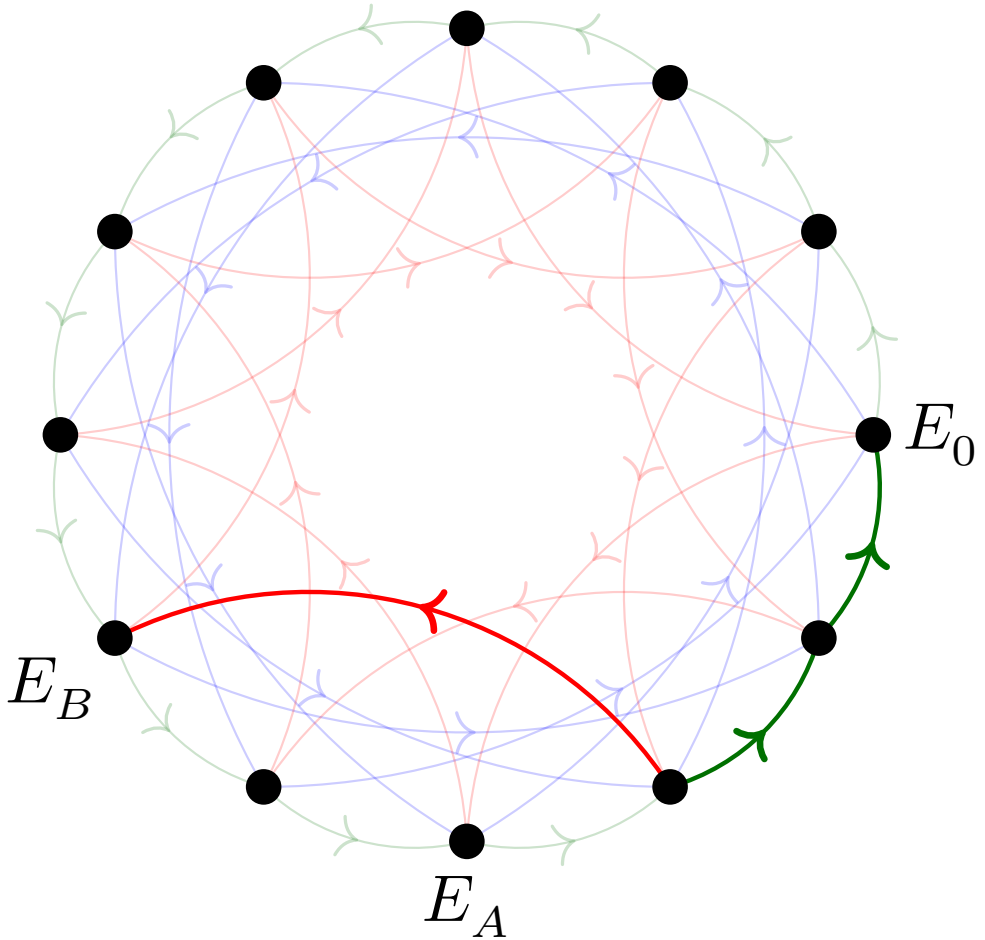
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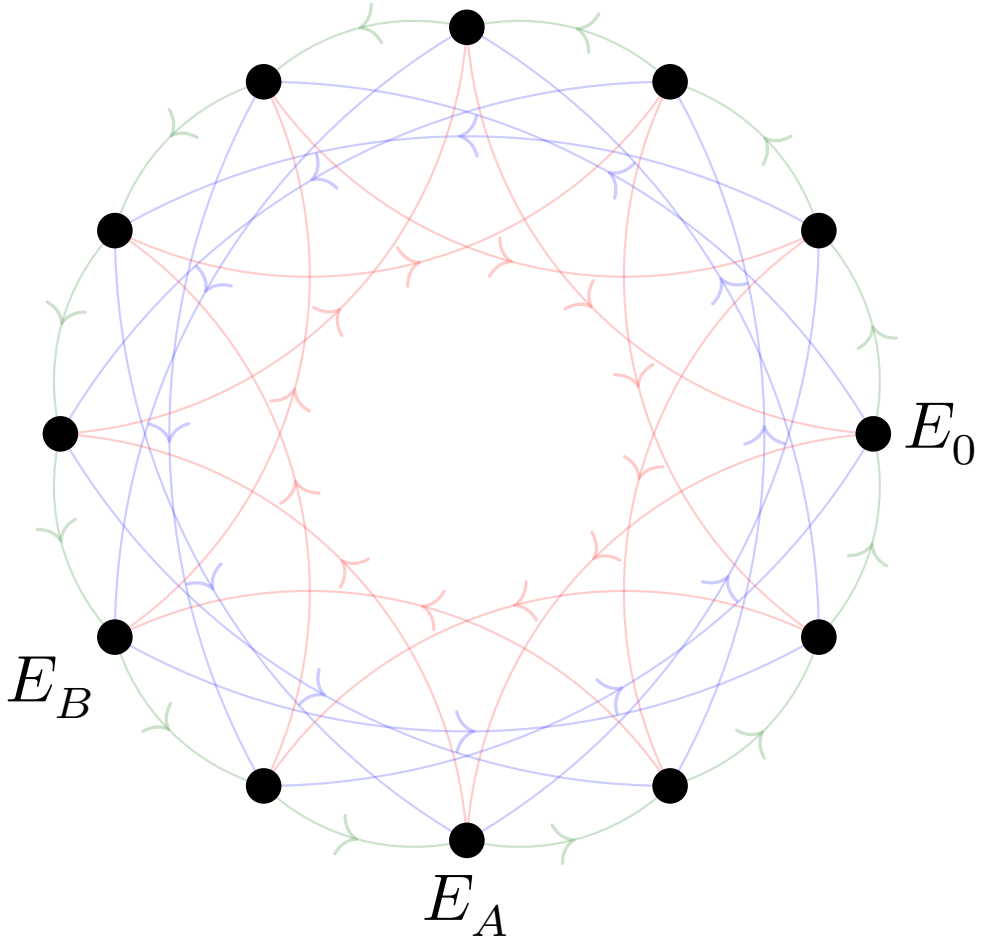
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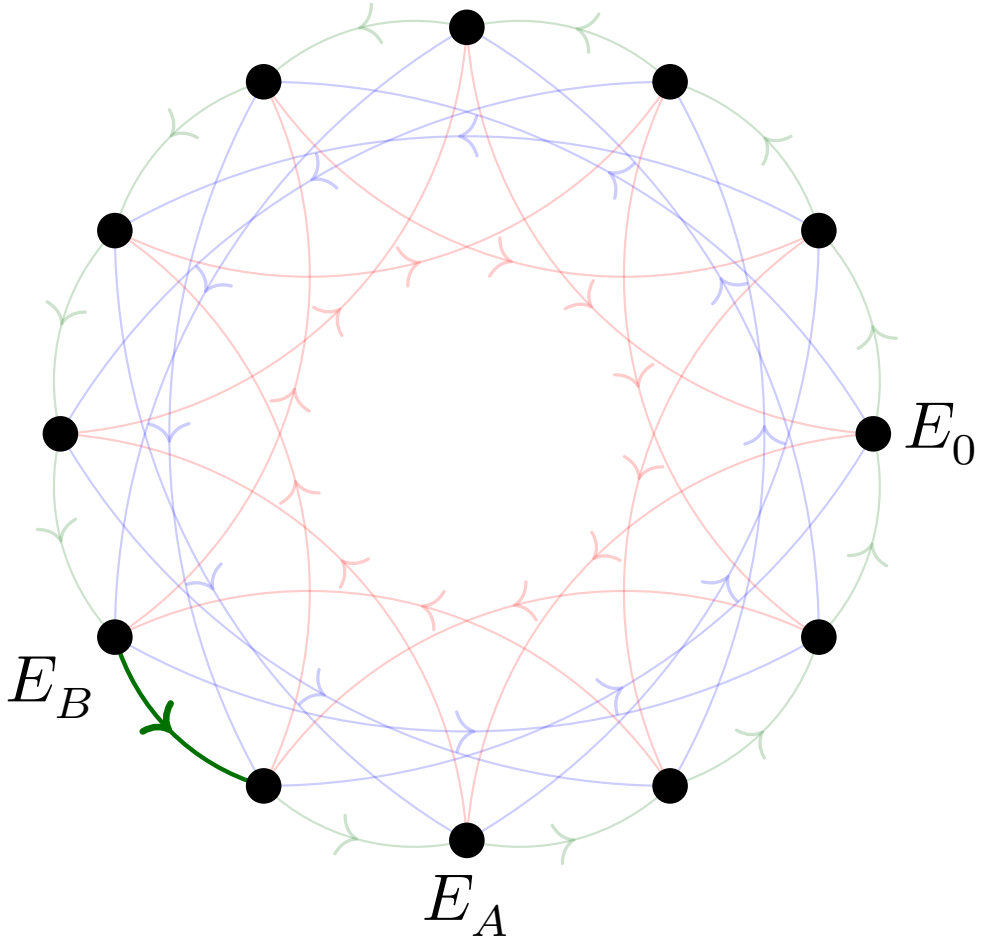
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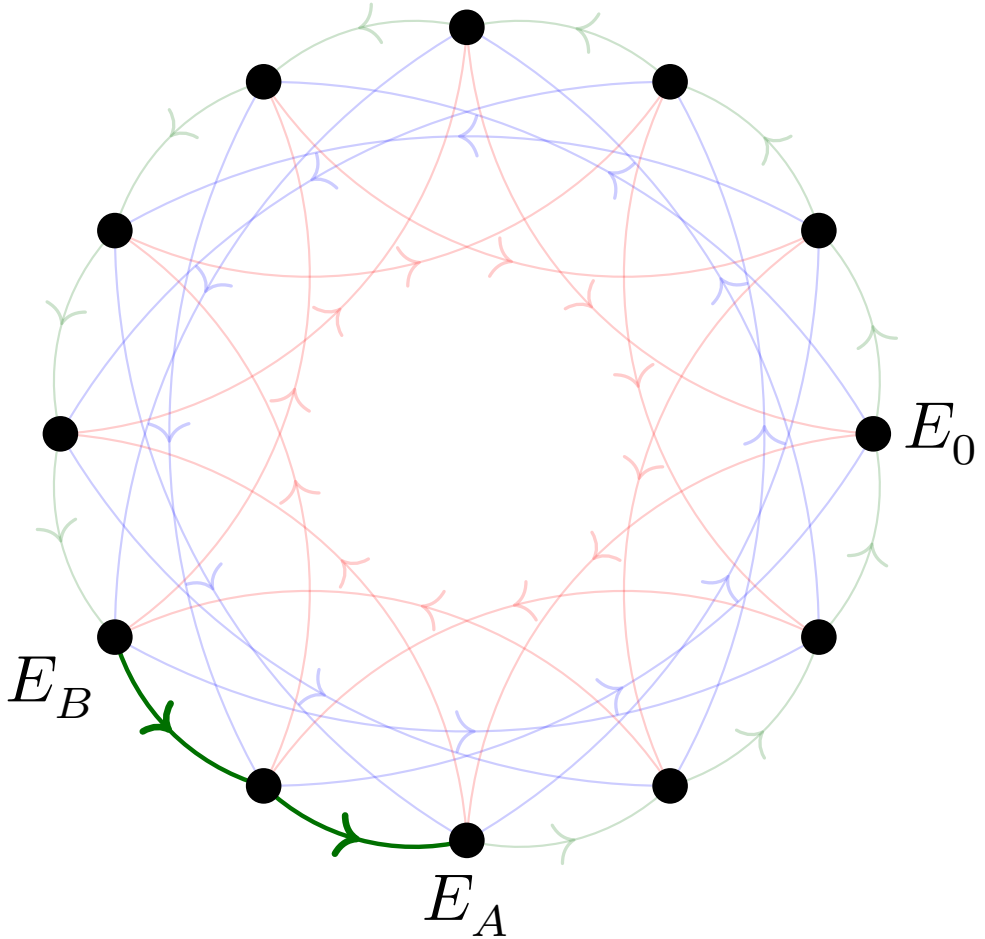
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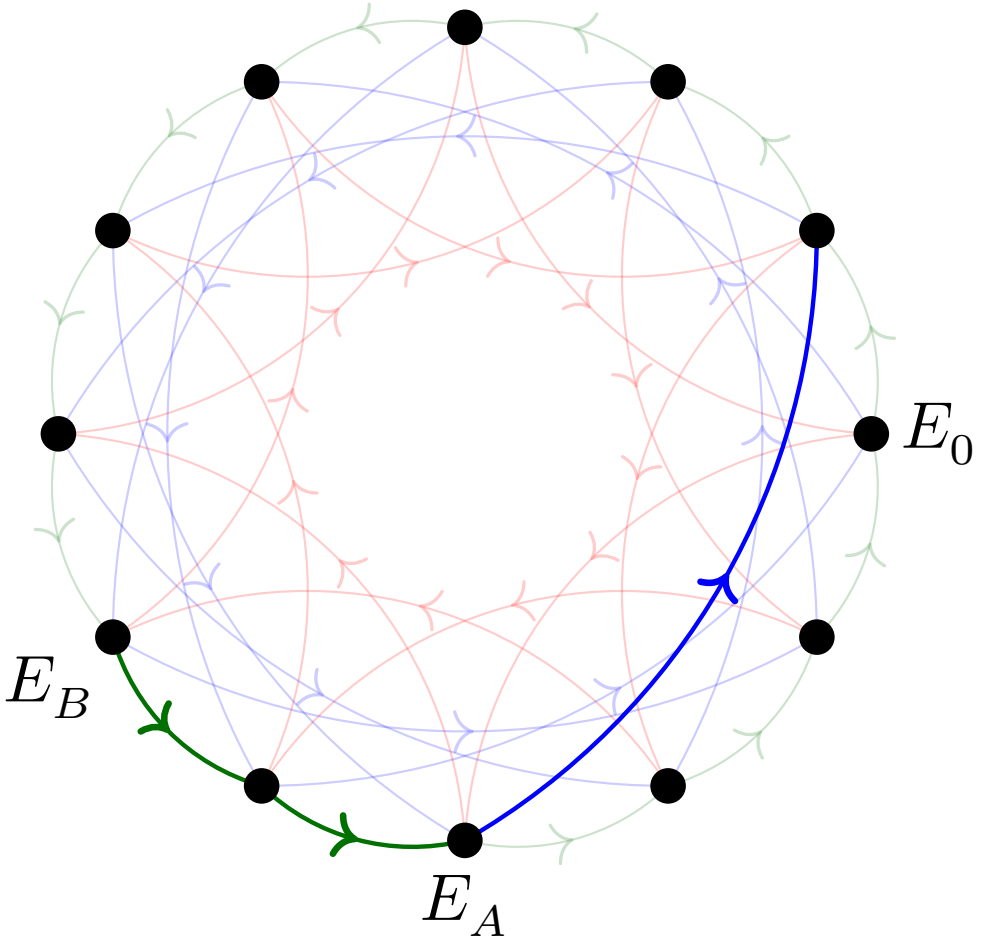
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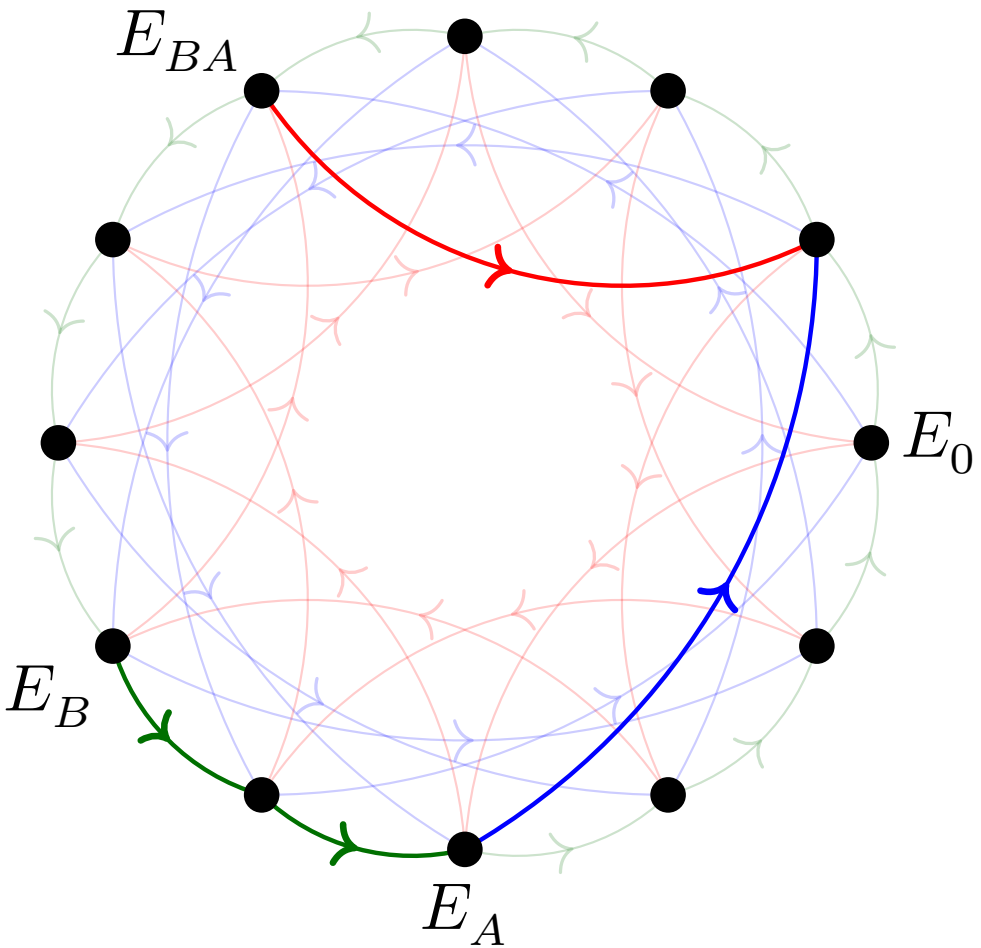
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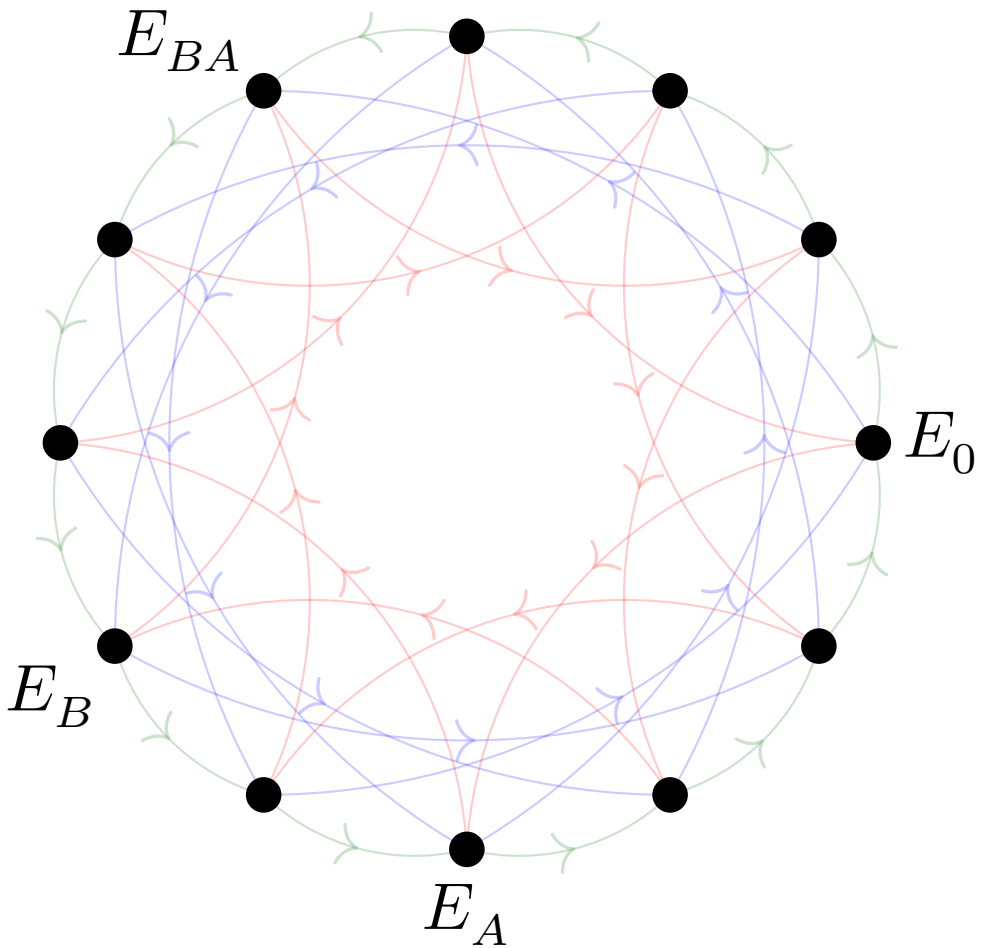
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Fix a large enough finite field $\mathbb{F}_q$ of large characteristic p and an ordinary elliptic curve $E_0/\mathbb{F}_q$ such that its Frobenius discriminant $D_\pi = t^2 - 4q$ contains a large enough prime factor.
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$\mathcal{L} = \{\mathfrak{l}_1, \mathfrak{l}_2, \mathfrak{l}_3\}$



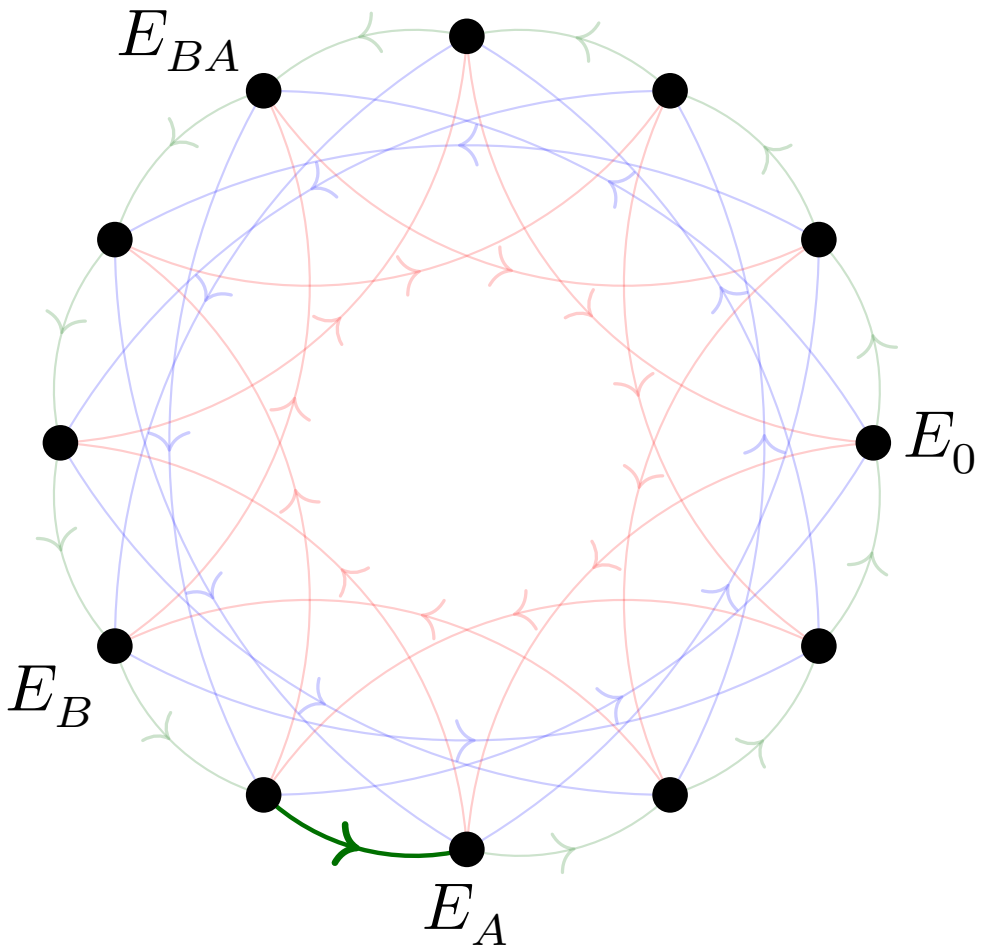
Alice
$\rho_A = (2, 1, -1)$
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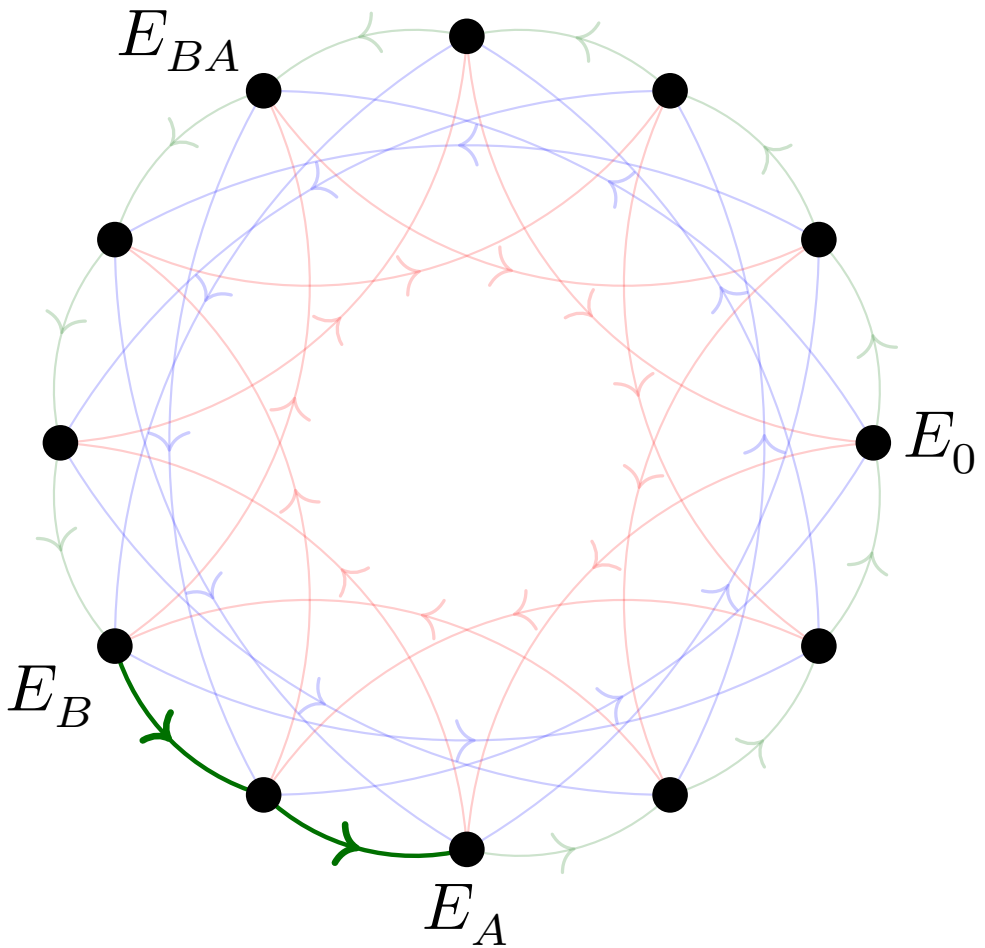
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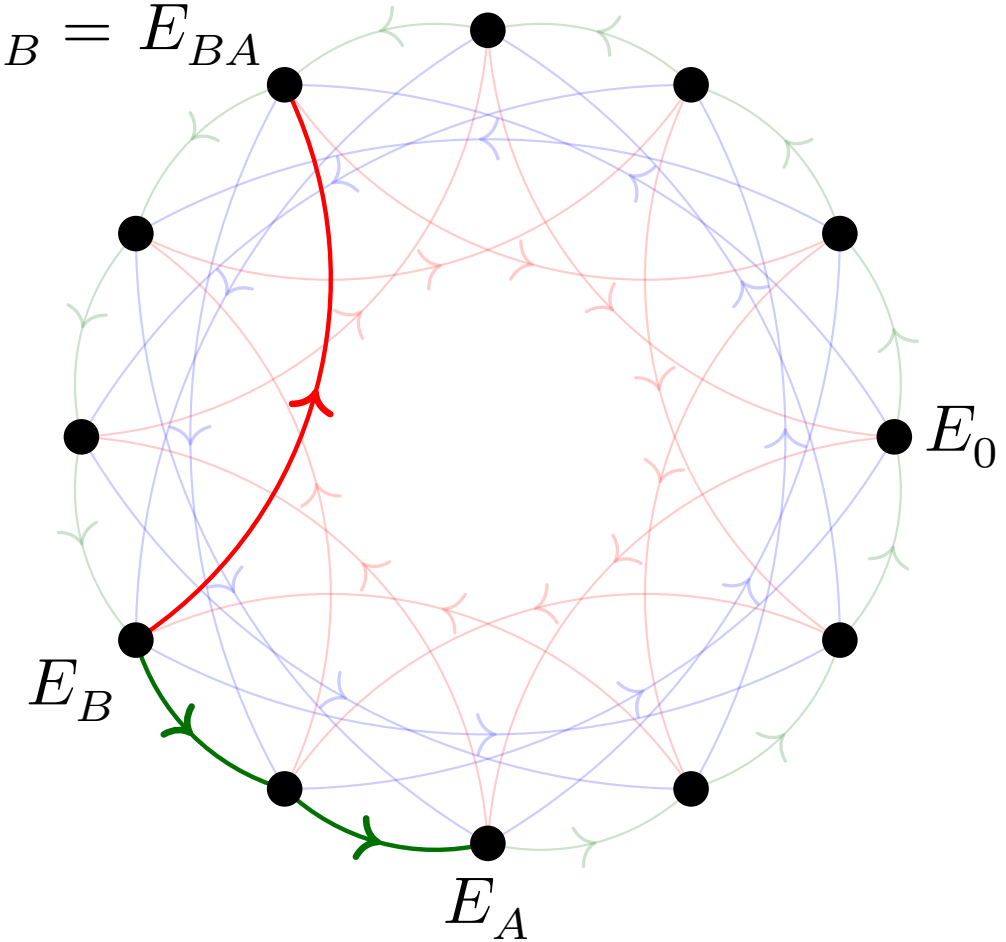
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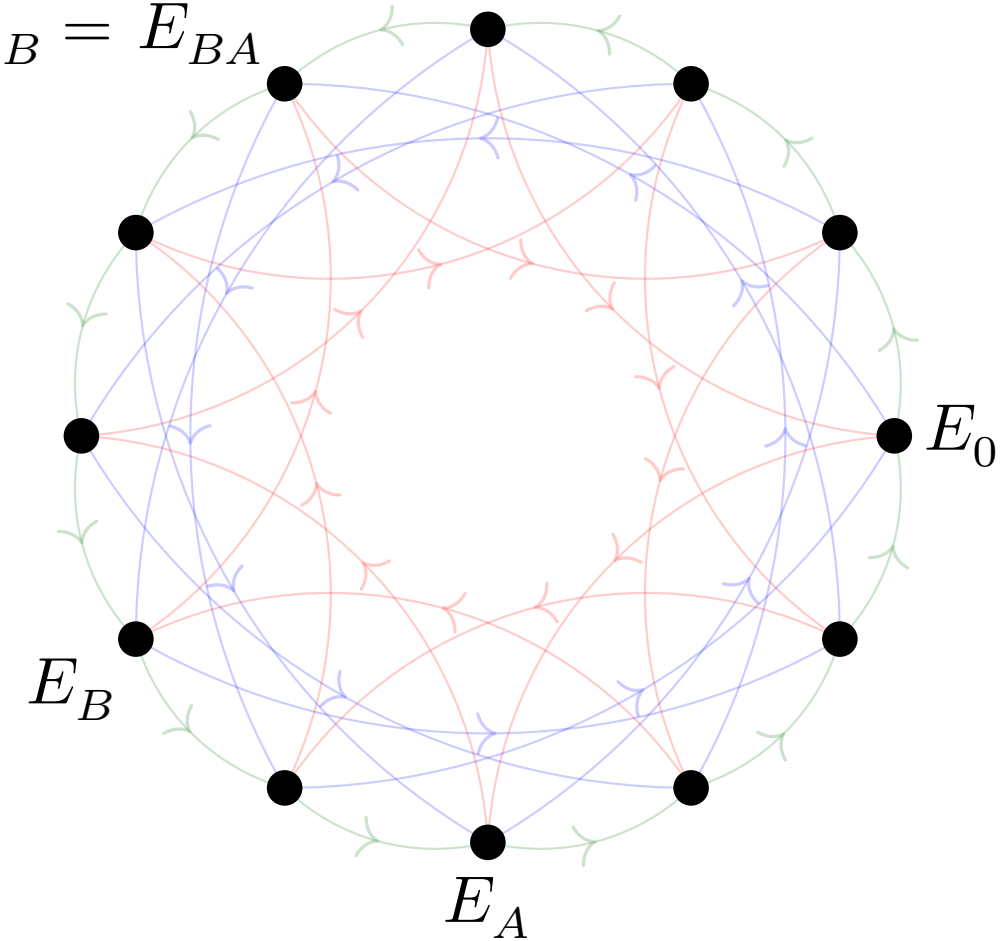
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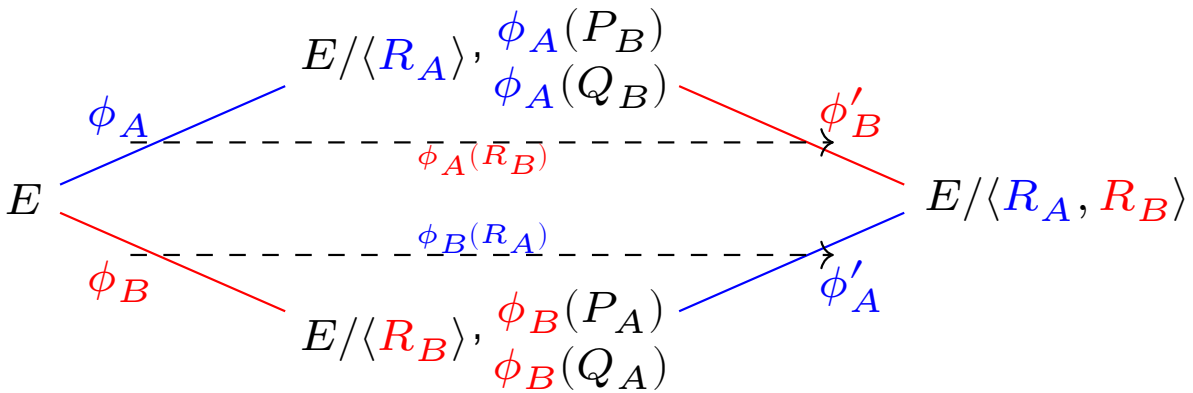
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Supersingular isogeny Diffie-Hellman

- Fix two small primes ℓ_A and ℓ_B;
- Choose a prime p such that $p+1 = \ell_A^a \ell_B^b f$ for a small correction term f;
- Pick a supersingular elliptic curve $E/\mathbb{F}_{p^2}$: $E\left(\mathbb{F}_{p^2}\right) \simeq \left(\frac{\mathbb{Z}}{(p+1)\mathbb{Z}}\right)^2$
- Alice consider $E\left[\ell_A^a\right] = \langle P_A, Q_A \rangle$ while Bob takes $E\left[\ell_B^b\right] = \langle P_B, Q_B \rangle$.
- **Secret Data:** $R_A = m_A P_A + n_A Q_A$ and $R_B = m_B P_B + n_B Q_B$.
- **Private Key:** isogenies $\phi_A : E \to E_A = E/\langle R_A \rangle$ and $\phi_B : E \to E_B = E/\langle R_B \rangle$.
- **Shared Data:** E_A, $\phi_A(P_B)$, $\phi_A(Q_B)$ and E_B, $\phi_B(P_A)$, $\phi_B(Q_A)$.
- **Shared Key:** $E/\langle R_A, R_B \rangle = E_B/\langle \phi_B(R_A) \rangle = E_A/\langle \phi_A(R_B) \rangle$.



CSIDH - CASTRYCK, LANGE, MARTINDALE, PANNY & RENES, 2018

It is an adaptation of the Couveignes–Rostovtsev–Stolbunov scheme to supersingular elliptic curves.

Commutative Supersingular isogeny Diffie-Hellman

- Fix a prime $p = 4 \cdot \ell_1 \cdot \ldots \cdot \ell_t - 1$ for small distinct odd primes ℓ_i.
- The elliptic curve $E_0 : y^2 = x^3 + x / \mathbb{F}_p$ is supersingular and its endomorphism ring over $\mathbb{F}_p$ is $\mathcal{O} = \mathbb{Z}[\pi]$ (commutative).
- A supersingular Montgomery curve $E_A : y^2 = x^3 + Ax^2 + x / \mathbb{F}_p$ appear in the $\mathcal{Cl}(\mathcal{O})$-orbit of E_0.
- **Private Key:** it is an n-tuple of integers $(e_1, \ldots, e_t)$ sampled in a range $\{-m, \ldots, m\}$ representing an ideal class $[\mathfrak{a}] = [\mathfrak{l}_1^{e_1} \cdot \ldots \cdot \mathfrak{l}_t^{e_t}] \in \mathcal{Cl}(\mathcal{O})$ where $\mathfrak{l}_i = (\ell_i, \pi - 1)$.
- **Public Key:** The Montgomery coefficients A of the elliptic curve $E_A = [\mathfrak{a}] \cdot E_0 : y^2 = x^3 + Ax^2 + x$.
- **Shared Key:** If Alice and Bob have private key $(\mathfrak{a}, A)$ and $(\mathfrak{b}, B)$ then they can compute the shared key $E_{AB} = [\mathfrak{a}][\mathfrak{b}] \cdot E_0 = [\mathfrak{b}][\mathfrak{a}] \cdot E_0$.

ORIENTATIONS AND CLASS GROUP ACTIONS

ORIENTATIONS

Let $\mathcal{O}$ be an order in an imaginary quadratic field K.
An *$\mathcal{O}$-orientation* on a supersingular elliptic curve E is an embedding

$$\iota : \mathcal{O} \hookrightarrow \mathsf{End}(E).$$

A *K-orientation* is an embedding

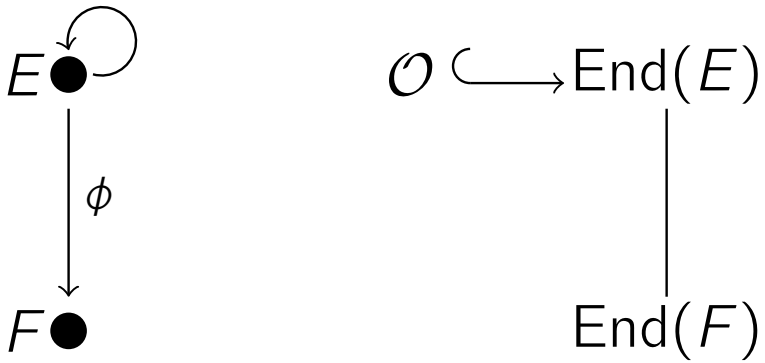
$$\iota : K \hookrightarrow \mathsf{End}^0(E) = \mathsf{End}(E) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

An $\mathcal{O}$-orientation is *primitive* if

$$\mathcal{O} \simeq \mathsf{End}(E) \cap \iota(K).$$

Theorem

The category of K-oriented supersingular elliptic curves (E, ι), whose morphisms are isogenies commuting with the K-orientations, is equivalent to the category of elliptic curves with CM by K.



Let $\phi : E \to F$ be an isogeny of degree ℓ. A K-orientation $\iota : K \hookrightarrow \mathbf{End}^0(E)$ determines a K-orientation $\phi_*(\iota) : K \hookrightarrow \mathbf{End}^0(F)$ on F, defined by

$$\phi_*(\iota)(\alpha) = \frac{1}{\ell}\, \phi \circ \iota(\alpha) \circ \hat{\phi}.$$

Conversely, given K-oriented elliptic curves (E, ι_E) and (F, ι_F) we say that an isogeny $\phi : E \to F$ is K-oriented if $\phi_*(\iota_E) = \iota_F$, i.e., if the orientation on F is induced by ϕ.

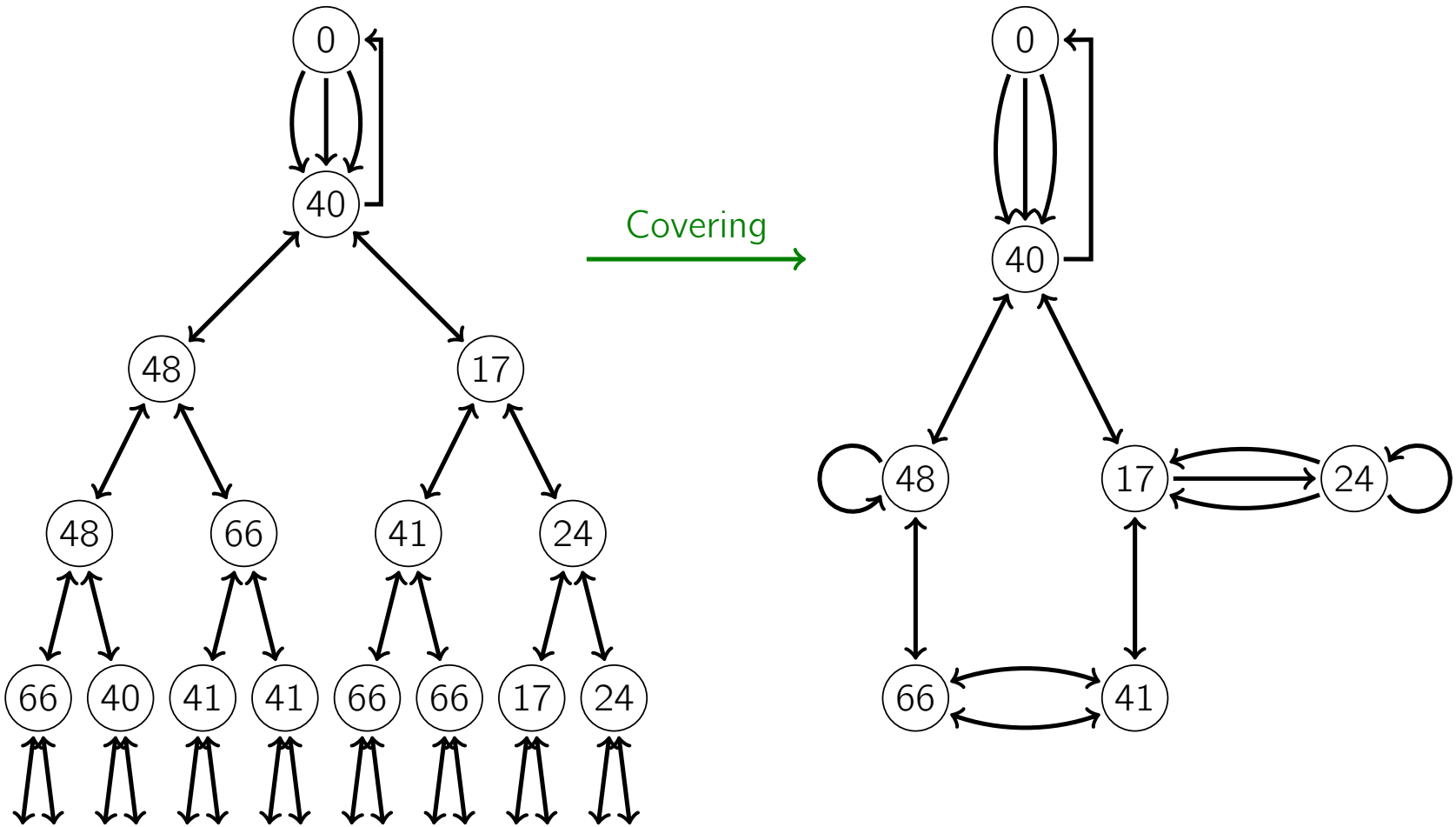
ORIENTED ISOGENY GRAPHS - VERTICES & EDGES

Two K-oriented curves are isomorphic if and only if there exists a K-oriented isomorphism between them.
We denote $G_S(E, K)$ the S-isogeny graph of K-oriented supersingular elliptic curves whose

- vertices are isomorphism classes of K-oriented supersingular elliptic curves
- edges are equivalence classes of K-oriented isogenies of degree in S.

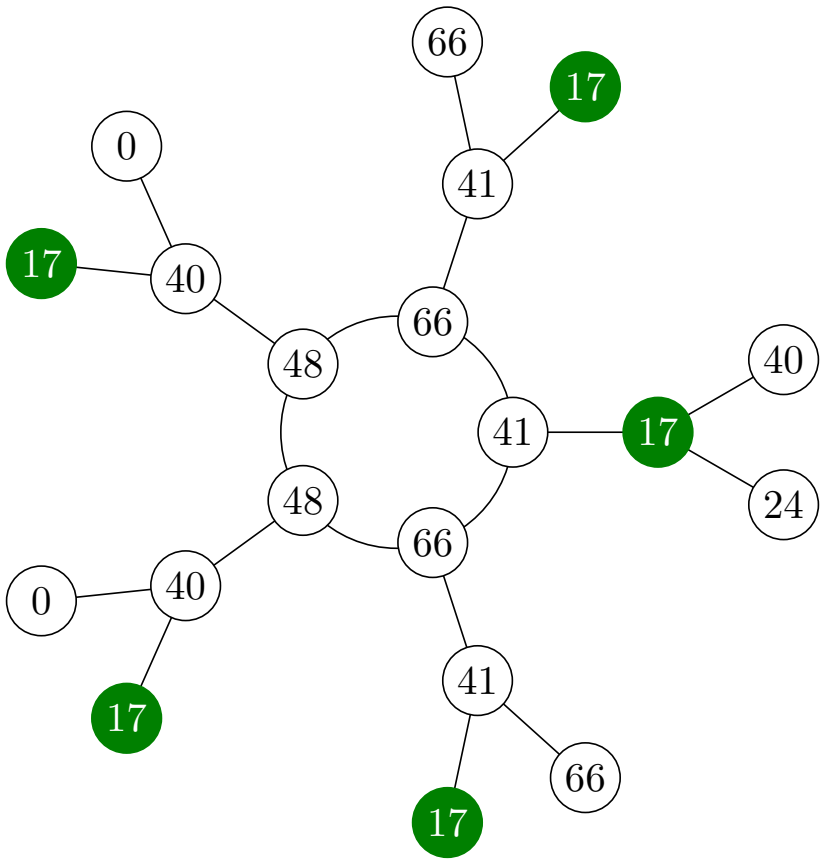
ORIENTED ISOGENY GRAPHS - AN EXAMPLE

Let $p = 71$ and $E_0/\mathbb{F}_{71}$ be the supersingular elliptic curve with $j(E) = 0$ oriented by the $\mathcal{O}_K = \mathbb{Z}[\omega]$, where $\omega^2 + \omega + 1 = 0$.
The orientation by $K = \mathbb{Q}[\omega]$ differentiates vertices in the descending paths from E_0, determining an infinite graph shown here to depth 4:



ORIENTED ISOGENY GRAPHS - YET ANOTHER EXAMPLE

We let again $p = 71$ and we consider the isogeny graph oriented by $\mathbb{Z}[\omega_{79}]$ where ω_{79} generates the ring of integers of $\mathbb{Q}(\sqrt{-79})$.



ISOGENY CHAINS

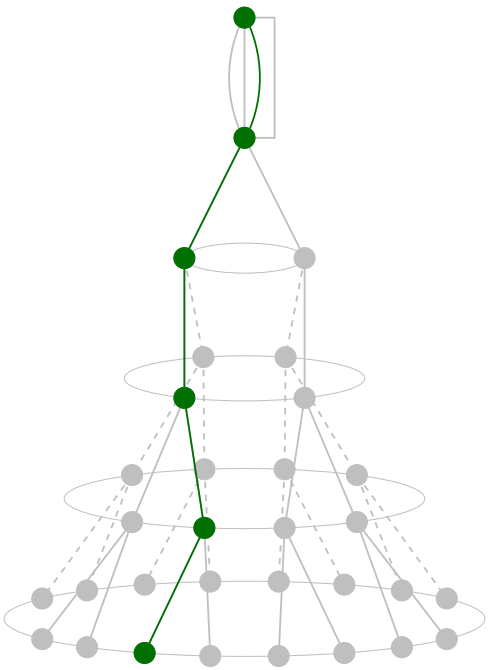
L.COLÒ 2M

Definition

An ℓ-isogeny chain of length n from E_0 to E is a sequence of isogenies of degree ℓ:

$$E_0 \xrightarrow{\phi_0} E_1 \xrightarrow{\phi_1} E_2 \xrightarrow{\phi_2} \ldots \xrightarrow{\phi_{n-1}} E_n = E.$$

The ℓ-isogeny chain is without backtracking if $\texttt{ker}\ (\phi_{i+1} \circ \phi_i) \neq E_i[\ell],\ \forall i$.
The isogeny chain is descending (or ascending, or horizontal) if each ϕ_i is descending (or ascending, or horizontal, respectively).



PRIMITIVE ORIENTATIONS

L.COLÒ 2M

- $\mathsf{SS}(p) = \{\text{supersingular elliptic curves over } \overline{\mathbb{F}}_p \text{ up to isomorphism}\}$.
- $\mathsf{SS}_{\mathcal{O}}(p) = \{\mathcal{O}\text{-oriented s.s. elliptic curves over } \overline{\mathbb{F}}_p \text{ up to } K\text{-isomorphism}\}$.
- $\mathsf{SS}_{\mathcal{O}}^{pr}(p) =$ subset of primitive $\mathcal{O}$-oriented curves.

An element of $\mathsf{SS}_{\mathcal{O}}^{pr}(p)$ consists of

- A supersingular elliptic curve $E/\overline{\mathbb{F}}_p$;
- a primitive orientation $\iota : \mathcal{O} \hookrightarrow \mathsf{End}(E)$;
- a structure of a p-orientation which is a homomorphism $\rho : \mathcal{O} \to \overline{\mathbb{F}}_p$.

Note that $\mathsf{End}(E)$ is equipped with a p-orientation $\rho : \mathsf{End}(E) \hookrightarrow \overline{\mathbb{F}}_p$.

We denote by $\mathsf{SS}_{\mathcal{O}}(\rho)$ the set of oriented supersingular elliptic curves with ρ induced by ι and $\mathsf{End}(E)/\mathfrak{P} \hookrightarrow \overline{\mathbb{F}}_p$, and $\mathsf{SS}_{\mathcal{O}}(\overline{\rho})$ the opposite p-orientation class.

THE SET OF PRIMITIVE p-ORIENTED CURVES

Since $K \hookrightarrow \mathfrak{A}$, then

- If p ramifies in K then $\rho = \overline{\rho}$ and then

$$\mathsf{SS}^{pr}_{\mathcal{O}}(p) = \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) = \mathsf{SS}^{pr}_{\mathcal{O}}(\overline{\rho}).$$

- If p is inert in K then $\mathsf{SS}_{\mathcal{O}}(p)$ decomposes in

$$\begin{aligned}\mathsf{SS}^{pr}_{\mathcal{O}}(p) &= \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \cup \mathsf{SS}^{pr}_{\mathcal{O}}(\overline{\rho}) \\ &= \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \cup \mathsf{SS}^{pr}_{\mathcal{O}}(\rho)^{\sigma}.\end{aligned}$$

CLASS GROUP ACTION

The set $\mathsf{SS}_{\mathcal{O}}(p)$ admits a transitive group action:

$$\begin{aligned} \mathcal{Cl}(\mathcal{O}) \times \mathsf{SS}_{\mathcal{O}}(p) &\longrightarrow \mathsf{SS}_{\mathcal{O}}(p) \\ ([\mathfrak{a}], E) &\longmapsto [\mathfrak{a}] \cdot E = E/E[\mathfrak{a}] \end{aligned}$$

Proposition

The set $\mathsf{SS}^{pr}_{\mathcal{O}}(\rho)$ is a torsor for the class group $\mathcal{Cl}(\mathcal{O})$.

In particular, for fixed primitive p-oriented supersingular elliptic curve E, we obtain a bijection of sets:

$$\begin{aligned} \mathcal{Cl}(\mathcal{O}) &\longrightarrow \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \\ [\mathfrak{a}] &\longmapsto [\mathfrak{a}] \cdot E \end{aligned}$$

EFFECTIVE CLASS GROUP ACTION

Suppose that (E_i, ϕ_i) is a descending ℓ-isogeny chain with

$$\mathcal{O}_K \subseteq \mathsf{End}(E_0), \dots, \mathcal{O} = \mathbb{Z} + \ell^n \mathcal{O}_K \subseteq \mathsf{End}(E_n).$$

If $\mathfrak{q}$ is a split prime in $\mathcal{O}_K$ over $q \neq \ell, p$, then the isogeny

$$\psi_0 : E_0 \to F_0 = E_0/E_0[\mathfrak{q}]$$

can be extended to the ℓ-isogeny chain by pushing forward the cyclic group $C_0 = E_0[\mathfrak{q}]$:

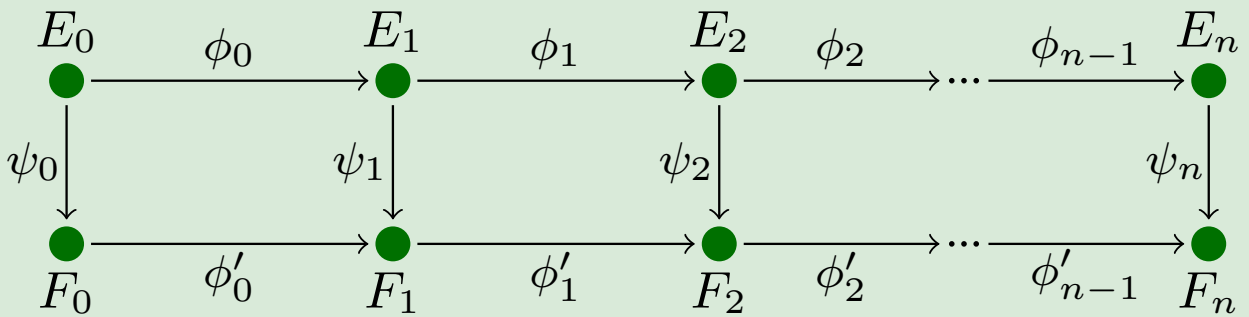
$$C_0 = E_0[\mathfrak{q}], C_1 = \phi_0(C_0), \dots, C_n = \phi_{n-1}(C_{n-1}),$$

and defining $F_i = E_i/C_i$.

This construction motivates the following definition.

Definition

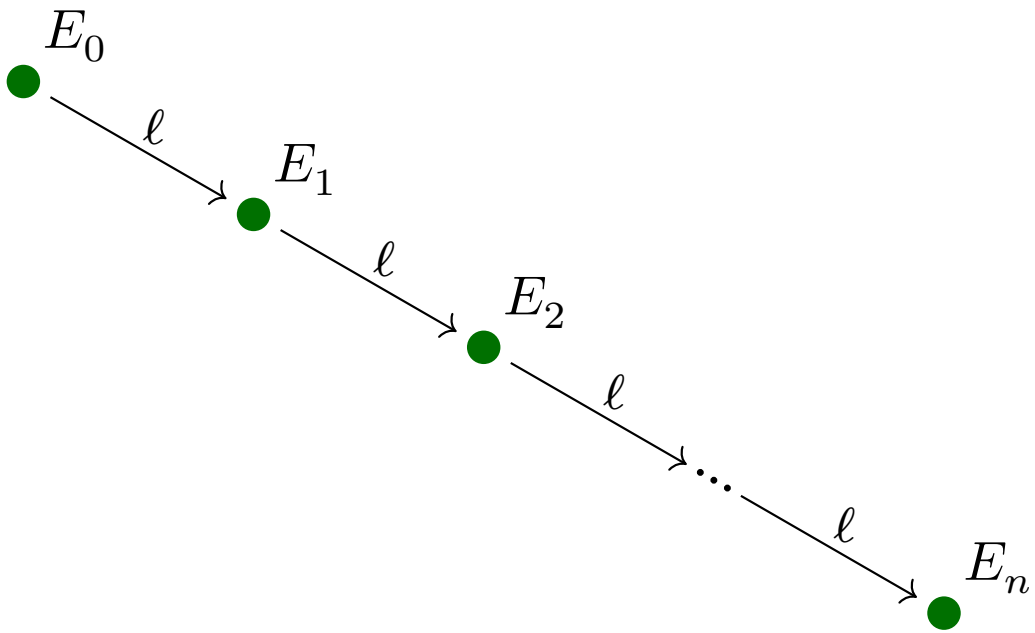
An ℓ-ladder of length n and degree q is a commutative diagram of ℓ-isogeny chains (E_i, ϕ_i), (F_i, ϕ'_i) of length n connected by q-isogenies $\psi_i : E_i \to F_i$.



If ψ_0 is as above ($(\psi_0) = E_0[\mathfrak{q}]$), the ladder encodes the action of $\mathcal{Cl}(\mathcal{O})$ on ℓ-isogeny chains, and consequently on elliptic curves at depth n.

LADDERS - AN EXAMPLE

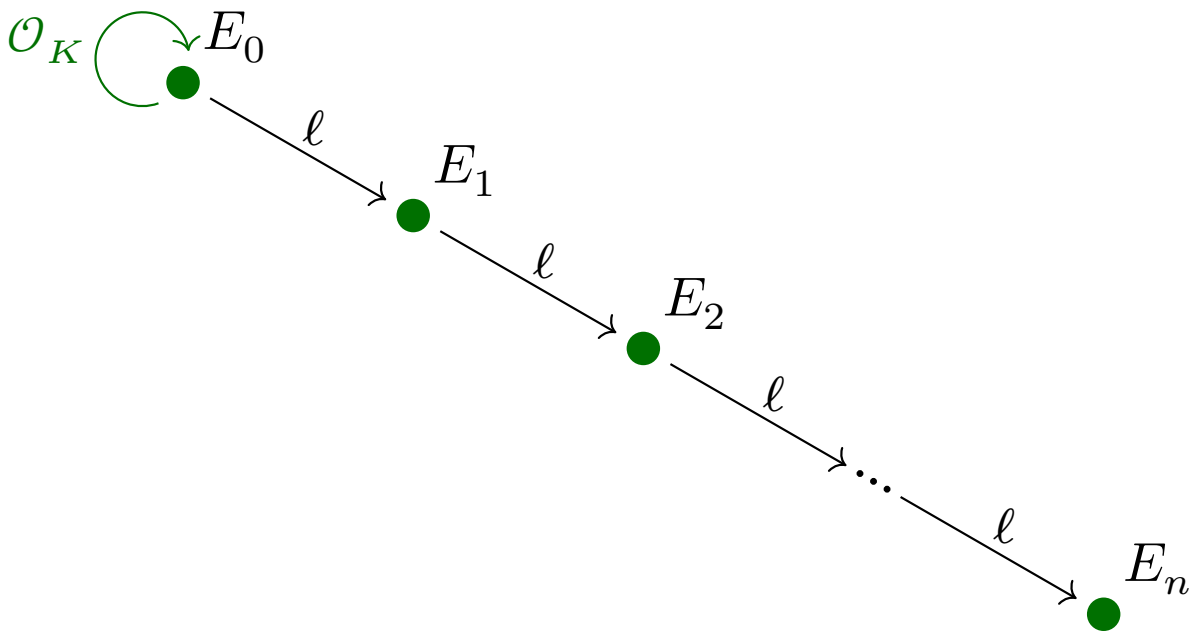
We consider an elliptic curve E_0 with an effective endomorphism ring (eg. $j_0 = 0, 1728$) and a chain of ℓ-isogenies.



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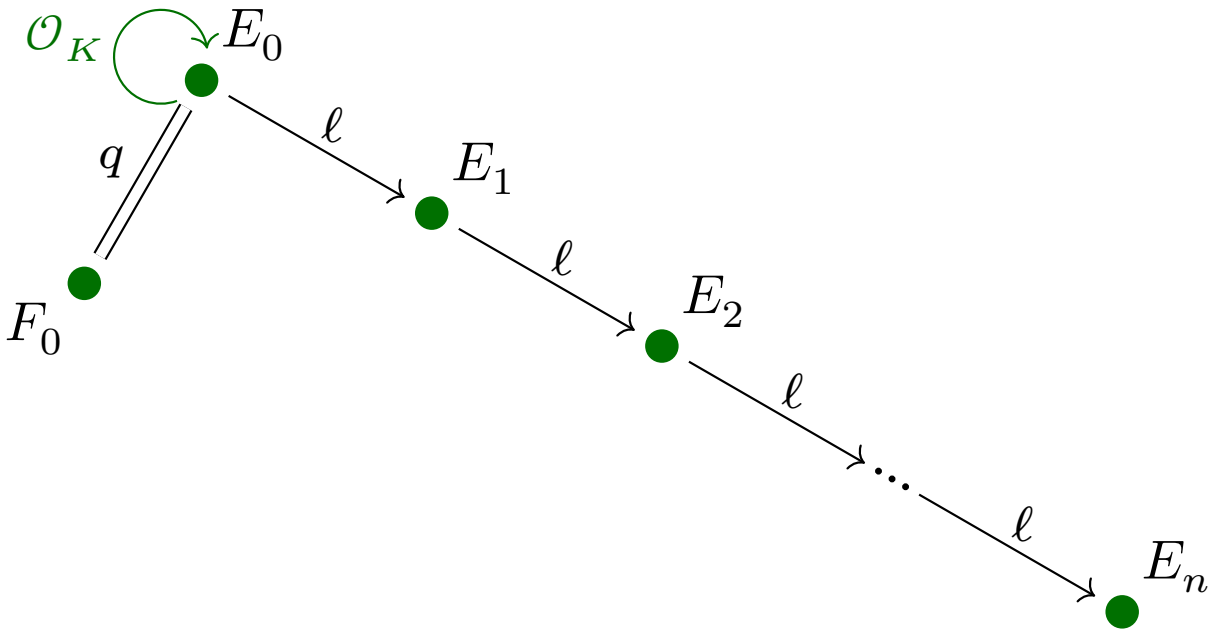
- For $\ell = 2$ (or 3) a suitable candidate for $\mathcal{O}_K$ could be the Gaussian integers $\mathbb{Z}[i]$ or the Eisenstein integers $\mathbb{Z}[\omega]$.



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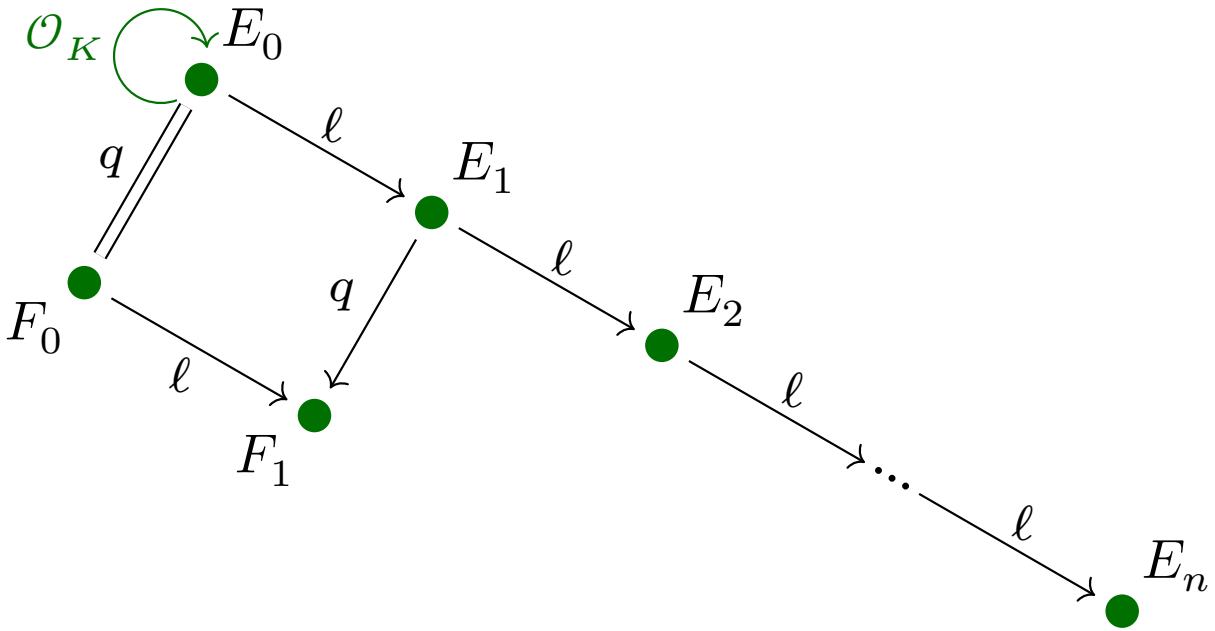
- Horizontal isogenies must be endomorphisms



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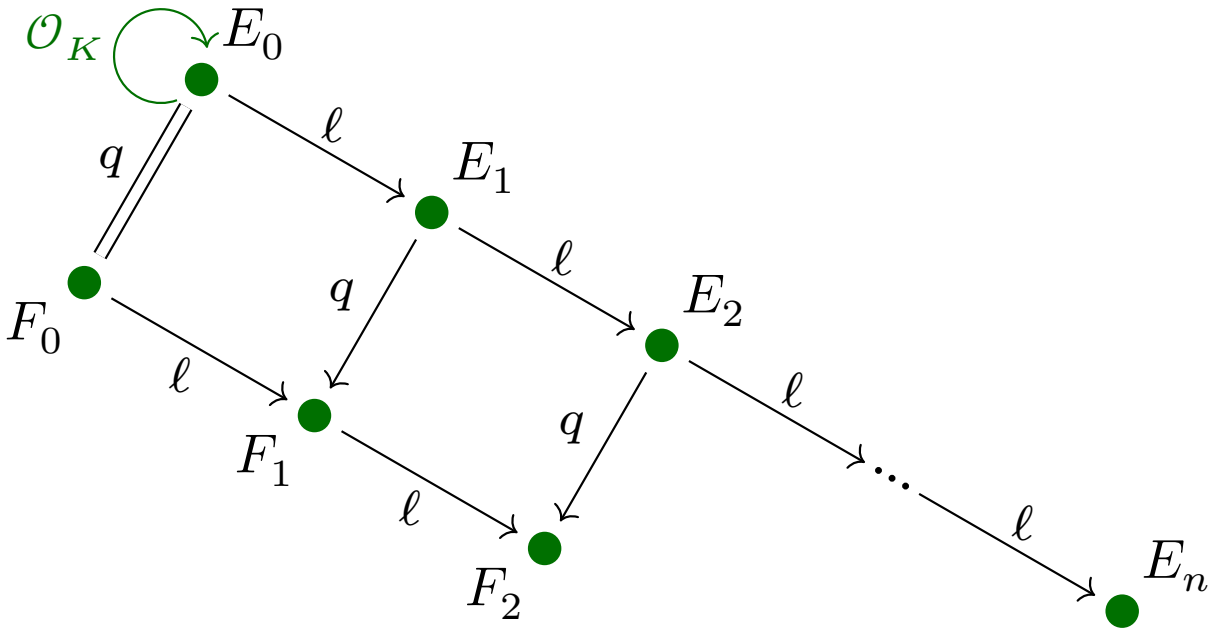
- We push forward our q-orientation obtaining F_1.



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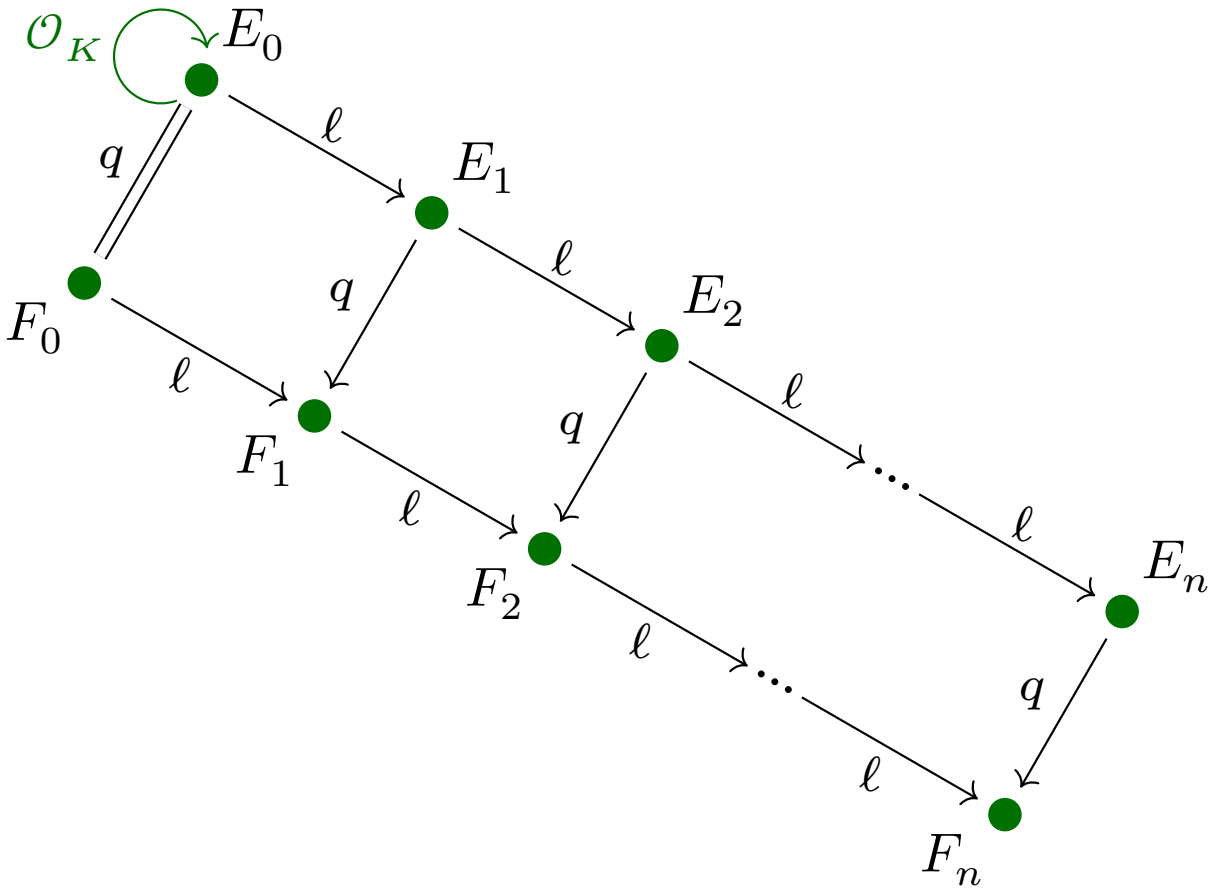
- We repeat the process for F_2.



LADDERS - AN EXAMPLE

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- And again till F_n.

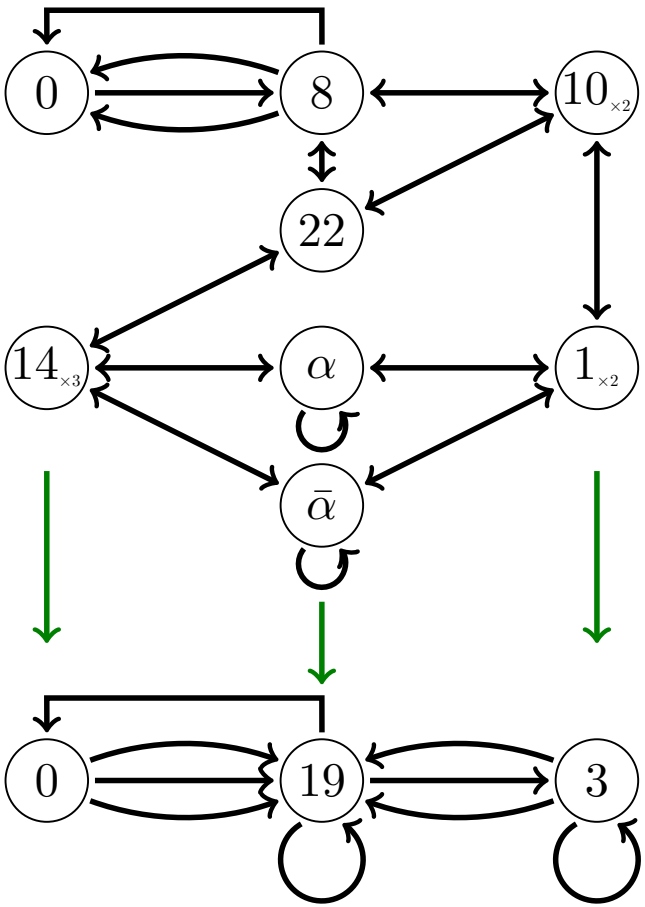


ISOGENY GRAPHS WITH LEVEL STRUCTURE

Let Γ be a congruence subgroup of level coprime to the characteristic.

We have a covering $G_S(E, \Gamma) \to G_S(E)$

- whose vertices are pairs $(E, \Gamma(P, Q))$ of curves $/\mathbb{F}_{p^2}$ and a Γ-level structure
- and edges are isogenies $\psi : (E, \Gamma(P, Q)) \to (E', \Gamma(P', Q'))$



Eg. On the left the $\Gamma_0(3)$ supersingular 2-isogeny graph.

$$14 \leftrightarrow \{(E_0, C_1), (E_0, C_2), (E_0, C_3)\}$$

The C_i's maps to each other under the automorphism of E_0; they define 3 isogenies to E_3.

ADDING LEVEL STRUCTURE - MOTIVATION

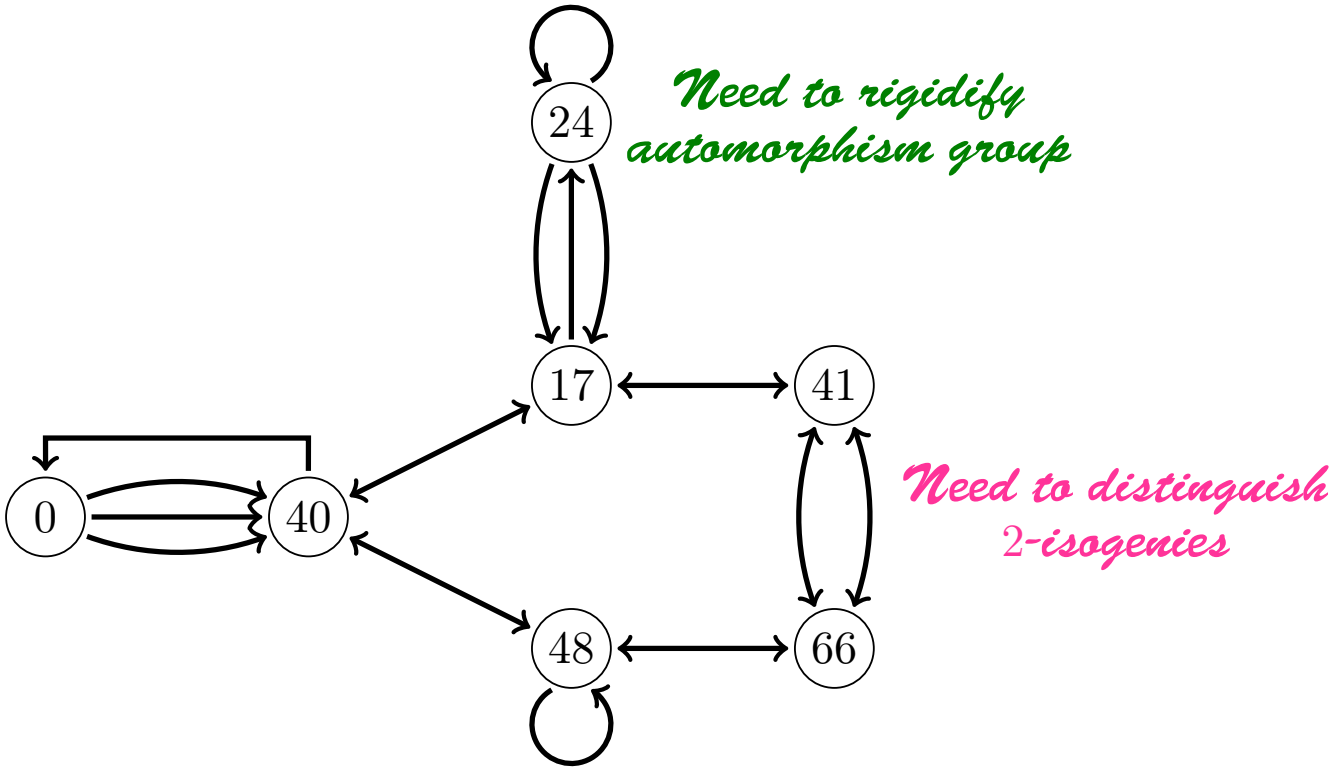
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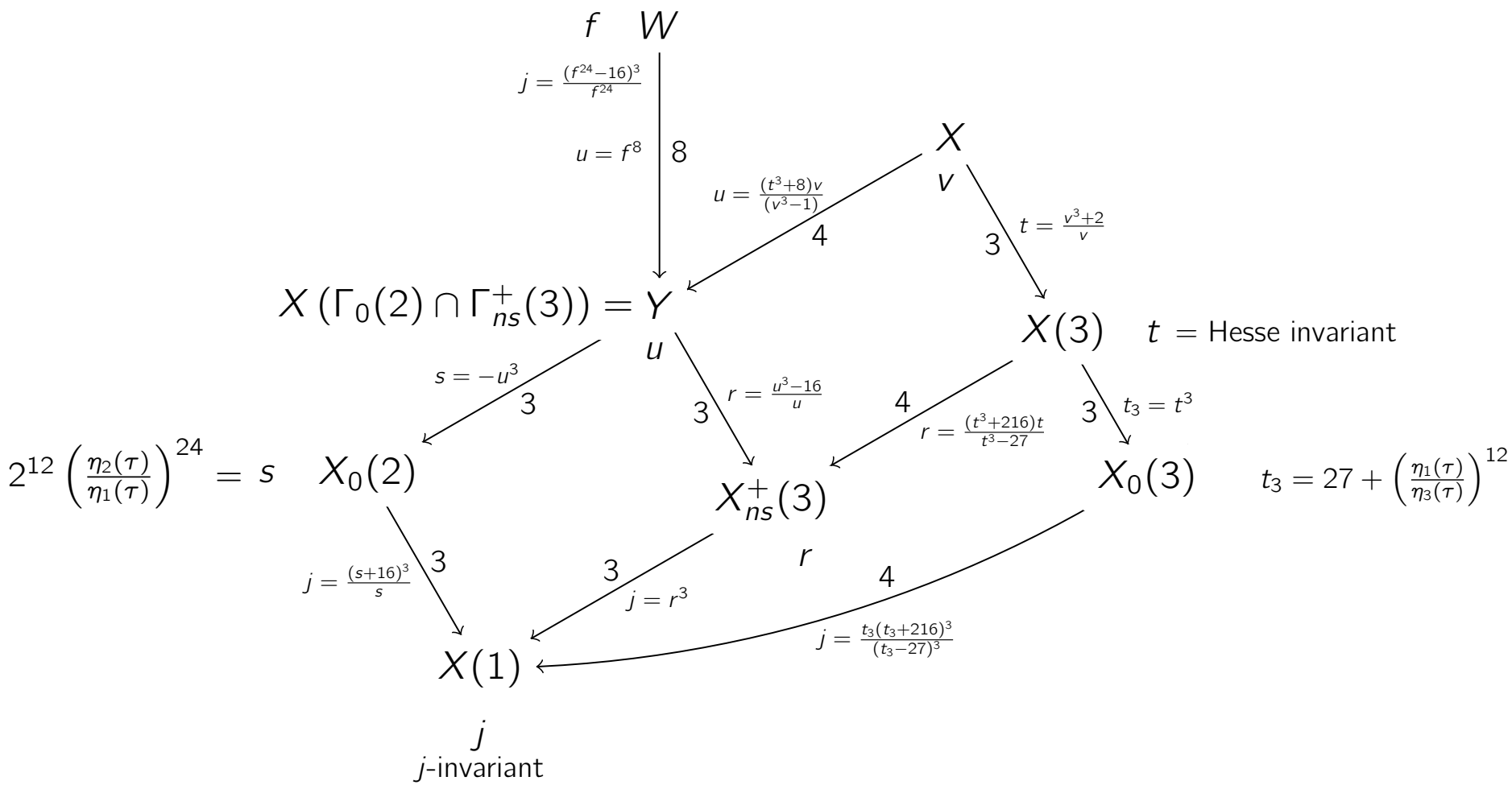
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- q-modular polynomial of higher level are smaller.

SOME MODULAR CURVES OF INTEREST



OSIDH

OSIDH PROTOCOL

PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$ and a set of splitting primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_t \subseteq \mathcal{O} \subseteq \mathsf{End}(E_n) \cap K \subseteq \mathcal{O}_K$

ALICE	**BOB**

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	ALICE	**BOB**
Choose integers in a bound $[-r, r]$	$(e_1, \ldots, e_t)$	$(d_1, \ldots, d_t)$
Construct an isogenous curve	$F_n = E_n/E_n\left[\mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}\right]$	$G_n = E_n/E_n\left[\mathfrak{p}_1^{d_1} \cdots \mathfrak{p}_t^{d_t}\right]$

OSIDH PROTOCOL

PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$ and a set of splitting primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_t \subseteq \mathcal{O} \subseteq \mathsf{End}(E_n) \cap K \subseteq \mathcal{O}_K$

	ALICE	**BOB**
Choose integers in a bound $[-r, r]$	$(e_1, \ldots, e_t)$	$(d_1, \ldots, d_t)$
Construct an isogenous curve	$F_n = E_n/E_n\left[\mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}\right]$	$G_n = E_n/E_n\left[\mathfrak{p}_1^{d_1} \cdots \mathfrak{p}_t^{d_t}\right]$
Precompute all directions $\forall i$	$F_{n,i}^{(-r)} \leftarrow F_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow F_{n,i}^{(1)} \leftarrow F_n$	$G_{n,i}^{(-r)} \leftarrow G_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow G_{n,i}^{(1)} \leftarrow G_n$

OSIDH PROTOCOL

PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$ and a set of splitting primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_t \subseteq \mathcal{O} \subseteq \mathsf{End}(E_n) \cap K \subseteq \mathcal{O}_K$

	ALICE	**BOB**
Choose integers in a bound $[-r, r]$	$(e_1, \ldots, e_t)$	$(d_1, \ldots, d_t)$
Construct an isogenous curve	$F_n = E_n/E_n\left[\mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}\right]$	$G_n = E_n/E_n\left[\mathfrak{p}_1^{d_1} \cdots \mathfrak{p}_t^{d_t}\right]$
Precompute all directions $\forall i$	$F_{n,i}^{(-r)} \leftarrow F_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow F_{n,i}^{(1)} \leftarrow F_n$	$G_{n,i}^{(-r)} \leftarrow G_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow G_{n,i}^{(1)} \leftarrow G_n$
... and their conjugates	$F_n \to F_{n,i}^{(1)} \to \ldots \to F_{n,i}^{(r-1)} \to F_{n,1}^{(r)}$	$G_n \to G_{n,i}^{(1)} \to \ldots \to G_{n,i}^{(r-1)} \to G_{n,1}^{(r)}$

OSIDH PROTOCOL

PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$ and a set of splitting primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_t \subseteq \mathcal{O} \subseteq \mathsf{End}(E_n) \cap K \subseteq \mathcal{O}_K$

Choose integers in a bound $[-r, r]$
Construct an isogenous curve
Precompute all directions $\forall i$
... and their conjugates
Exchange data

ALICE	**BOB**
$(e_1, \ldots, e_t)$	$(d_1, \ldots, d_t)$
$F_n = E_n/E_n\left[\mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}\right]$	$G_n = E_n/E_n\left[\mathfrak{p}_1^{d_1} \cdots \mathfrak{p}_t^{d_t}\right]$
$F_{n,i}^{(-r)} \leftarrow F_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow F_{n,i}^{(1)} \leftarrow F_n$	$G_{n,i}^{(-r)} \leftarrow G_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow G_{n,i}^{(1)} \leftarrow G_n$
$F_n \to F_{n,i}^{(1)} \to \ldots \to F_{n,i}^{(r-1)} \to F_{n,1}^{(r)}$	$G_n \to G_{n,i}^{(1)} \to \ldots \to G_{n,i}^{(r-1)} \to G_{n,1}^{(r)}$
G_n+directions	F_n+directions

OSIDH PROTOCOL

PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$ and a set of splitting primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_t \subseteq \mathcal{O} \subseteq \mathsf{End}(E_n) \cap K \subseteq \mathcal{O}_K$

	ALICE	**BOB**
Choose integers in a bound $[-r, r]$	$(e_1, \ldots, e_t)$	$(d_1, \ldots, d_t)$
Construct an isogenous curve	$F_n = E_n/E_n\left[\mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}\right]$	$G_n = E_n/E_n\left[\mathfrak{p}_1^{d_1} \cdots \mathfrak{p}_t^{d_t}\right]$
Precompute all directions $\forall i$	$F_{n,i}^{(-r)} \leftarrow F_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow F_{n,i}^{(1)} \leftarrow F_n$	$G_{n,i}^{(-r)} \leftarrow G_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow G_{n,i}^{(1)} \leftarrow G_n$
... and their conjugates	$F_n \to F_{n,i}^{(1)} \to \ldots \to F_{n,i}^{(r-1)} \to F_{n,1}^{(r)}$	$G_n \to G_{n,i}^{(1)} \to \ldots \to G_{n,i}^{(r-1)} \to G_{n,1}^{(r)}$
Exchange data	G_n+directions	F_n+directions
Compute shared data	Takes e_i steps in $\mathfrak{p}_i$-isogeny chain & push forward information for $j > i$.	Takes d_i steps in $\mathfrak{p}_i$-isogeny chain & push forward information for $j > i$.

OSIDH PROTOCOL

PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$ and a set of splitting primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_t \subseteq \mathcal{O} \subseteq \mathsf{End}(E_n) \cap K \subseteq \mathcal{O}_K$

	ALICE	**BOB**
Choose integers in a bound $[-r, r]$	$(e_1, \ldots, e_t)$	$(d_1, \ldots, d_t)$
Construct an isogenous curve	$F_n = E_n/E_n\left[\mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}\right]$	$G_n = E_n/E_n\left[\mathfrak{p}_1^{d_1} \cdots \mathfrak{p}_t^{d_t}\right]$
Precompute all directions $\forall i$	$F_{n,i}^{(-r)} \leftarrow F_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow F_{n,i}^{(1)} \leftarrow F_n$	$G_{n,i}^{(-r)} \leftarrow G_{n,i}^{(-r+1)} \leftarrow \ldots \leftarrow G_{n,i}^{(1)} \leftarrow G_n$
... and their conjugates	$F_n \to F_{n,i}^{(1)} \to \ldots \to F_{n,i}^{(r-1)} \to F_{n,1}^{(r)}$	$G_n \to G_{n,i}^{(1)} \to \ldots \to G_{n,i}^{(r-1)} \to G_{n,1}^{(r)}$
Exchange data	G_n+directions	F_n+directions
Compute shared data	Takes e_i steps in $\mathfrak{p}_i$-isogeny chain & push forward information for $j > i$.	Takes d_i steps in $\mathfrak{p}_i$-isogeny chain & push forward information for $j > i$.

In the end, they share $H_n = E_n/E_n\left[\mathfrak{p}_1^{e_1+d_1} \cdot \ldots \cdot \mathfrak{p}_t^{e_t+d_t}\right]$

OSIDH PROTOCOL - AN EXAMPLE

$p = 10007$

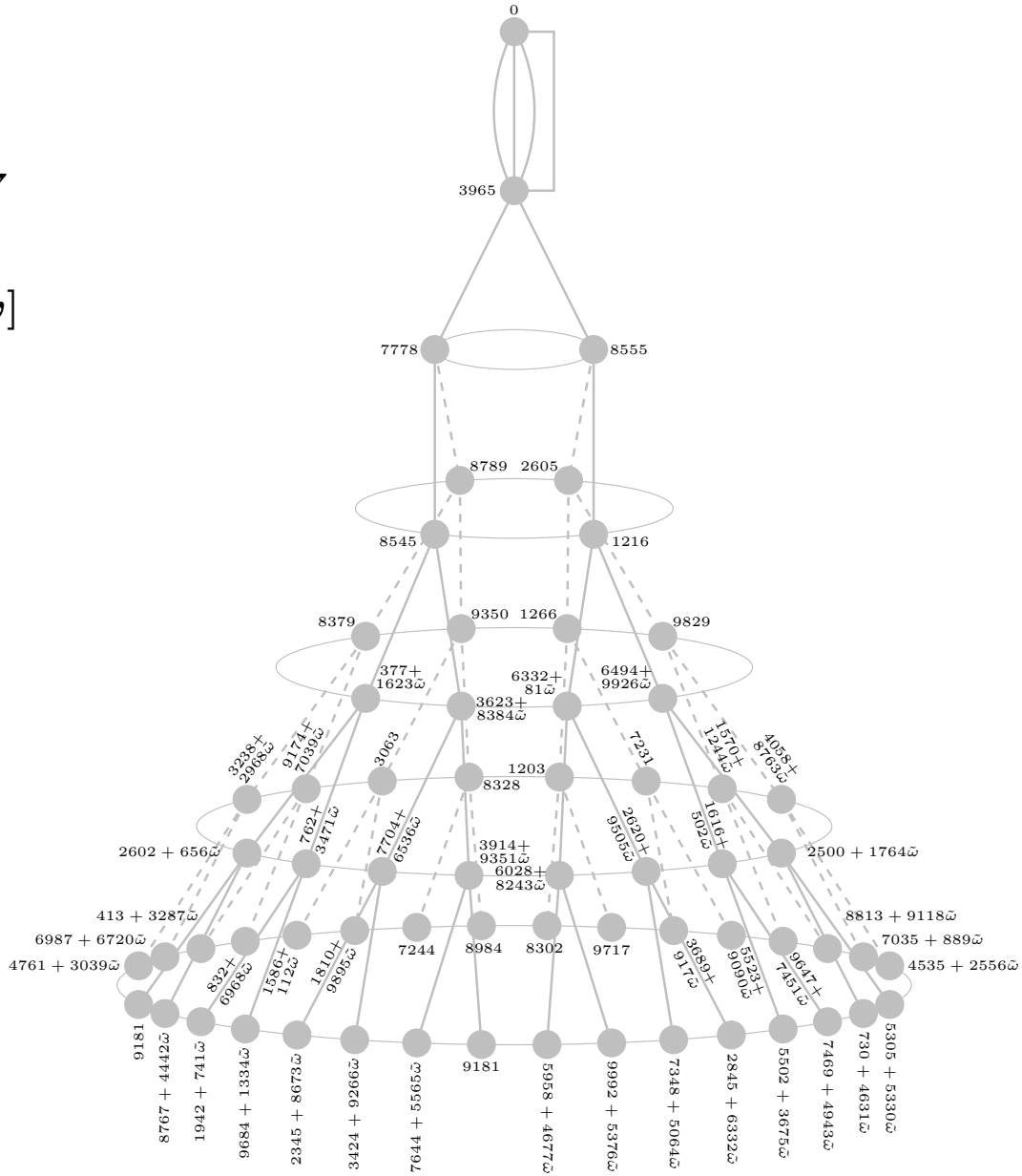
$\ell = 2$

$\mathcal{O}_K = \mathbb{Z}[\omega]$

$\ell_1 = 13$

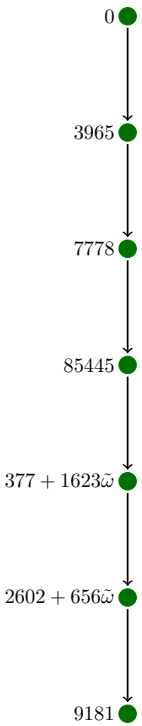
$\ell_2 = 31$

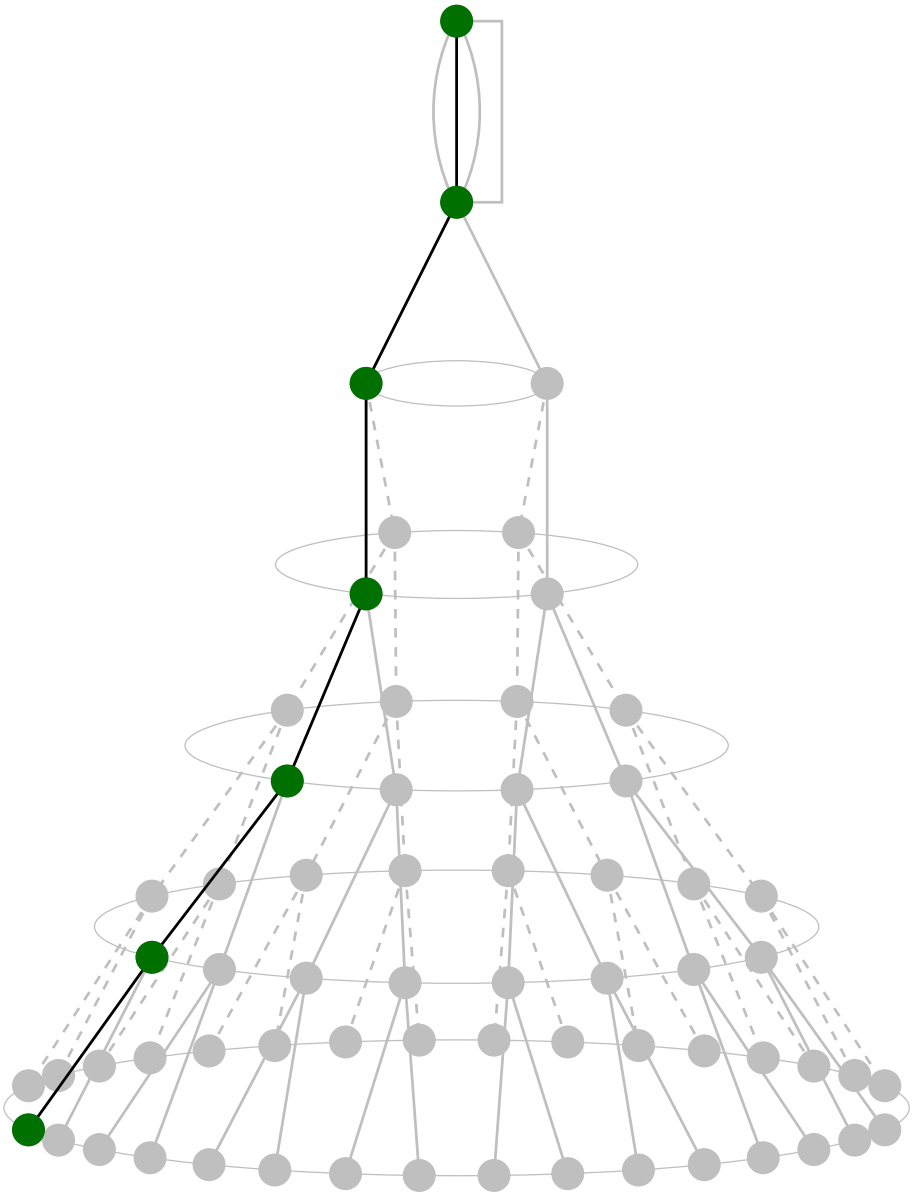
$\ell_3 = 43$



OSIDH PROTOCOL - AN EXAMPLE

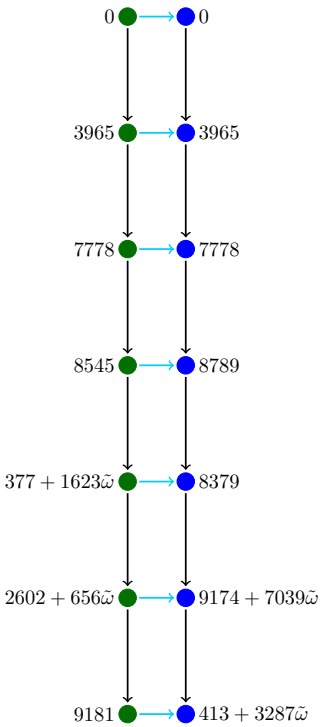
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

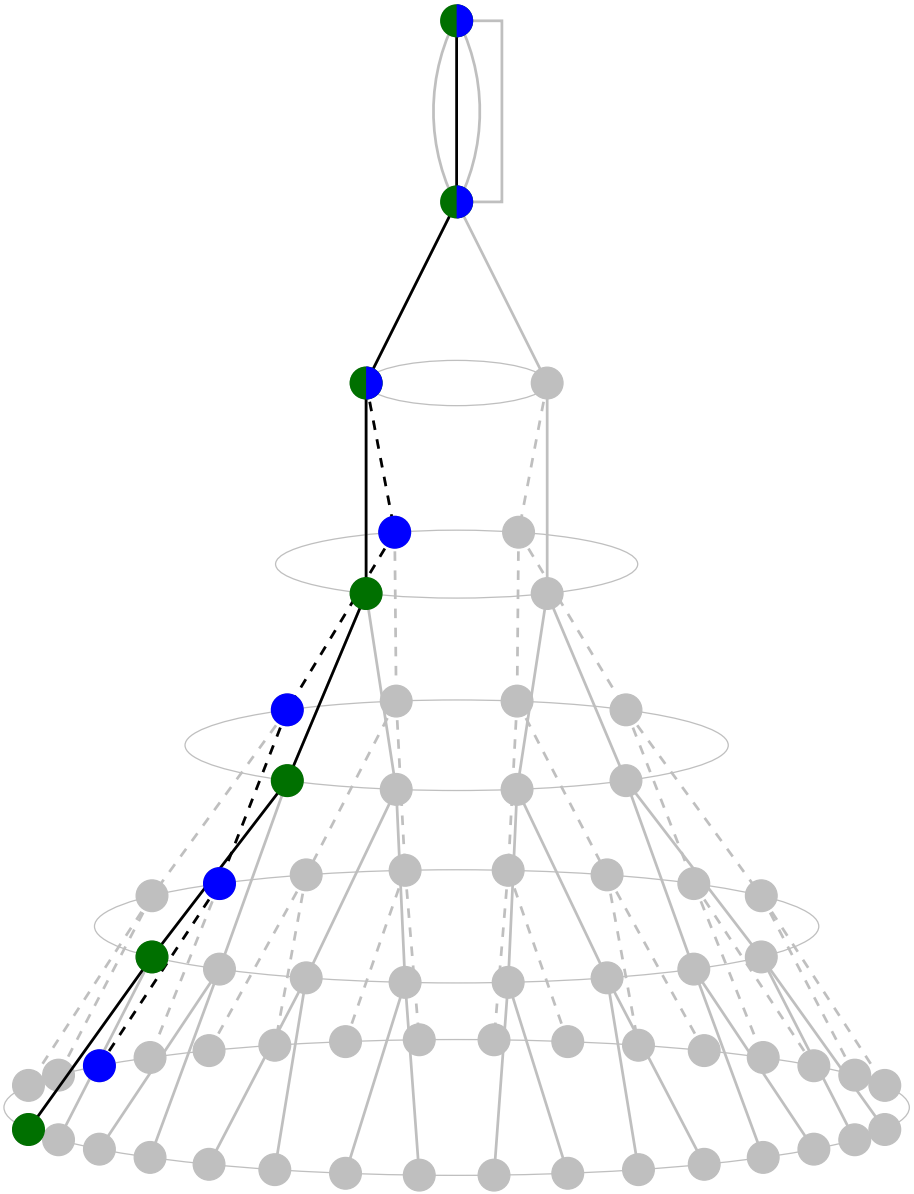




OSIDH PROTOCOL - AN EXAMPLE

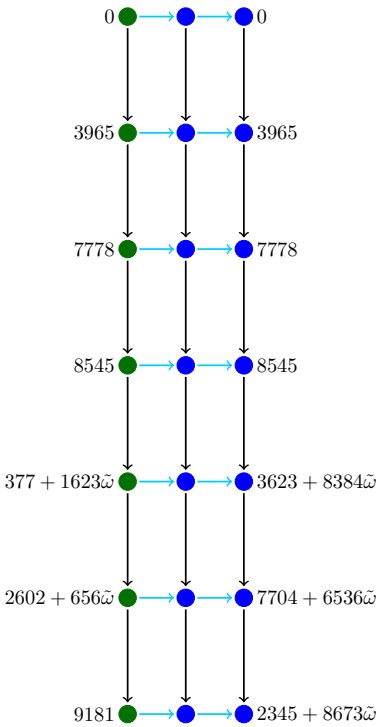
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

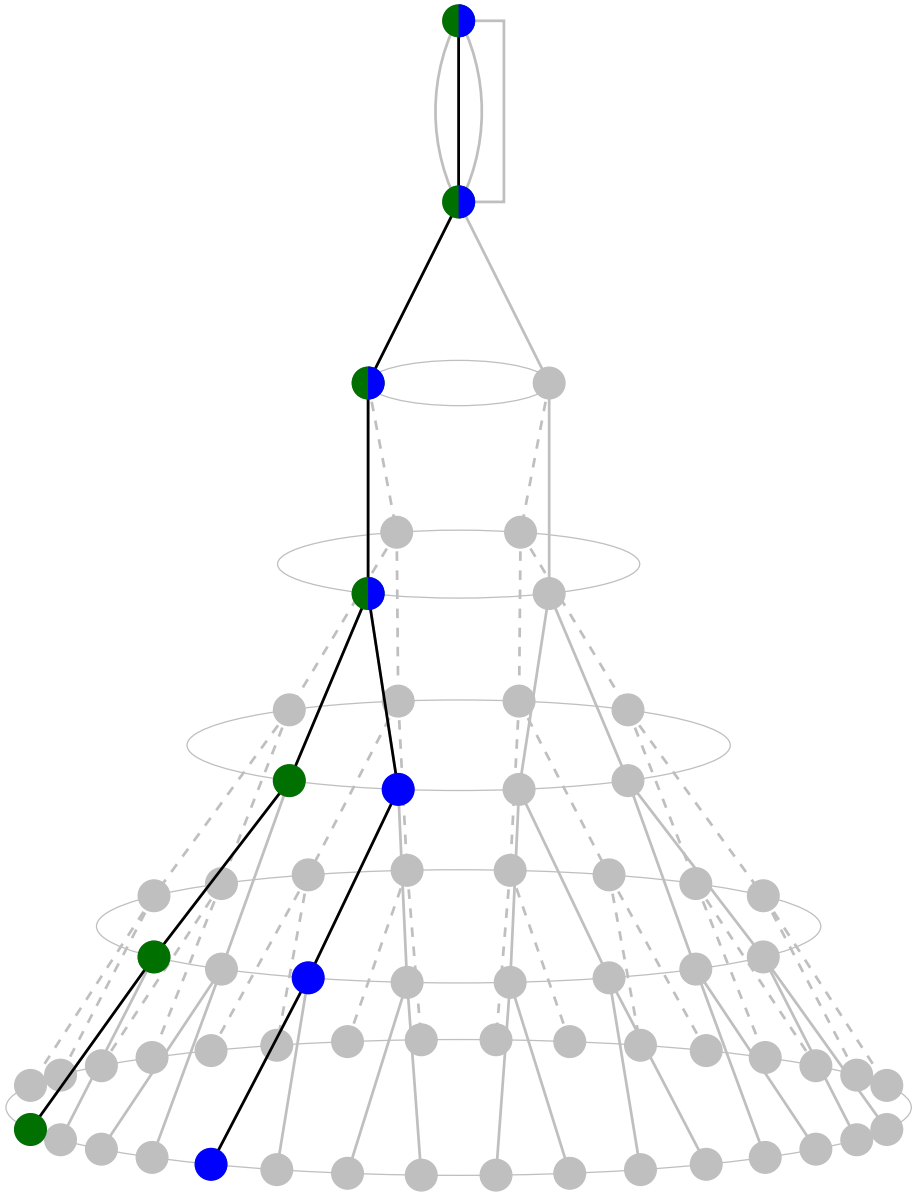




OSIDH PROTOCOL - AN EXAMPLE

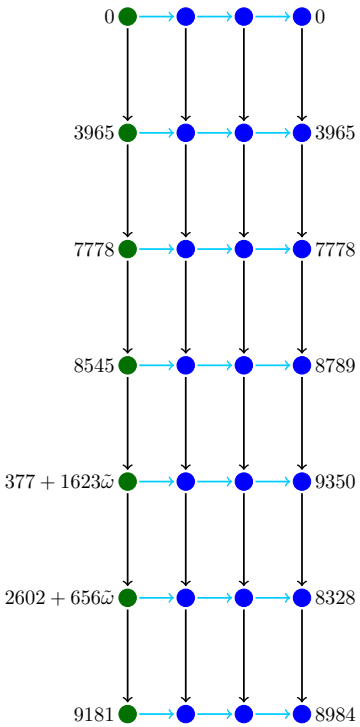
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

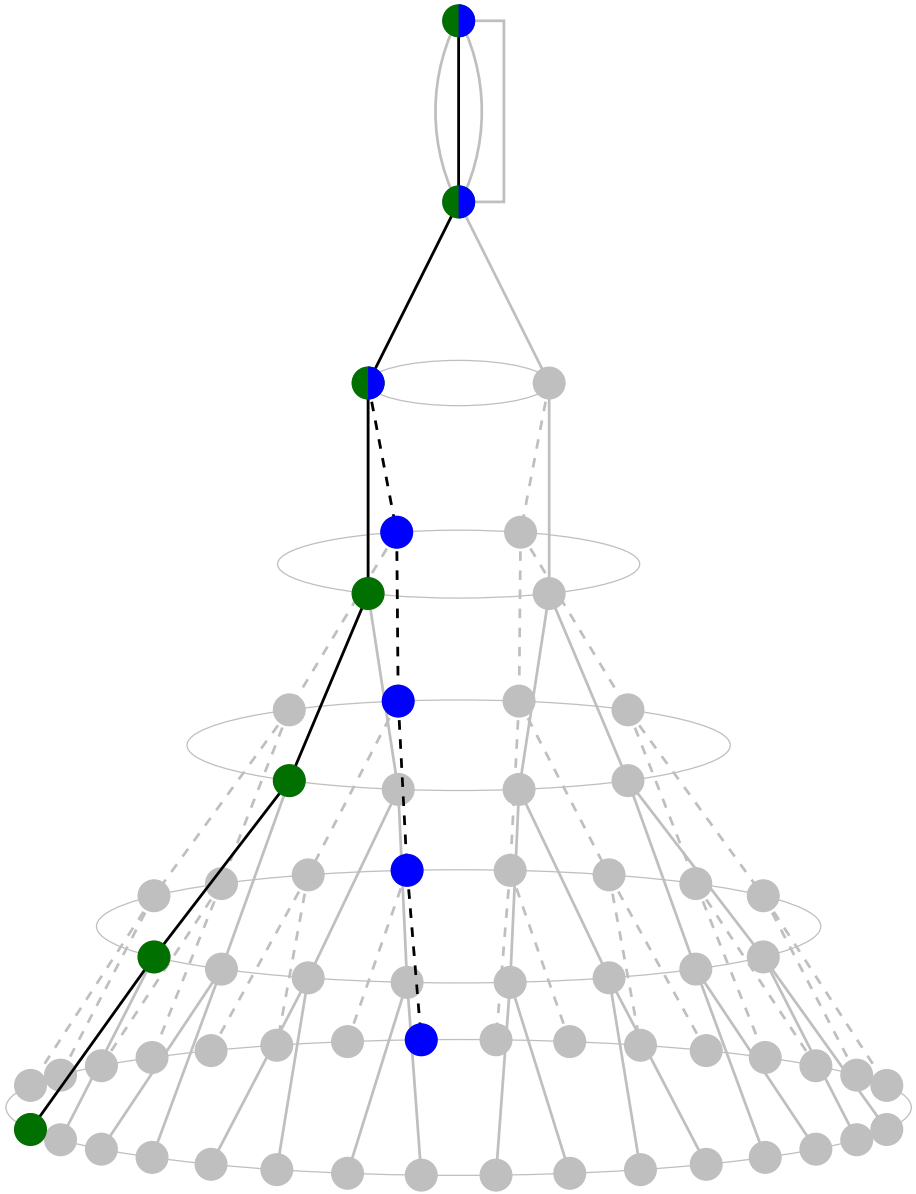




OSIDH PROTOCOL - AN EXAMPLE

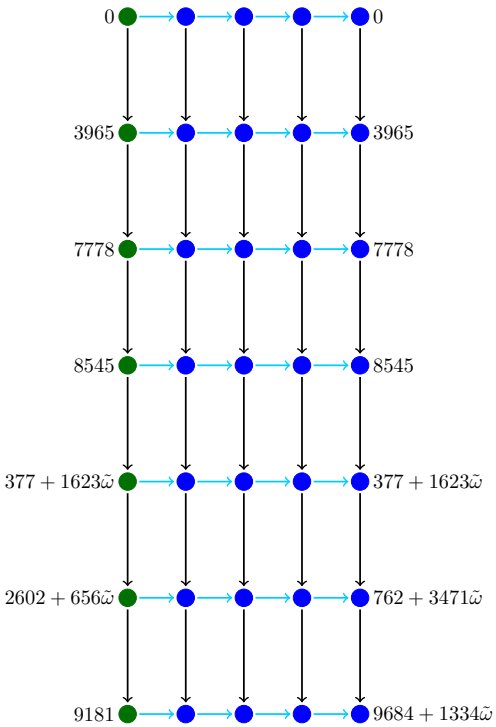
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

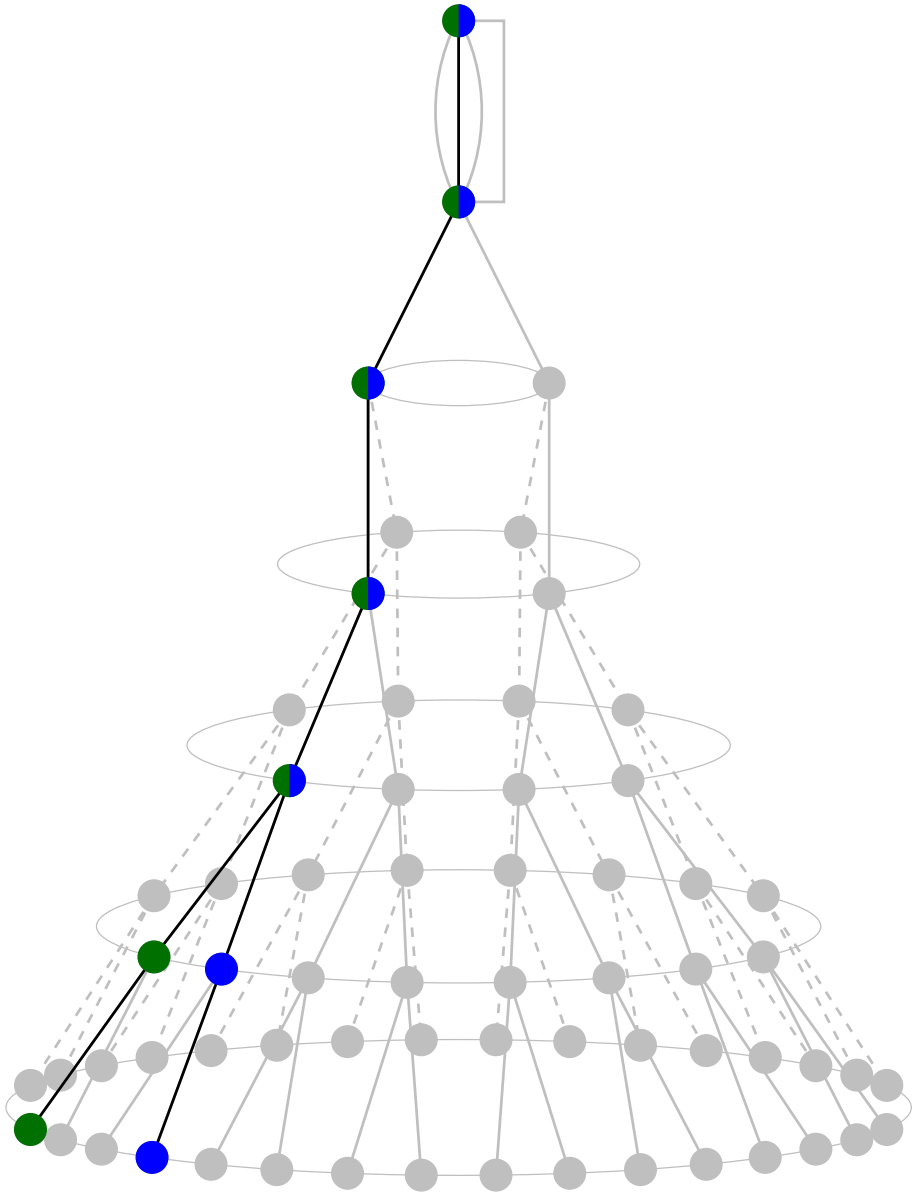




OSIDH PROTOCOL - AN EXAMPLE

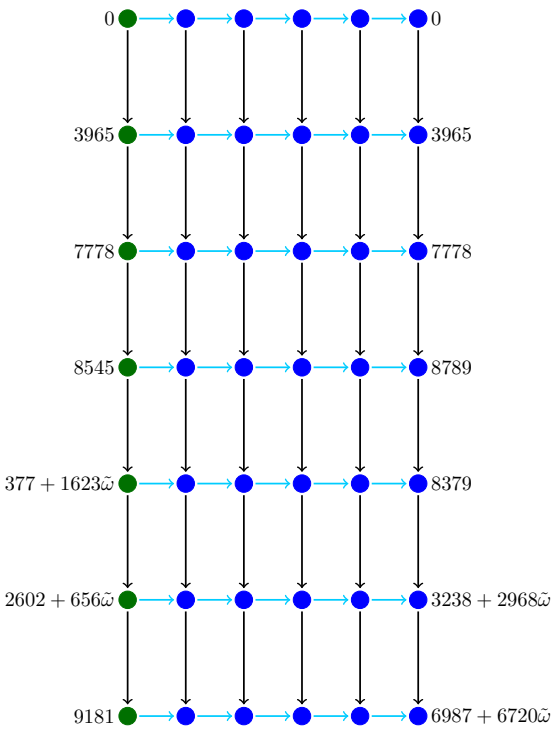
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

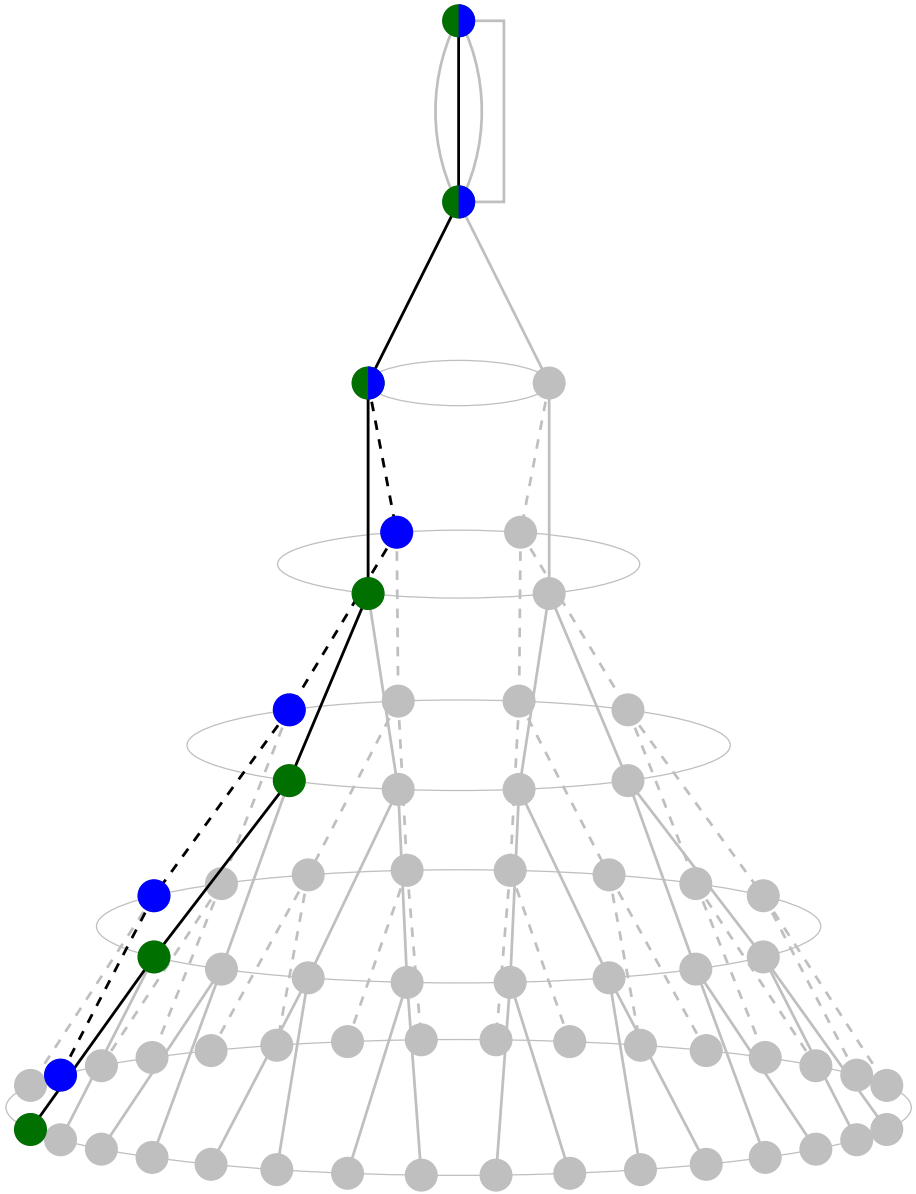




OSIDH PROTOCOL - AN EXAMPLE

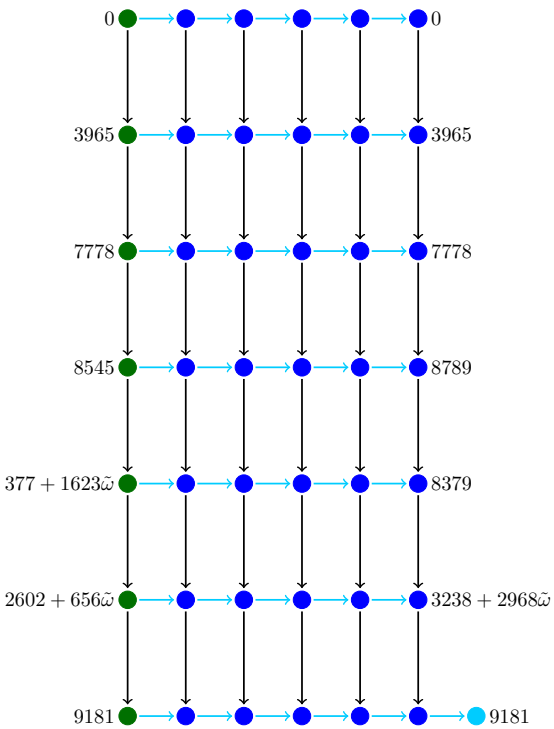
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

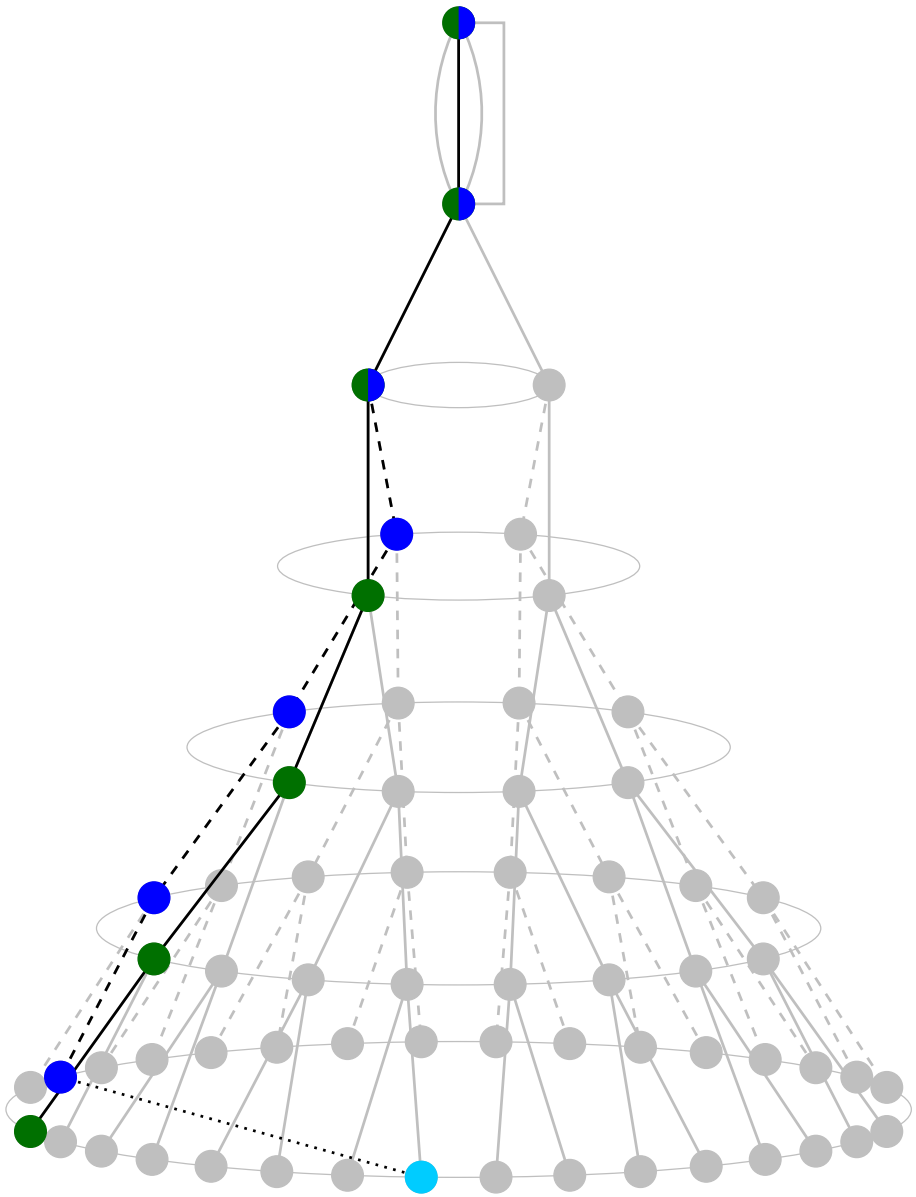




OSIDH PROTOCOL - AN EXAMPLE

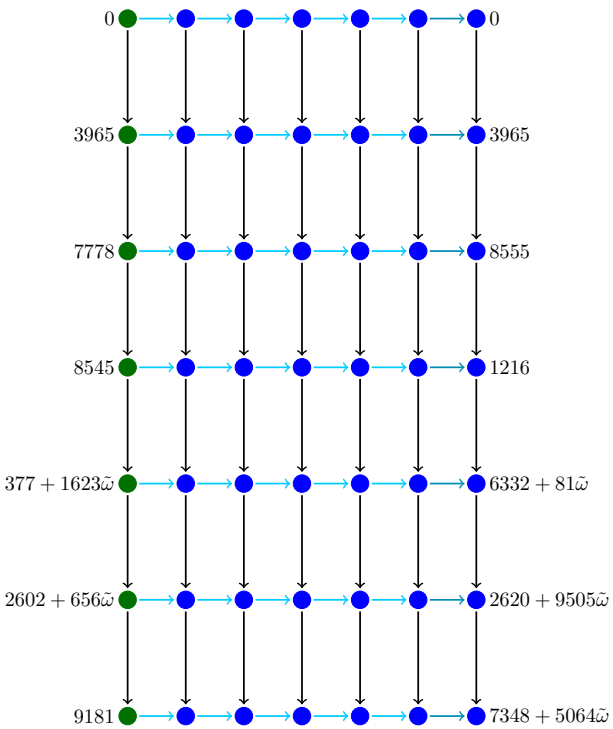
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

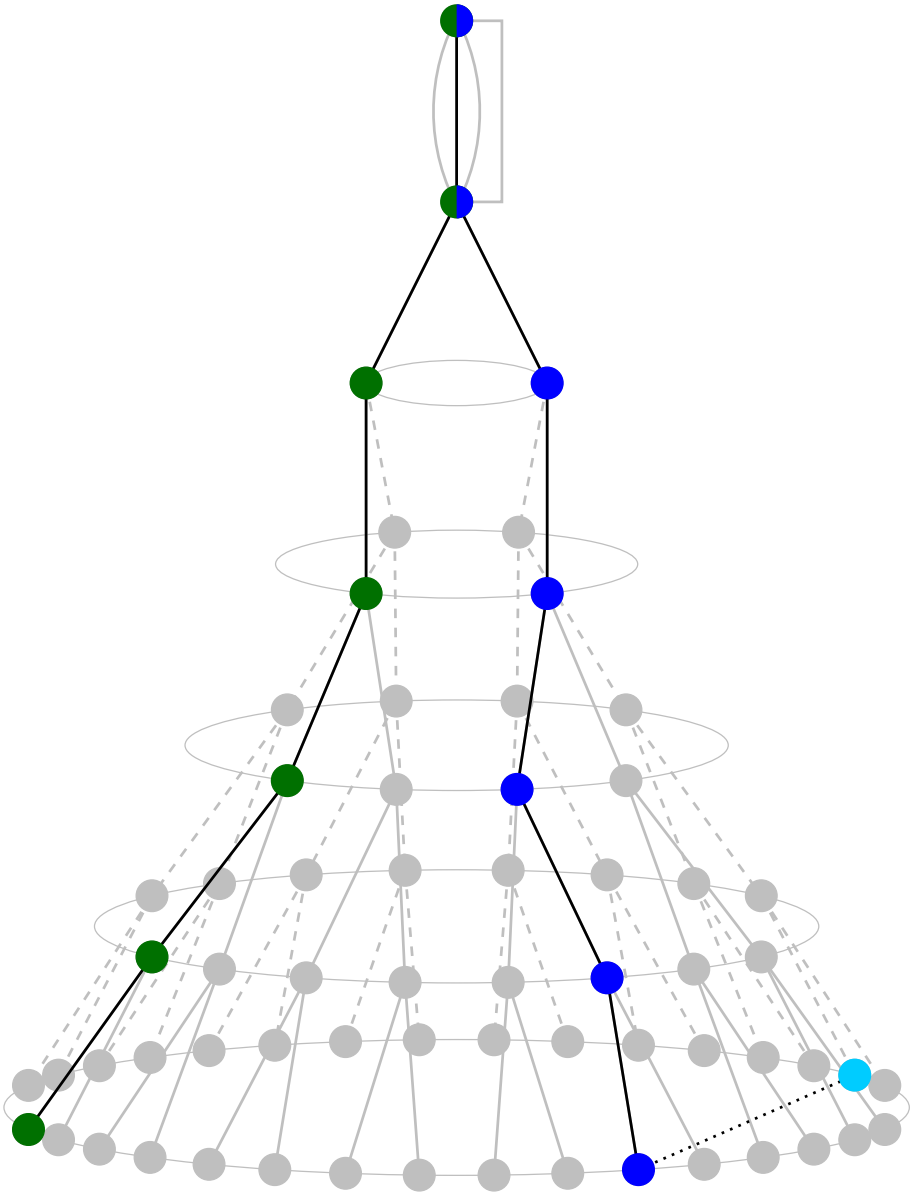




OSIDH PROTOCOL - AN EXAMPLE

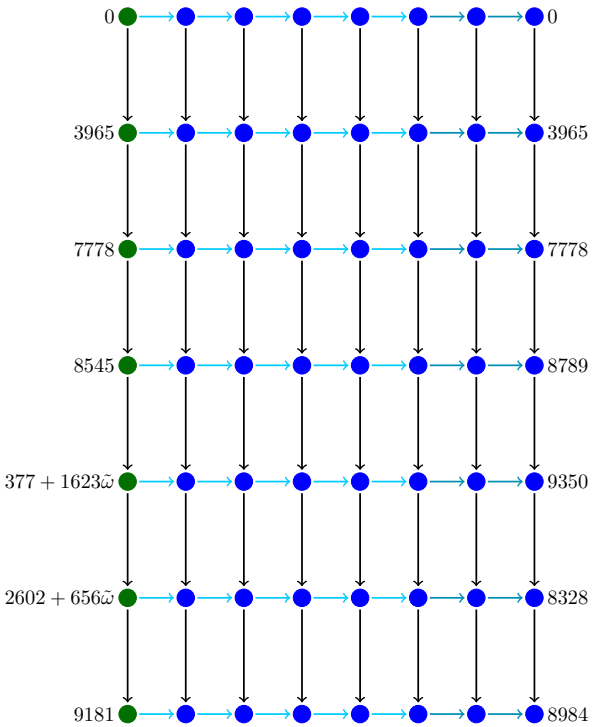
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

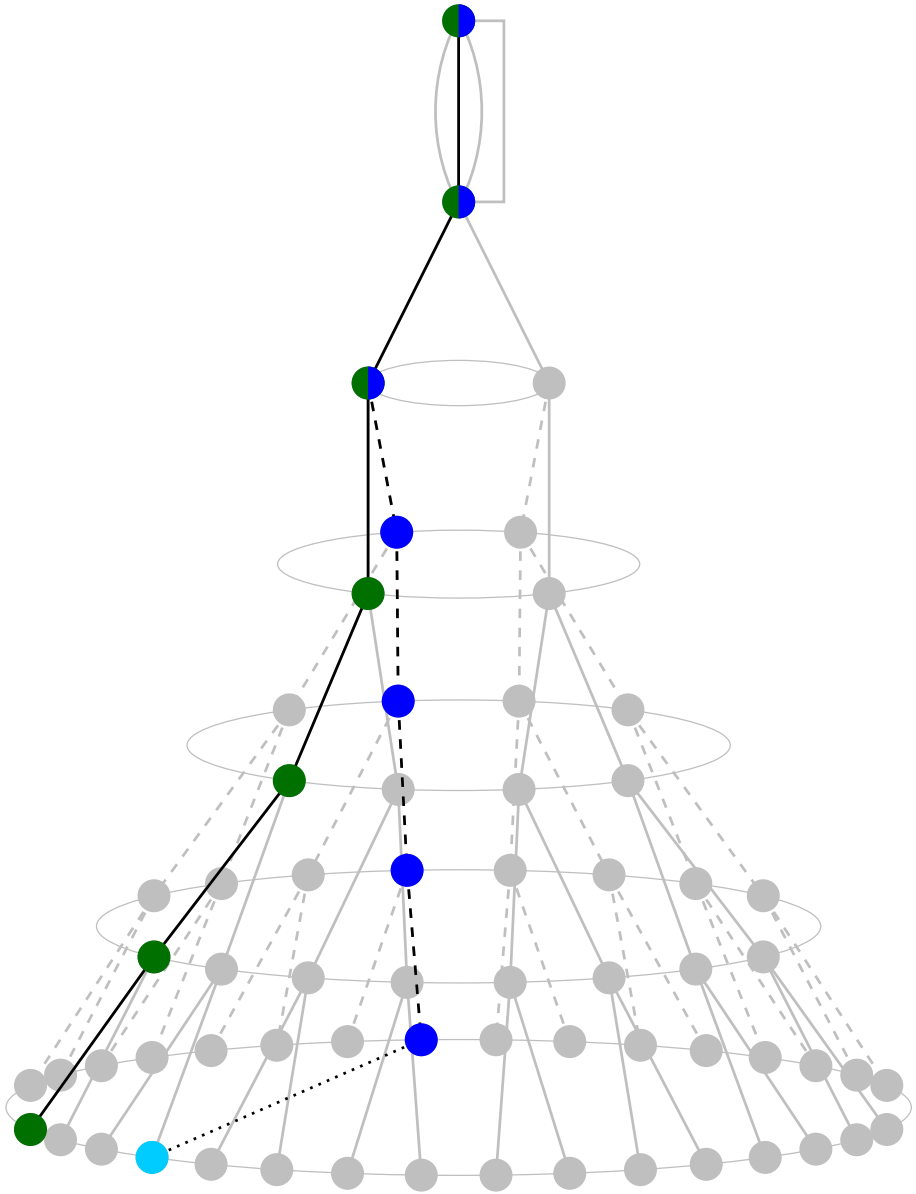




OSIDH PROTOCOL - AN EXAMPLE

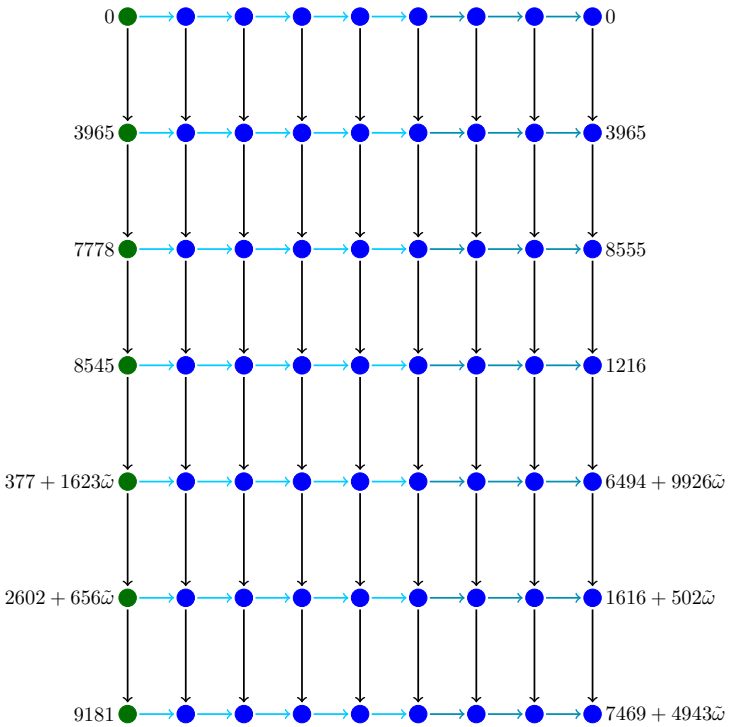
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

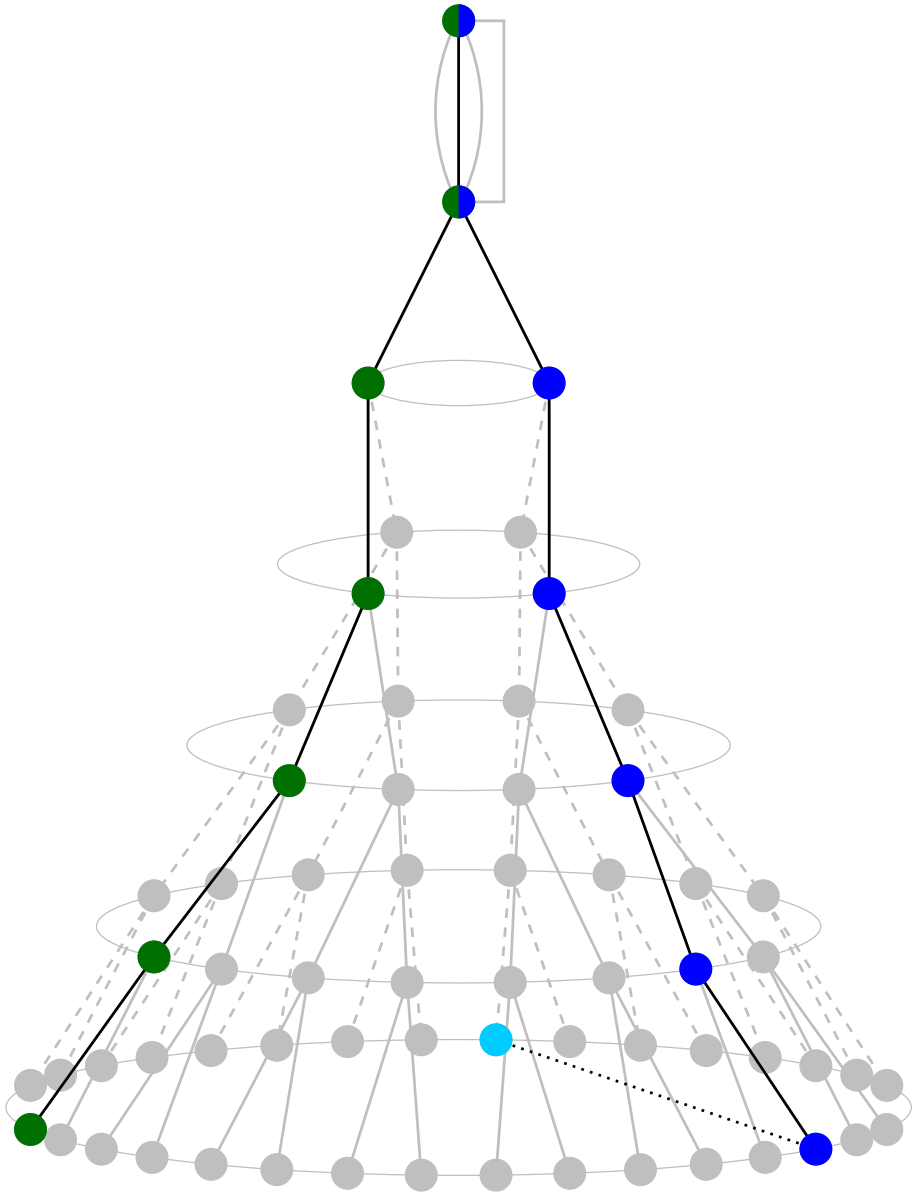




OSIDH PROTOCOL - AN EXAMPLE

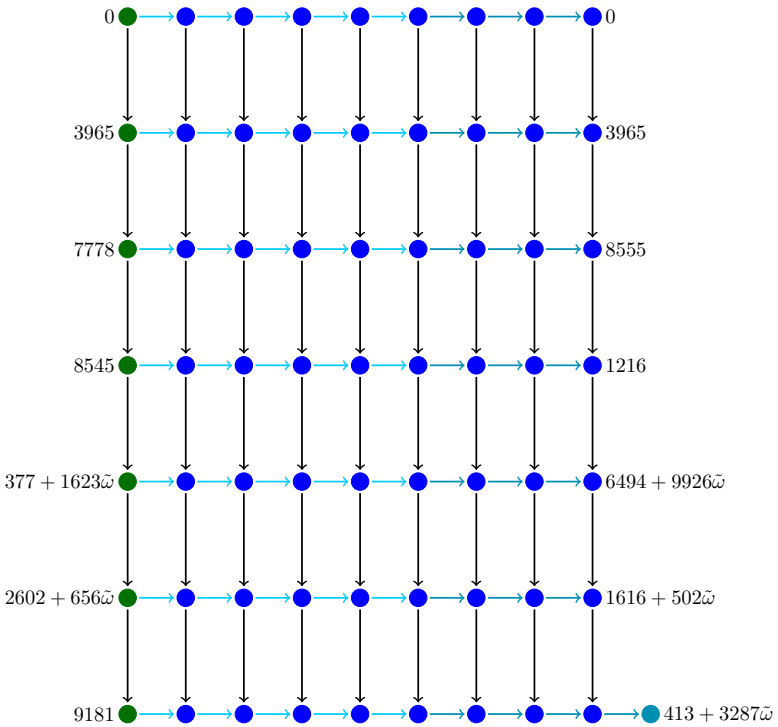
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

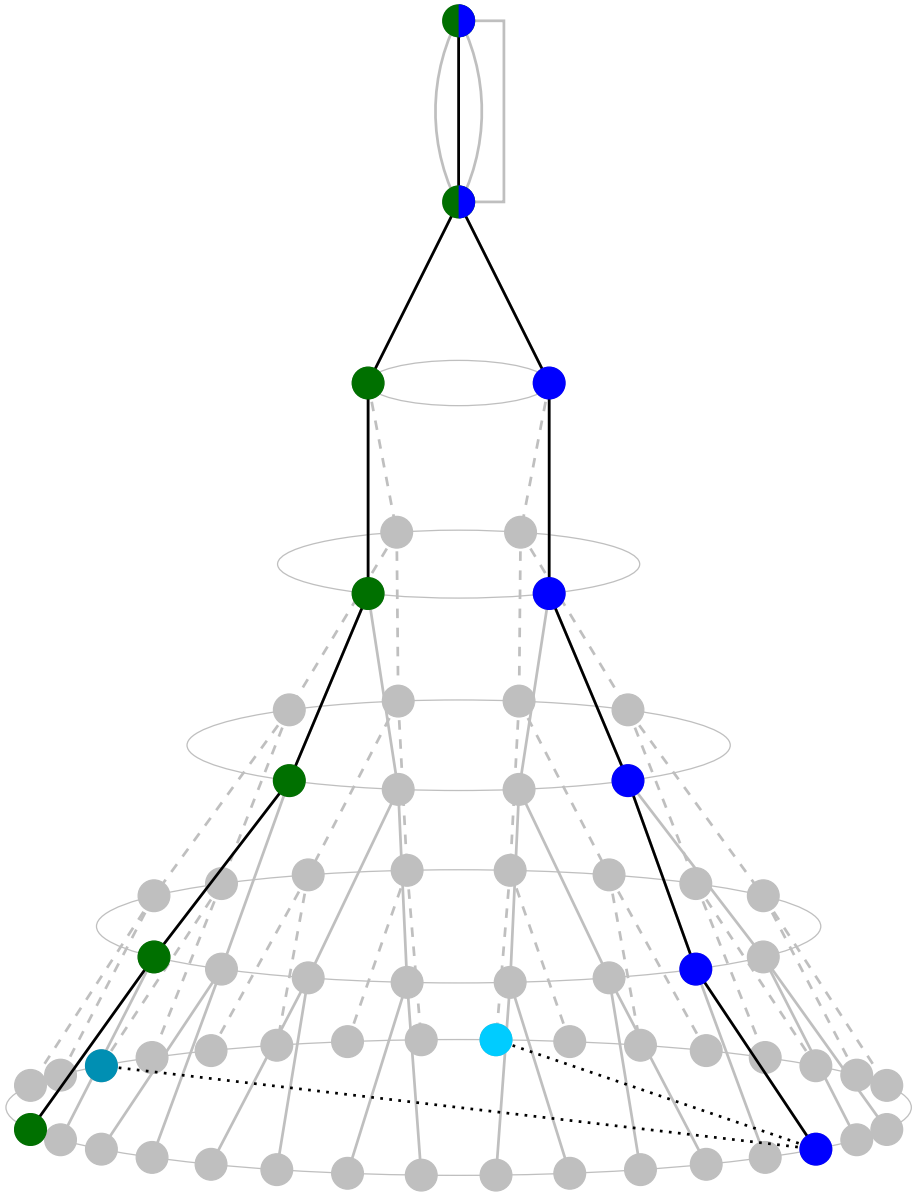




OSIDH PROTOCOL - AN EXAMPLE

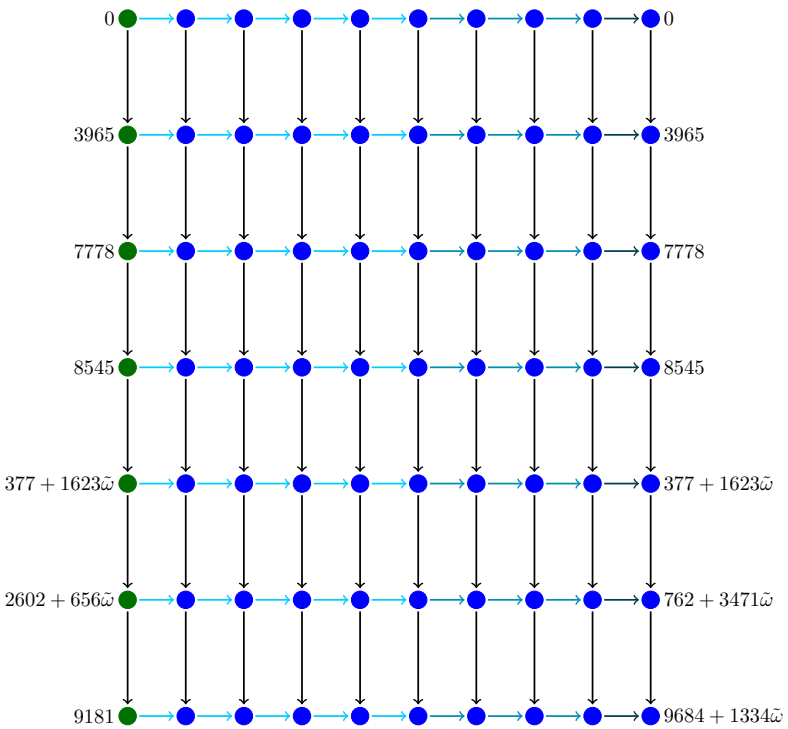
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

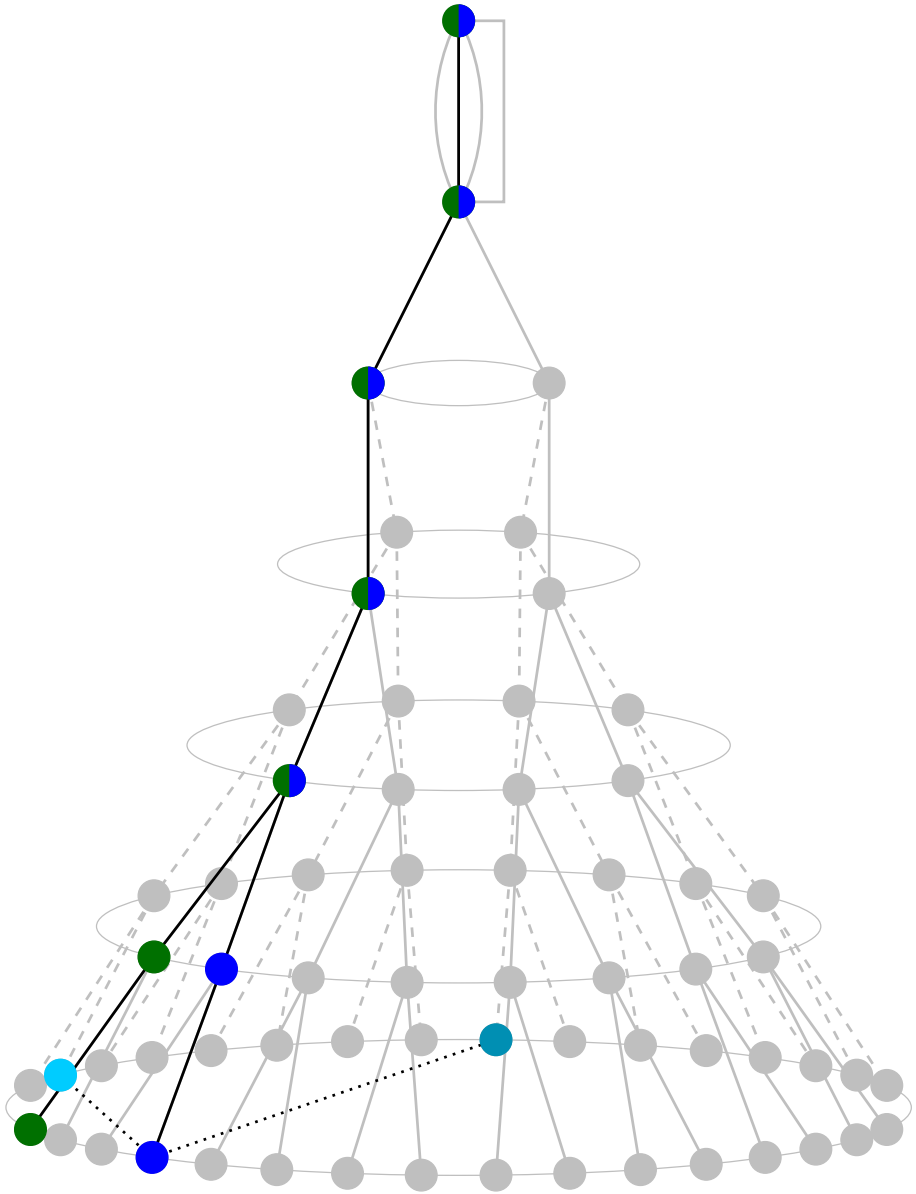




OSIDH PROTOCOL - AN EXAMPLE

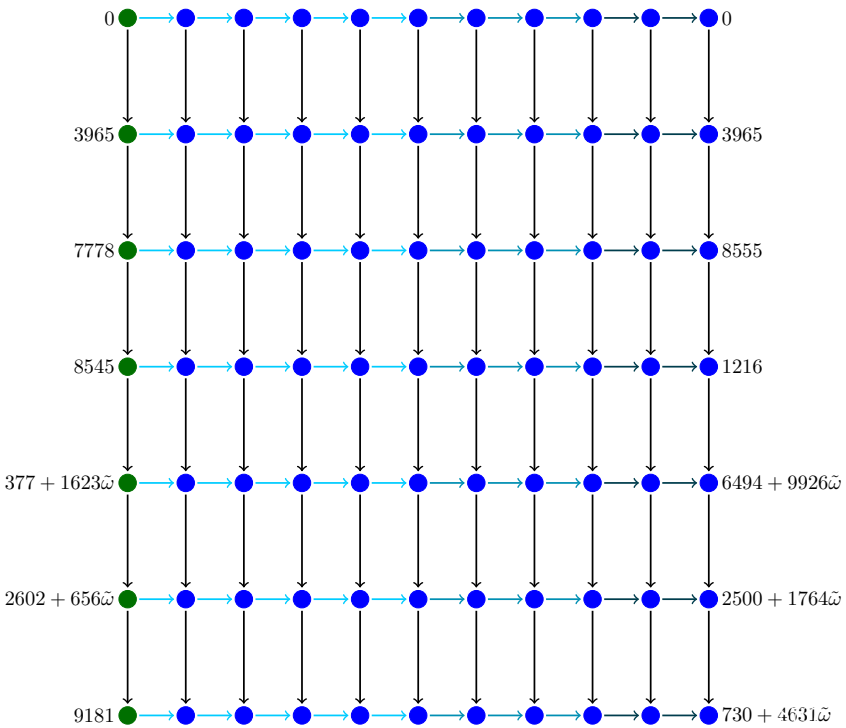
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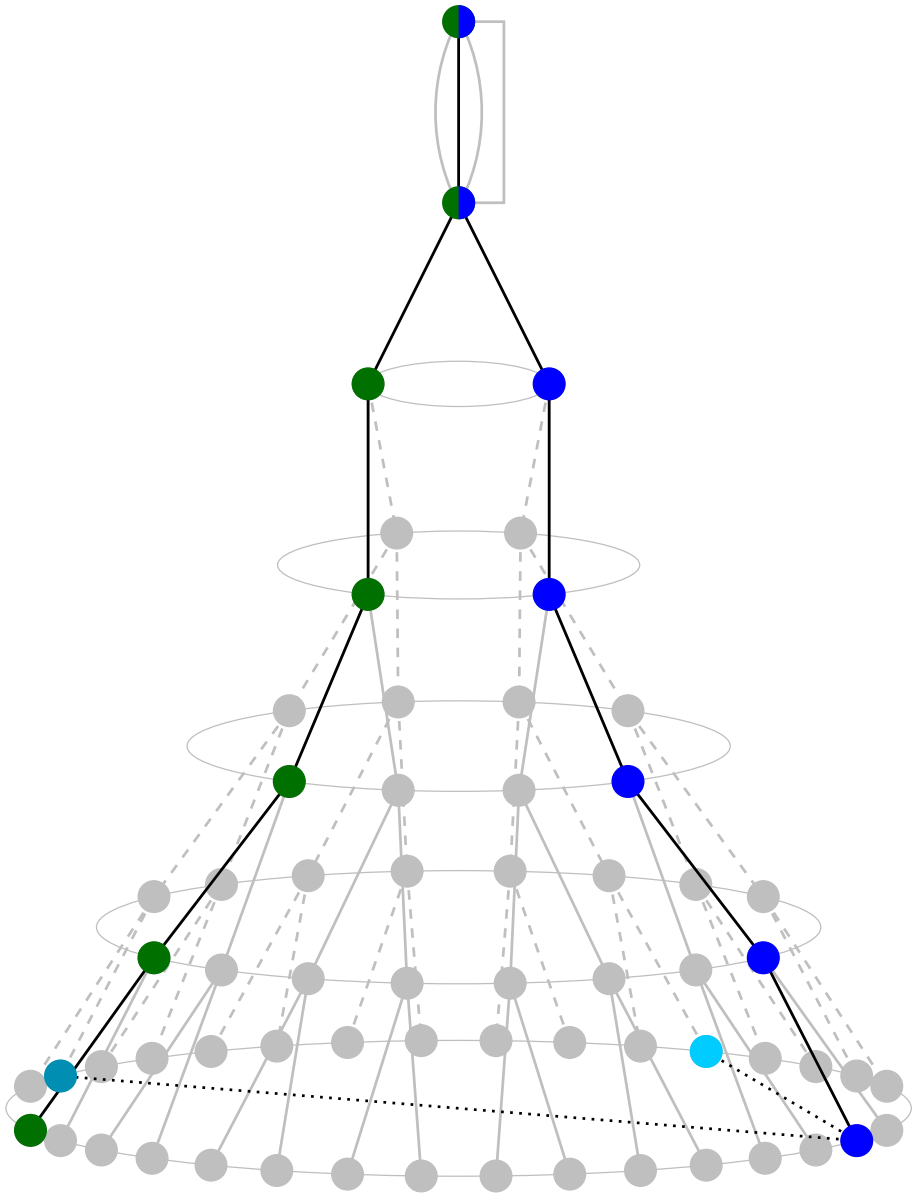




OSIDH PROTOCOL - AN EXAMPLE

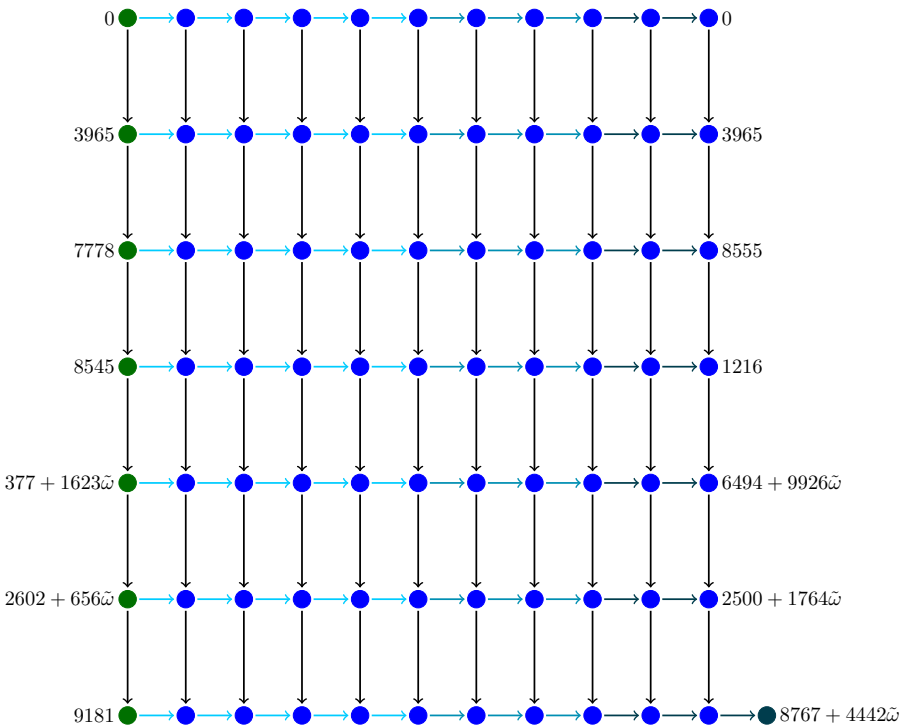
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

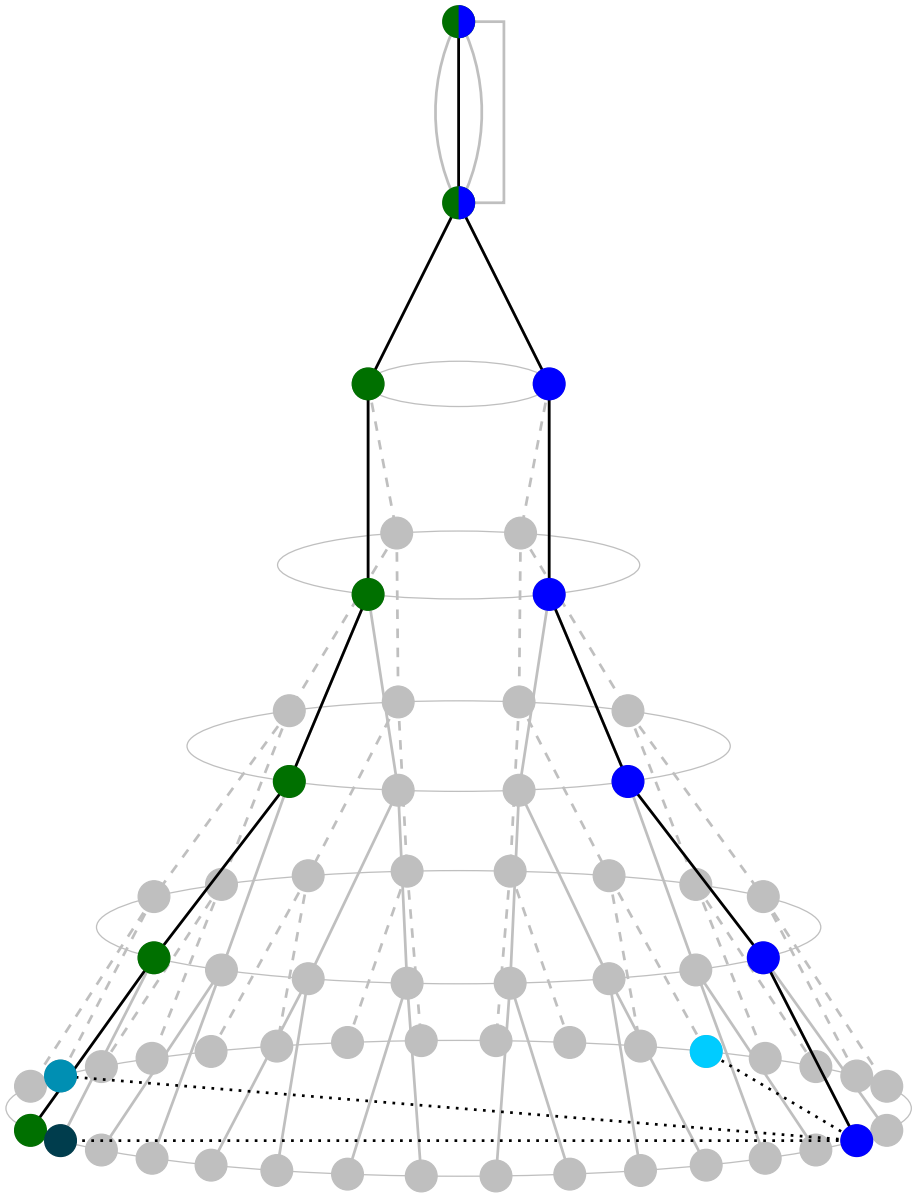




OSIDH PROTOCOL - AN EXAMPLE

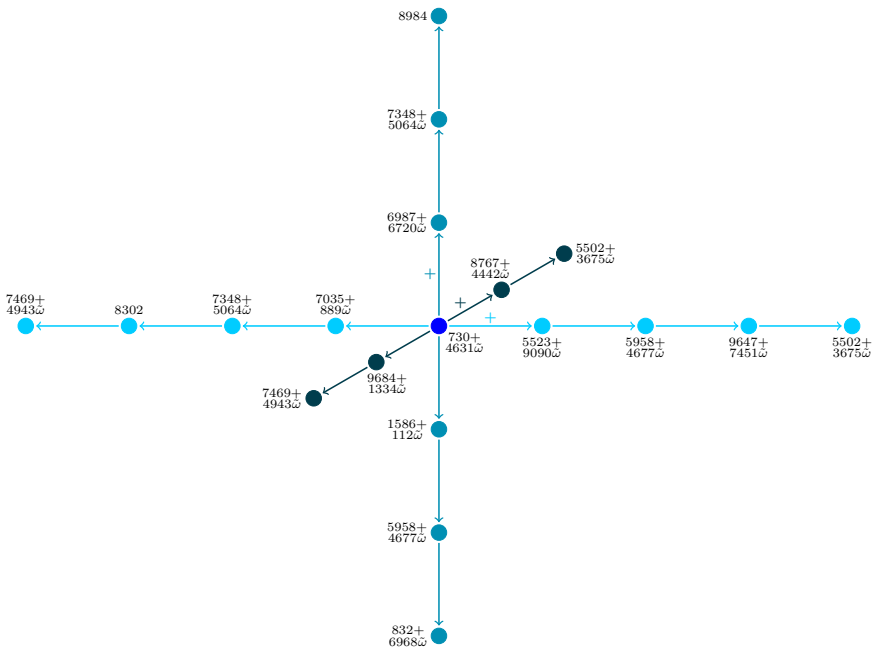
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

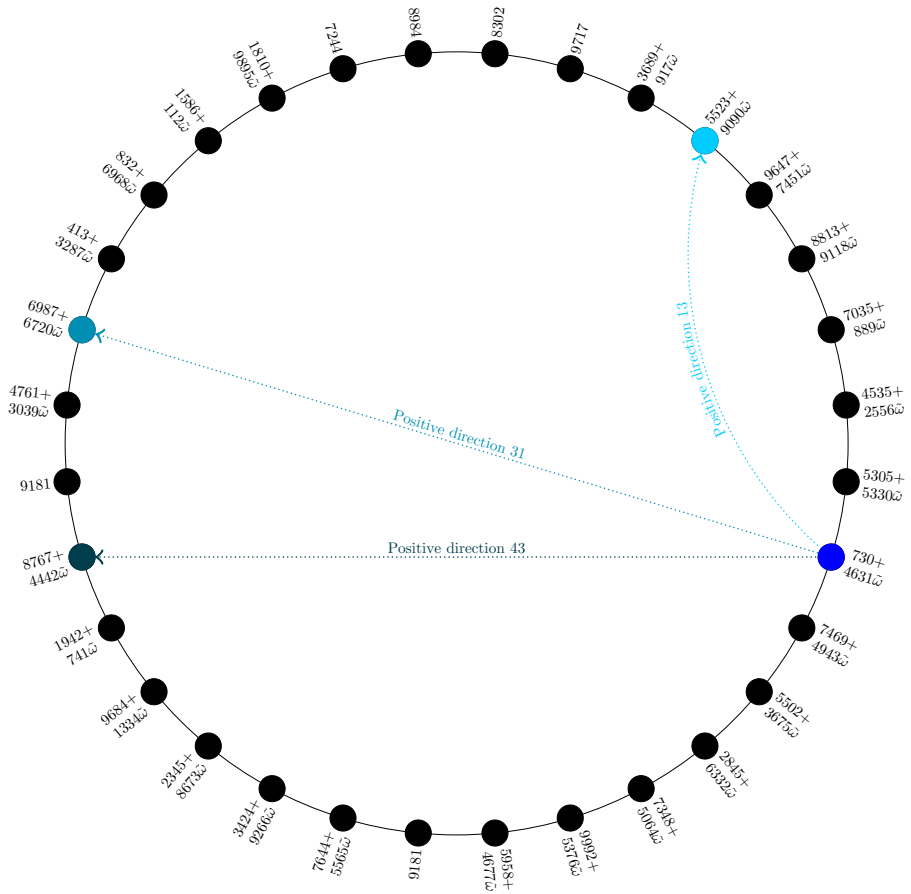




OSIDH PROTOCOL - AN EXAMPLE

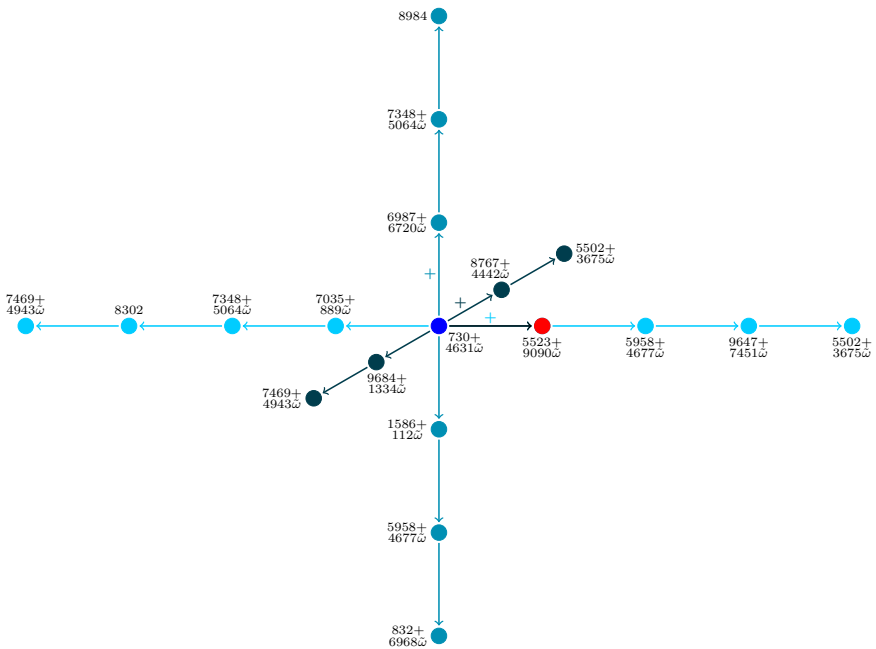
Bob secret key: $\mathfrak{l}_1^3 \mathfrak{l}_2 \mathfrak{l}_3^2$

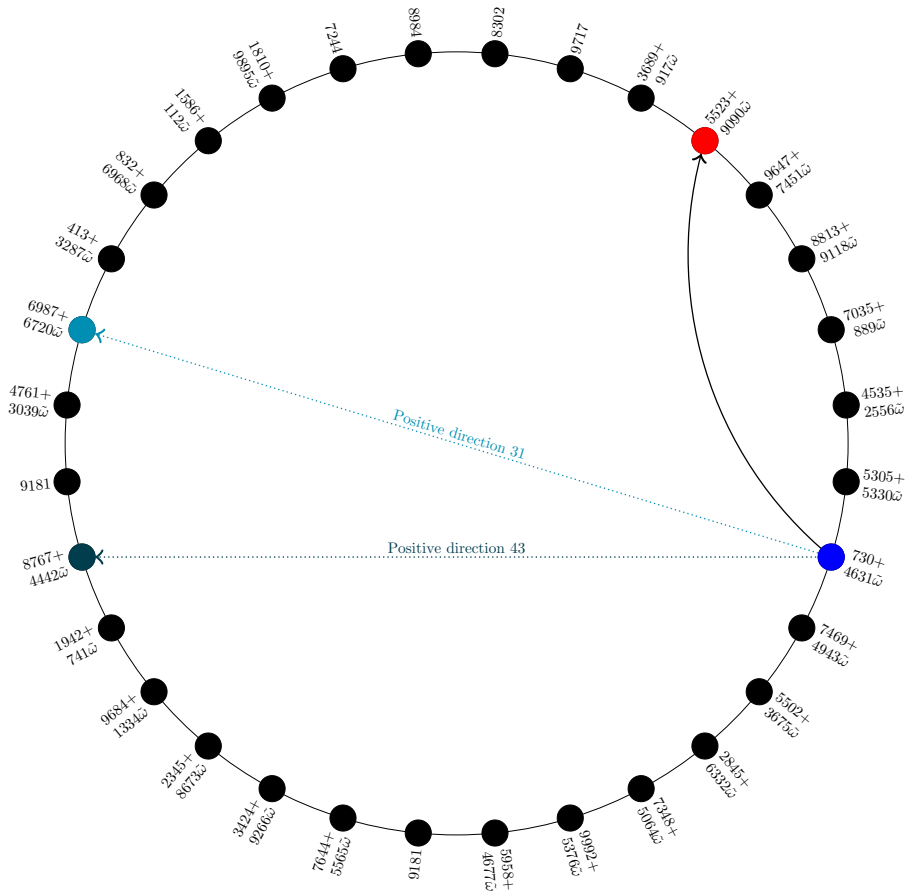




OSIDH PROTOCOL - AN EXAMPLE

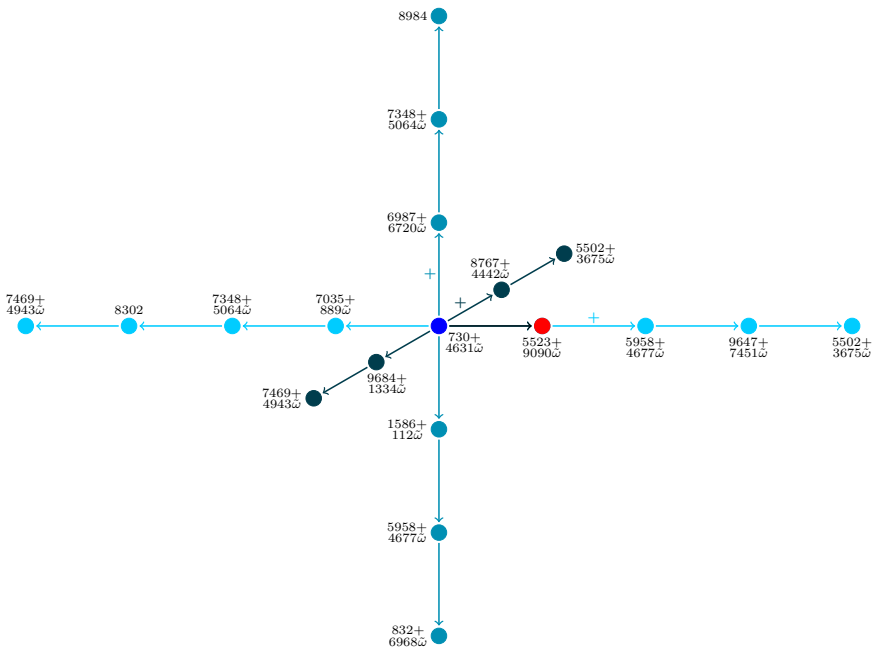
Bob secret key: $\mathfrak{l}_1^3 \mathfrak{l}_2 \mathfrak{l}_3^2$

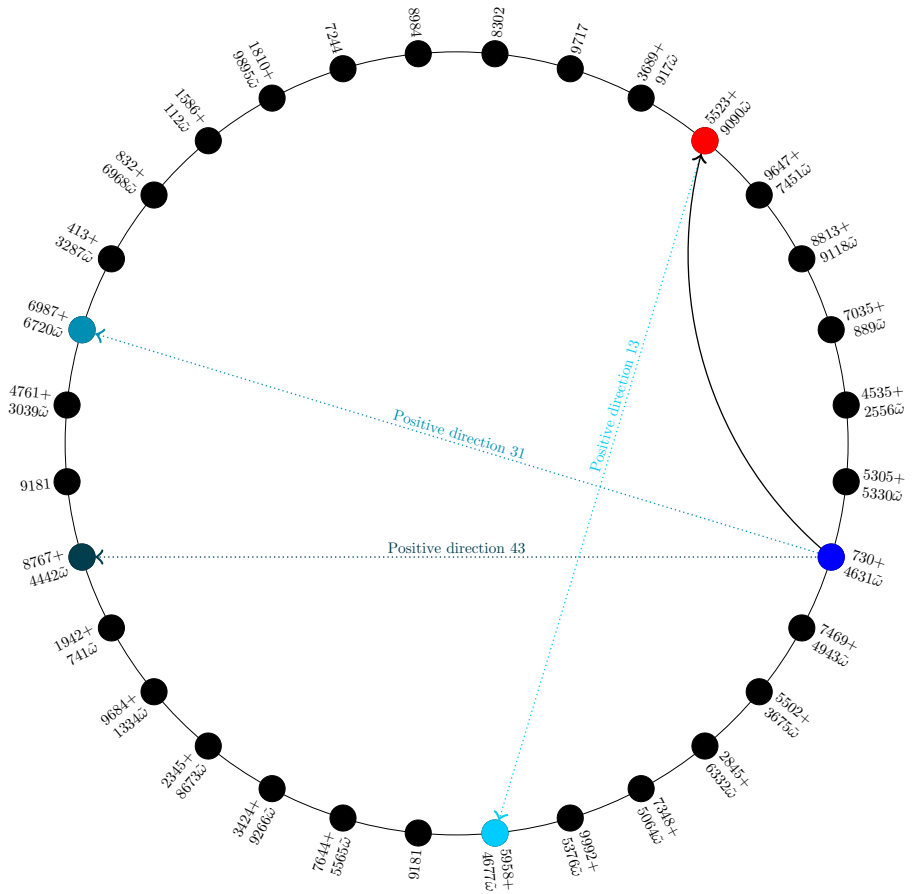




OSIDH PROTOCOL - AN EXAMPLE

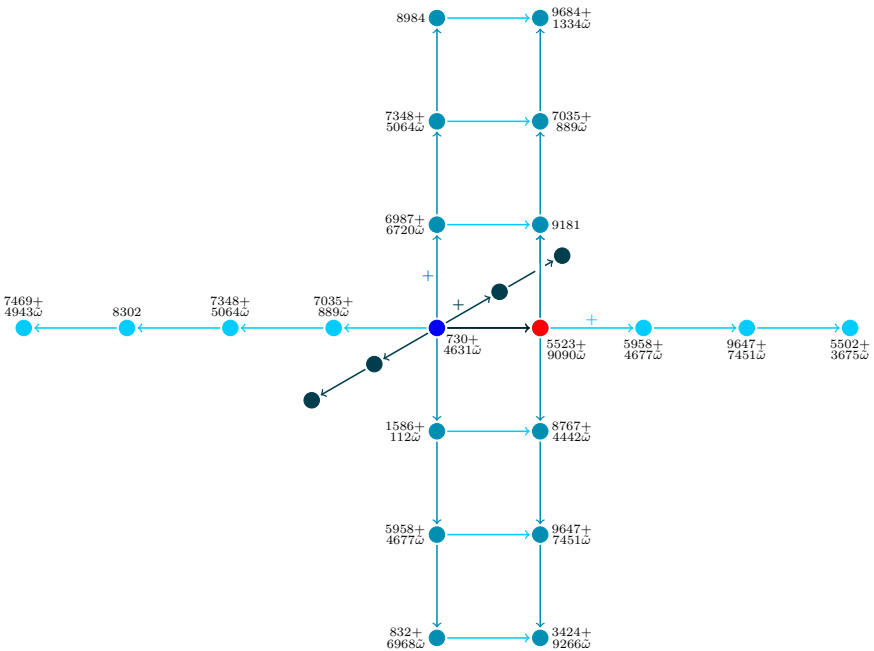
Bob secret key: $\mathfrak{l}_1^3 \mathfrak{l}_2 \mathfrak{l}_3^2$

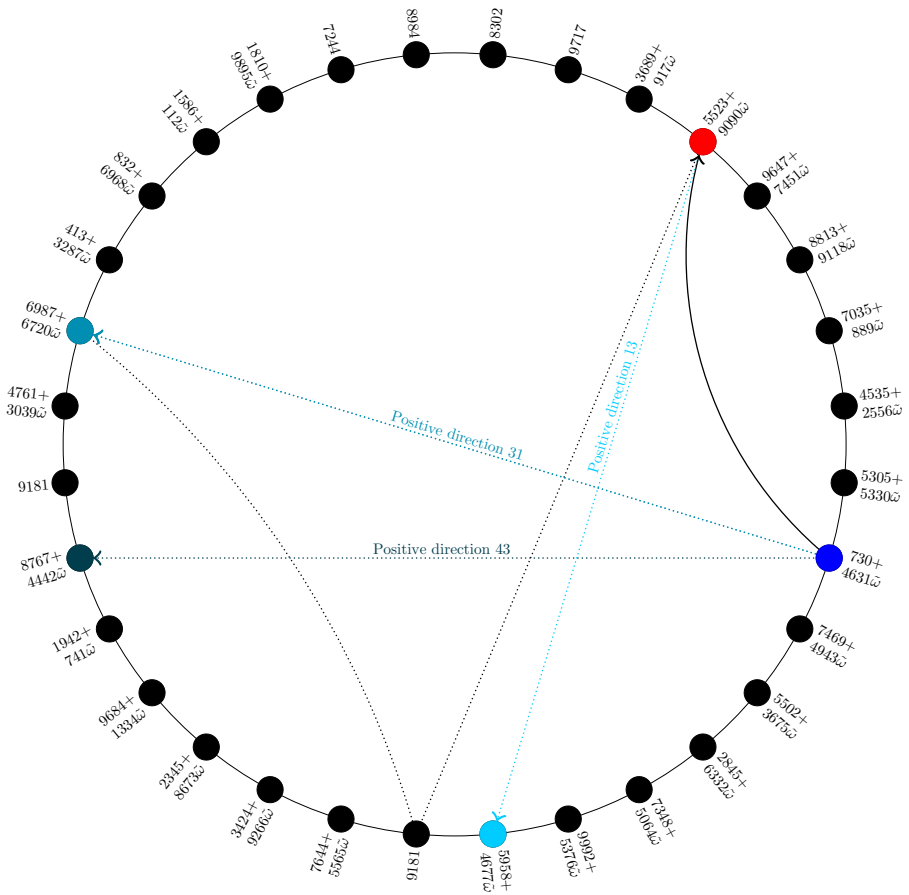




OSIDH PROTOCOL - AN EXAMPLE

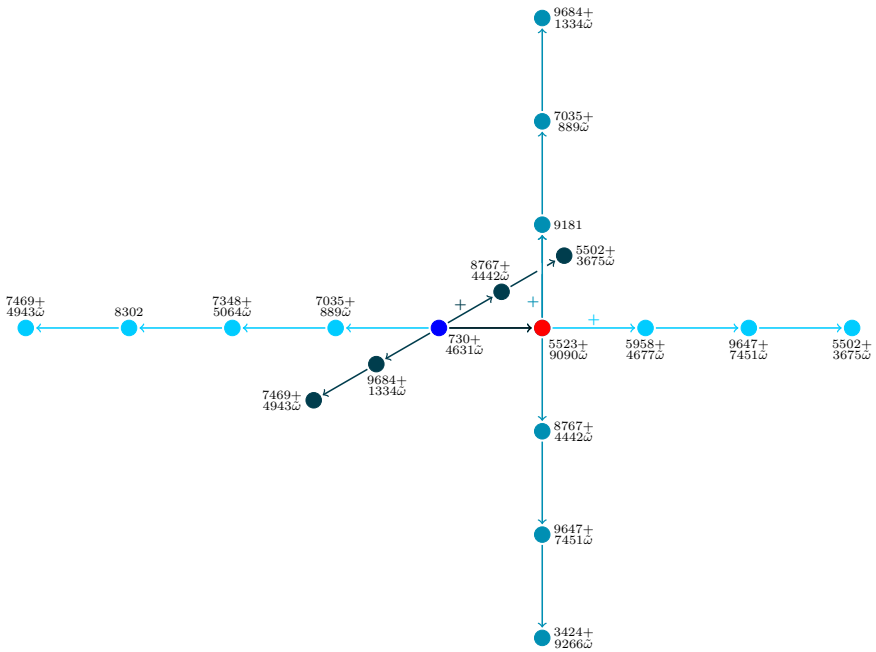
Bob secret key: $\mathfrak{l}_1^3 \mathfrak{l}_2 \mathfrak{l}_3^2$

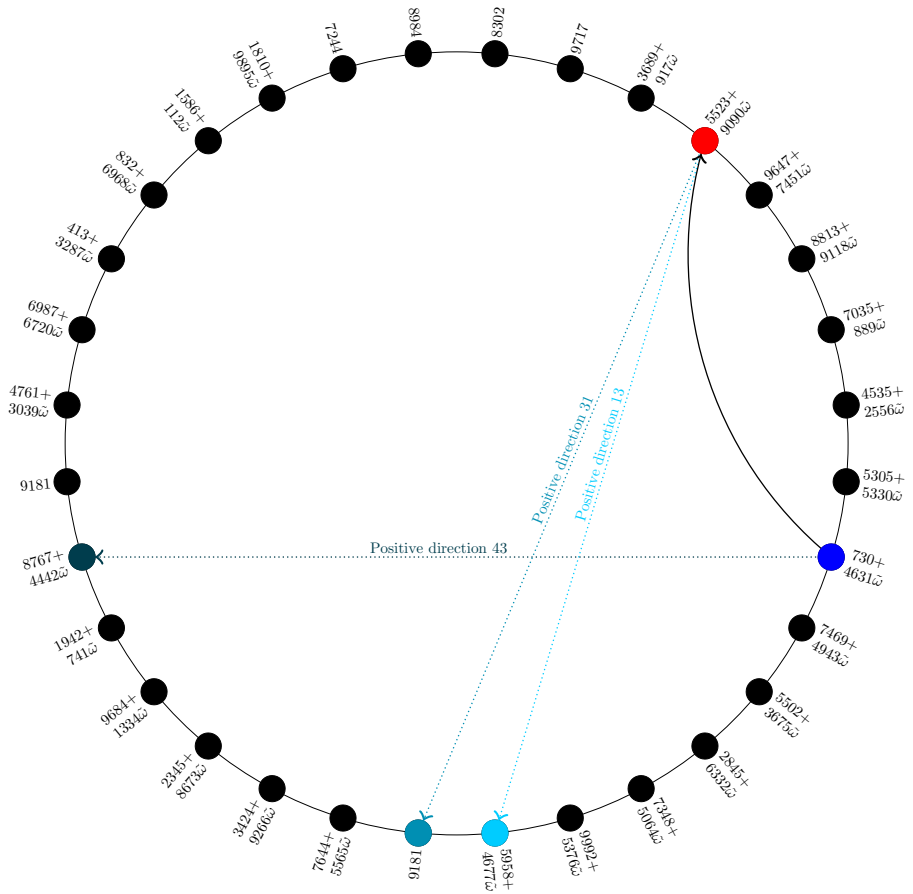




OSIDH PROTOCOL - AN EXAMPLE

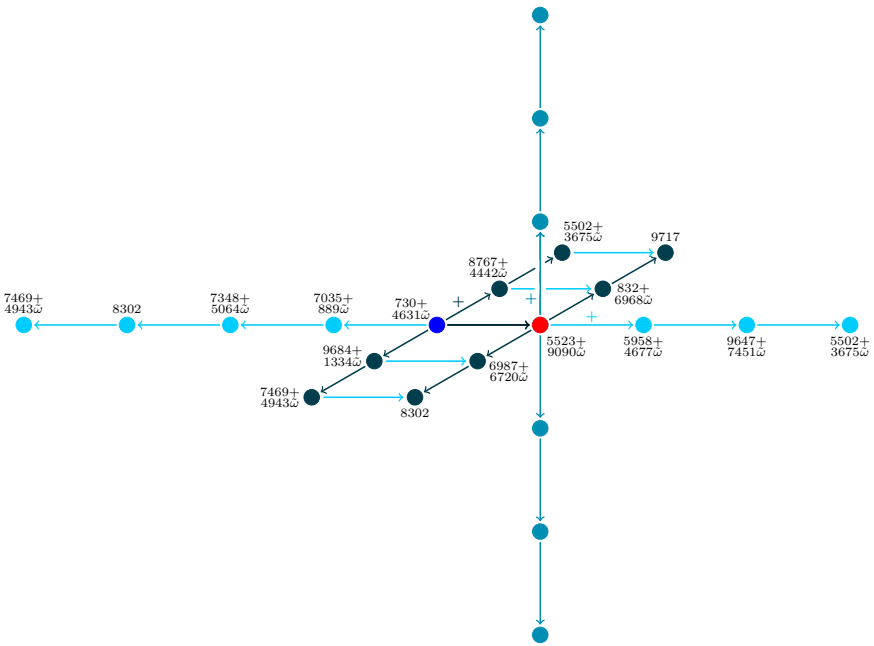
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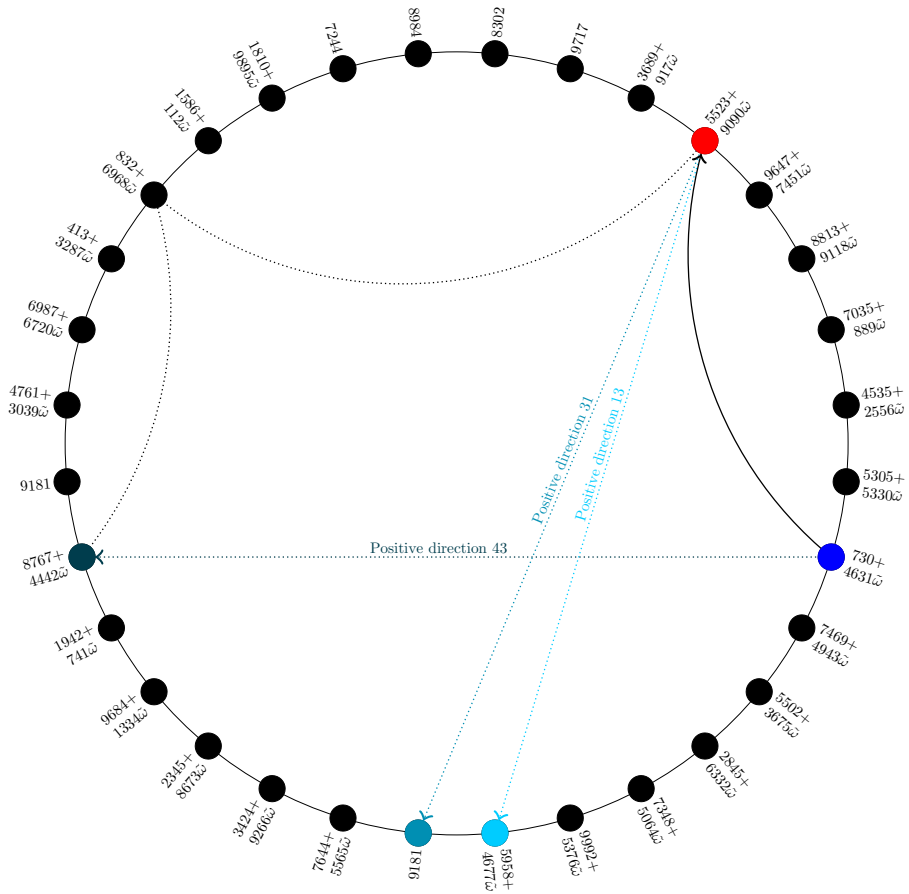




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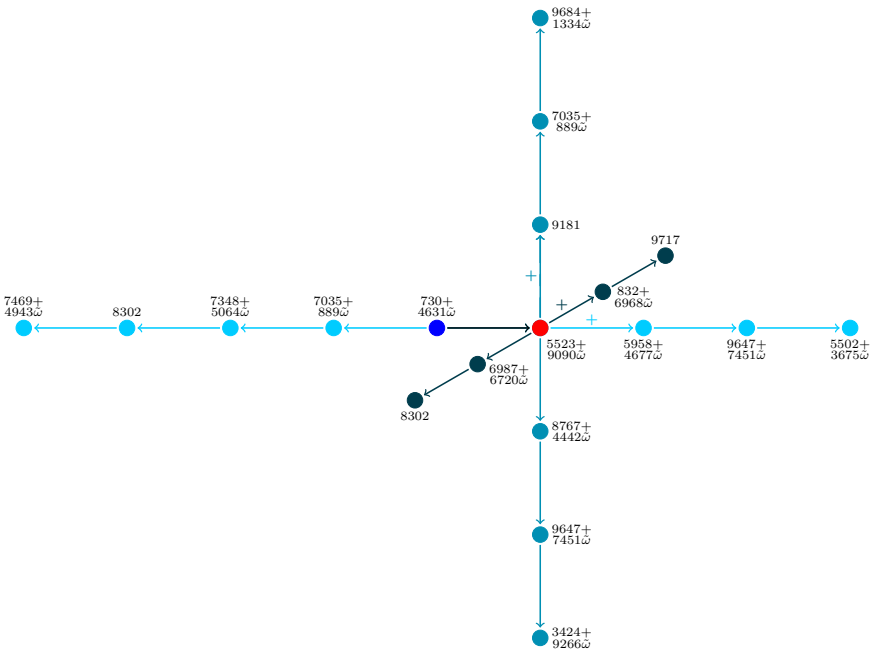
Bob secret key: $\mathfrak{l}_1^3\mathfrak{l}_2\mathfrak{l}_3^2$

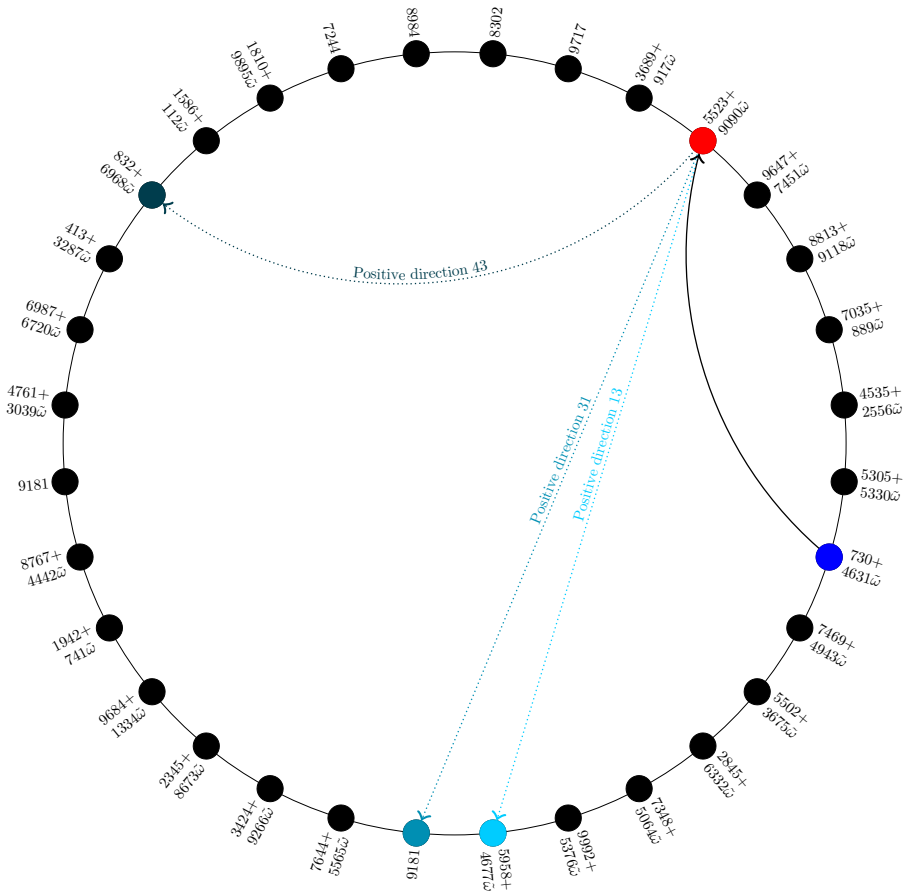




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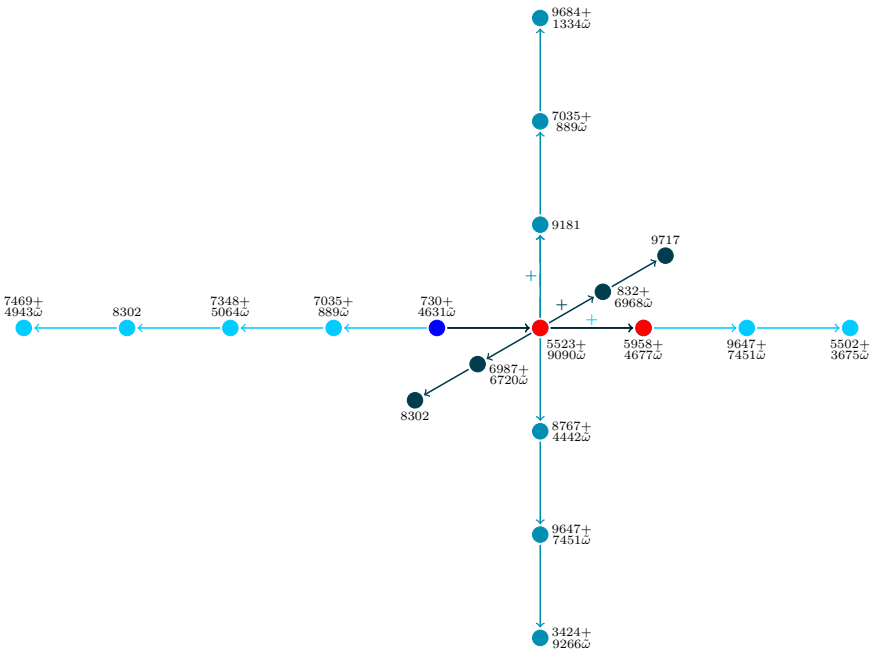
Bob secret key: $\mathfrak{l}_1^3 \mathfrak{l}_2 \mathfrak{l}_3^2$

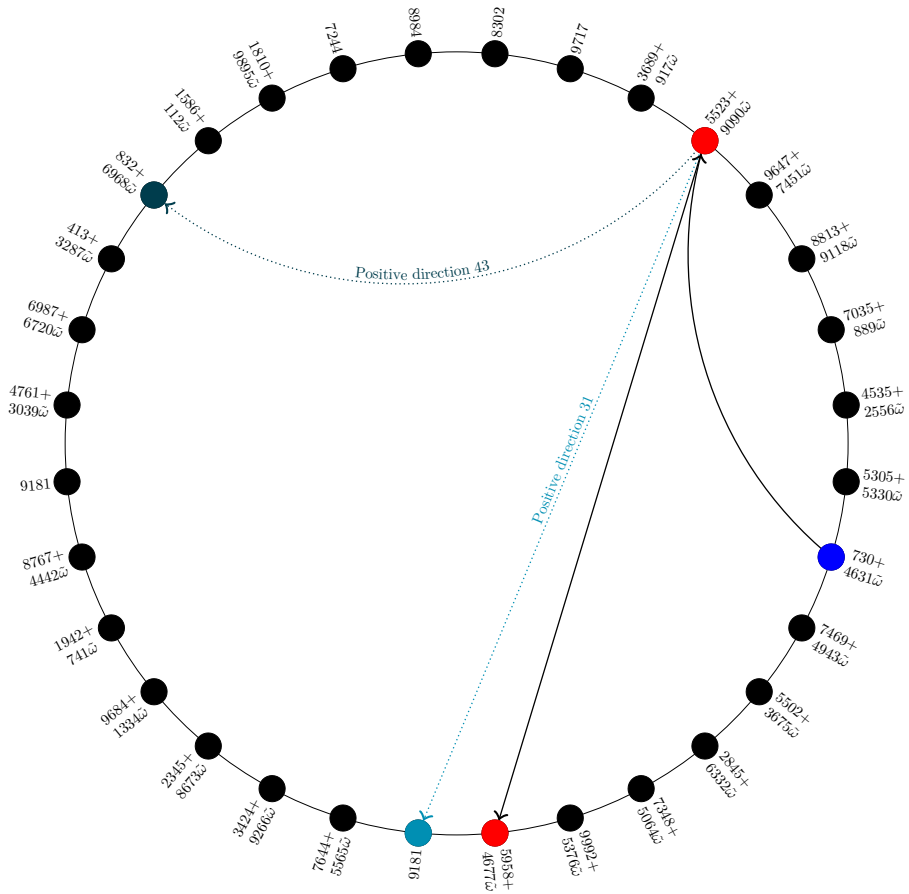




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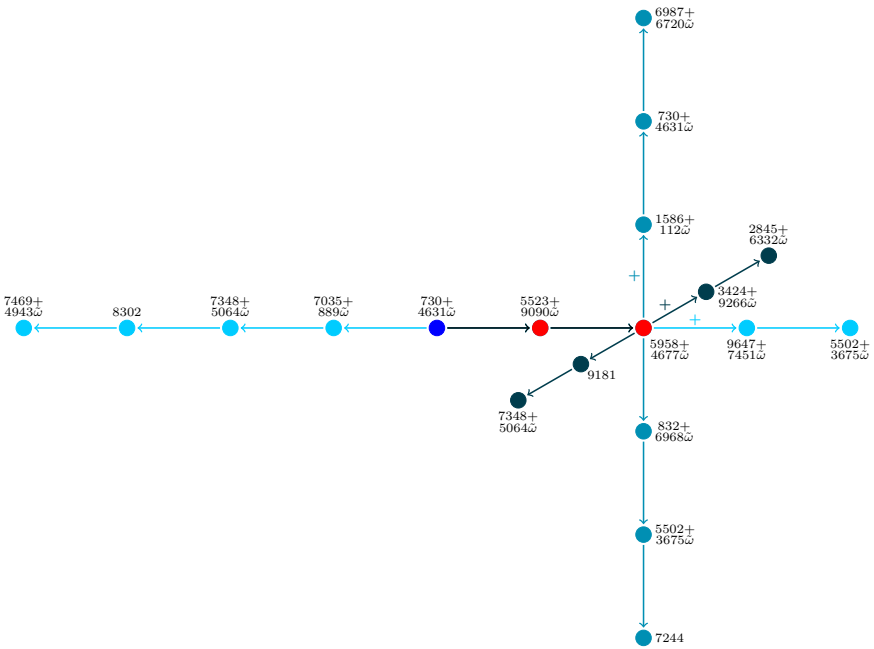
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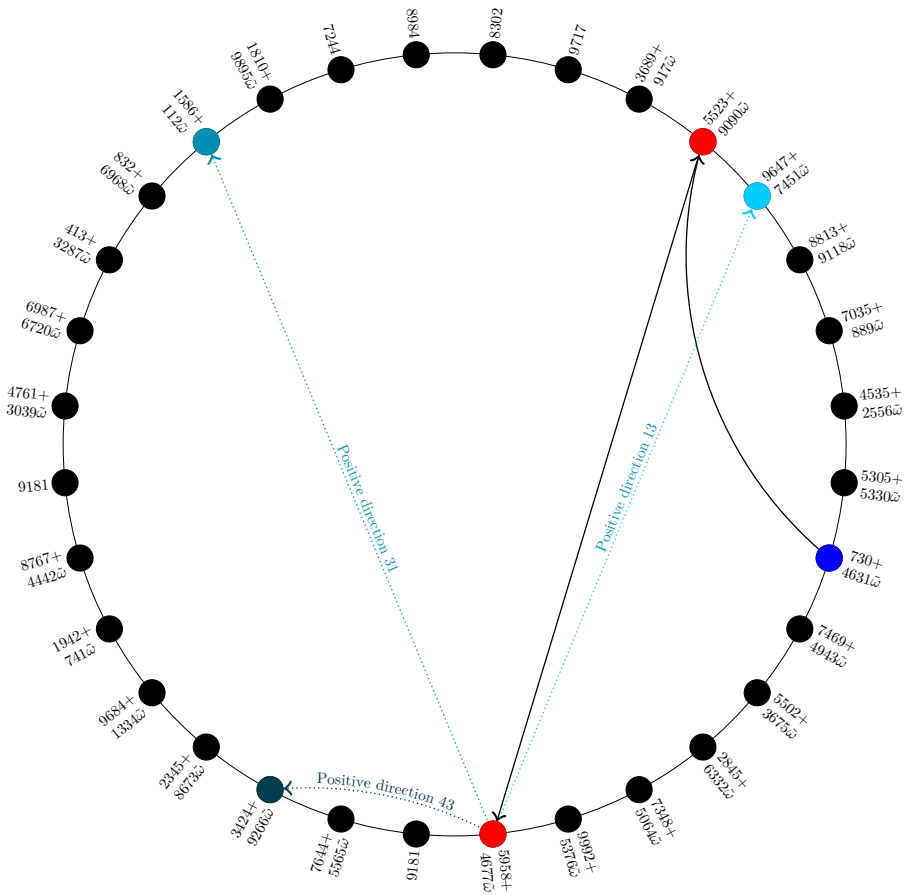




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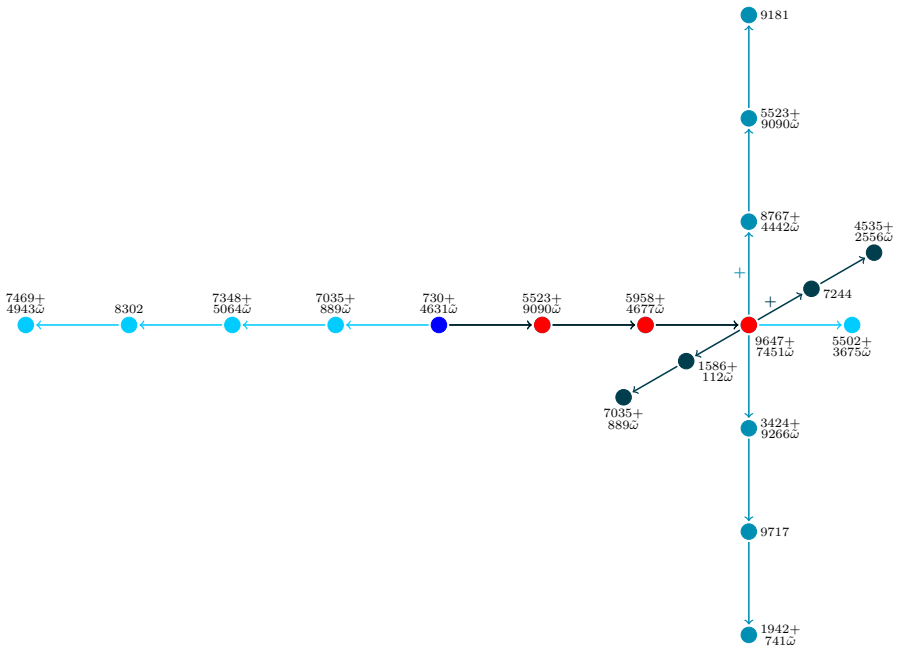
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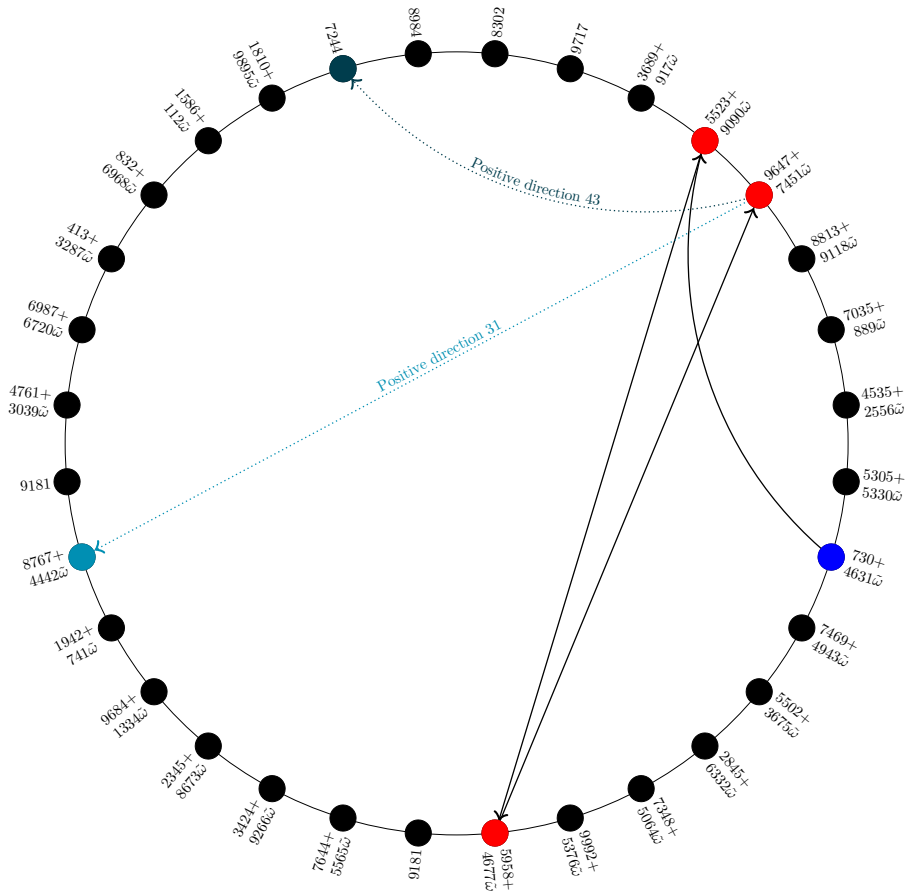




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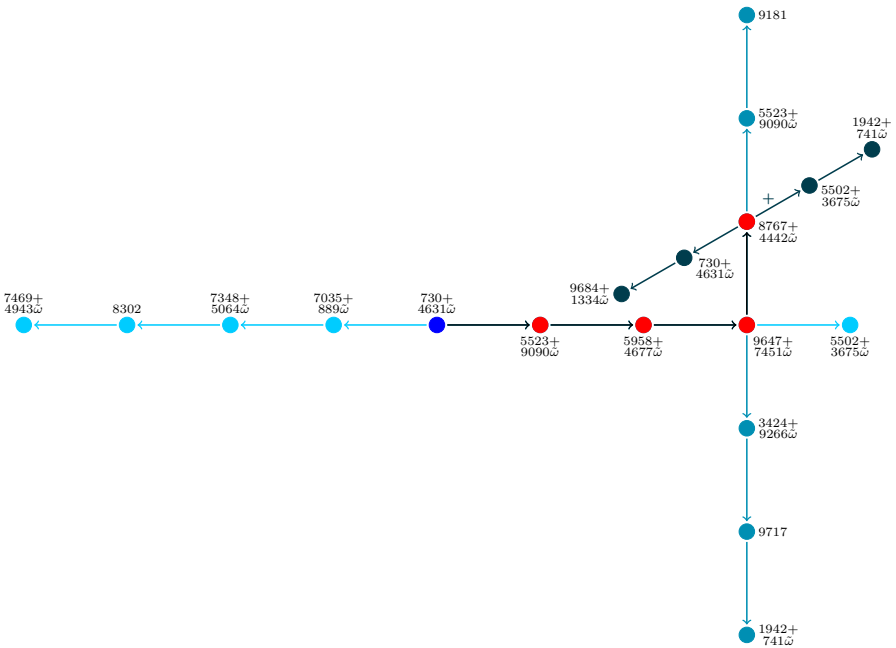
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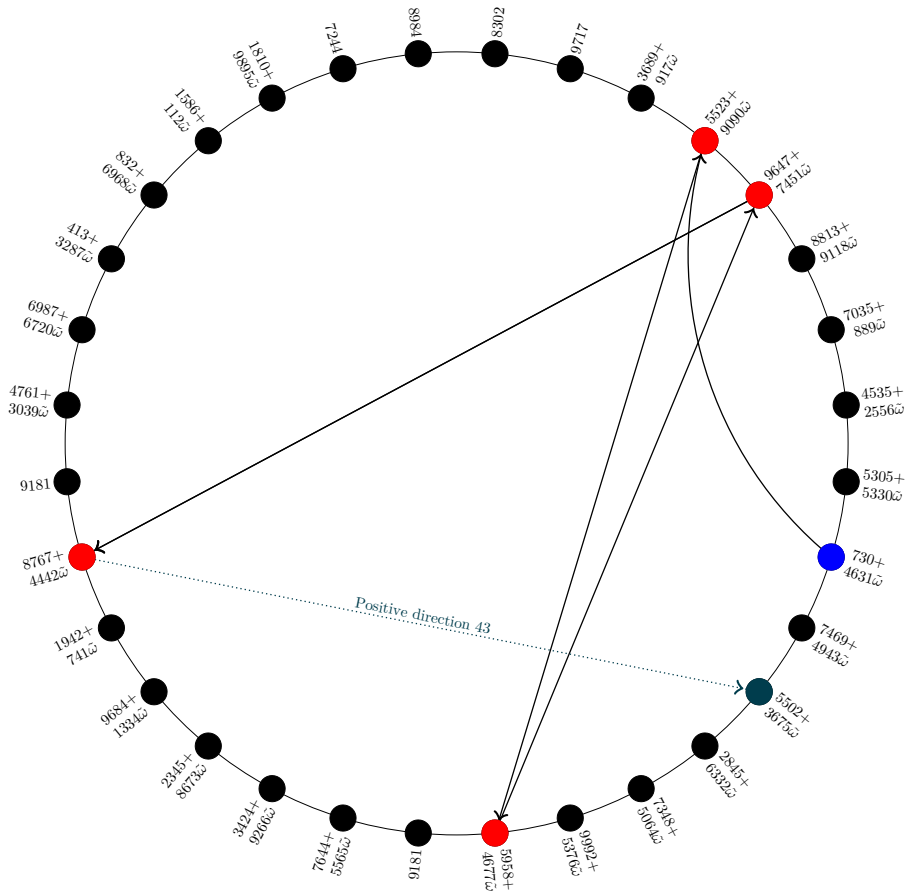




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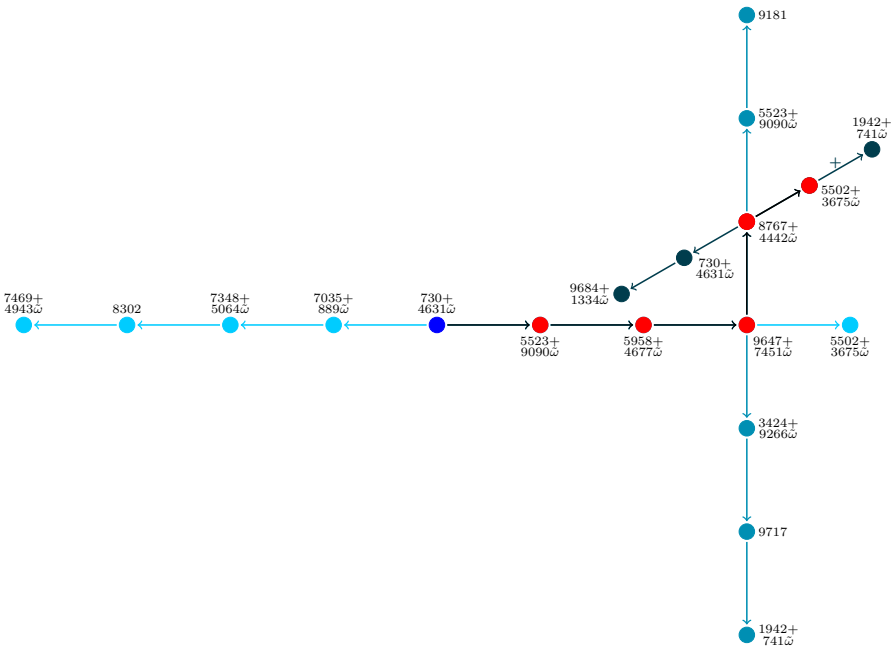
Bob secret key: $\mathfrak{l}_1^3\mathfrak{l}_2\mathfrak{l}_3^2$

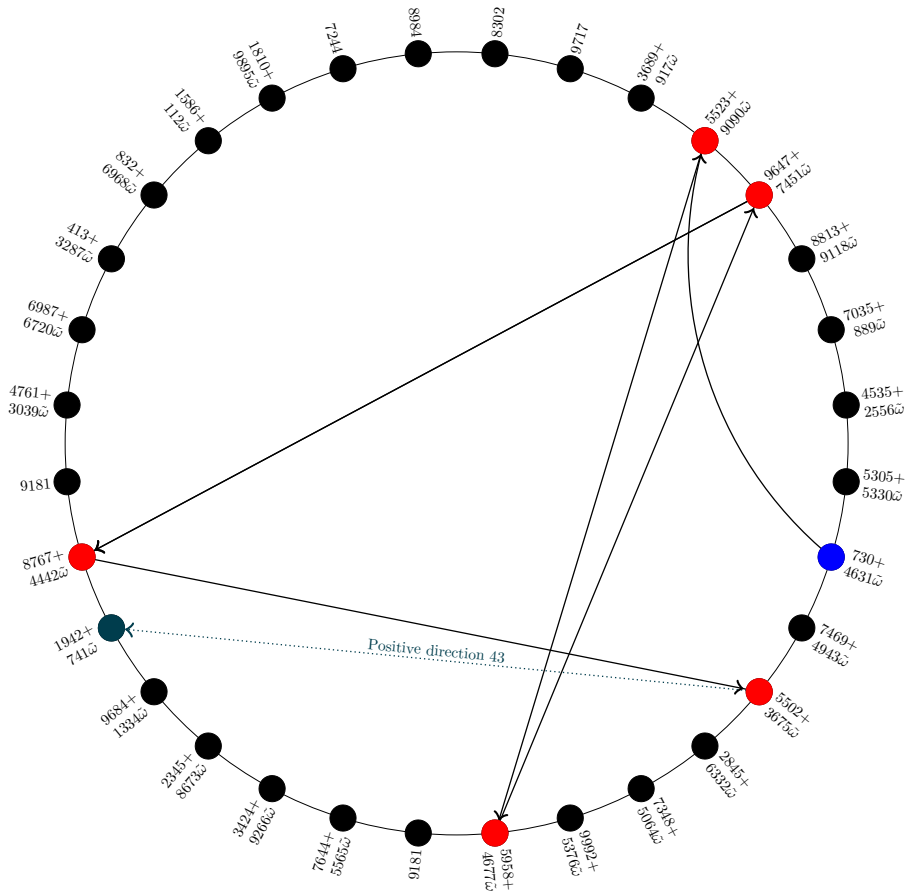




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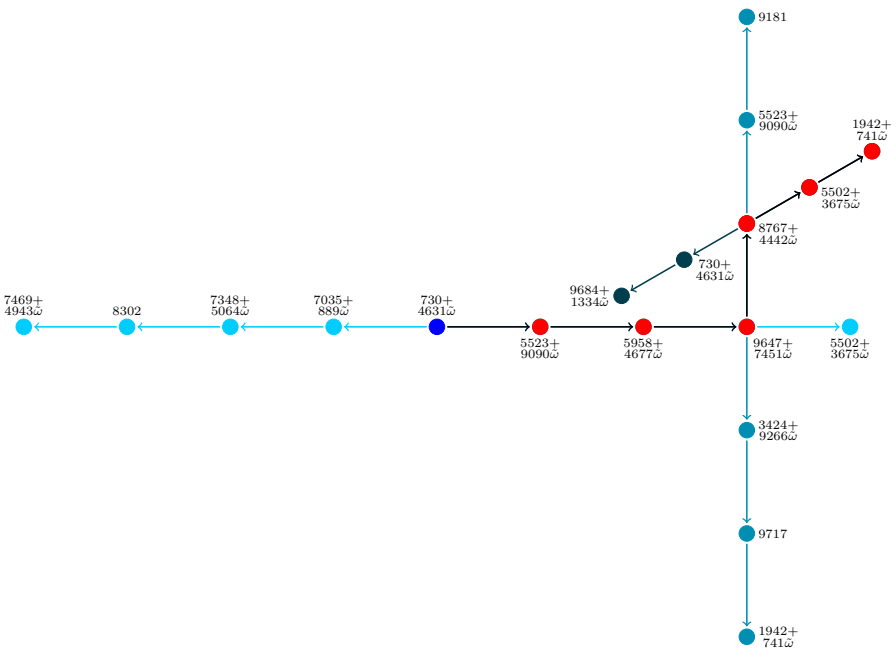
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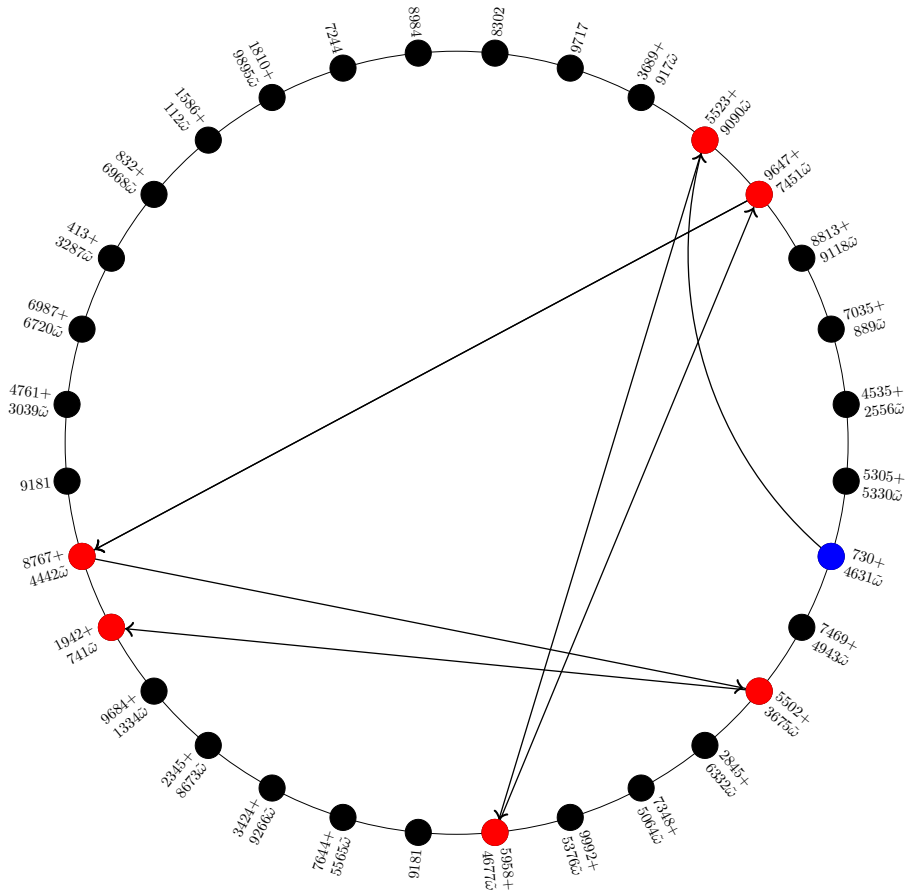




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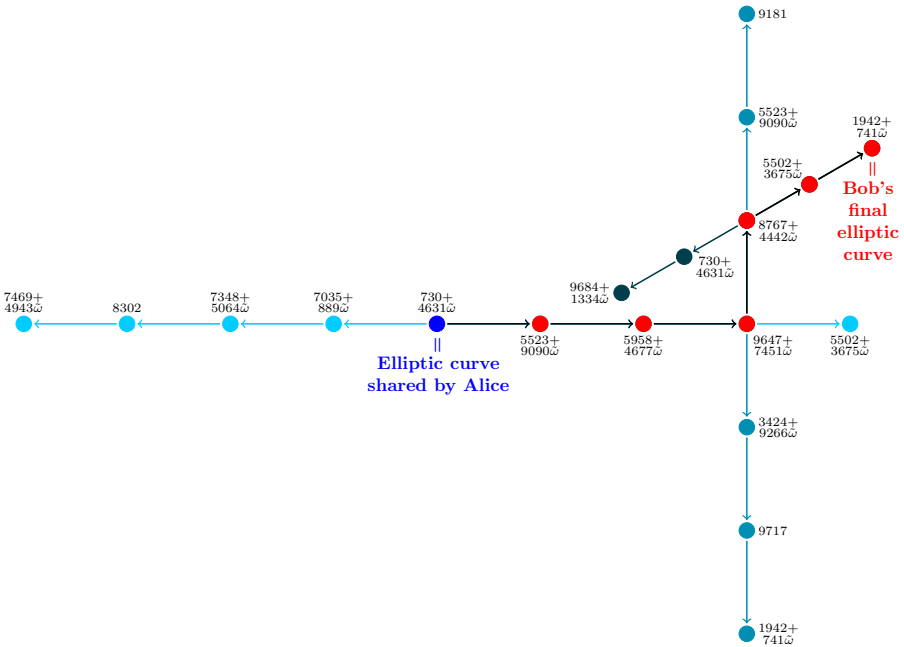
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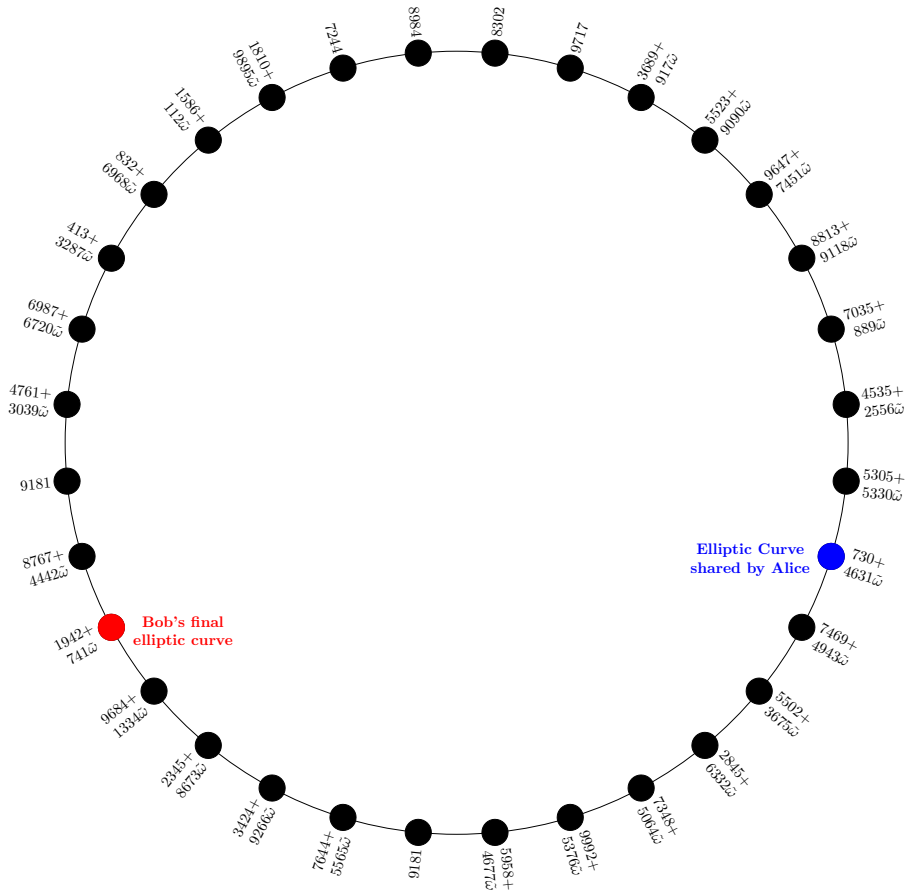




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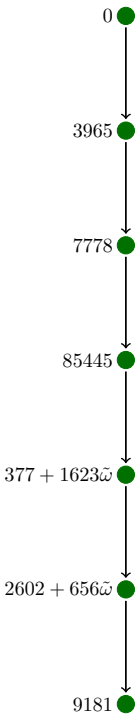
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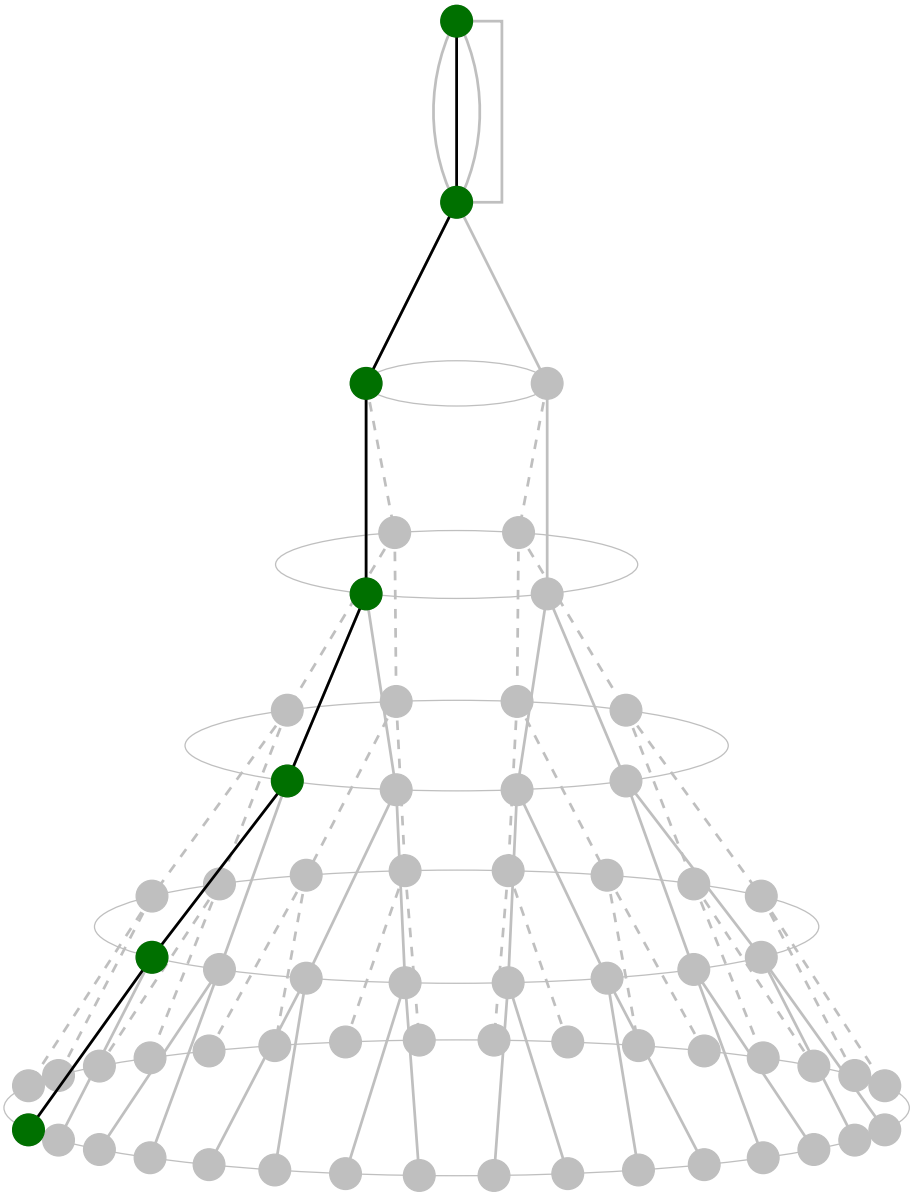




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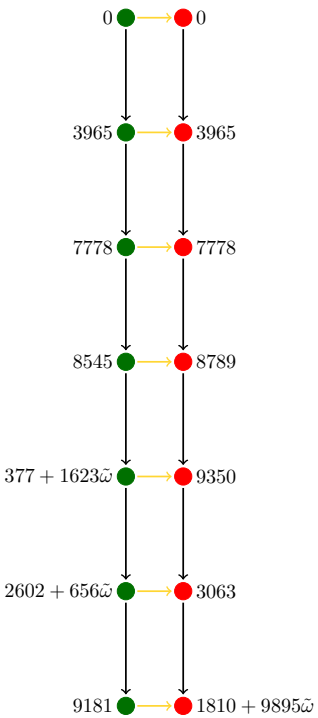
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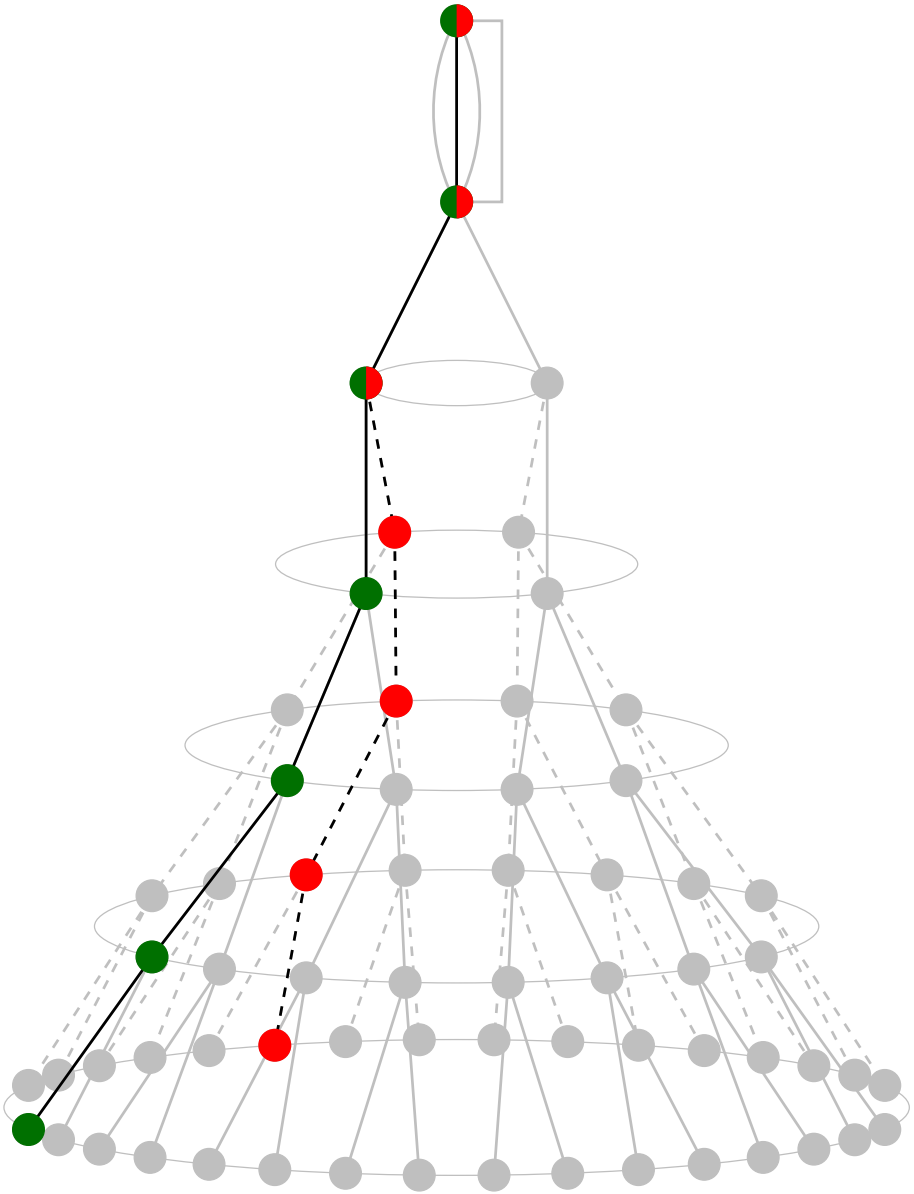




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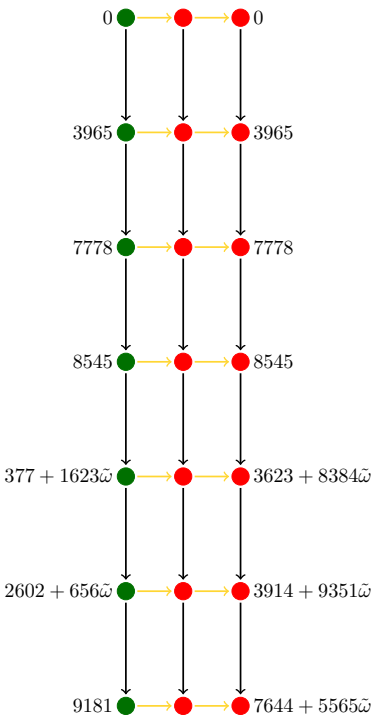
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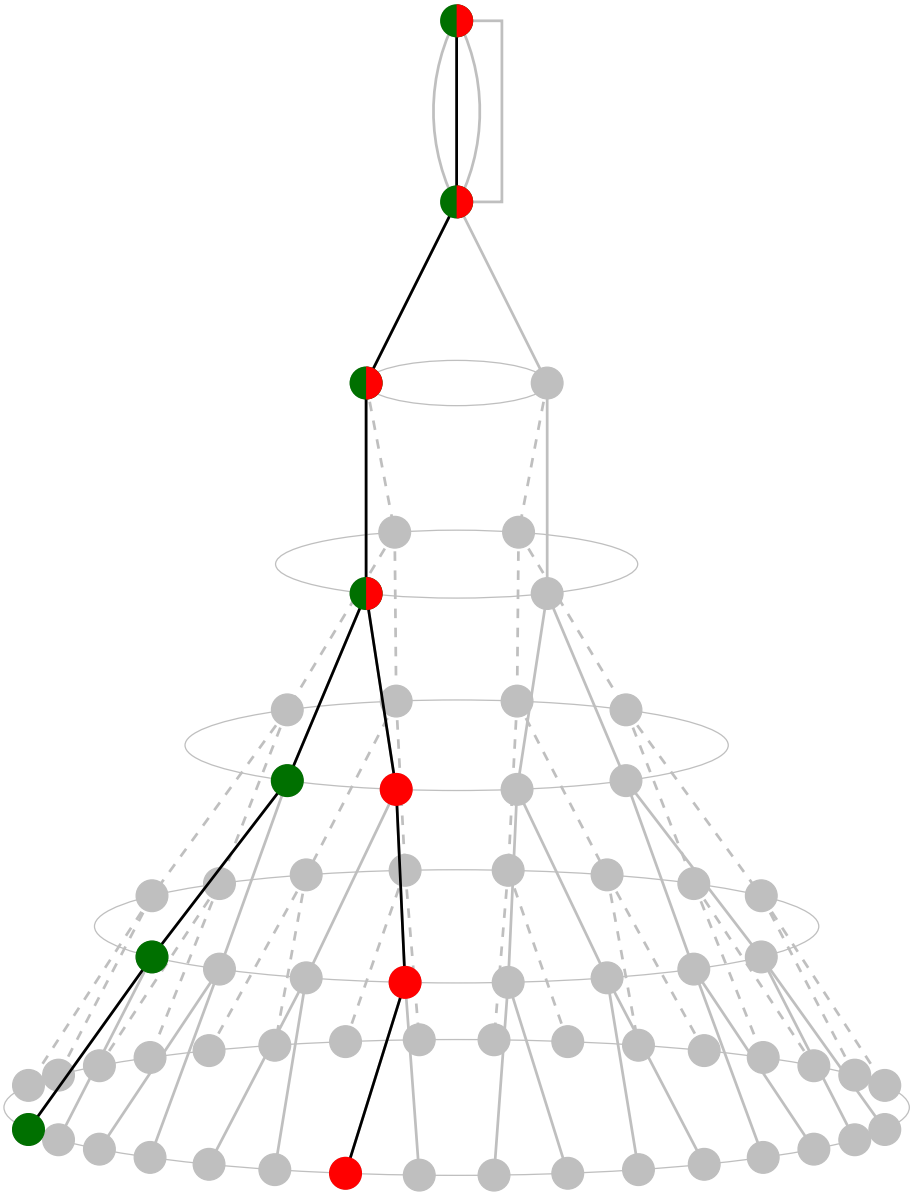




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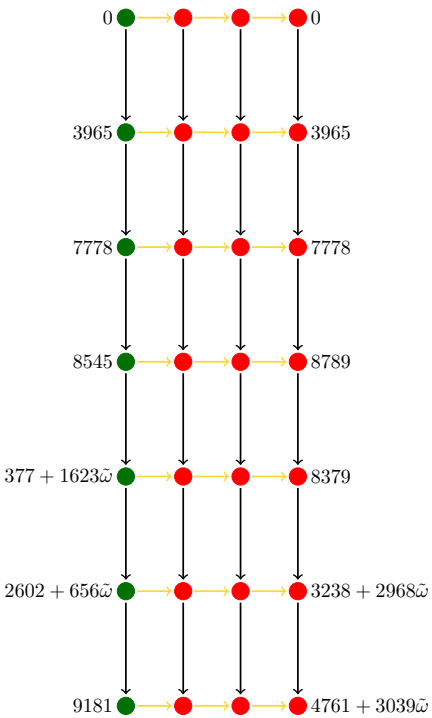
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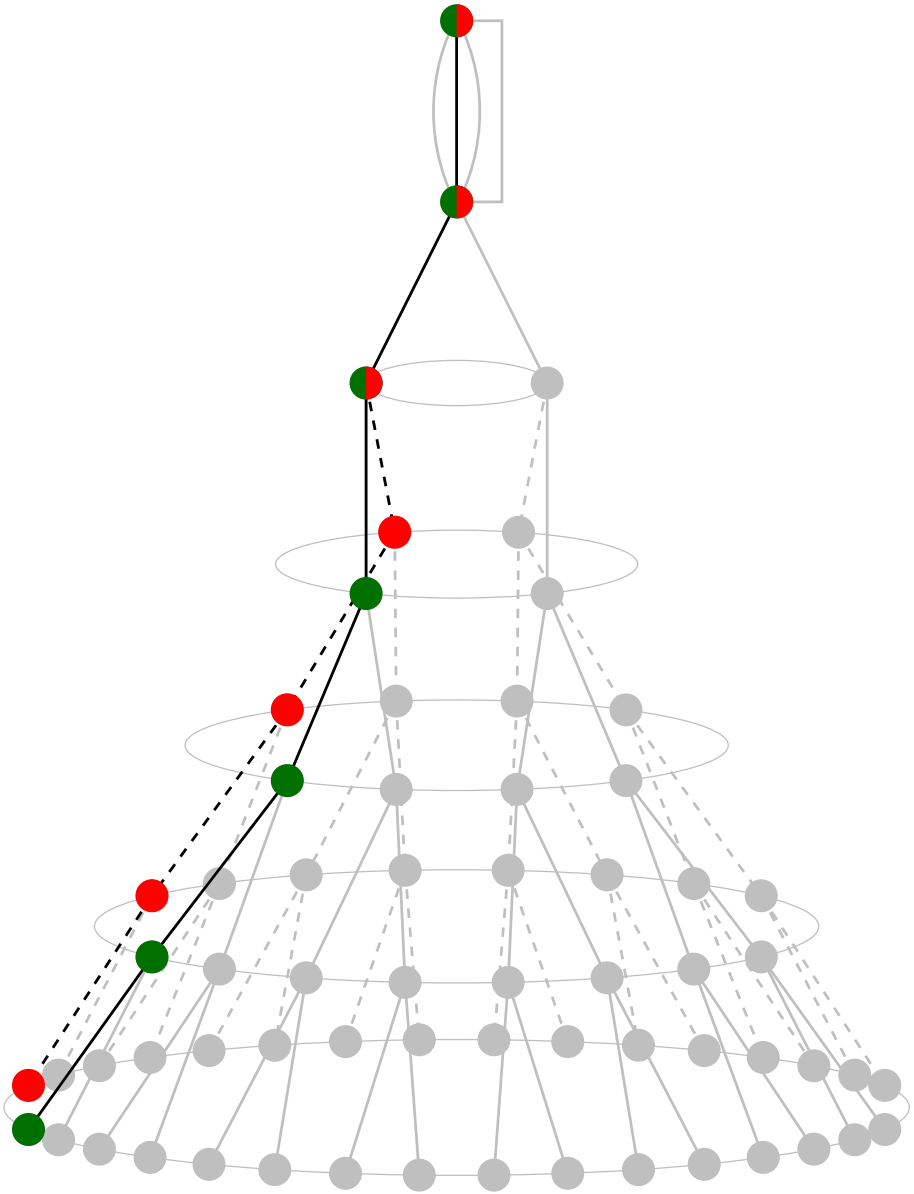




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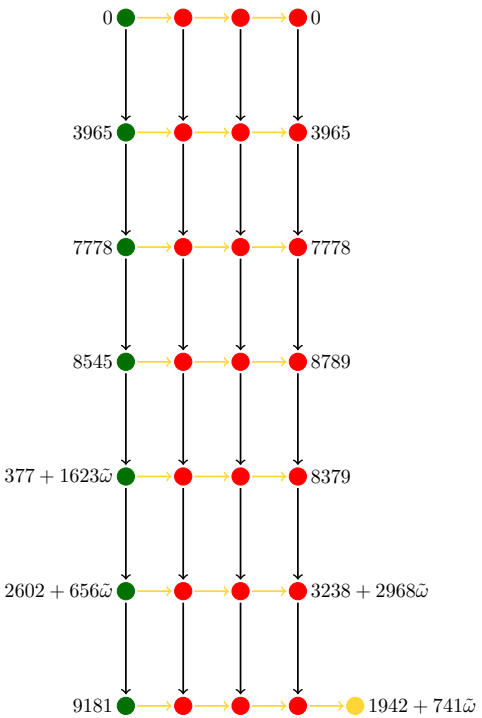
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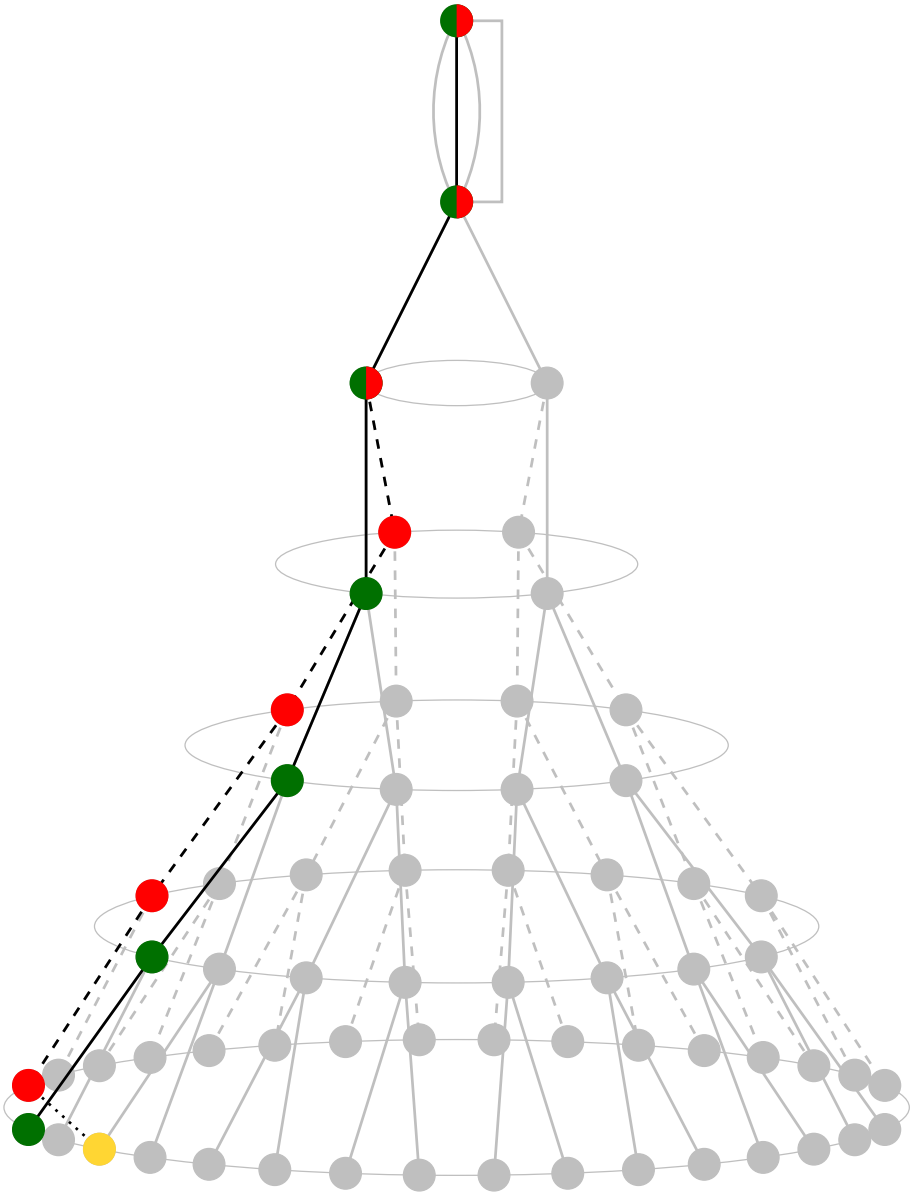




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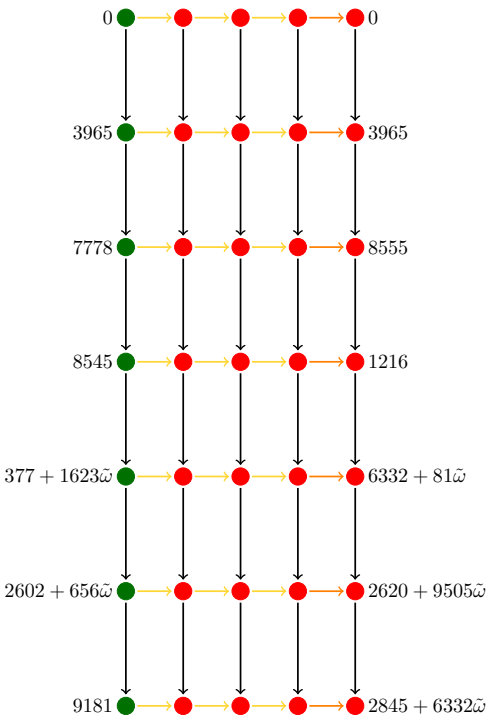
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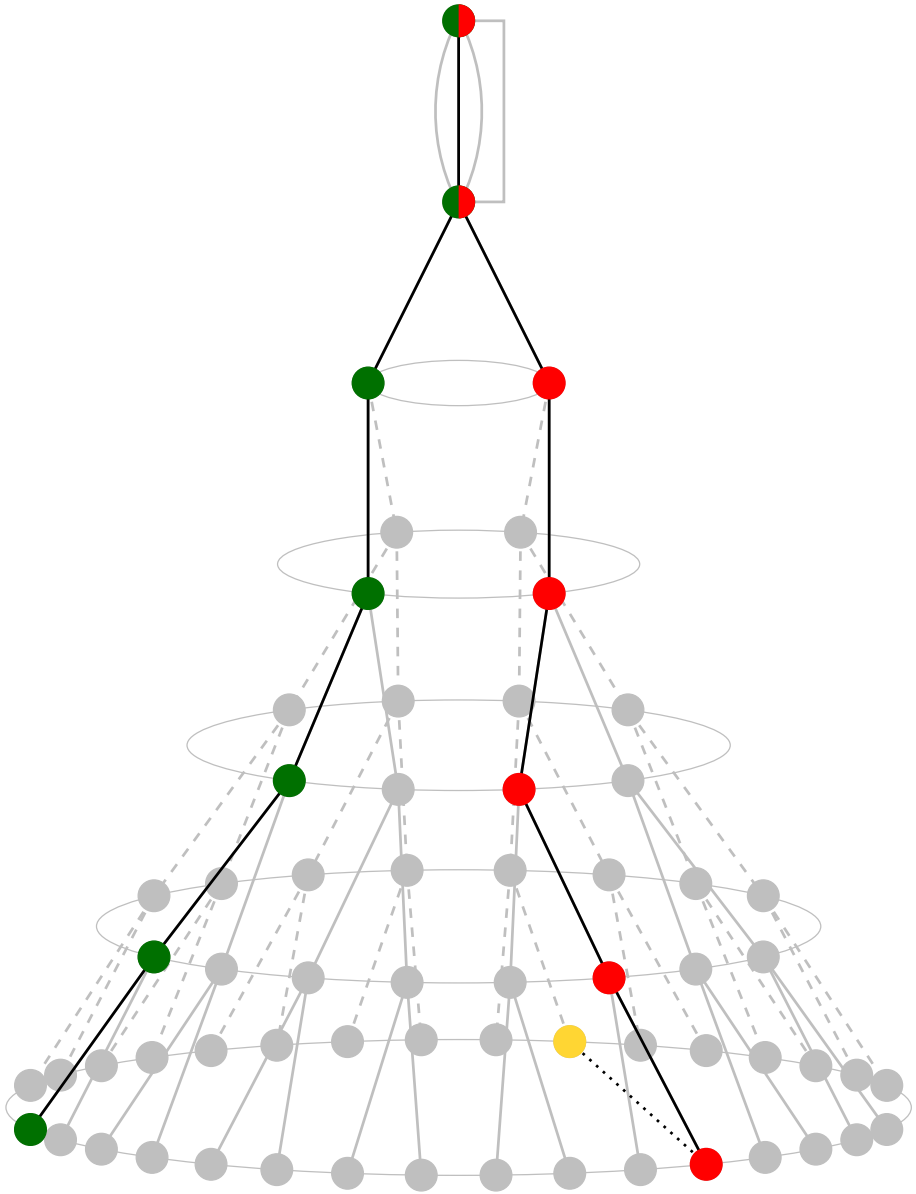




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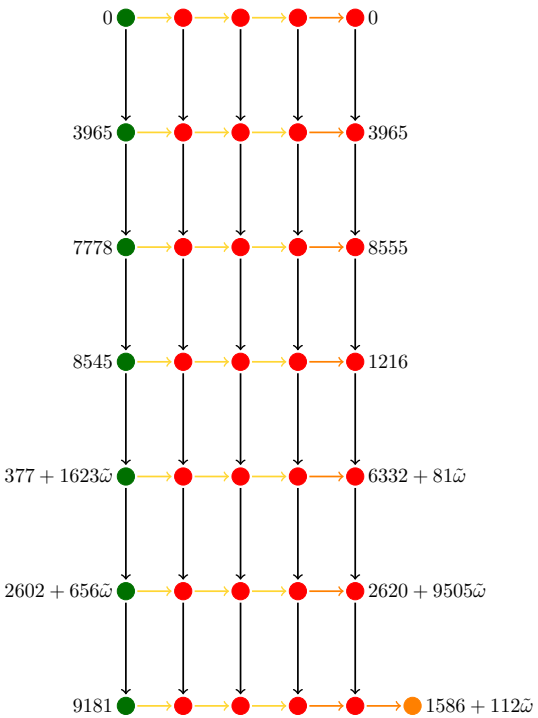
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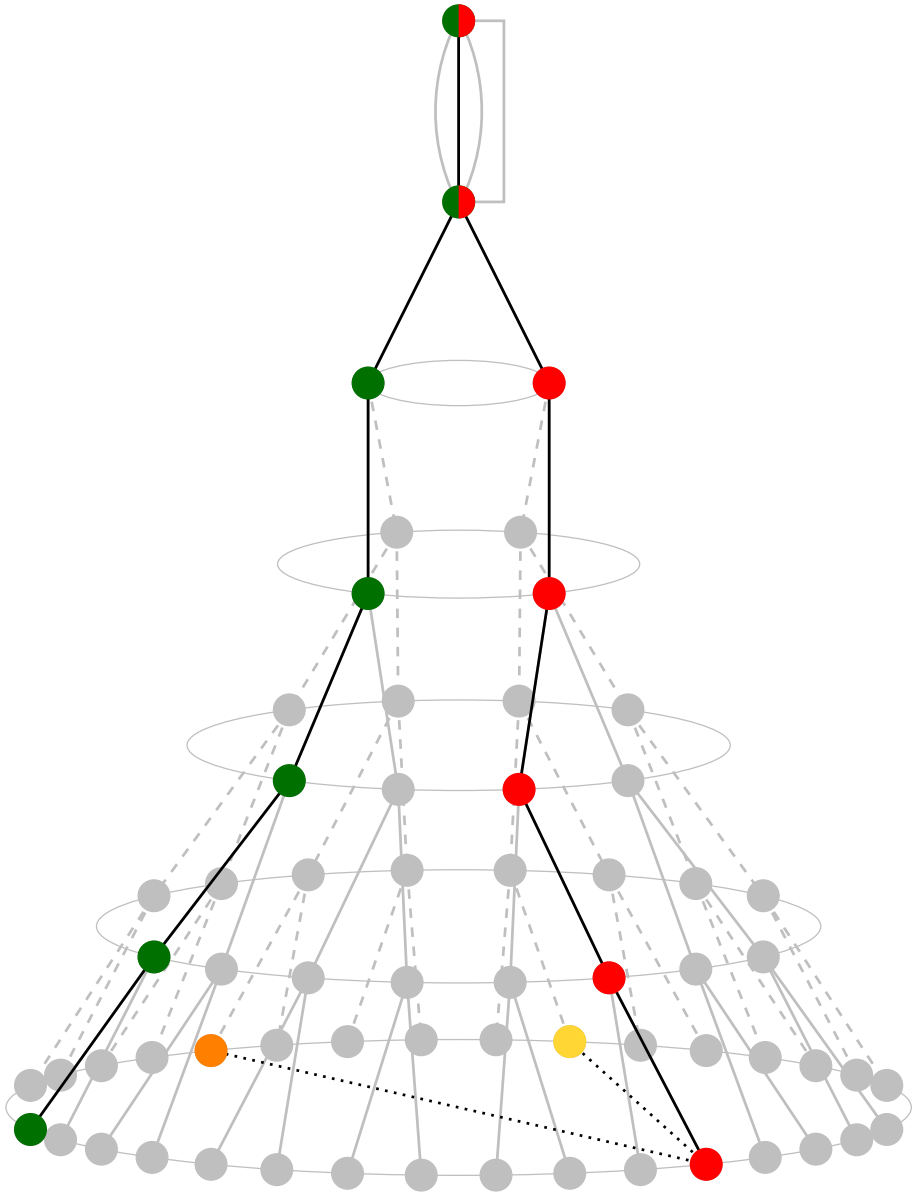




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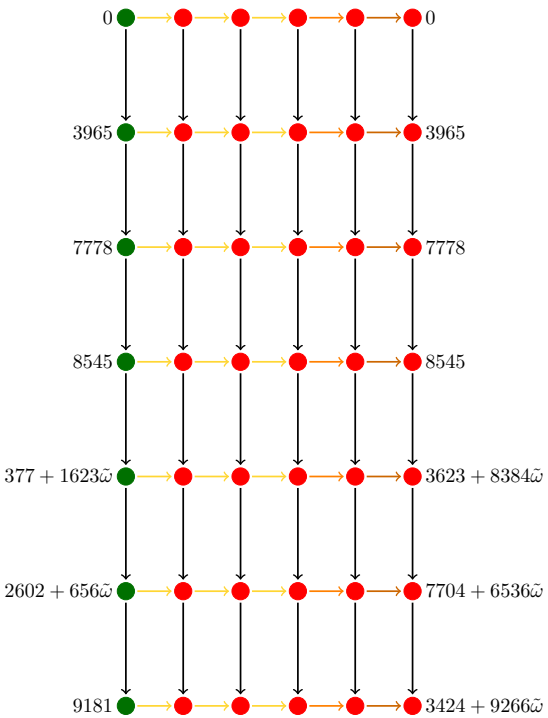
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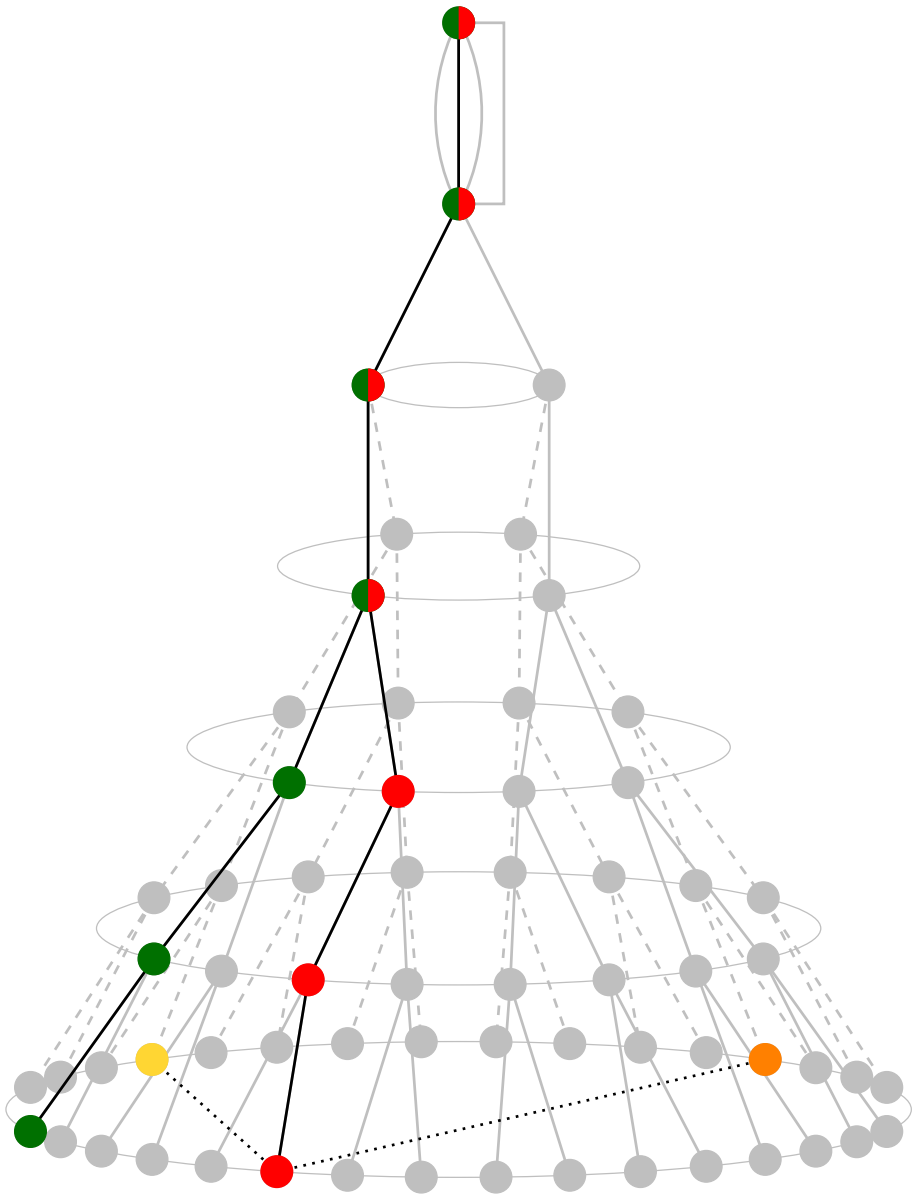




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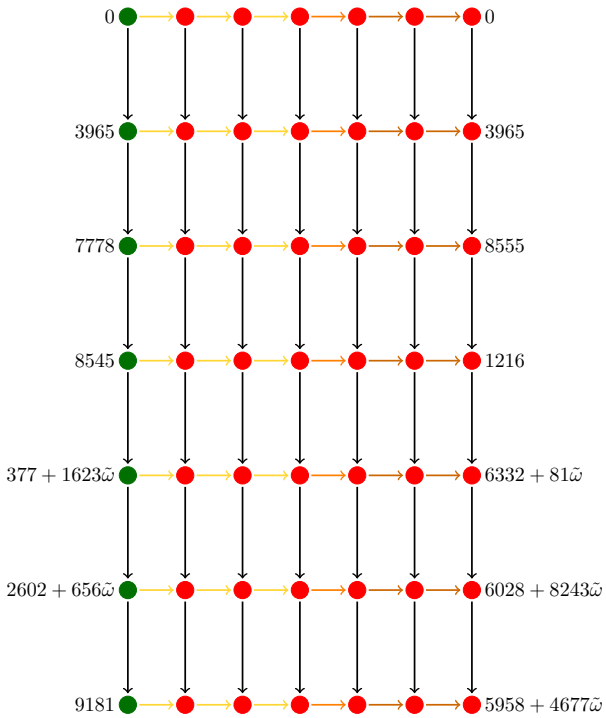
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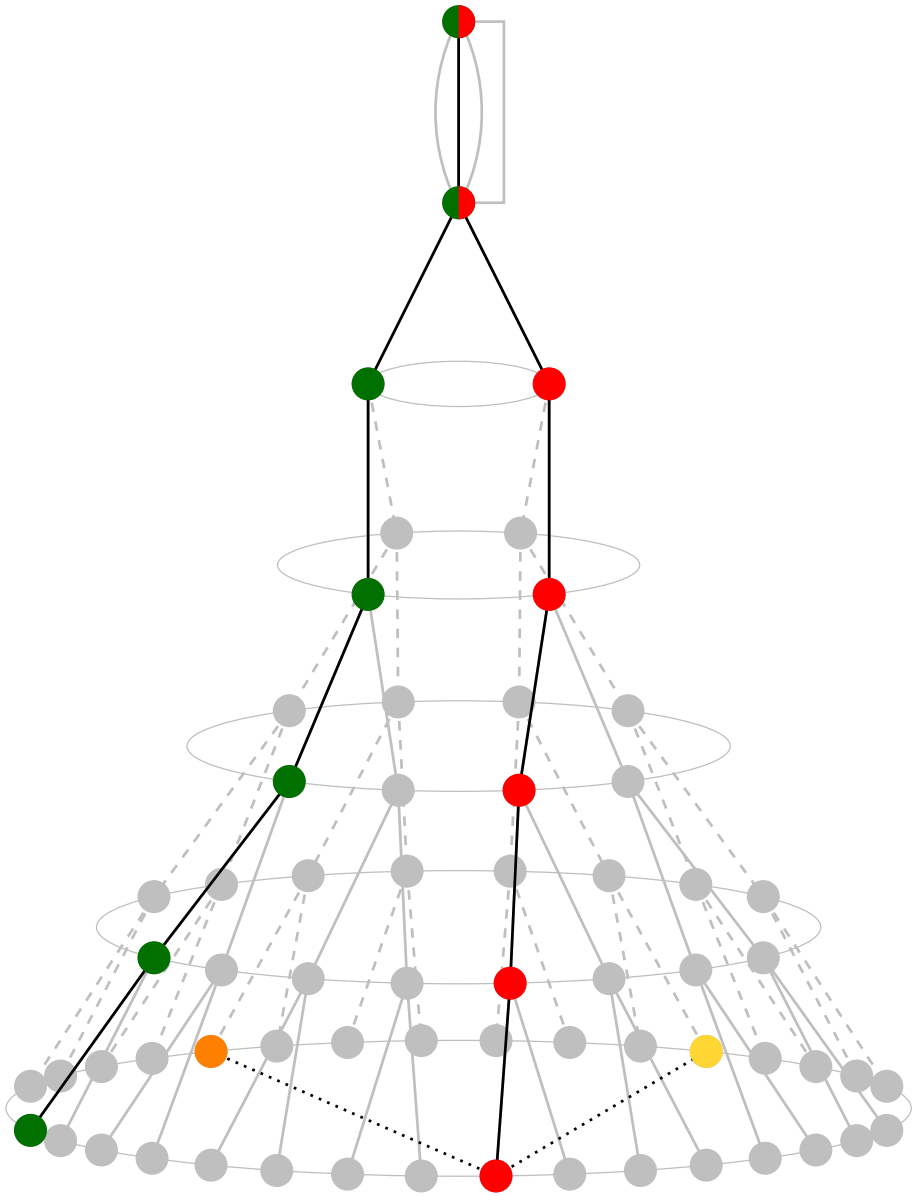




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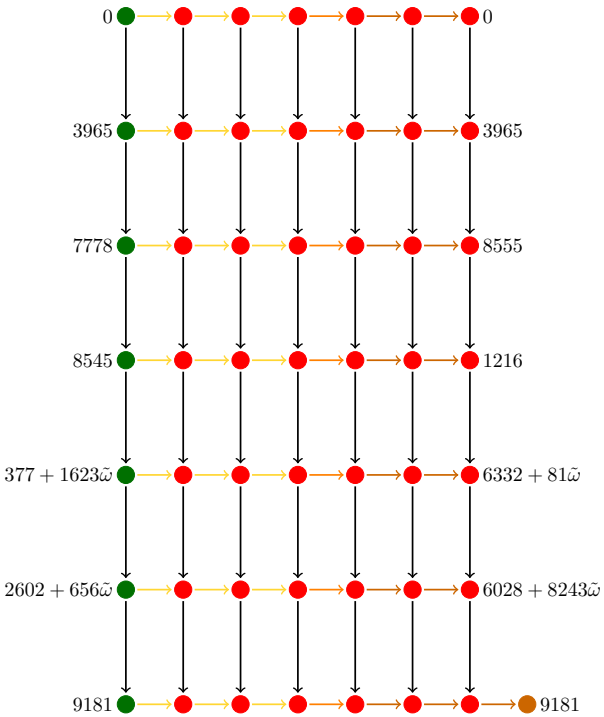
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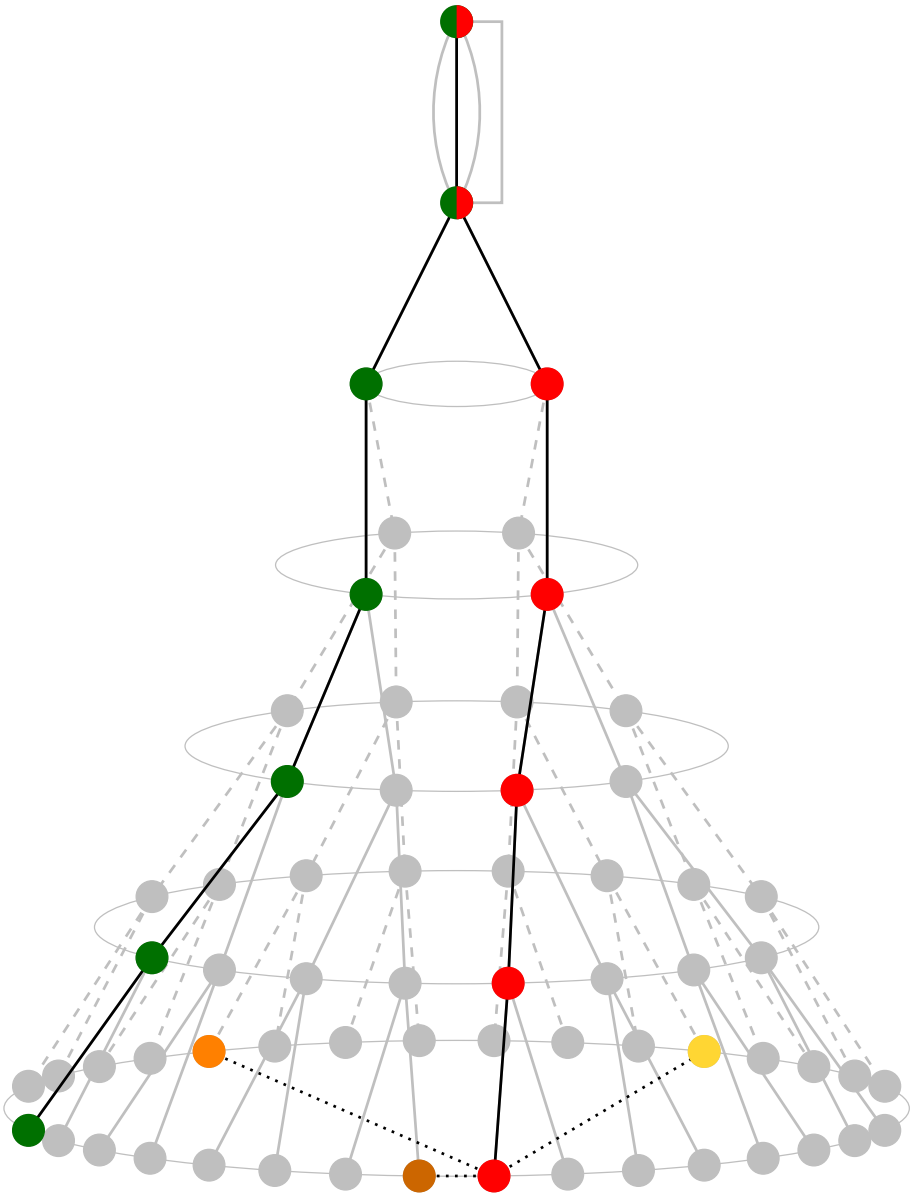




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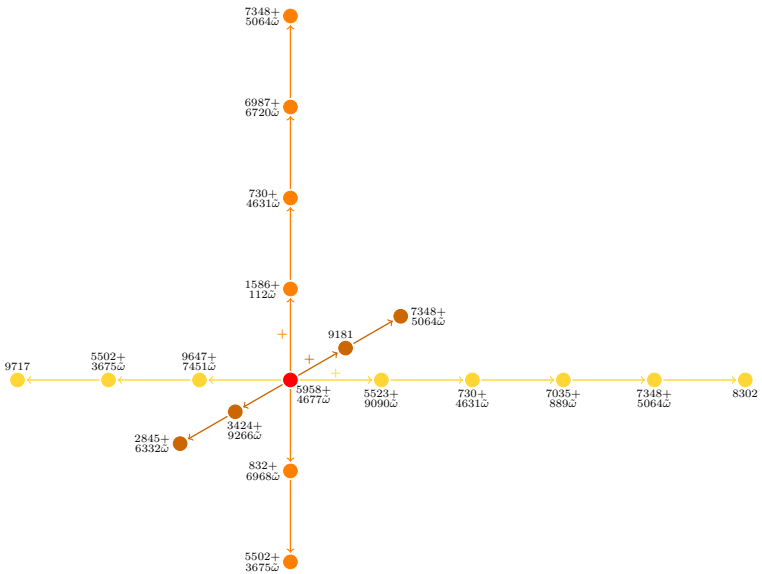
Bob secret key: $\mathfrak{l}_1^3\mathfrak{l}_2\mathfrak{l}_3^2$

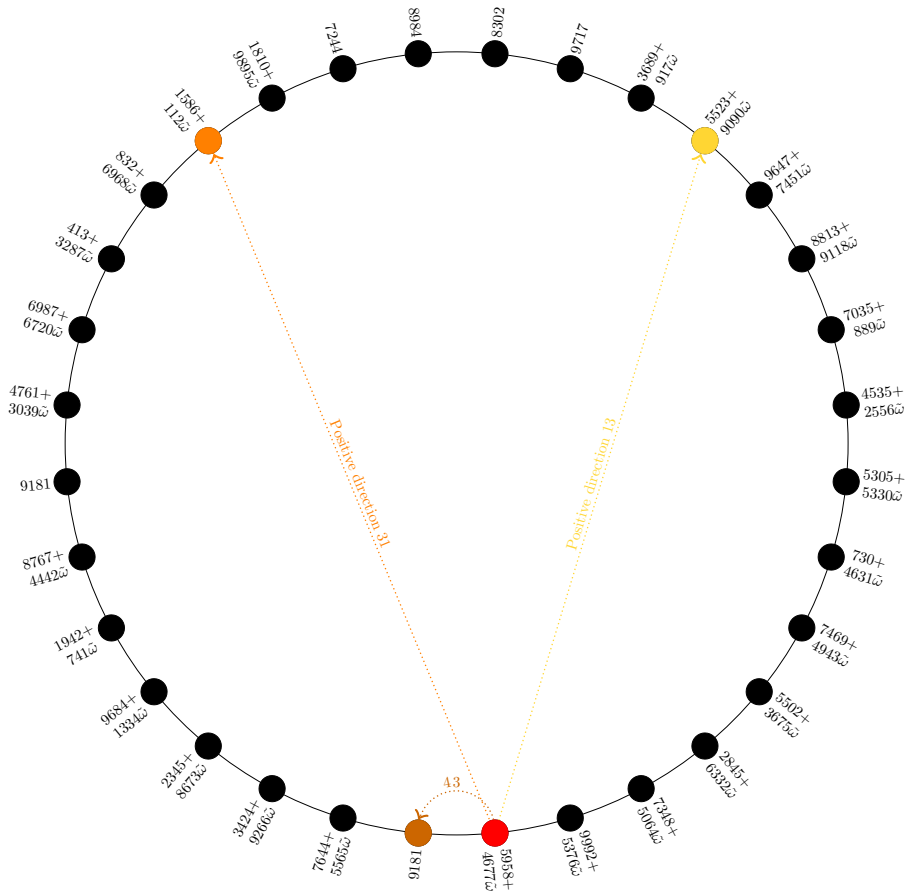




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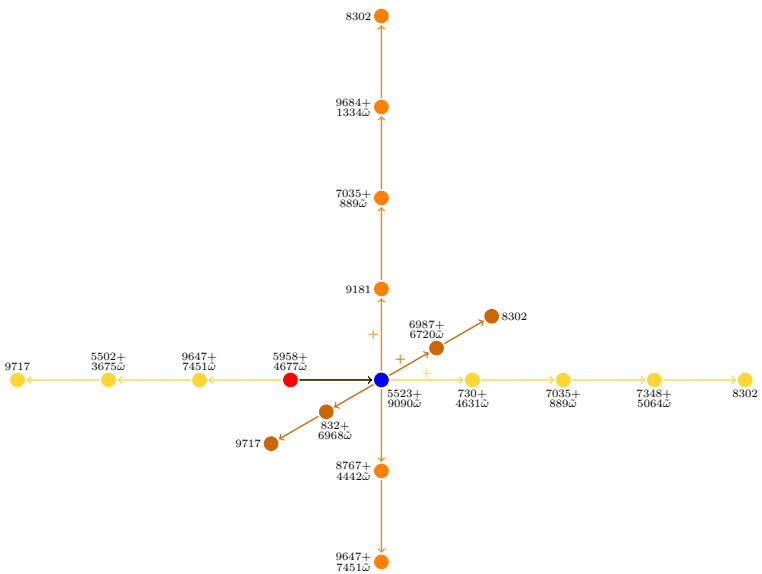
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

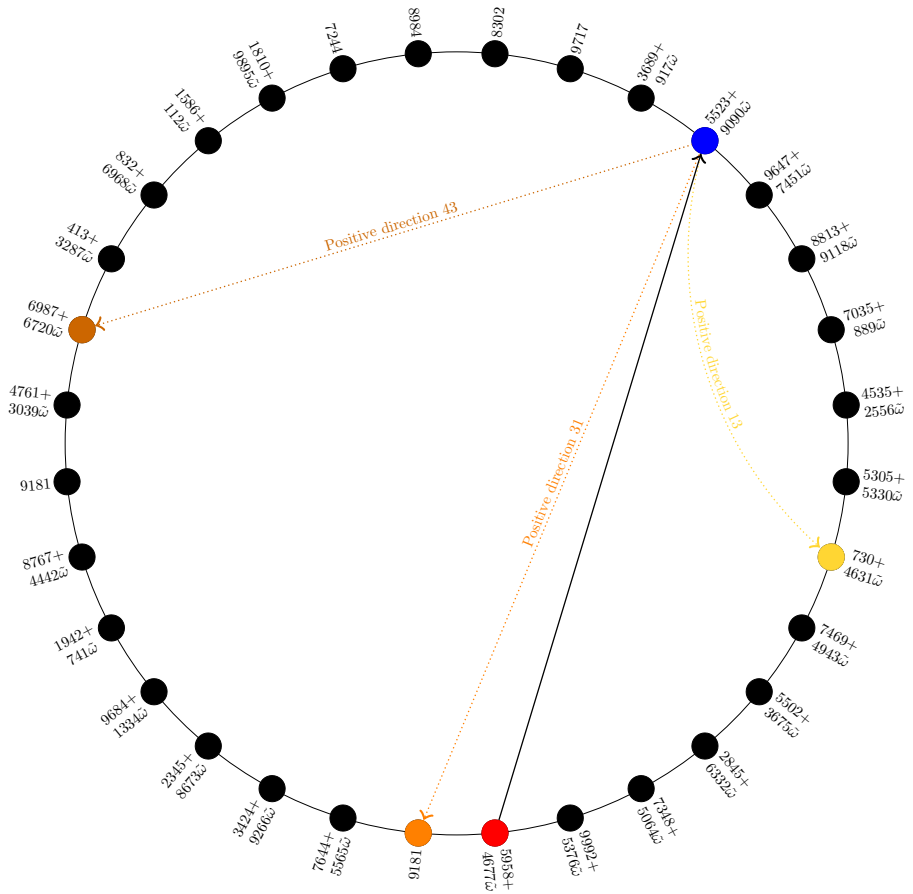




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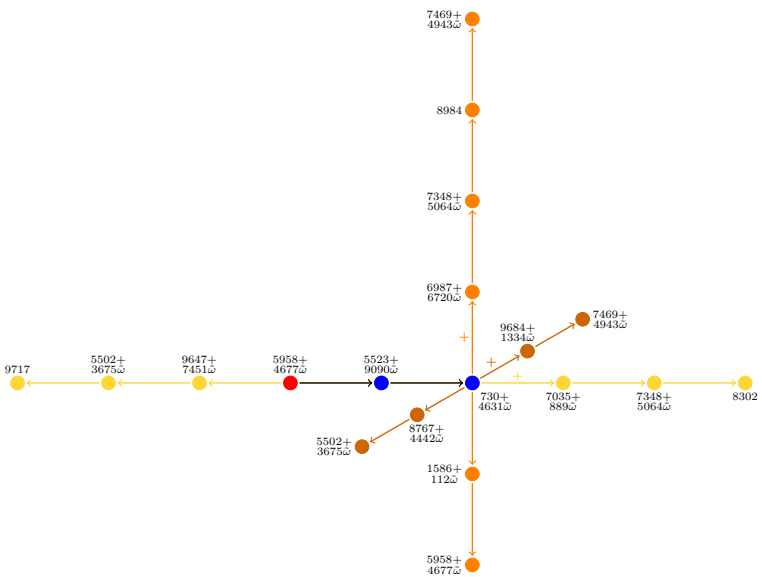
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

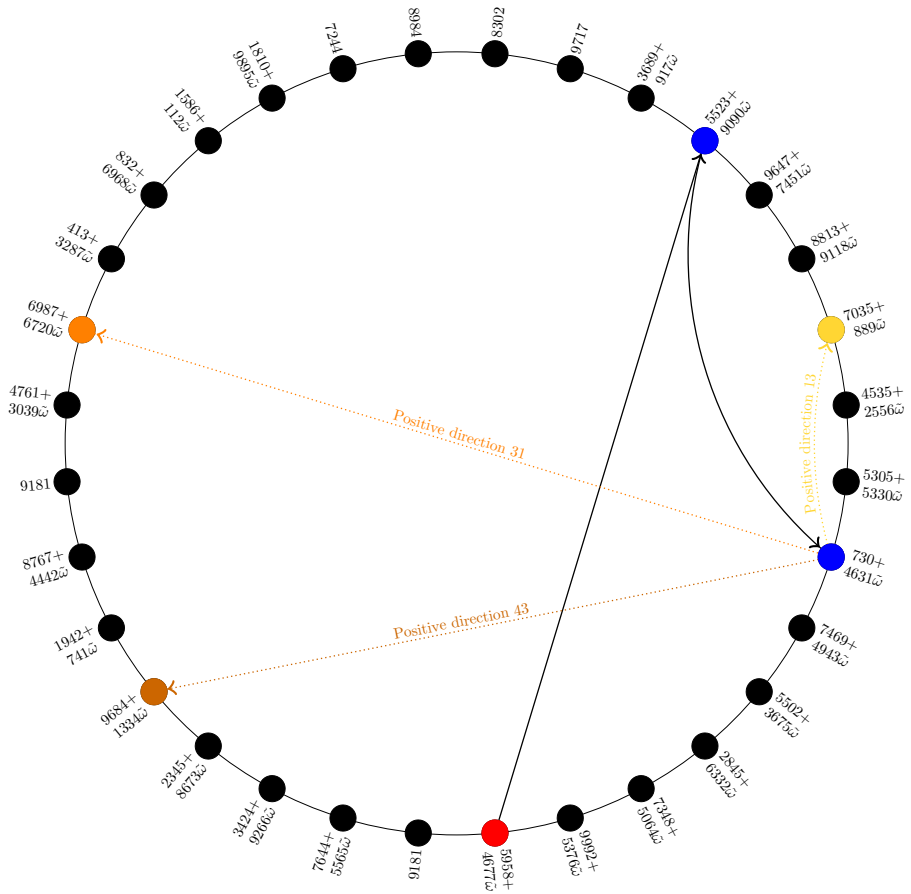




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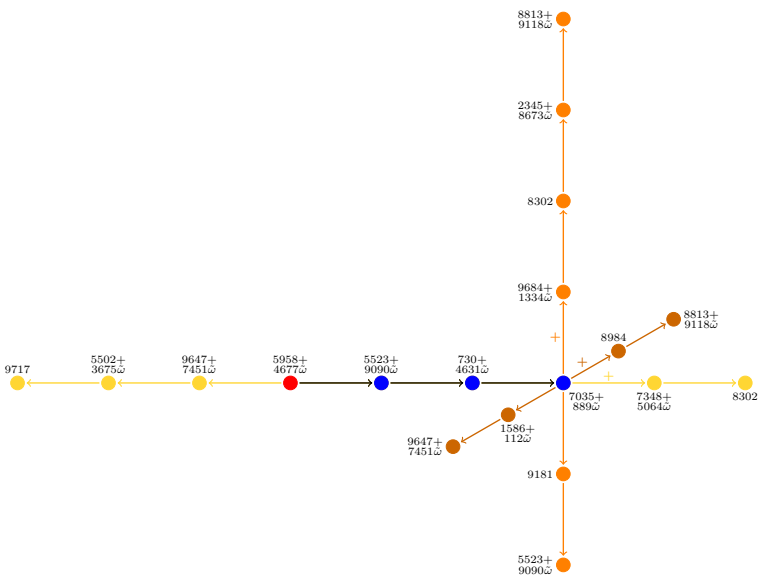
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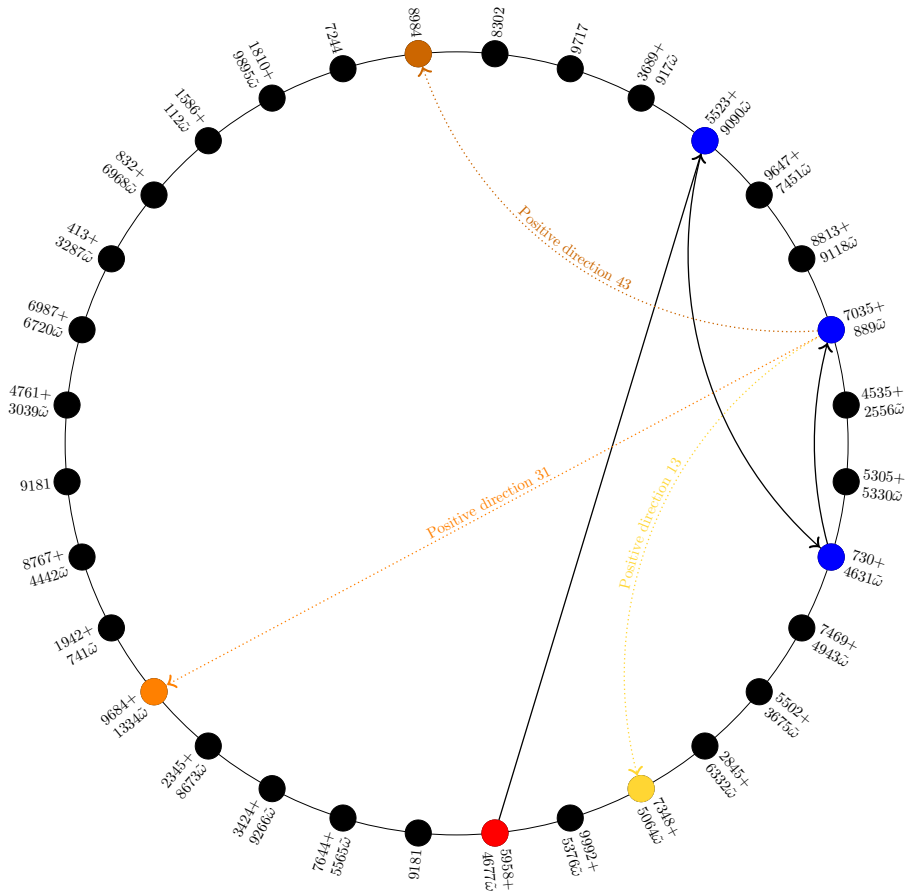




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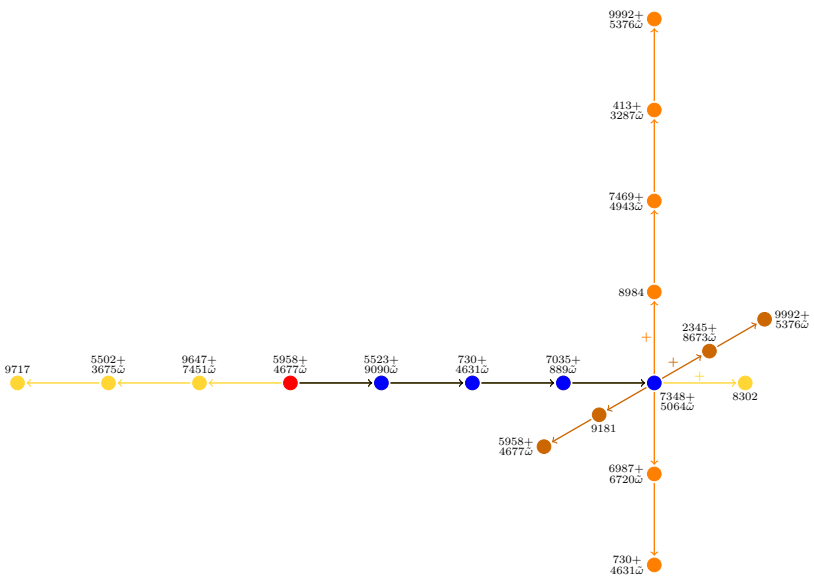
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

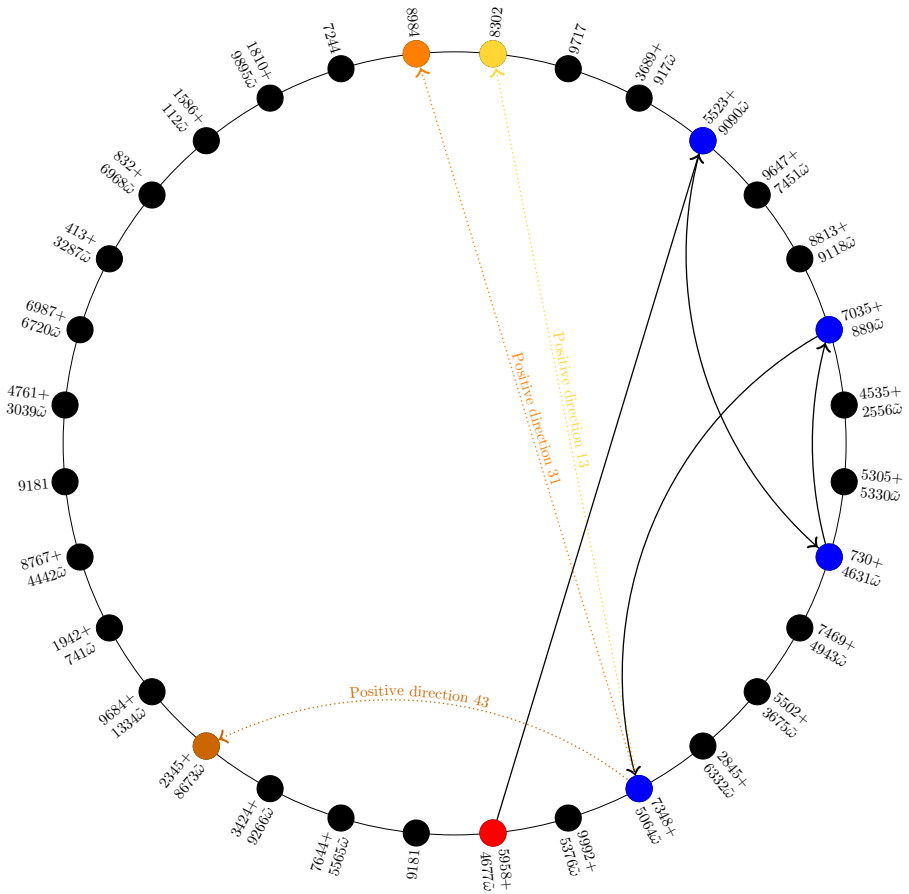




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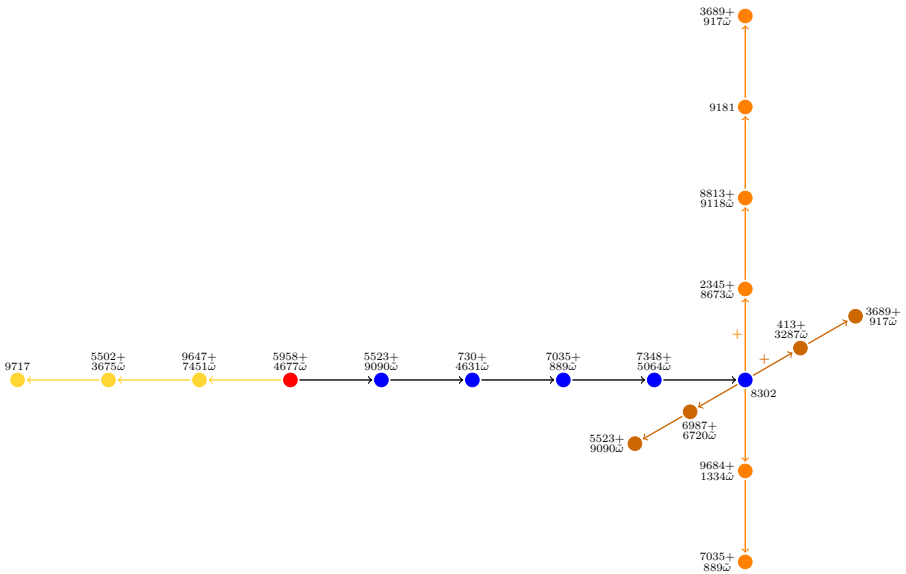
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

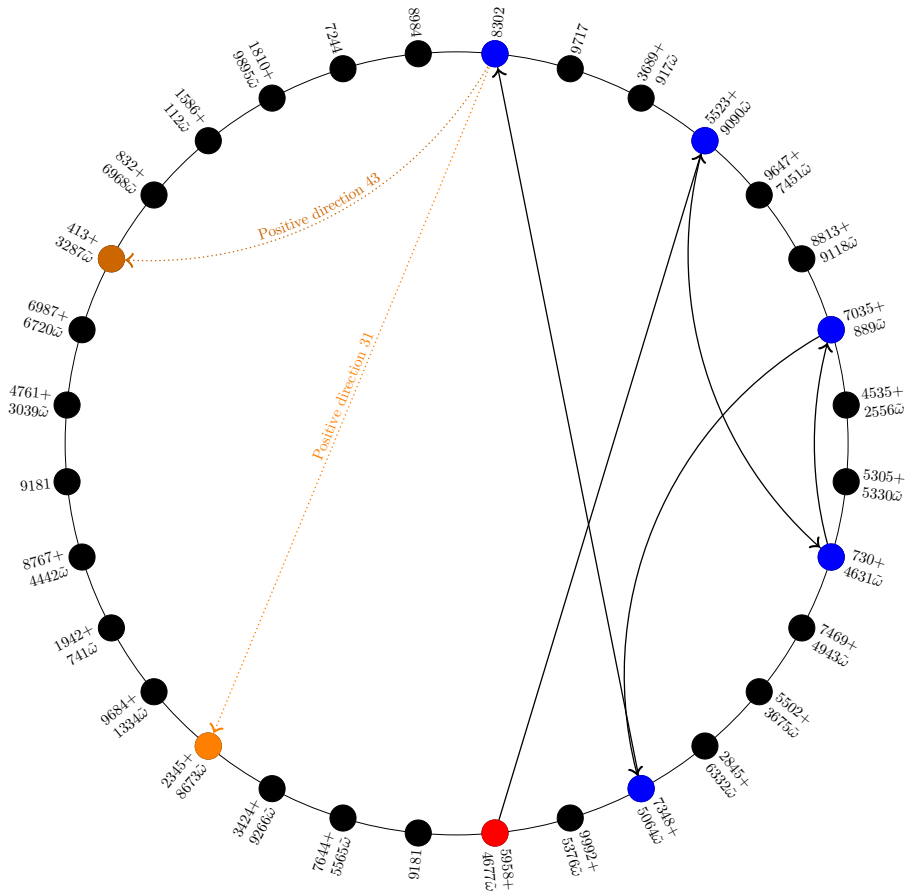




OSIDH PROTOCOL - AN EXAMPLE

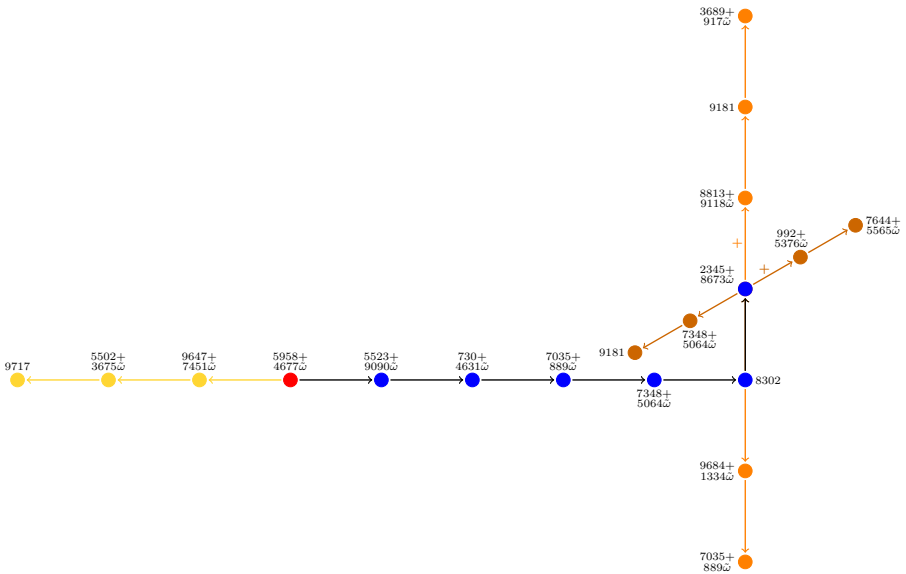
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

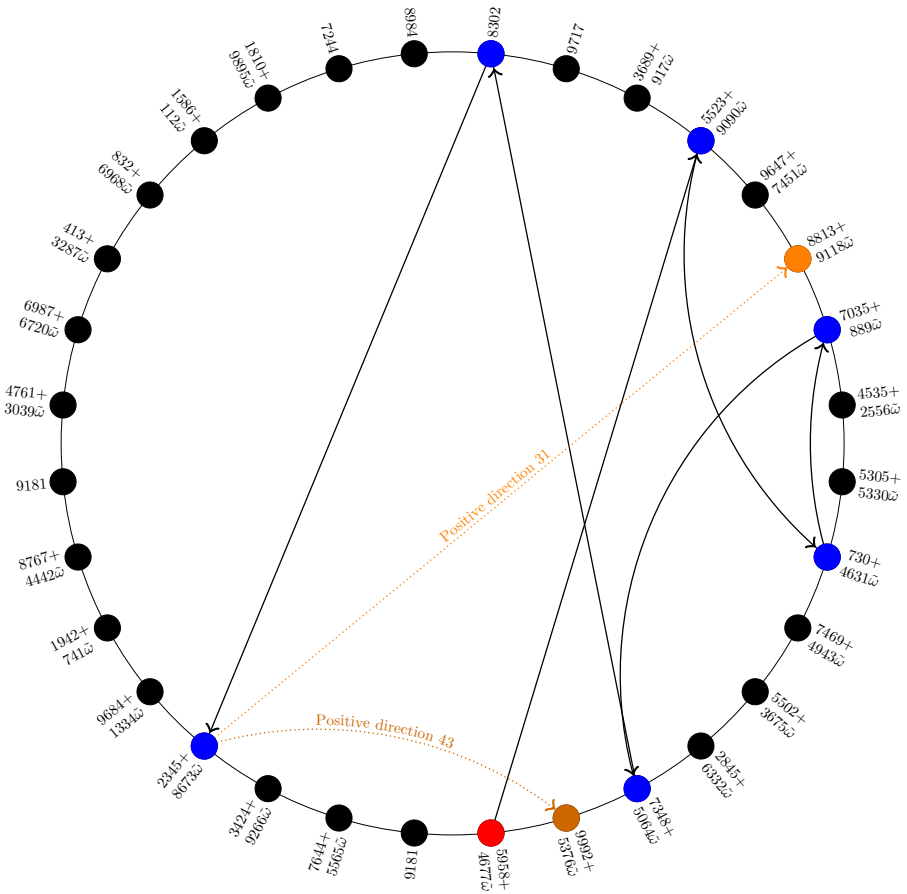




OSIDH PROTOCOL - AN EXAMPLE

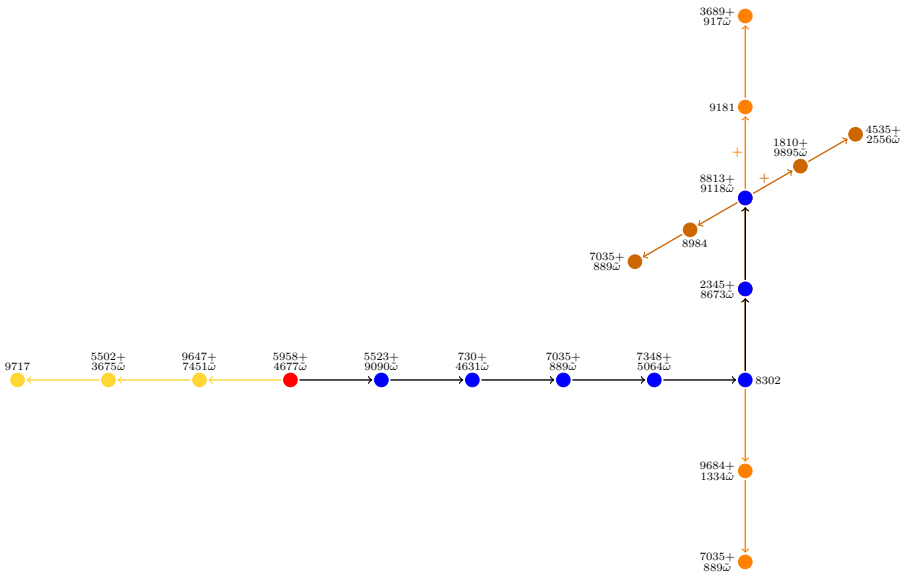
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

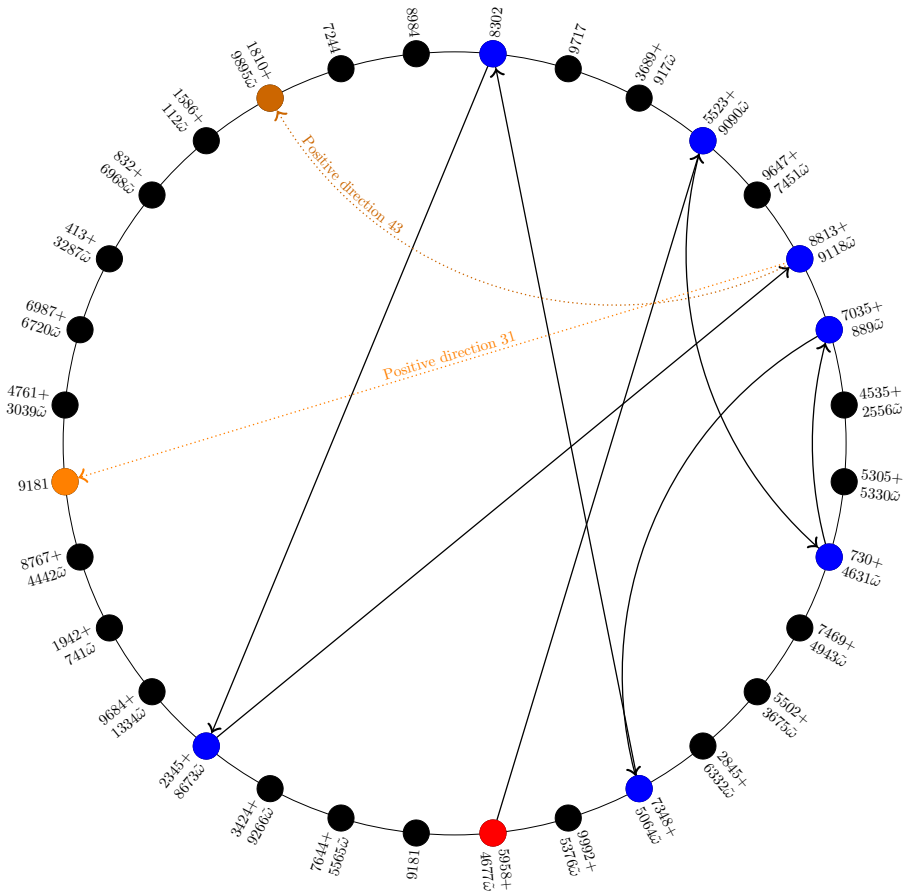




OSIDH PROTOCOL - AN EXAMPLE

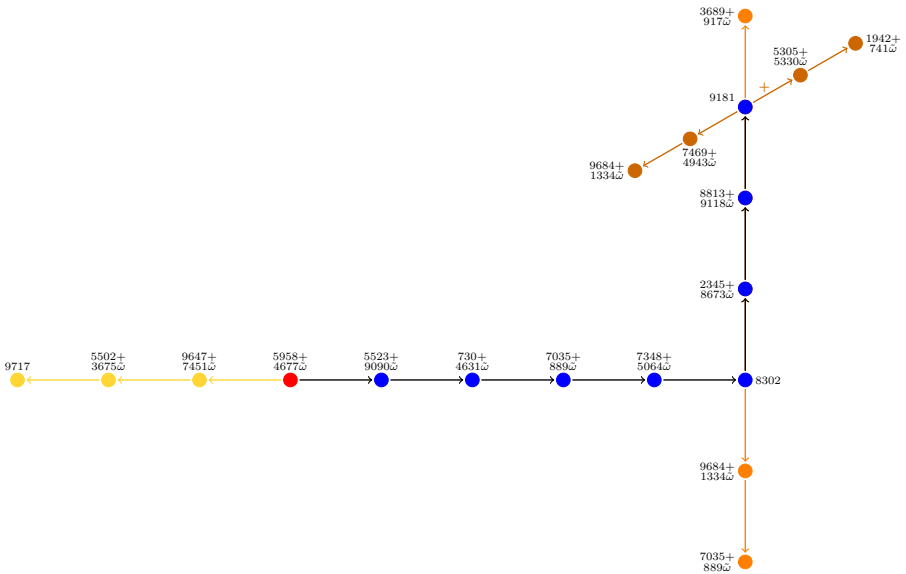
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

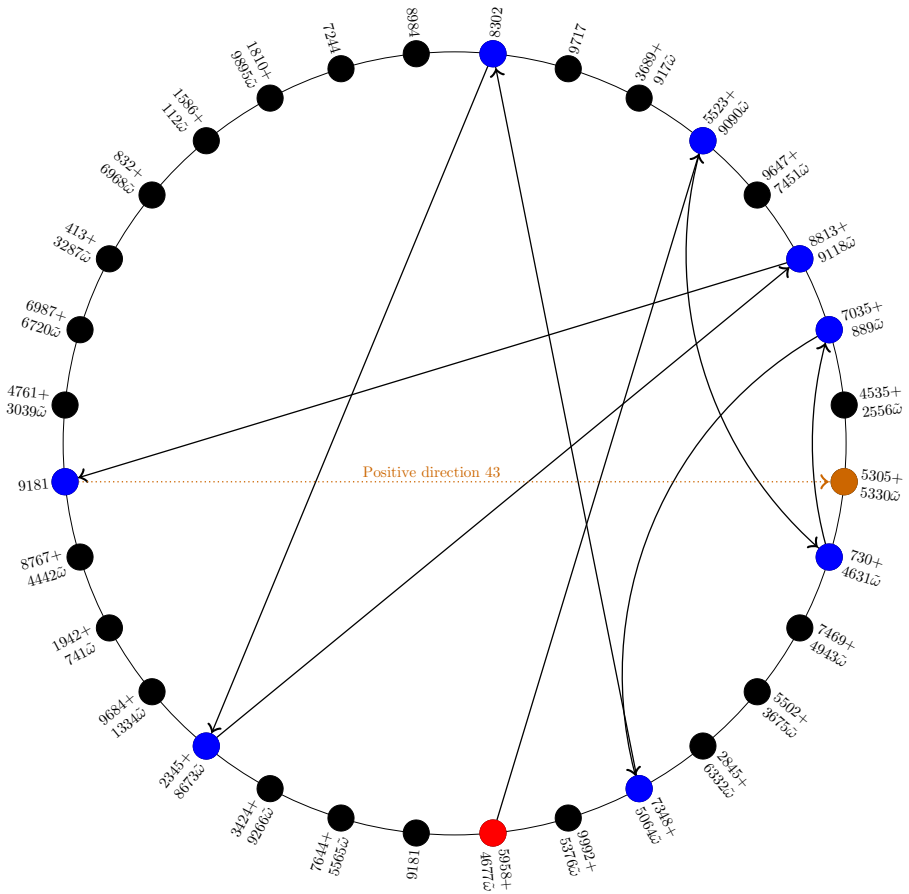




OSIDH PROTOCOL - AN EXAMPLE

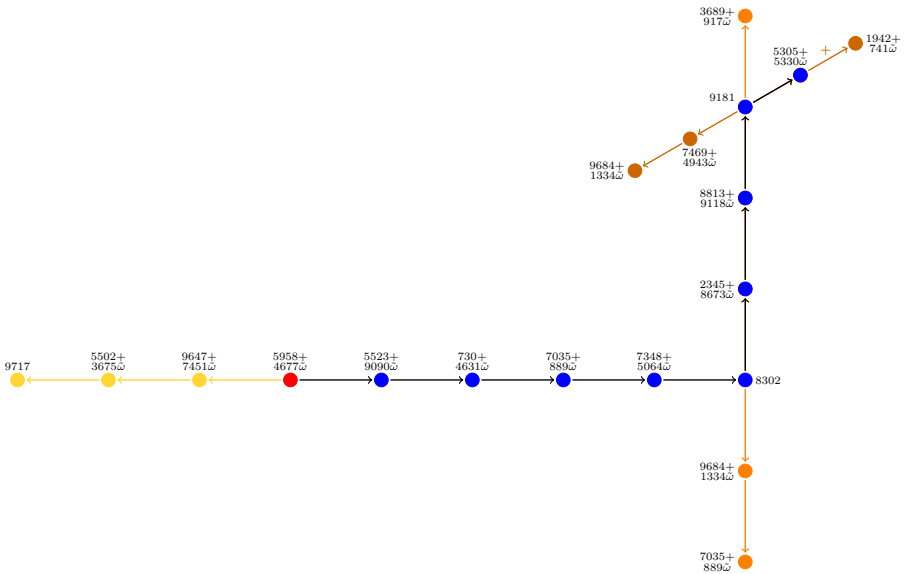
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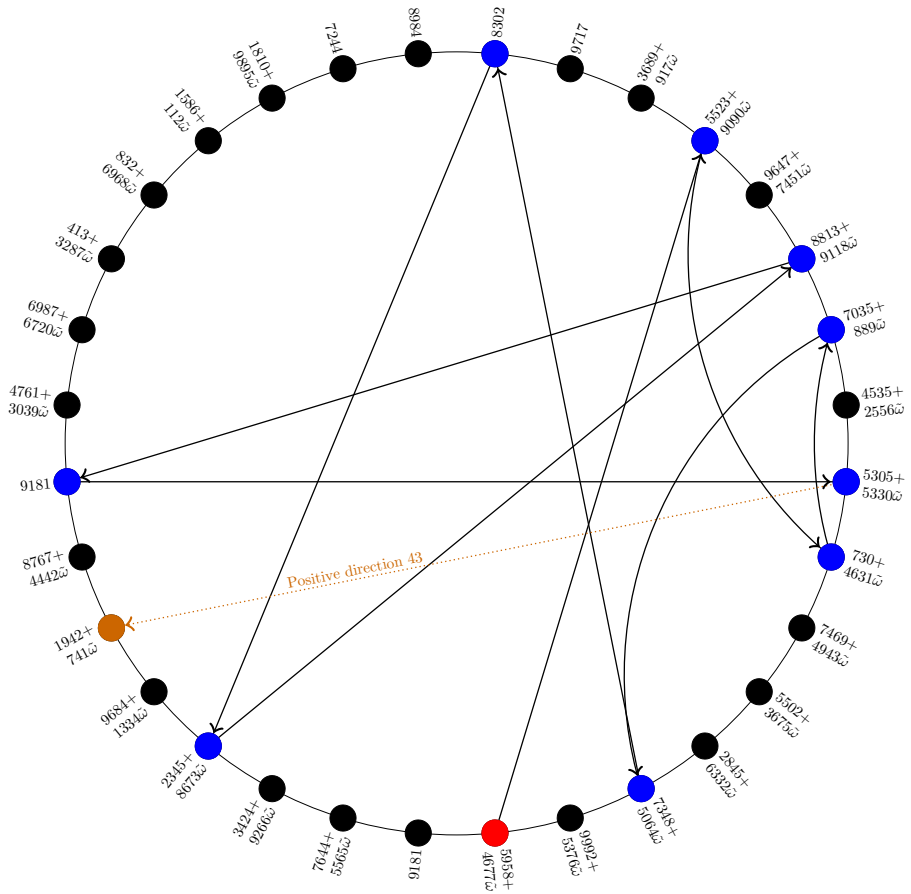




OSIDH PROTOCOL - AN EXAMPLE

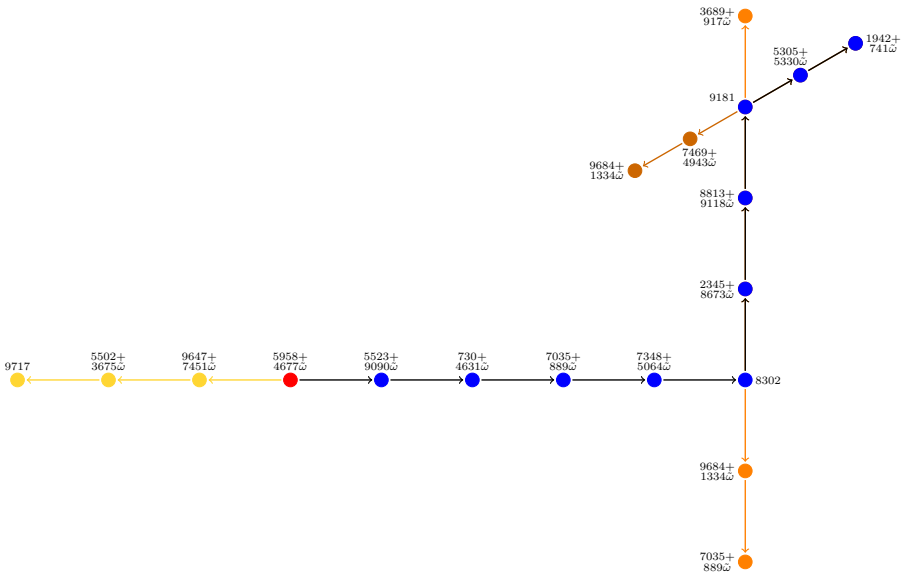
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

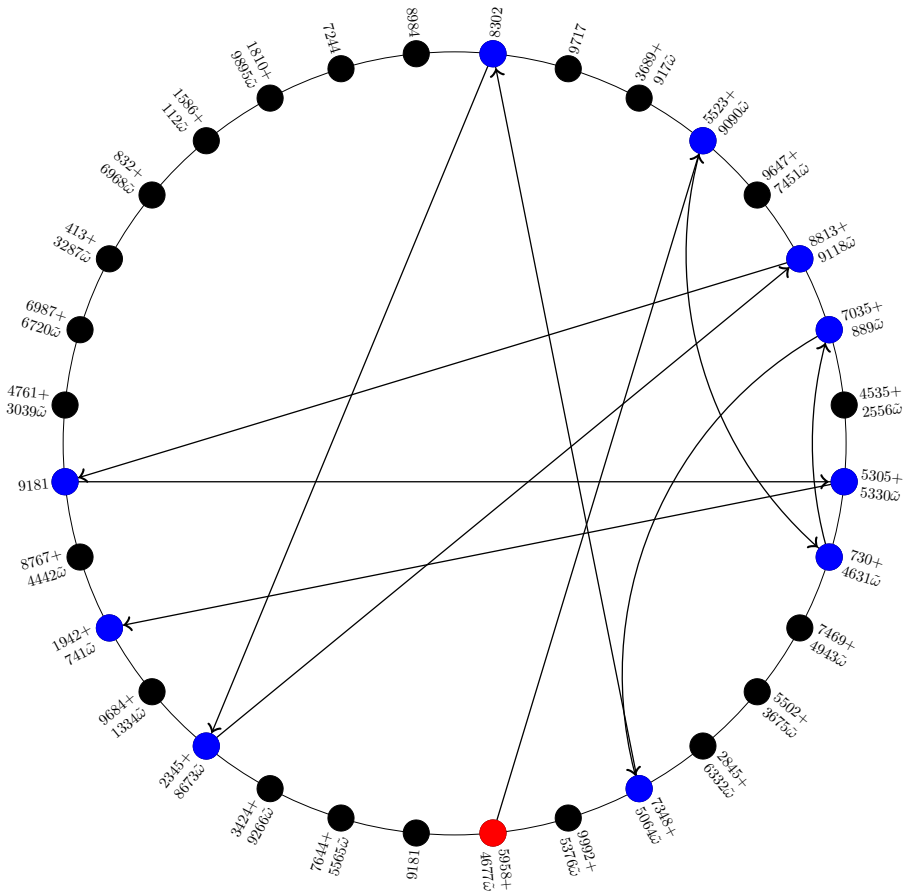




OSIDH PROTOCOL - AN EXAMPLE

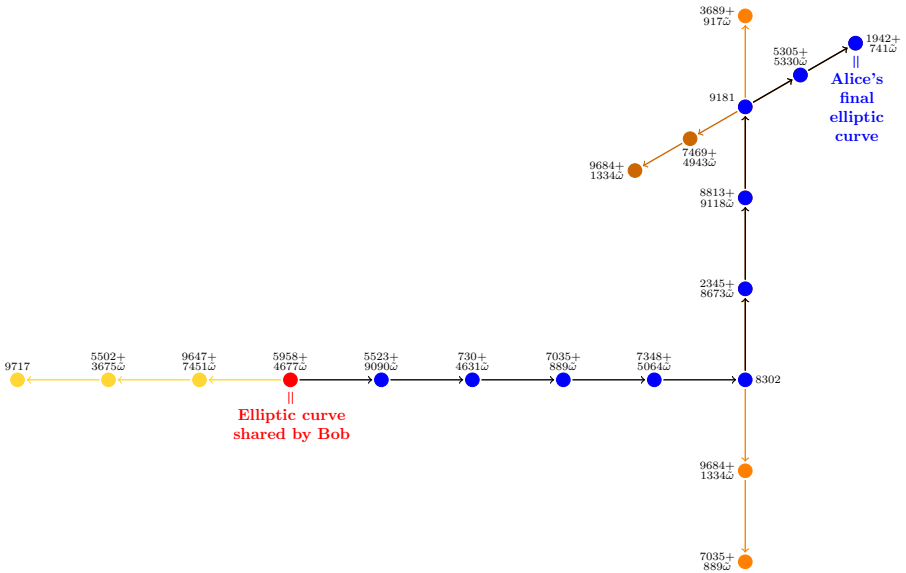
Alice secret key: $\mathfrak{l}_1^5 \mathfrak{l}_2^3 \mathfrak{l}_3^2$

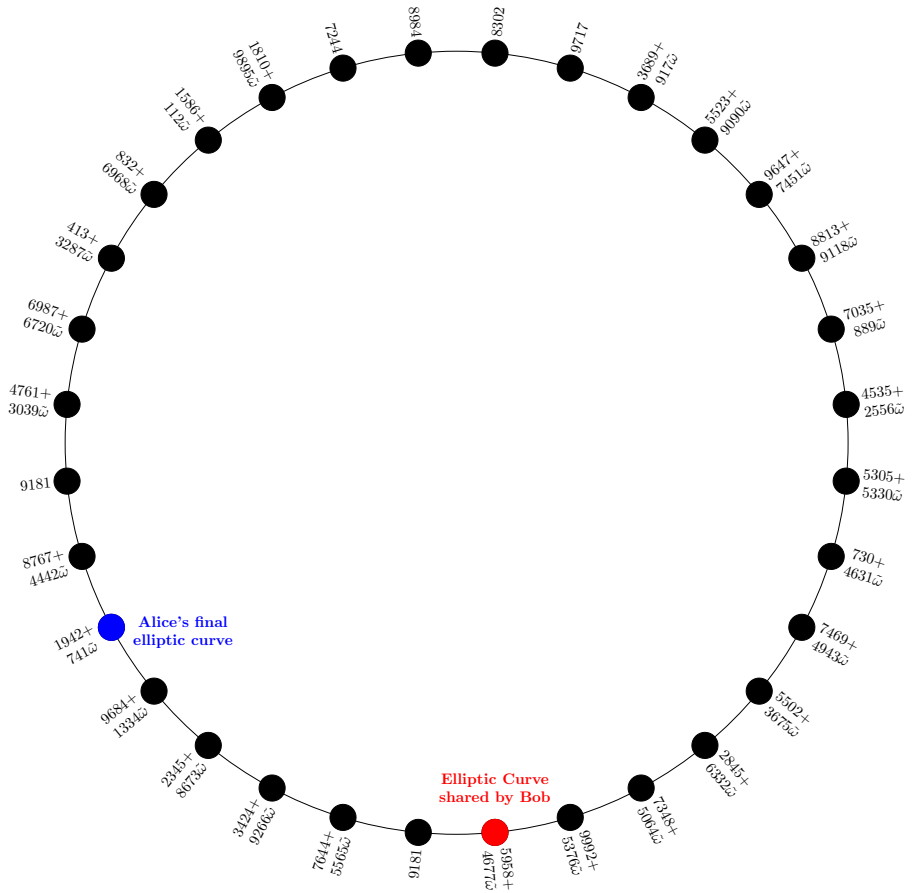




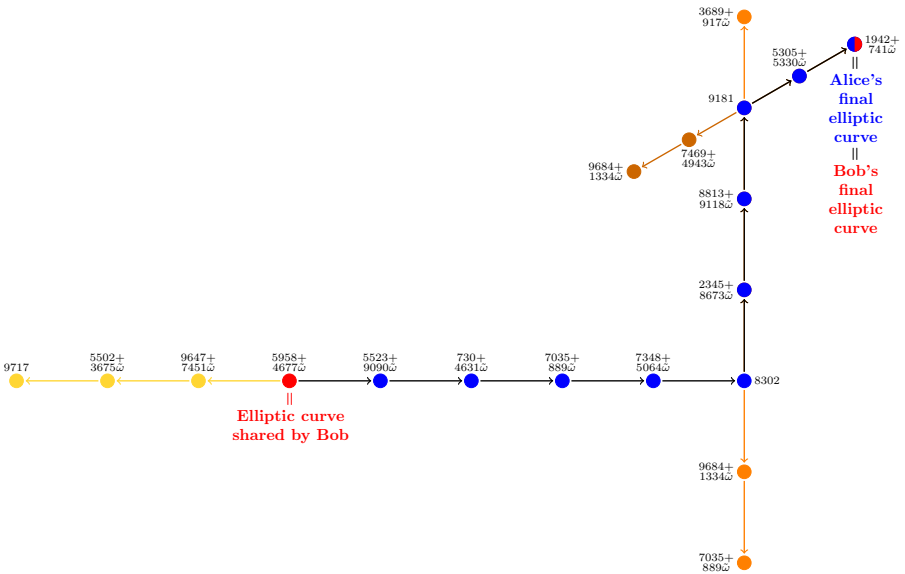
OSIDH PROTOCOL - AN EXAMPLE

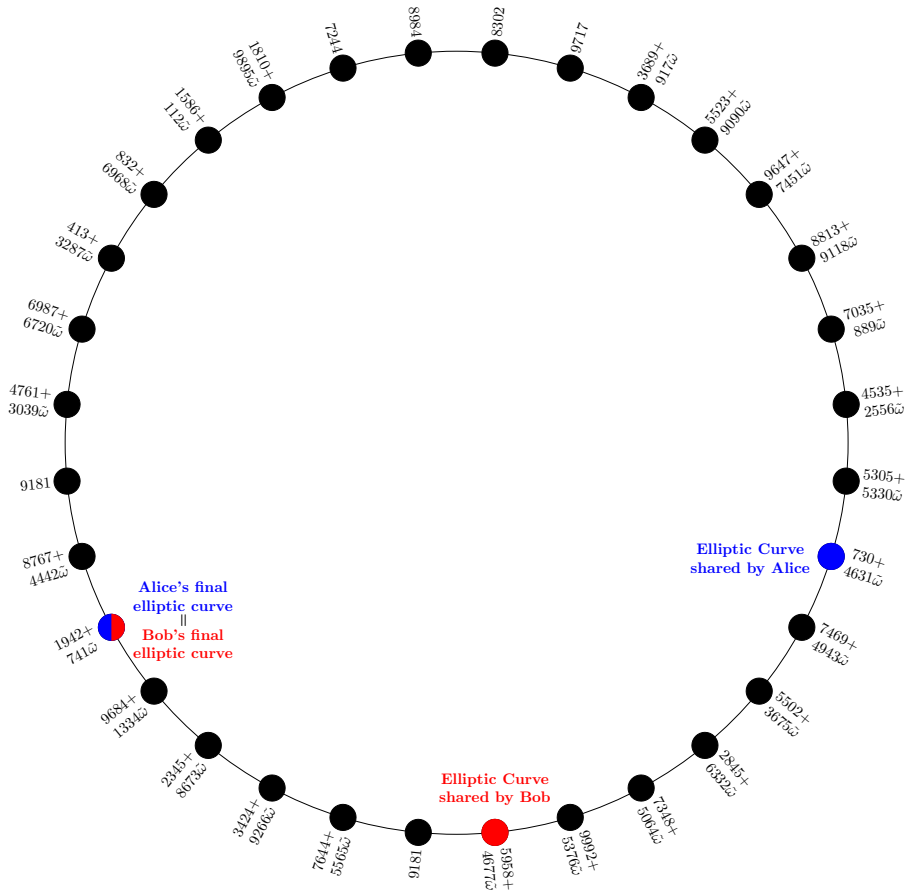
Alice secret key: $\mathfrak{l}_1^5\mathfrak{l}_2^3\mathfrak{l}_3^2$





OSIDH PROTOCOL - AN EXAMPLE





OSIDH PROTOCOL - SECURITY CONSIDERATIONS

L.COLÒ 2M

For an order $\mathcal{O}$ of conductor $\ell^n M$, we note that $\mathcal{Cl}(\mathcal{O}) \simeq \mathsf{SS}^{pr}_{\mathcal{O}}(\rho)$ and define

$$I = I_1 \times ... \times I_t \subseteq \mathbb{Z}^t \quad \text{where } I_j = [-r_j, r_j].$$

The security of OSIDH depends on the following maps

$$I = \prod_{i=1}^{t} [-r_i, r_i] \longrightarrow \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \longrightarrow \mathsf{SS}(p)$$

Supersingular covering bound

We say that the map $\mathcal{Cl}(\mathcal{O}) \simeq \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \longrightarrow \mathsf{SS}(p)$ is λ-surjective if

$$\#\mathcal{Cl}(\mathcal{O}) \geq p^{\lambda}$$

where λ is the logarithmic covering radius. We get

$$\lambda \log_{\ell}(p) \leq n + \log_{\ell}(M) + \log_{\ell}\left(h(\mathcal{O}_K)\right)$$

OSIDH PROTOCOL - SECURITY CONSIDERATIONS

For an order $\mathcal{O}$ of conductor $\ell^n M$, we note that $\mathcal{C}\ell(\mathcal{O}) \simeq \mathsf{SS}^{pr}_{\mathcal{O}}(\rho)$ and define

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The security of OSIDH depends on the following maps

$$I = \prod_{i=1}^{t} [-r_i, r_i] \longrightarrow \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \longrightarrow \mathsf{SS}(p)$$

Supersingular injectivity bound

How can one insure the injectivity of the map $\mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \to \mathsf{SS}(p)$? We set

$$n + \log_\ell(M) + \frac{1}{2}\log_\ell(|\Delta_K|) \leq \frac{1}{2}\log_\ell(p)$$

If (SIB) holds, then the map $\mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \to SS(p)$ is injective.

OSIDH PROTOCOL - SECURITY CONSIDERATIONS

For an order $\mathcal{O}$ of conductor $\ell^n M$, we note that $\mathcal{Cl}(\mathcal{O}) \simeq \mathsf{SS}^{pr}_{\mathcal{O}}(\rho)$ and define

$$I = I_1 \times ... \times I_t \subseteq \mathbb{Z}^t \quad \text{where } I_j = [-r_j, r_j].$$

The security of OSIDH depends on the following maps

$$I = \prod_{i=1}^{t} [-r_i, r_i] \longrightarrow \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \longrightarrow \mathsf{SS}(p)$$

Minkowski norm bound

The set of elements obtained by random walks should contain no cycle; thus,

$$\sum_{i=1}^{t} r_i \log_\ell(q_i) \leq n + \log_\ell(M) + \frac{1}{2}\log_\ell(|\Delta_K|/4)$$

The attack of Dartois and De Feo exploits the non-injectivity of the map $I \to \mathsf{SS}^{pr}_{\mathcal{O}}(\rho)$ to recover an endomorphism of E.

OSIDH PROTOCOL - SECURITY CONSIDERATIONS

For an order $\mathcal{O}$ of conductor $\ell^n M$, we note that $\mathcal{Cl}(\mathcal{O}) \simeq \mathsf{SS}^{pr}_{\mathcal{O}}(\rho)$ and define

$$I = I_1 \times ... \times I_t \subseteq \mathbb{Z}^t \quad \text{where } I_j = [-r_j, r_j].$$

The security of OSIDH depends on the following maps

$$I = \prod_{i=1}^{t} [-r_i, r_i] \longrightarrow \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \longrightarrow \mathsf{SS}(p)$$

Class group covering bound

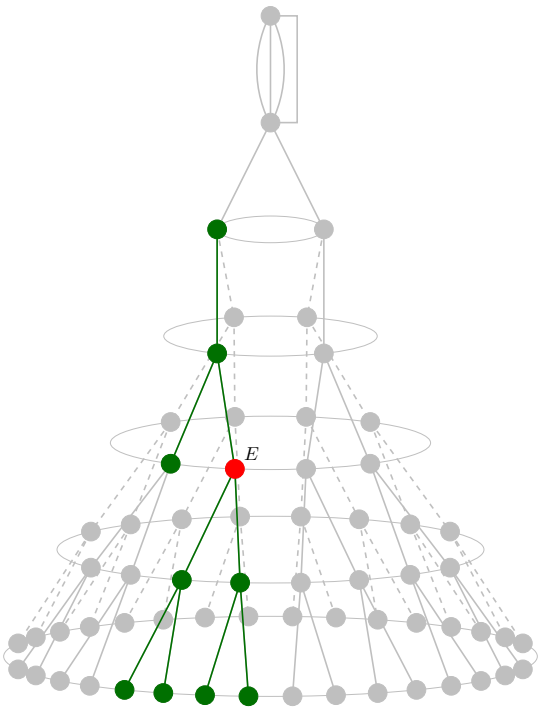
In order to have a uniform element of $\mathcal{Cl}(\mathcal{O})$ it is desirable to be able to reach all elements of $\mathcal{Cl}(\mathcal{O})$.

$$\sum_{i=1}^{t} \log_\ell(2r_i + 1) \geq \lambda \left(n + \log_\ell(M) + \log_\ell(h(\mathcal{O}_K)) \right)$$

COUNTERMEASURES- THE USE OF NON-SPLIT PRIME

Disjoint prime sets: $\mathcal{P}_s$ (split primes), and $\mathcal{P}_A$ and $\mathcal{P}_B$ (non-split primes).

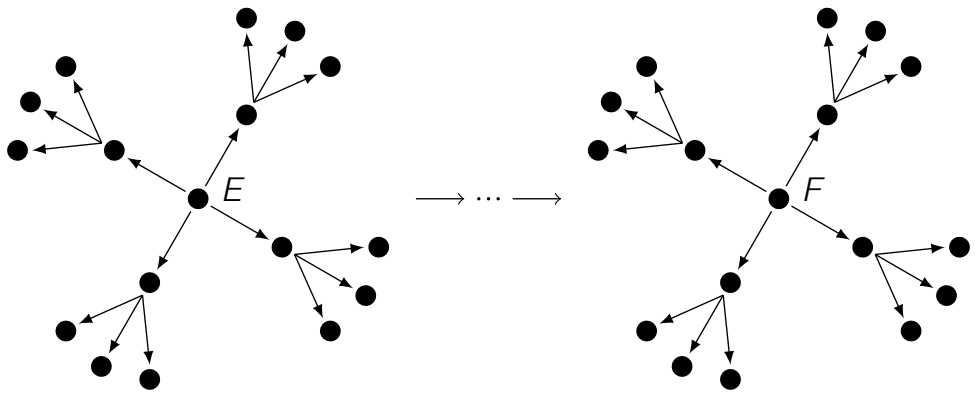
- Split primes: precompute initial directions in $\mathcal{C}\ell(\mathcal{O}_K)$ and push forward to E.
- Non-split primes: precompute labeled clouds and push forward to E.



COUNTERMEASURES- THE USE OF NON-SPLIT PRIME

Disjoint prime sets: $\mathcal{P}_s$ (split primes), and $\mathcal{P}_A$ and $\mathcal{P}_B$ (non-split primes).

- Split primes: precompute initial directions in $\mathcal{C}\ell(\mathcal{O}_K)$ and push forward to E.
- Non-split primes: precompute labeled clouds and push forward to E.



Future directions:

- Generalization to higher level structure.
- Formal orientations.

THANK YOU FOR YOUR ATTENTION

APPENDICES

- Forgetful maps $\mathsf{SS}_{\mathcal{O}}^{pr}(\rho) \to \mathsf{SS}(p)$ - (29).
- Naive OSIDH protocol - (31).
- Geometric definitions - (37).
- Initializing the ladder - (42).
- Extending the ladder - (47).
- Weber Modular Functions (49), Polynomials (51), Curves (53) and Groups (55).
- Weber initialization of a ladder - (58).
- Dartois-De Feo attack on OSIDH (62) and countermeasures (63).

FORGETFUL MAP

We have a bijection (isomorphism of sets with $\mathcal{Cl}(\mathcal{O})$-action):

$$\mathcal{Cl}(\mathcal{O}) \cong \mathsf{SS}^{pr}_{\mathcal{O}}(\rho) \subseteq \mathsf{SS}_{\mathcal{O}}(p)$$

On the other hand, the inclusion $\mathcal{O}_i \subset \mathcal{O}_{i-1}$ determines an inclusion

$$\begin{gathered}\mathsf{SS}_{\mathcal{O}_i}(p) \subset \mathsf{SS}_{\mathcal{O}_{i+1}}(p) = \mathsf{SS}_{\mathcal{O}_i}(p) \cup \mathsf{SS}^{pr}_{\mathcal{O}_{i+1}}(p) \\ \Downarrow \\ \mathsf{SS}_{\mathcal{O}_K}(p) \subset \mathsf{SS}_{\mathcal{O}_1}(p) \subset \cdots \subset \mathsf{SS}_{\mathcal{O}_i}(p) \subset \cdots\end{gathered}$$

equipped with forgetful maps

$$\begin{aligned}\mathsf{SS}_{\mathcal{O}_i}(p) &\to \mathsf{SS}(p) \\ [(E, \mathcal{O}_i)] &\to j(E)\end{aligned}$$

Question

When the map $\mathsf{SS}_{\mathcal{O}_i}(p) \to \mathsf{SS}(p)$ and its restriction to $\mathsf{SS}^{pr}_{\mathcal{O}_i}(\rho)$ are injective? When are they surjective?

FORGETFUL MAP - FIRST RESULTS

L.COLÒ 2M

Proposition

Let $\mathcal{O}$ be an imaginary quadratic order of discriminant Δ and p a prime which is inert in $\mathcal{O}$. If $|\Delta| < p$, then the map $\mathsf{SS}_{\mathcal{O}}(p) \to \mathsf{SS}(p)$ is injective.

$p = 1013$

i	$h(O_i)$	$\lvert Y_i \rvert$	$\lvert X_i \rvert$	$H(p)$	λ_i
1	1	1	1	85	0.3590
2	2	2	3	85	0.5593
3	4	4	7	85	0.7596
4	8	8	15	85	0.9599
5	16	16	29	85	1.1603
6	32	26	47	85	1.3606
7	64	43	66	85	1.5609
8	128	70	82	85	1.7612
9	256	79	85	85	1.9615
10	512	83	85	85	2.1618

$p = 1019$

i	$h(O_i)$	$\lvert Y_i \rvert$	$\lvert X_i \rvert$	$H(p)$	λ_i
1	1	1	1	86	0.3587
2	2	2	3	86	0.5588
3	4	4	7	86	0.7590
4	8	8	15	86	0.9591
5	16	15	30	86	1.1593
6	32	29	49	86	1.3594
7	64	46	69	86	1.5595
8	128	64	81	86	1.7597
9	256	83	84	86	1.9598
10	512	86	86	86	2.1600

A FIRST NAIVE PROTOCOL

L.COLÒ 2M

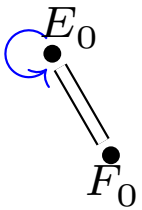
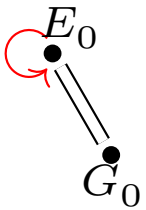
PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$

ALICE	**BOB**

A FIRST NAIVE PROTOCOL

L.COLÒ 2M

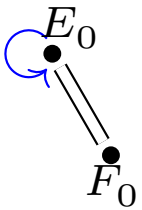
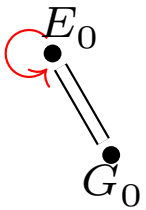
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	ALICE	**BOB**
Choose a primitive $\mathcal{O}_K$-orientation of E_0		

A FIRST NAIVE PROTOCOL

L.COLÒ 2M

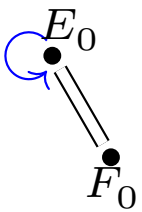
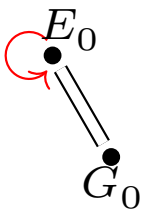
PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$

	ALICE	**BOB**
Choose a primitive $\mathcal{O}_K$-orientation of E_0		
Push it forward to depth n	$\underbrace{E_0 = F_0 \to F_1 \to \ldots \to F_n}_{\phi_A}$	$\underbrace{E_0 = G_0 \to G_1 \to \ldots \to G_n}_{\phi_B}$

A FIRST NAIVE PROTOCOL

L.COLÒ 2M

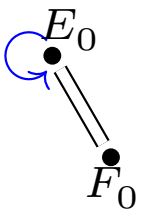
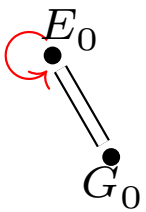
PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$

	ALICE	**BOB**
Choose a primitive $\mathcal{O}_K$-orientation of E_0		
Push it forward to depth n	$\underbrace{E_0 = F_0 \to F_1 \to \ldots \to F_n}_{\phi_A}$	$\underbrace{E_0 = G_0 \to G_1 \to \ldots \to G_n}_{\phi_B}$
Exchange data	$\{G_i\}_{i=1}^n$	$\{F_i\}_{i=1}^n$

A FIRST NAIVE PROTOCOL

L.COLÒ 2M

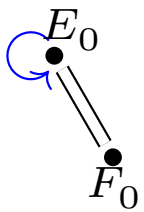
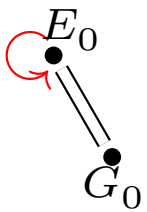
PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$

	ALICE	**BOB**
Choose a primitive $\mathcal{O}_K$-orientation of E_0		
Push it forward to depth n	$\underbrace{E_0 = F_0 \to F_1 \to \ldots \to F_n}_{\phi_A}$	$\underbrace{E_0 = G_0 \to G_1 \to \ldots \to G_n}_{\phi_B}$
Exchange data	$\{G_i\}_{i=1}^n$	$\{F_i\}_{i=1}^n$
Compute shared secret	Compute $\phi_A \cdot \{G_i\}$	Compute $\phi_B \cdot \{F_i\}$

A FIRST NAIVE PROTOCOL

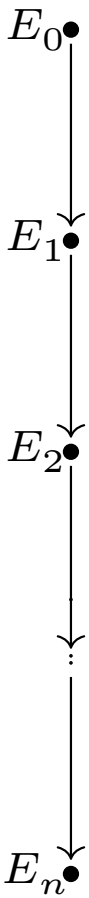
L.COLÒ 2M

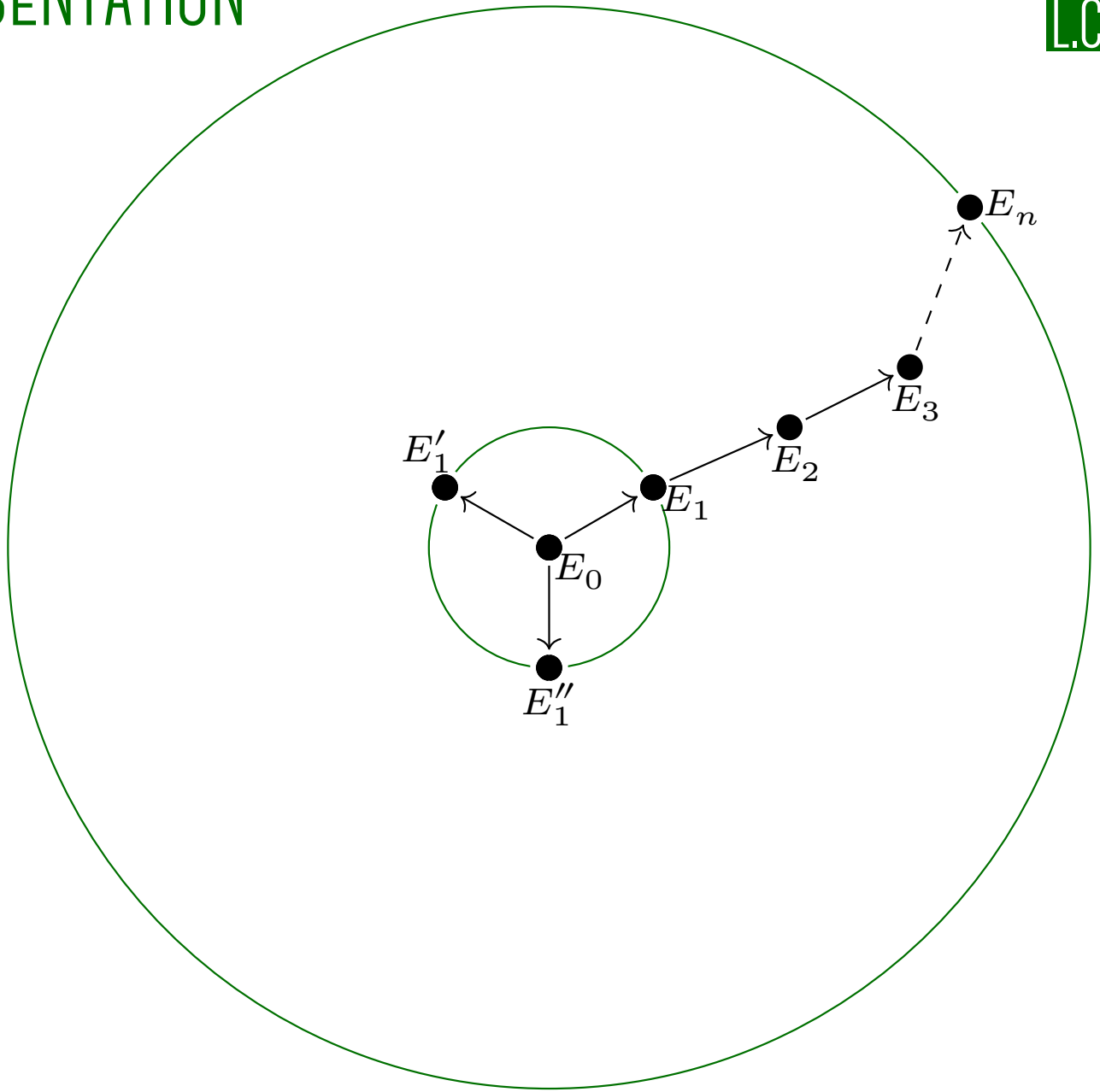
PUBLIC DATA: A chain of ℓ-isogenies $E_0 \to E_1 \to \ldots \to E_n$

	ALICE	**BOB**
Choose a primitive $\mathcal{O}_K$-orientation of E_0		
Push it forward to depth n	$\underbrace{E_0 = F_0 \to F_1 \to \ldots \to F_n}_{\phi_A}$	$\underbrace{E_0 = G_0 \to G_1 \to \ldots \to G_n}_{\phi_B}$
Exchange data	$\{G_i\}_{i=1}^n$	$\{F_i\}_{i=1}^n$
Compute shared secret	Compute $\phi_A \cdot \{G_i\}$	Compute $\phi_B \cdot \{F_i\}$

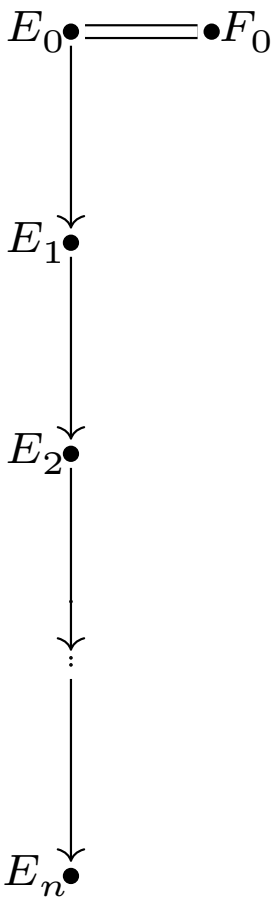
In the end, Alice and Bob will share a new chain $E_0 \to H_1 \to \ldots \to H_n$

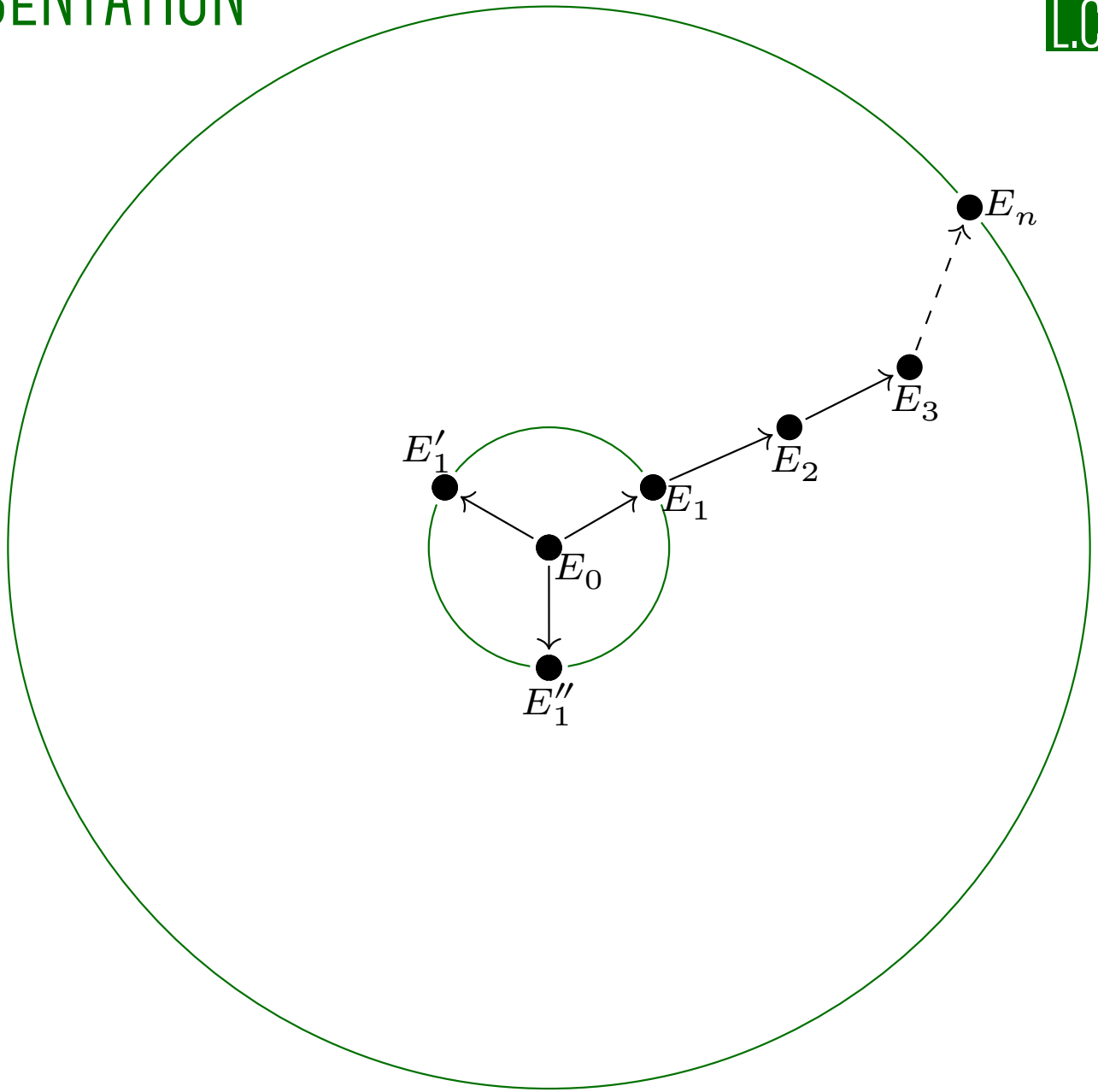
GRAPHIC REPRESENTATION



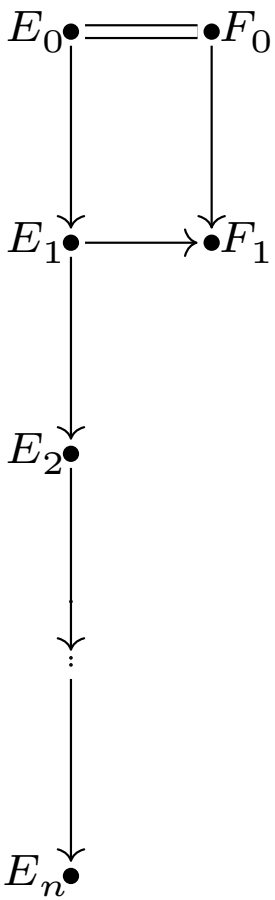


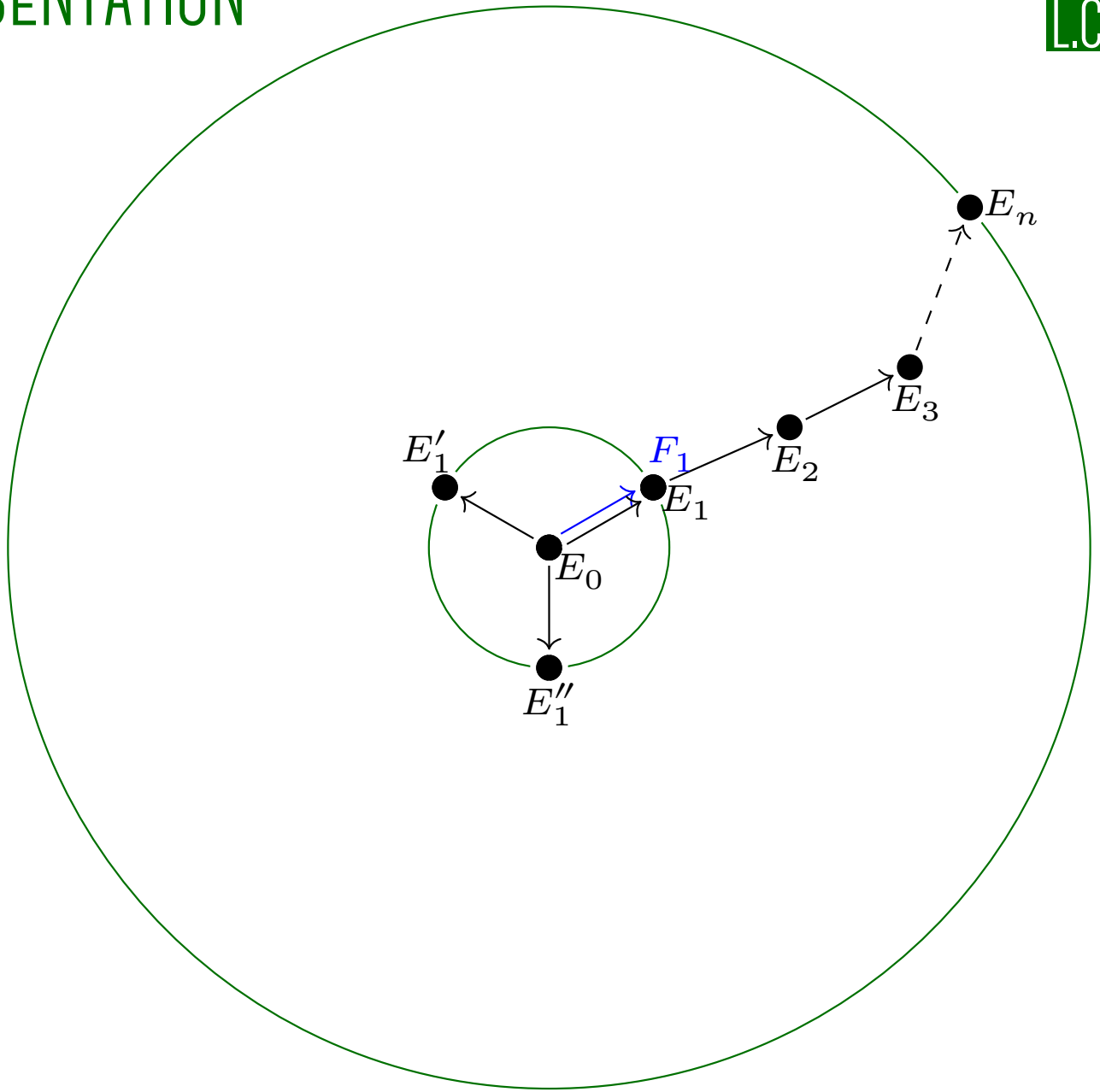
GRAPHIC REPRESENTATION



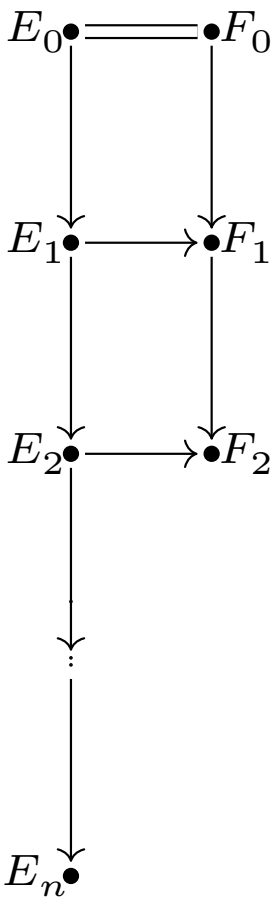


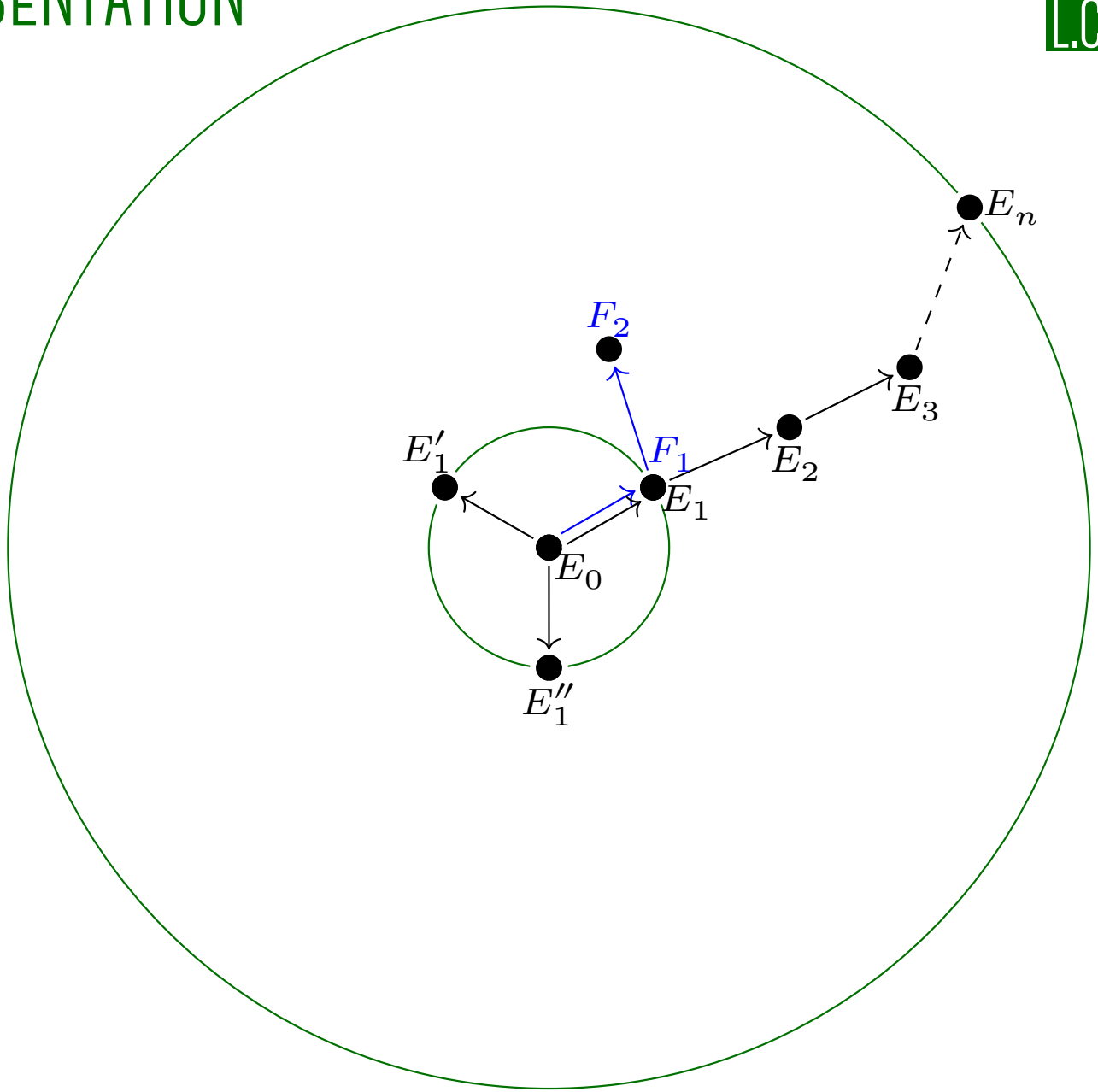
GRAPHIC REPRESENTATION



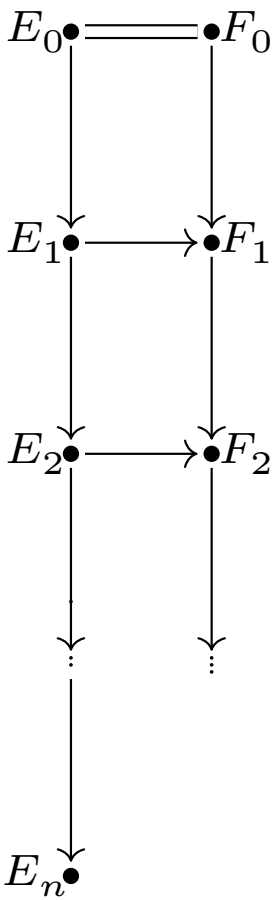


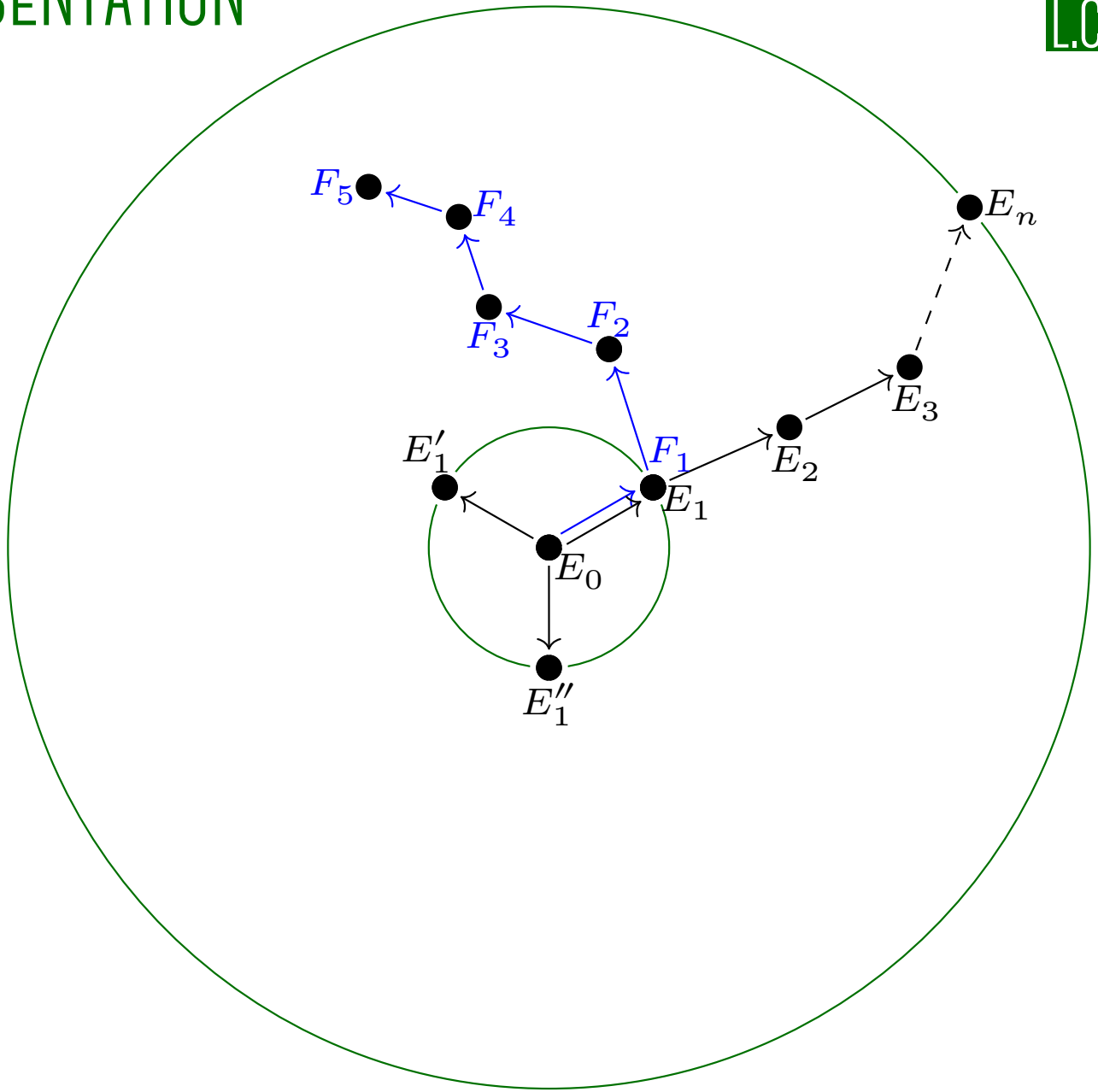
GRAPHIC REPRESENTATION



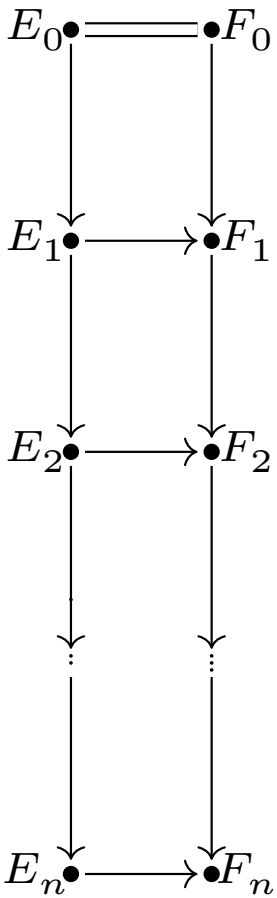


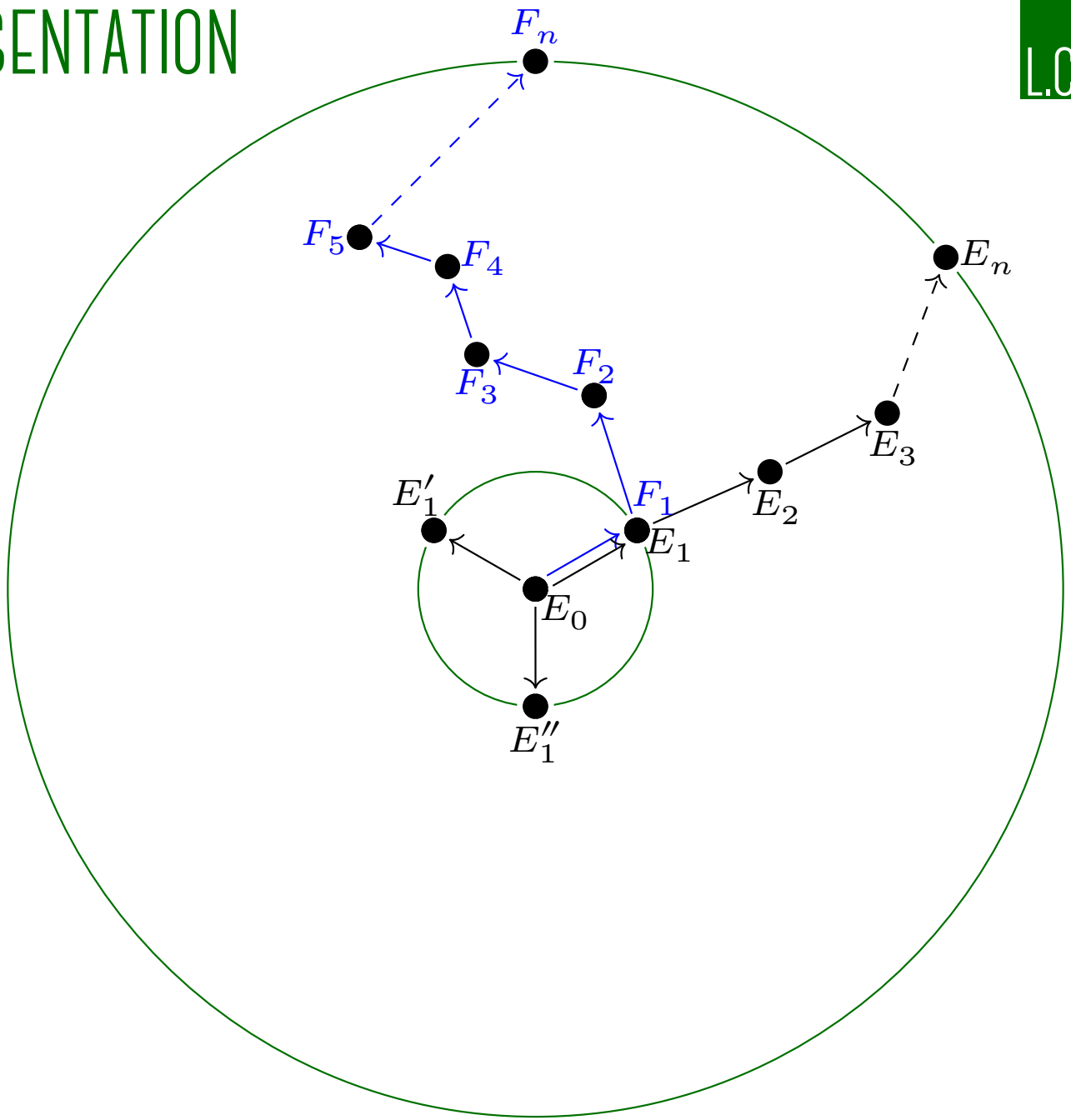
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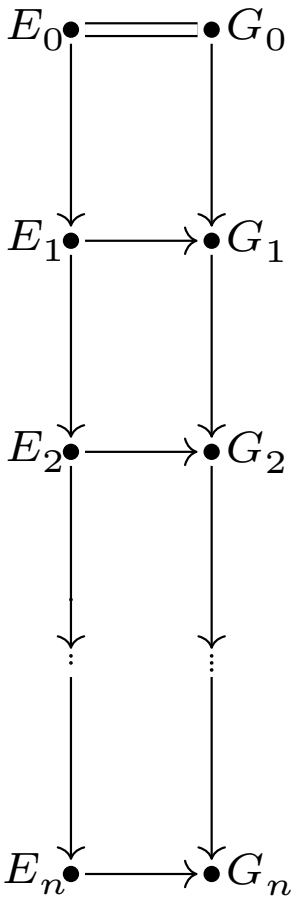


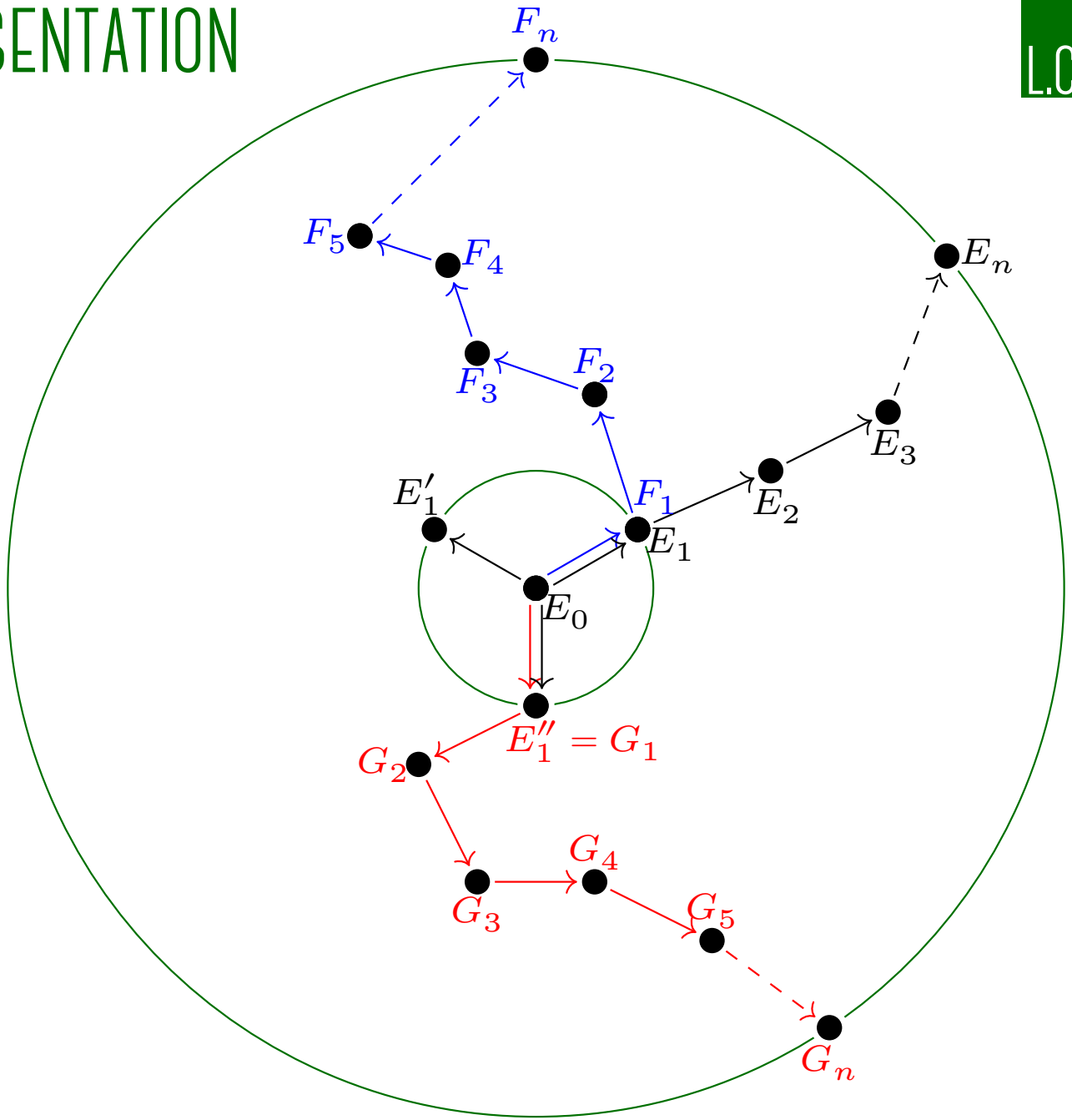
GRAPHIC REPRESENTATION



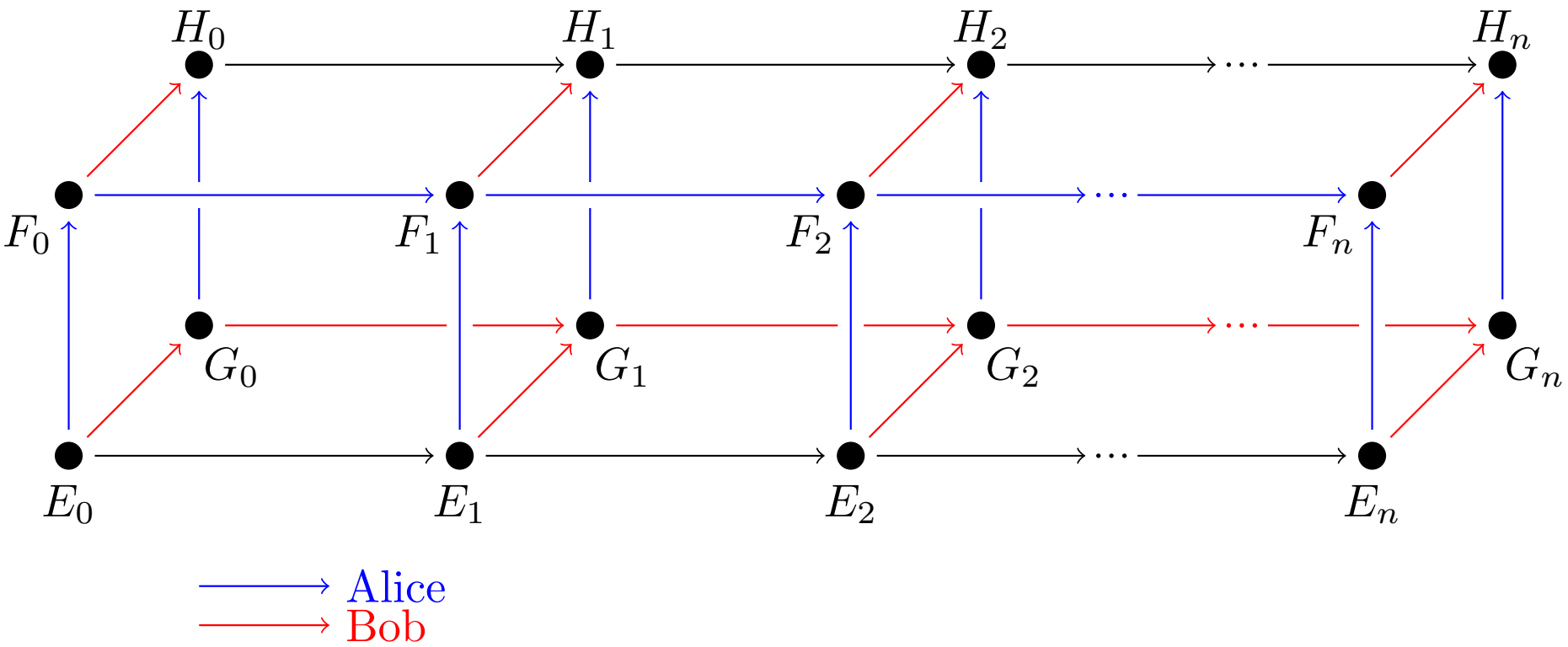


GRAPHIC REPRESENTATION





GRAPHIC REPRESENTATION

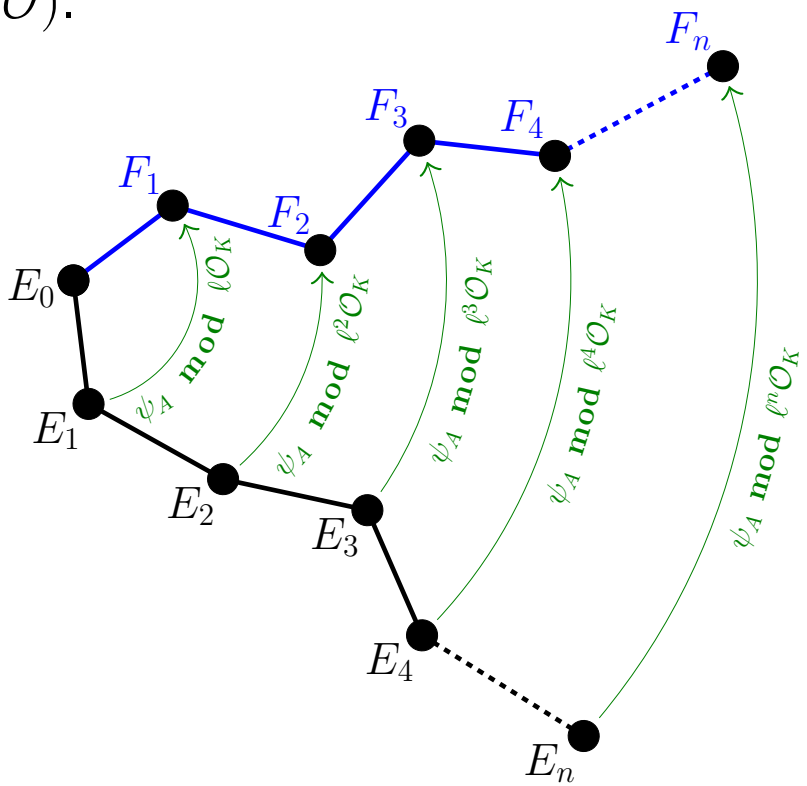


A FIRST NAIVE PROTOCOL - WEAKNESS

From the short exact sequence of class groups:

$$1 \to \frac{(\mathcal{O}_K/\ell^n\mathcal{O}_K)^\times}{\mathcal{O}_K^\times(\mathbb{Z}/\ell^n\mathbb{Z})^\times} \to \mathcal{Cl}(\mathcal{O}) \to \mathcal{Cl}(\mathcal{O}_K) \to 1$$

an adversary can compute successive approximations (mod ℓ^i) to ϕ_A and ϕ_B modulo ℓ^n hence in $\mathcal{Cl}(\mathcal{O})$.



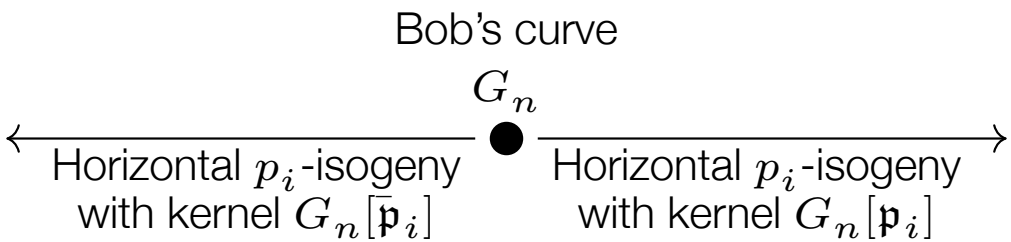
An example in the Appendix (??)

TOWARDS A MORE SECURE OSIDH PROTOCOL

How can we avoid this while still giving the other enough information?

Instead Alice and Bob can send only $F = F_n$ and $G = G_n$.

Problem Once Alice receives the unoriented curve G_n computed by Bob she also needs additional information for each prime $\mathfrak{p}_i$:



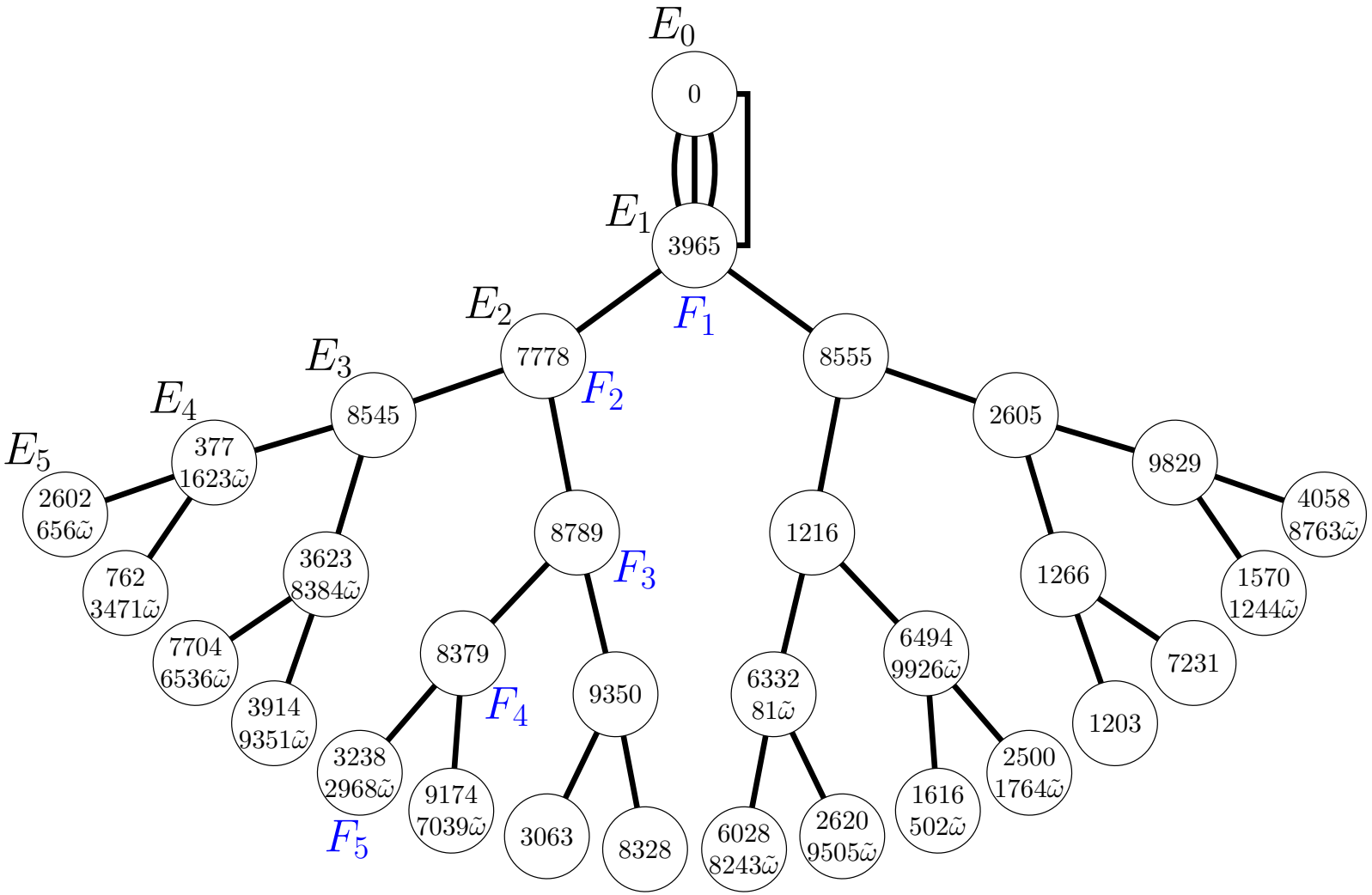
In fact, she has no information as to which directions — out of $p_i + 1$ total p_i-isogenies — to take as $\mathfrak{p}_i$ and $\bar{\mathfrak{p}}_i$.

Solution They share a collection of local isogeny data $(F_n[\mathfrak{q}_j])$ and $(G_n[\mathfrak{q}_j])$ which identifies the isogeny directions (out of $q_i + 1$) for a system of small split primes $(\mathfrak{q}_i)$ in $\mathcal{O}_K$.

COMPUTING SUCCESSIVE APPROXIMATIONS: AN EXAMPLE

L.COLÒ 2M

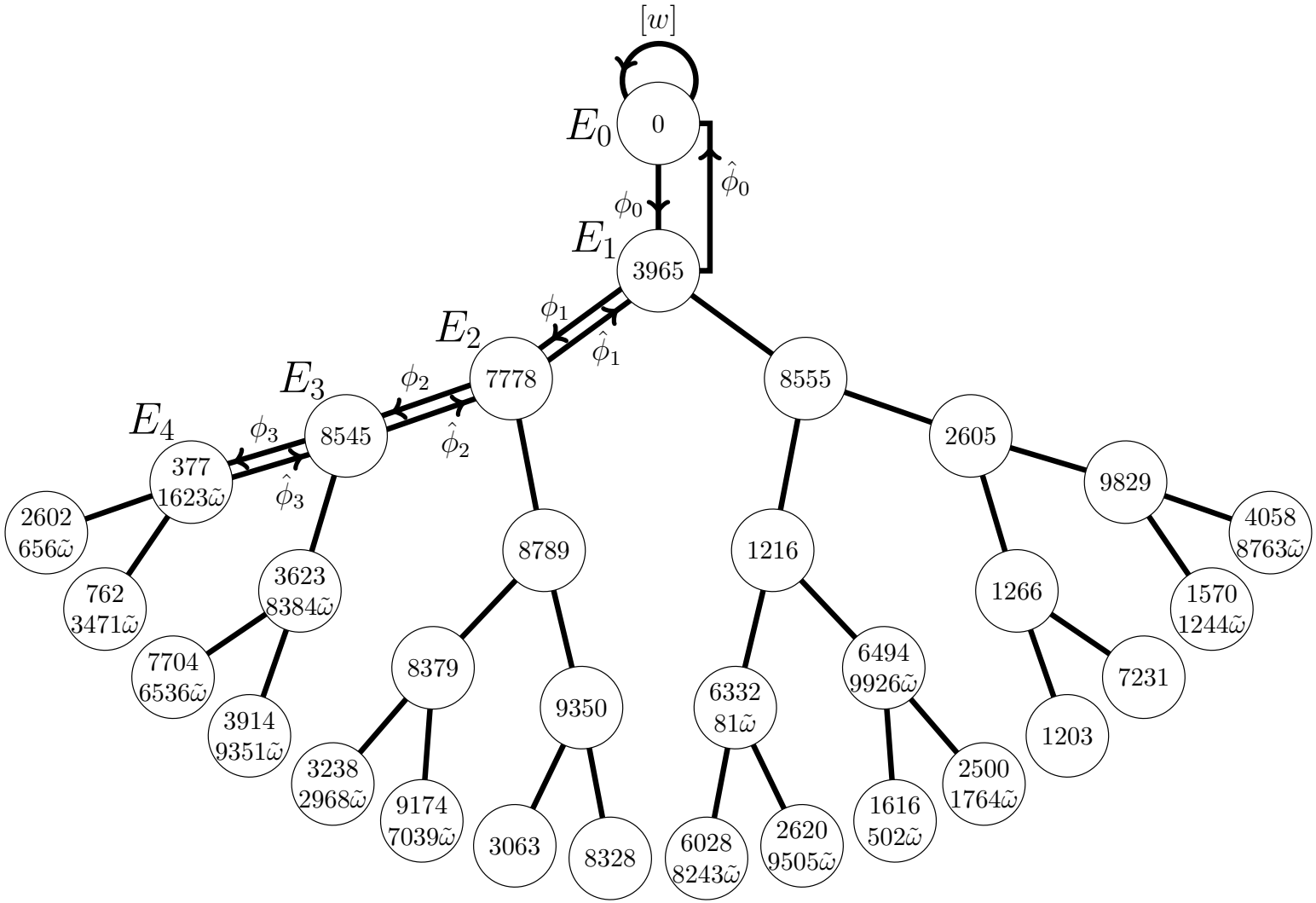
Take $q = p^2 = 10007^2$. $E_0 : y^2 = x^3 + 1$ of j-invariant 0 is supersingular over $\mathbb{F}_q$.
We orient E_0 by $\mathcal{O}_K = \mathbb{Z}[\omega] \hookrightarrow \mathsf{End}(E_0)$ where $w^2 + w + 1$.



COMPUTING SUCCESSIVE APPROXIMATIONS: AN EXAMPLE

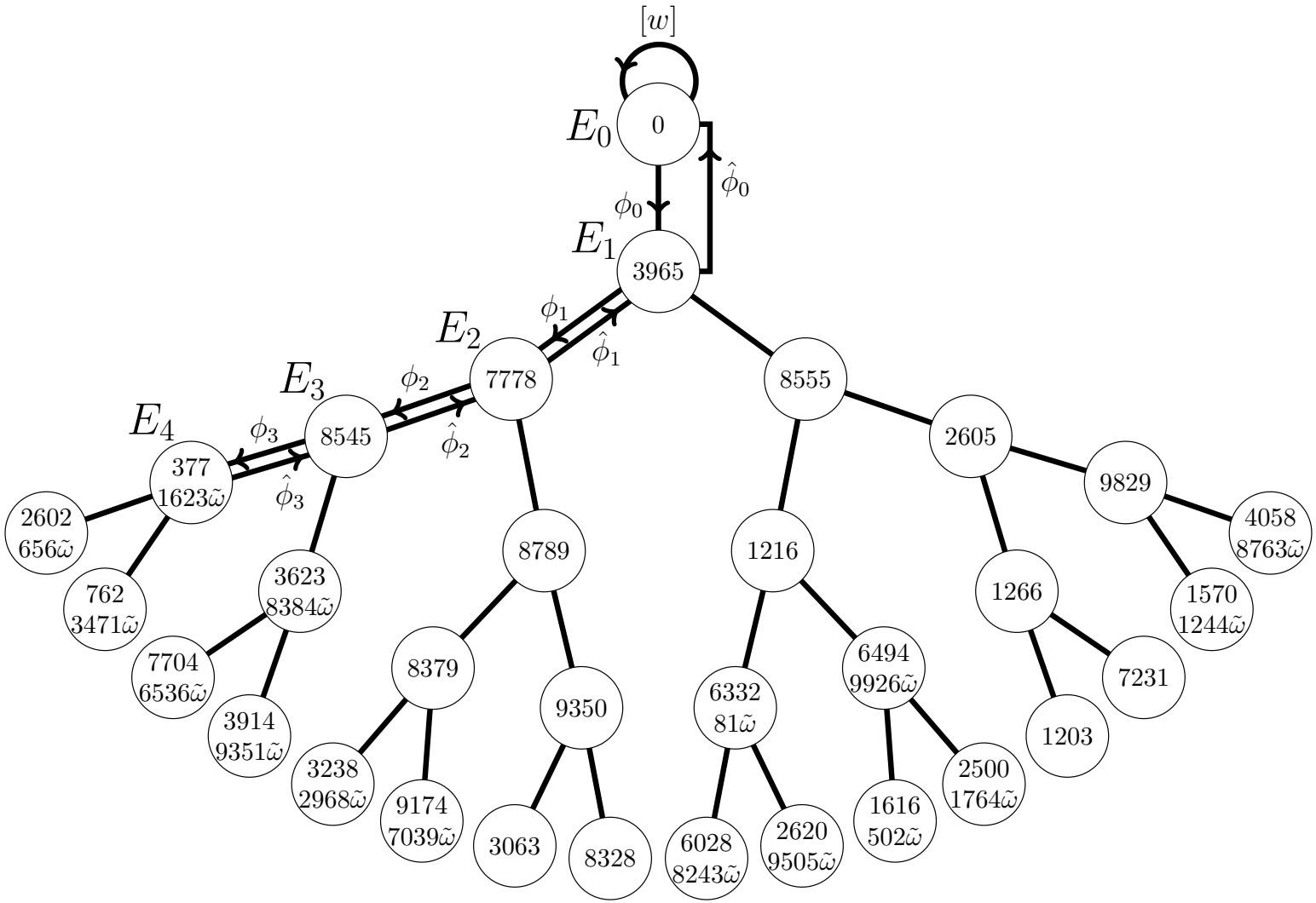
The action of $\ell^i\omega$ on E_i will be given by the composition

$$\phi_{i-1} \circ \cdots \circ \phi_2 \circ \phi_1 \circ \phi_0 \circ [\omega] \circ \hat{\phi}_0 \circ \hat{\phi}_1 \circ \hat{\phi}_2 \circ \cdots \circ \hat{\phi}_{i-1}$$



COMPUTING SUCCESSIVE APPROXIMATIONS: AN EXAMPLE

Observe that this is exactly the definition of orientation by $\mathcal{O}_i$ transmitted to E_i along the isogeny $E_0 \to E_1 \to E_2 \to \ldots \to E_i$.



Computing successive approximations

We are given two sequences $\{E_i\}_{i=0}^n$ and $\{F_i\}_{i=0}^n$. Suppose that $E_i = F_i$ for all $i \leq m$; there are l possibilities for F_{m+1}, and we need to find $\beta \in \mathsf{End}(\mathcal{O}_K)$ such that

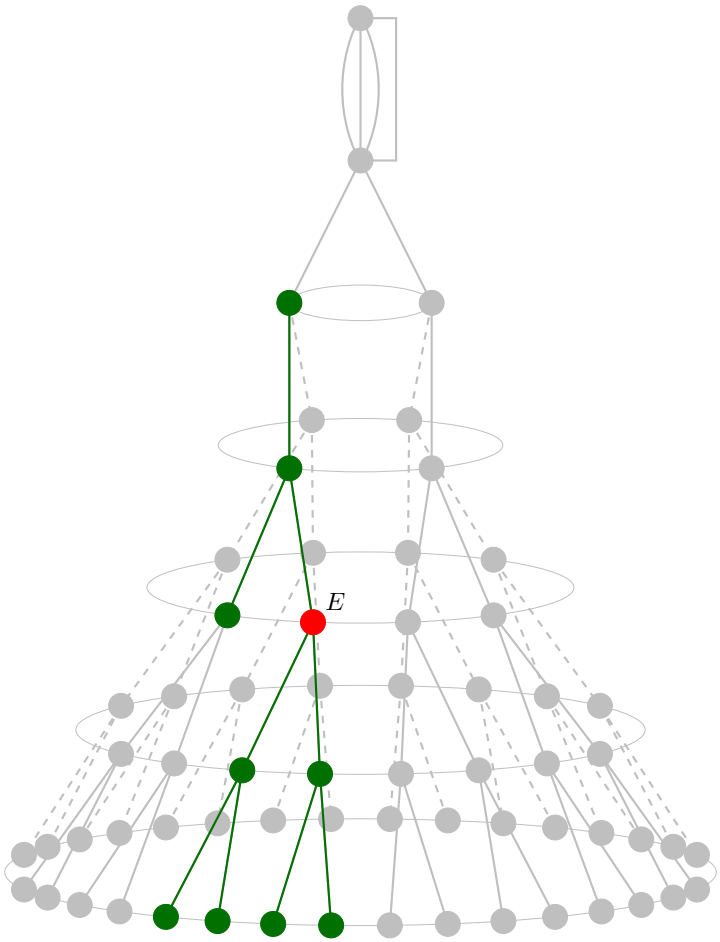
1. $\beta \equiv 1 \bmod \ell^m$ so that $\beta_* E_i = F_i = E_i$ for all $i \leq m$;
2. $\beta_* E_{m+1} = F_{m+1}$;
3. β is smooth with small exponents (n order to determine the action of β modulo ℓ^{m+1} effectively).

Once that we have constructed α such that $\alpha_* E_i = F_i$ for all $m < i \leq k$, then we can substitute **1** with

1'. $\beta \equiv \alpha \bmod \ell^k$ so that $\beta_* E_{k+1} = F_{k+1}$.

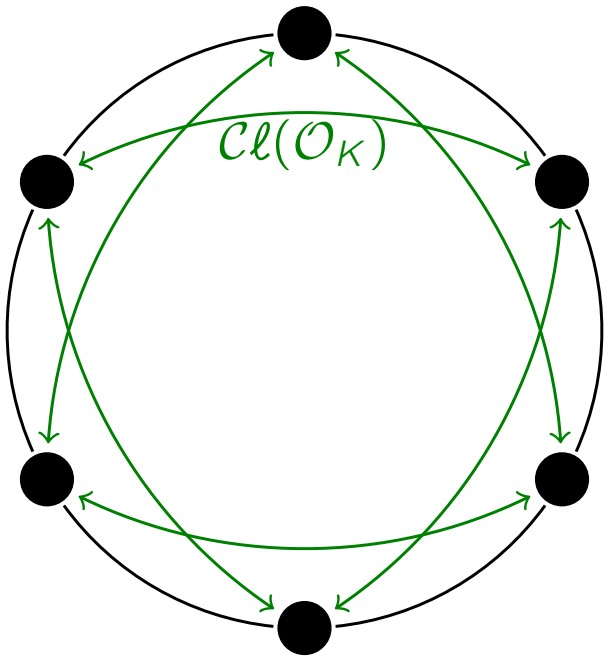
CLOUDS

In order to discuss the local neighborhood of a graph, we introduce the notion of an ℓ-isogeny cloud around E: this is a subgraph of $G_\ell(E)$, whose paths from E extend to length r.



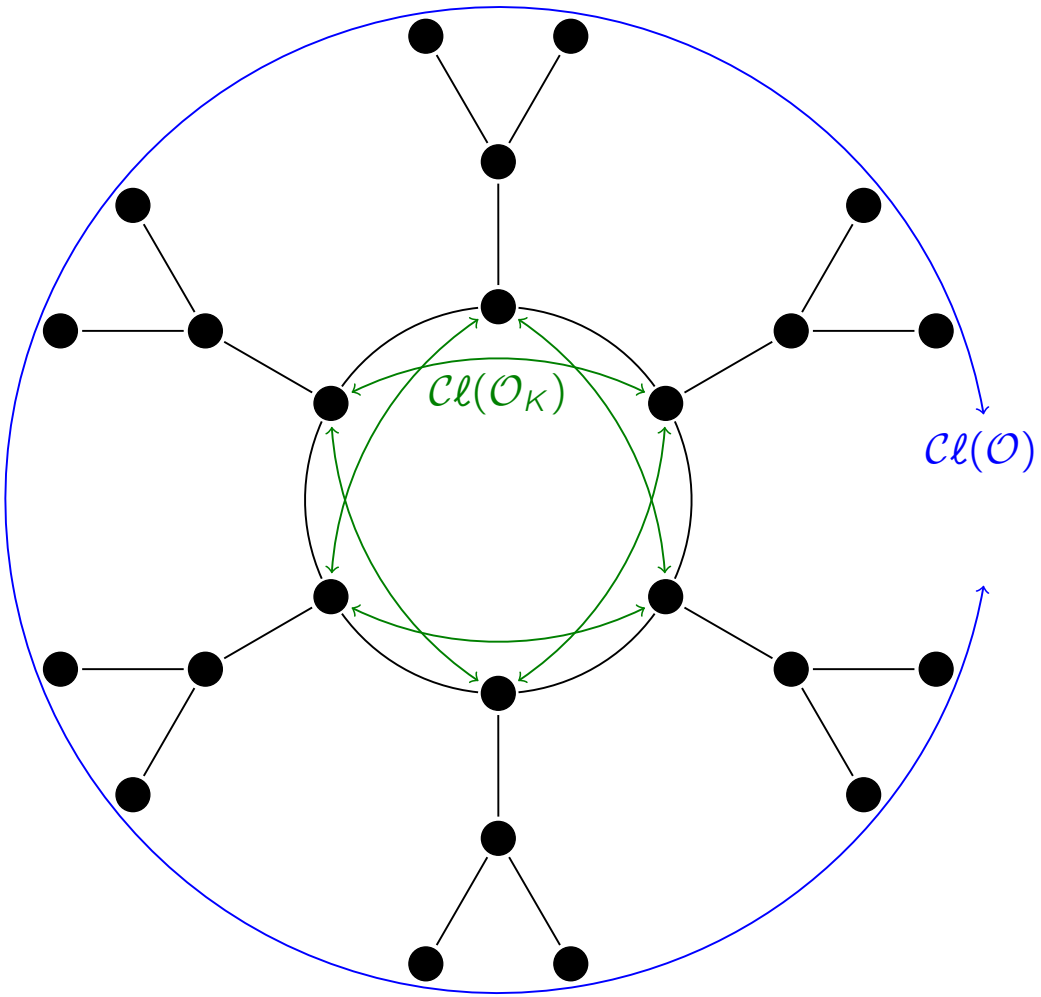
VORTEX

We define a vortex to be the ℓ-isogeny subgraph whose vertices are isomorphism classes of $\mathcal{O}$-oriented elliptic curves with ℓ-maximal endomorphism ring, equipped with an action of $\mathcal{Cl}(\mathcal{O})$.



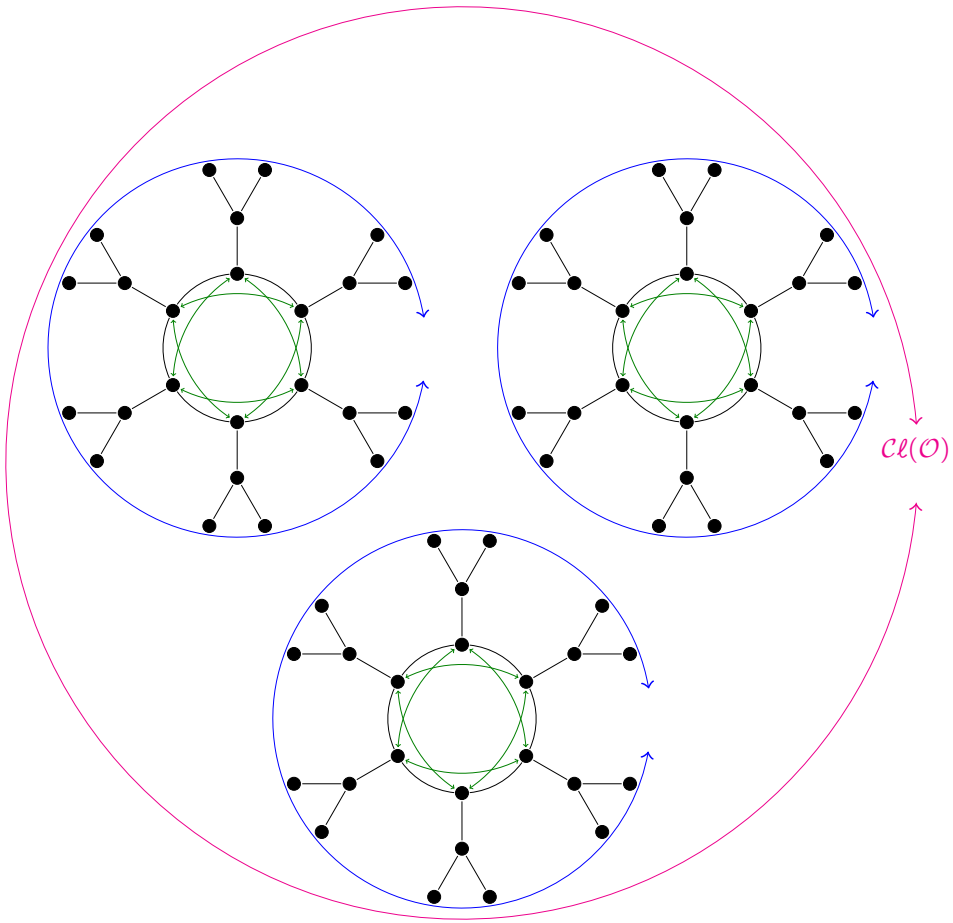
WHIRLPOOL

We define a *whirlpool* to be a complete isogeny volcano acted on by the class group. We would like to think at isogeny graphs as moving objects.



WHIRLPOOL

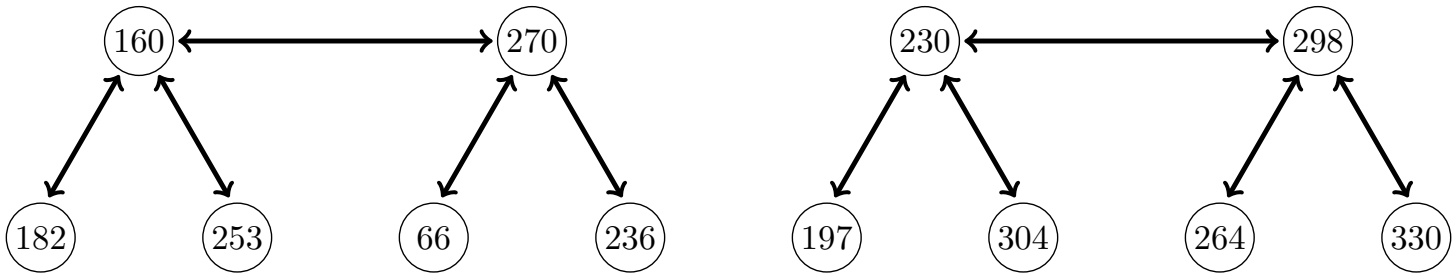
Actually, we would like to take the ℓ-isogeny graph on the full $\mathcal{C}\ell(\mathcal{O}_K)$-orbit. This might be composed of several ℓ-isogeny orbits (craters), although the class group is transitive.



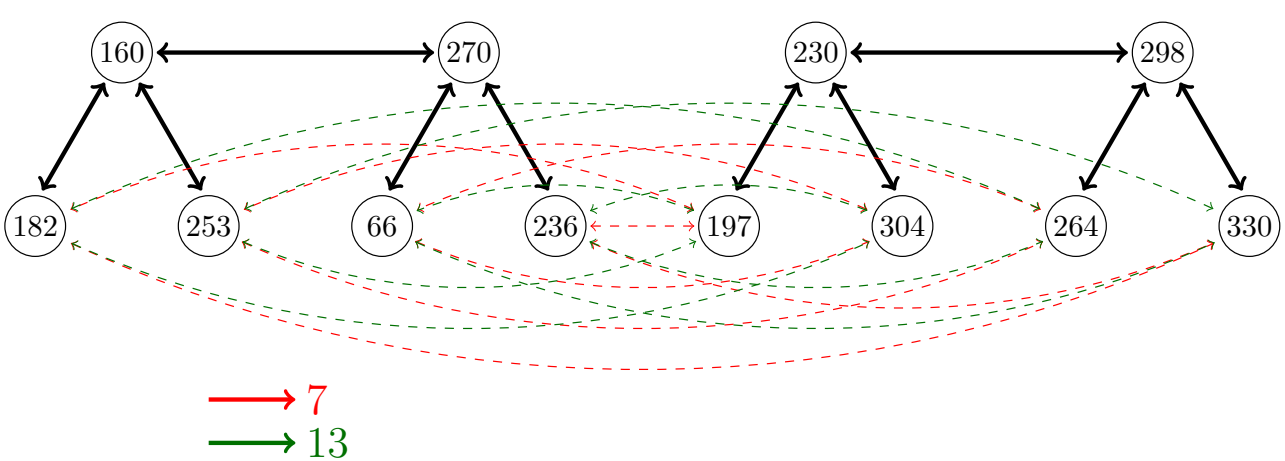
WHIRLPOOL - AN EXAMPLE

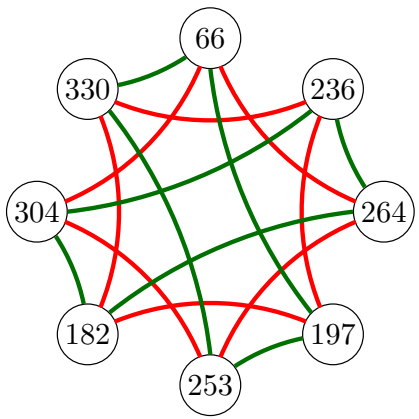
The set of multiple ℓ-volcanoes is called ℓ-cordillera.

Example. $p = 353, \ell = 2$, elliptic curves with 344 $\mathbb{F}_{353}$-rational points.



A whirlpool is the union of the two, shuffled by the class group of $\mathbb{Z}[2\sqrt{-82}]$.





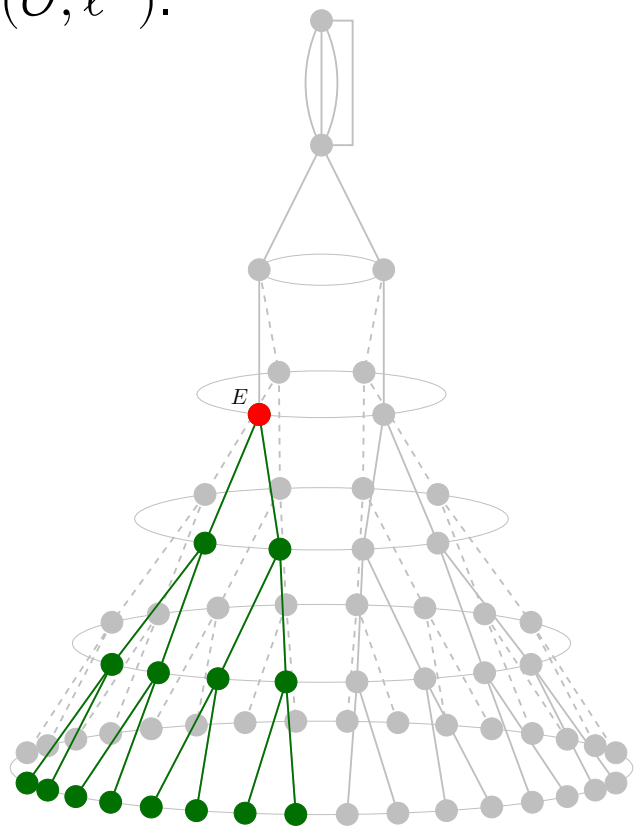
EDDIES

Given an order $\mathcal{O}$, we write $\mathcal{O}(M) = \mathbb{Z} + M\mathcal{O}$ - the order of index M, and $O_n = \mathcal{O}(\ell^n)$. Moreover, we denote the kernel

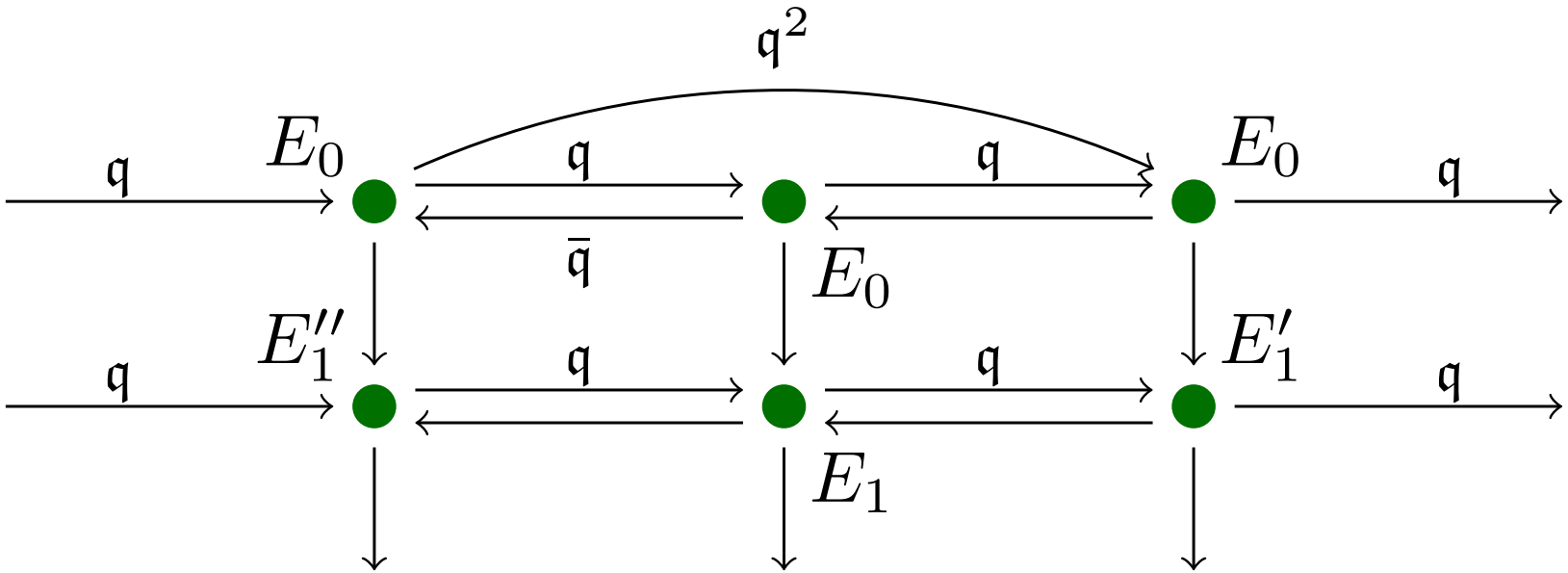
$$U(\mathcal{O}, M) = \mathsf{ker}(\mathcal{C}\ell(\mathcal{O}(M)) \longrightarrow \mathcal{C}\ell(\mathcal{O}))$$

which is the stabilizer of an isomorphism class of a curve oriented by $\mathcal{O}$.

An Eddy at E is the subgroup of ℓ-isogenies descending from E, equipped with the compatible action of $U(\mathcal{O}, \ell^n)$.



DIFFERENTIATING CONJUGATE IDEAL CLASSES



We write $\mathcal{O}_K = \mathbb{Z}[\omega]$ and we observe that if $\mathfrak{q}^2$ was principal, then

$$q^2 = \mathsf{N}(\mathfrak{q}^2) = \mathsf{N}(a + b\ell^i\omega)$$

since it would be generated by an element of $\mathcal{O}_i = \mathbb{Z} + \ell^i\mathcal{O}_K$. Now

$$\mathsf{N}(a + b\ell^i) = a^2 \pm abt\ell^i + b^2 s\ell^{2i} \quad \text{where} \quad \omega^2 + t\omega + s = 0$$

Thus, as soon as $\ell^{2i} \gg q^2$, we are guaranteed that $\mathfrak{q}^2$ is not principal.

AN EQUIVALENCE OF CATEGORIES

L.COLÒ 2M

Quadratic forms (a, b, c)	Ideal classes $\left[(a, \frac{-b+\sqrt{\Delta}}{2})\right]$	K-lattices $a\mathbb{Z} + \frac{-b\sqrt{\Delta}}{2}\mathbb{Z}$
Ideal classes $[\mathfrak{a}] = [(\alpha, \beta)]$	Quadratic forms $\frac{N(\alpha x - \beta y)}{N(\mathfrak{a})}$	K-lattices $\alpha\mathbb{Z} + \beta\mathbb{Z}$
K-lattices $\omega_1\mathbb{Z} + \omega_2\mathbb{Z}$	Quadratic forms $\frac{N(\omega_1 x - \omega_2 y)}{N(\omega_1)}$	Ideal classes (ω_1, ω_2)

We can associate to any such object a point in $\mathbb{H}$ (the upper half plane) and therefore a CM point on the Poincaré plane.

INITIALIZING THE LADDER - AN EXAMPLE

Suppose $D_K = -3$, and $\ell = 2$; we note that for all $n \geq 3$, that

$$\mathcal{Cl}(\mathcal{O}_n) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{n-2}\mathbb{Z}$$

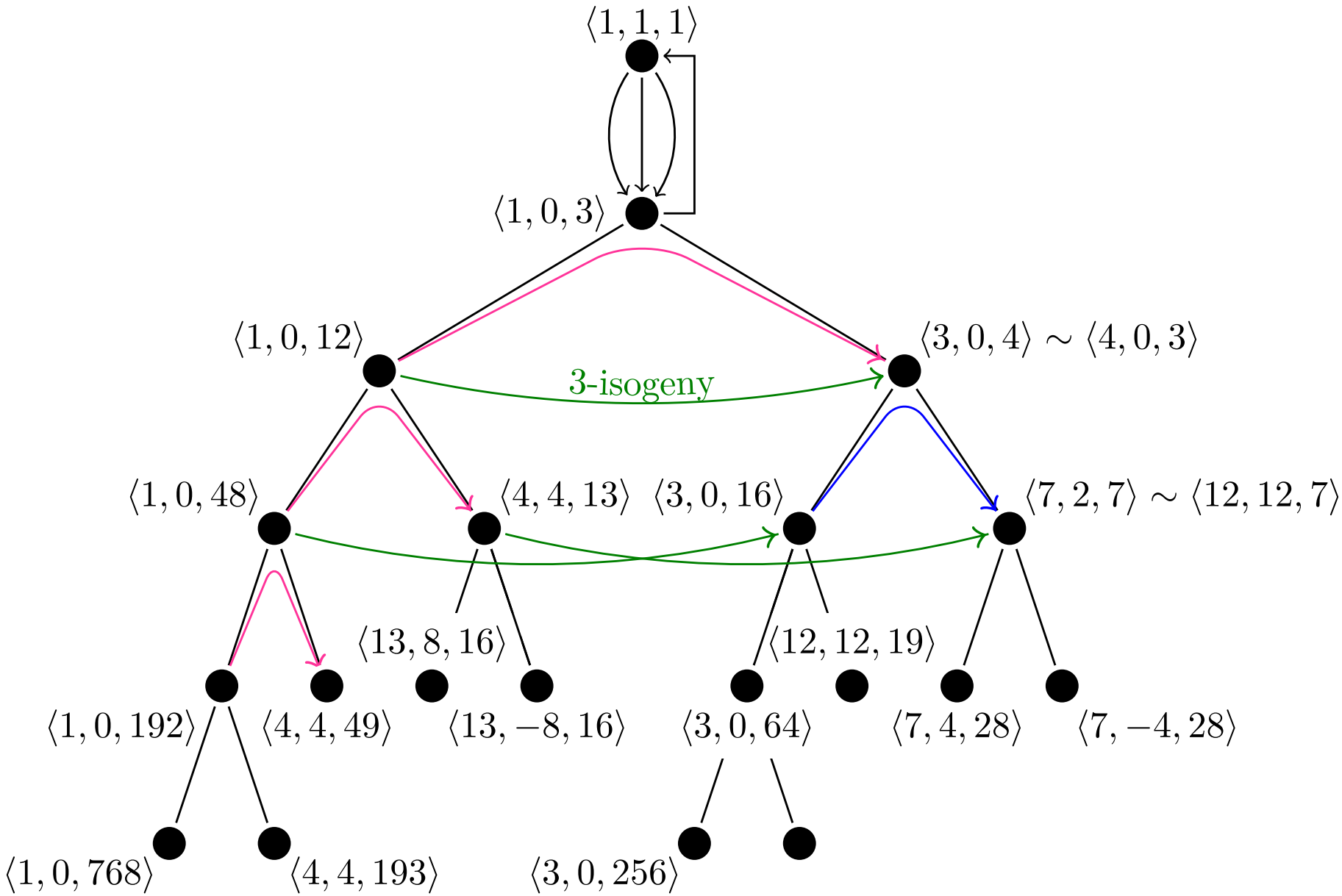
and in particular, $\mathcal{Cl}(\mathcal{O}_n)[2]$ consist of the classes of binary quadratic forms

$$\{\langle 1, 0, |D_K|\ell^{2(n-1)}\rangle, \langle |D_K|, 0, \ell^{2(n-1)}\rangle, \langle \ell^2, \ell^2, n_1\rangle, \langle \ell^2|D_K|, \ell^2|D_K|, n_2\rangle\}.$$

where $\ell^4 - 4\ell^2 n_1 = \ell^4|D_K|^2 - 4\ell^2|D_K|n_2 = -\ell^{2n}|D_K|$, whence

$$n_1 = 1 + \ell^{2(n-2)}|D_K| \text{ and } n_2 = |D_K| + \ell^{2(n-2)}.$$

INITIALIZING THE LADDER - A PICTURE



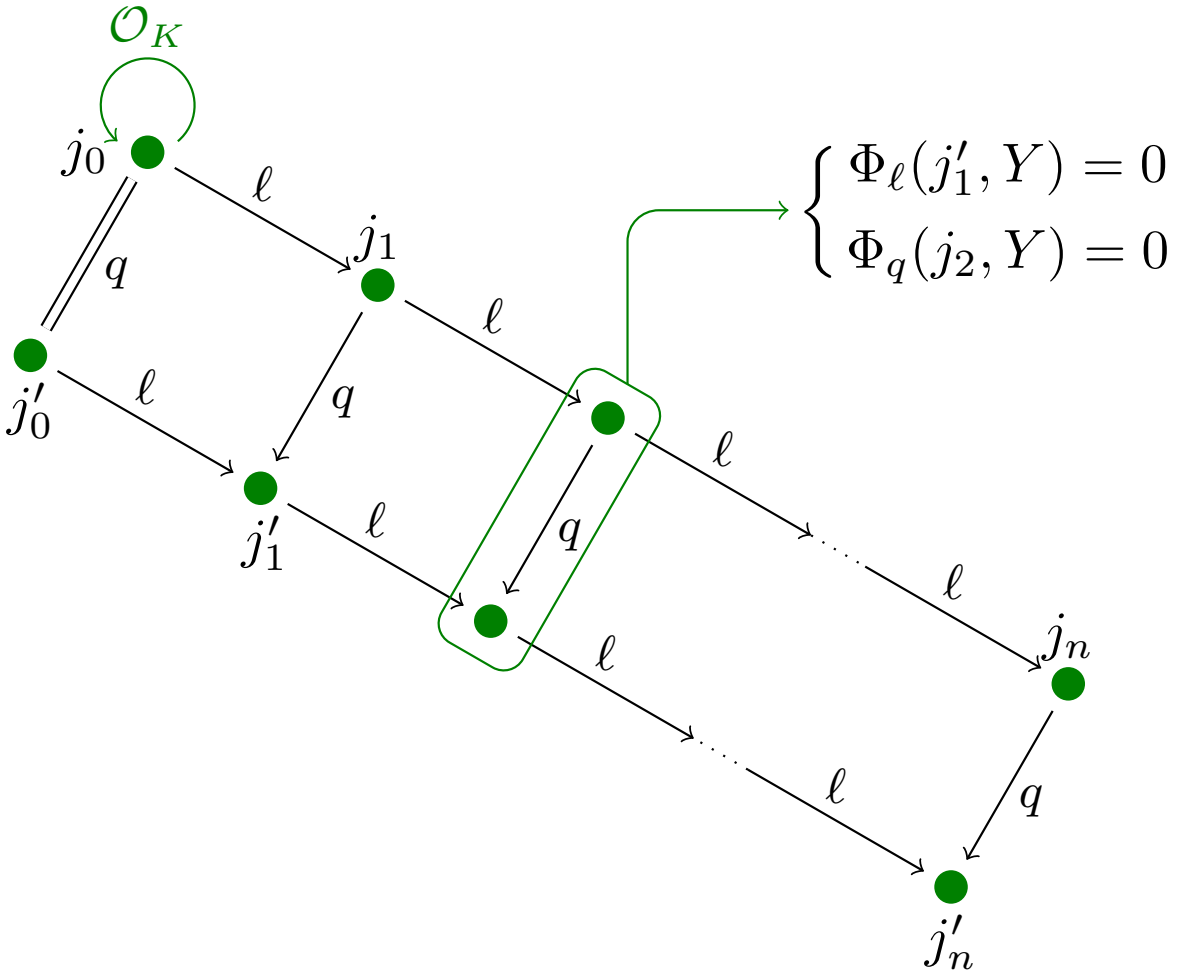
INITIALIZING THE LADDER - A TABLE

L.COLÒ 2M

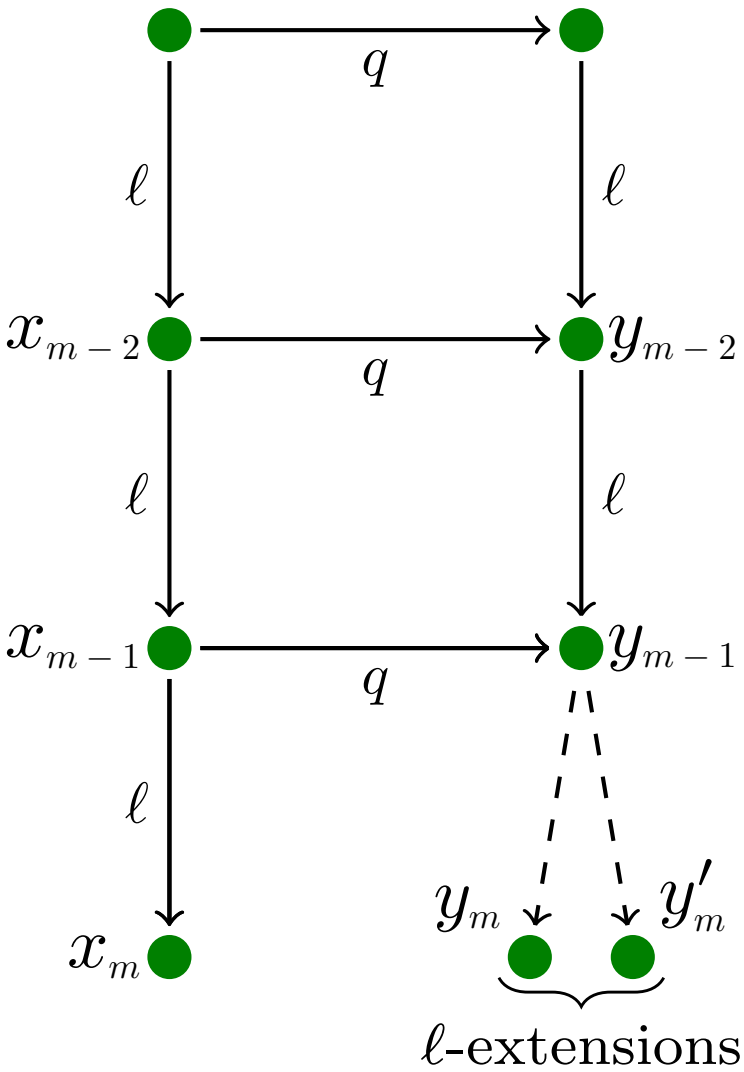
q	m	f_m	$[f_m]$	$[f_{m-1}]$
7	4	$\langle 7, 4, 28\rangle$	$[\langle 7, 4, 28\rangle]$	$[\langle 7, 2, 7\rangle]$
13	4	$\langle 13, 8, 16\rangle$	$[\langle 13, 8, 16\rangle]$	$[\langle 4, 4, 13\rangle]$
19	5	$\langle 19, 14, 43\rangle$	$[\langle 19, 14, 43\rangle]$	$[\langle 12, 12, 19\rangle]$
31	4	$\langle 31, 10, 7\rangle$	$[\langle 7, 4, 28\rangle]$	$[\langle 7, 2, 7\rangle]$
37	4	$\langle 37, 34, 13\rangle$	$[\langle 13, -8, 16\rangle]$	$[\langle 4, 4, 13\rangle]$
43	5	$\langle 43, 14, 19\rangle$	$[\langle 19, -14, 43\rangle]$	$[\langle 12, 12, 19\rangle]$
61	4	$\langle 61, 56, 16\rangle$	$[\langle 13, -8, 16\rangle]$	$[\langle 4, 4, 13\rangle]$
67	6	$\langle 67, 24, 48\rangle$	$[\langle 48, -24, 67\rangle]$	$[\langle 12, 12, 67\rangle]$
73	5	$\langle 73, 40, 16\rangle$	$[\langle 16, -8, 49\rangle]$	$[\langle 4, 4, 49\rangle]$
79	4	$\langle 79, 38, 7\rangle$	$[\langle 7, 4, 28\rangle]$	$[\langle 7, 2, 7\rangle]$
97	5	$\langle 97, 56, 16\rangle$	$[\langle 16, 8, 49\rangle]$	$[\langle 4, 4, 49\rangle]$
103	4	$\langle 103, 46, 7\rangle$	$[\langle 7, -4, 28\rangle]$	$[\langle 7, 2, 7\rangle]$
109	4	$\langle 109, 70, 13\rangle$	$[\langle 13, 8, 16\rangle]$	$[\langle 4, 4, 13\rangle]$
127	4	$\langle 127, 116, 28\rangle$	$[\langle 7, 4, 28\rangle]$	$[\langle 7, 2, 7\rangle]$

COMPLETING SQUARES OF ISOGENIES

Once the ladder is initialized we have to complete it. This amounts to extend the initial q-isogeny o the whole chain.



EXTENDING THE LADDER



Let $\ell = 2$.

- The two ℓ-extensions are determined by a quadratic polynomial (deduced from y_{m-1}, y_{m-2}:

$$\phi_\ell(y) = 0$$

  We can solve for y_m, y'_m, its roots.

- We have a degree $q+1$ polynomial $\phi_q(y) = 0$ determined by x_m but we do note need to compute it. It suffices

$$\phi_q(y) \mod \phi_\ell(y)$$

  Indeed

$$\Phi_q(x, y) \equiv \phi_q(y) \mod (x - x_m, \phi_\ell(y))$$

WEBER MODULAR FUNCTIONS

Introduced by Weber, they are

$$\mathfrak{f}(\tau) = \zeta_{48}^{-1} \frac{\eta\left(\frac{\tau+1}{2}\right)}{\eta(\tau)} = q^{-\frac{1}{48}} \prod_{n=1}^{+\infty}(1+q^{n-\frac{1}{2}})$$

$$\mathfrak{f}_1(\tau) = \frac{\eta\left(\frac{\tau}{2}\right)}{\eta(\tau)} = q^{-\frac{1}{48}} \prod_{n=1}^{+\infty}(1-q^{n-\frac{1}{2}})$$

$$\mathfrak{f}_2(\tau) = \sqrt{2}\frac{\eta\left(2\tau\right)}{\eta(\tau)} = \sqrt{2}q^{\frac{1}{24}} \prod_{n=1}^{+\infty}(1+q^{n})$$

Historically, $\mathfrak{f}_2$ was the first to be studied by Weber, who eventually introduced the others such that

$$(X+\mathfrak{f}^8)(X-\mathfrak{f}_1^8)(X-\mathfrak{f}_2^8) = X^3 - \gamma_2 X + 16$$

WEBER MODULAR FUNCTIONS - PROPERTIES

The previous relation $(X+\mathfrak{f}^8)(X-\mathfrak{f}_1^8)(X-\mathfrak{f}_2^8) = X^3 - \gamma_2 X + 16$ yields

- $\mathfrak{f}^8 = \mathfrak{f}_1^8 + \mathfrak{f}_2^8$

WEBER MODULAR FUNCTIONS - PROPERTIES

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- $\mathfrak{f}_1(2\tau)\mathfrak{f}_2(\tau) = \mathfrak{f}(\tau)\mathfrak{f}_1(\tau)\mathfrak{f}_2(\tau) = \sqrt{2}$

WEBER MODULAR FUNCTIONS - PROPERTIES

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- $\mathfrak{f}_1(2\tau)\mathfrak{f}_2(\tau) = \mathfrak{f}(\tau)\mathfrak{f}_1(\tau)\mathfrak{f}_2(\tau) = \sqrt{2}$

which gives

- $\zeta_{48}^{-1}\eta\left(\frac{\tau+1}{2}\right)\eta\left(\frac{\tau}{2}\right)\eta(2\tau) = \eta(\tau)^3$

WEBER MODULAR FUNCTIONS - PROPERTIES

The previous relation $(X+\mathfrak{f}^8)(X-\mathfrak{f}_1^8)(X-\mathfrak{f}_2^8) = X^3 - \gamma_2 X + 16$ yields

- $\mathfrak{f}^8 = \mathfrak{f}_1^8 + \mathfrak{f}_2^8$
- $\mathfrak{f}_1(2\tau)\mathfrak{f}_2(\tau) = \mathfrak{f}(\tau)\mathfrak{f}_1(\tau)\mathfrak{f}_2(\tau) = \sqrt{2}$

which gives

- $\zeta_{48}^{-1}\eta\left(\frac{\tau+1}{2}\right)\eta\left(\frac{\tau}{2}\right)\eta(2\tau) = \eta(\tau)^3$

They have transformation formulæ

- $(\mathfrak{f}, \mathfrak{f}_1, \mathfrak{f}_2) \circ S = (\mathfrak{f}, \mathfrak{f}_2, \mathfrak{f}_1)$
- $(\mathfrak{f}, \mathfrak{f}_1, \mathfrak{f}_2) \circ T = (\zeta_{48}^{-1}\mathfrak{f}_1, \zeta_{48}^{-1}\mathfrak{f}, \zeta_{24}\mathfrak{f}_2)$

WEBER MODULAR FUNCTIONS - PROPERTIES

The previous relation $(X+\mathfrak{f}^8)(X-\mathfrak{f}_1^8)(X-\mathfrak{f}_2^8) = X^3 - \gamma_2 X + 16$ yields

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- $\mathfrak{f}_1(2\tau)\mathfrak{f}_2(\tau) = \mathfrak{f}(\tau)\mathfrak{f}_1(\tau)\mathfrak{f}_2(\tau) = \sqrt{2}$

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- $\zeta_{48}^{-1}\eta\left(\frac{\tau+1}{2}\right)\eta\left(\frac{\tau}{2}\right)\eta(2\tau) = \eta(\tau)^3$

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- $(\mathfrak{f}, \mathfrak{f}_1, \mathfrak{f}_2) \circ T = (\zeta_{48}^{-1}\mathfrak{f}_1, \zeta_{48}^{-1}\mathfrak{f}, \zeta_{24}\mathfrak{f}_2)$

and relations with the j-invariant

- $j = \frac{(\mathfrak{f}^{24}-16)^3}{\mathfrak{f}^{24}} = \frac{(\mathfrak{f}_1^{24}+16)^3}{\mathfrak{f}_1^{24}} = \frac{(\mathfrak{f}_2^{24}+16)^3}{\mathfrak{f}_2^{24}}$

WEBER MODULAR POLYNOMIALS

$\mathfrak{f}(\tau)$ is a modular function of level 48 giving a degree 72 cover of the j-line.

The modular polynomials with respect to $\mathfrak{f}$ are the new (Weber) integral polynomials $\Phi_q(x, y)$ such that

$$\Phi_q\left(\mathfrak{f}(\tau), \mathfrak{f}(q\tau)\right) = 0$$

Division Polynomials

The symmetry $\Phi_q(\zeta_{24}x, \zeta_{24}^q y) = \zeta_{24}^{q+1}(x, y)$ yields a great sparseness:

$$\Phi_5(x, y) = x^6 - x^5y^5 + 4xy + y^6$$
$$\Phi_7(x, y) = x^8 - x^7y^7 + 7x^4y^4 - 8xy + y^8$$
$$\Phi_{11}(x, y) = x^{12} - x^{11}y^{11} + 11x^9y^9 - 44x^7y^7 + 88x^5y^5 - 88x^3y^3 + 32xy + y^{12}$$

WEBER MODULAR POLYNOMIALS - FURTHER REDUCTIONS

For $\ell = 2$ or $\ell = 3$, the 48-level structure gives the modular polynomials $\Phi_2(x, y)$ and $\Phi_3(x, y)$ a particular form.

- We descend the 2-level structure by setting $t = -\mathfrak{f}^8$, so that $j = \left(\frac{t^3+16}{t}\right)^3$. We obtain the modular polynomial:

$$\Psi_2(x, y) = (x^2 - y)y + 16x$$

  and the Weber modular polynomial $\Phi_2(x, y) = -\Psi_2(-x^8, -y^8)$ remains irreducible
- A similar descent of the 3-level to the function $r = \mathfrak{f}^3$, gives the modular polynomial

$$\Psi_3(x, y) = x^4 - x^3y^3 + 8xy + y^4,$$

  such that $\Psi_3(r(\tau), r(3\tau)) = 0$, for which $\Phi_3(x, y) = \Psi_3(x^3, y^3)$ is irreducible.

WEBER CURVES

We fix the normalization $(\mathfrak{u}_0, \mathfrak{u}_1, \mathfrak{u}_2) = (\mathfrak{f}, \zeta_{16}\mathfrak{f}_1, \zeta_{16}^{-1}\mathfrak{f}_2)$.
Notice that $\{\zeta_{24}^j \mathfrak{u}_i \mid j \in \mathbb{Z}/24\mathbb{Z}\}$ are the 72 roots of

$$(X^{24} - 16)^3 - j(q)X \in \mathbb{Q}(\zeta_{24})[[q^{1/24}]]$$

The map determined by the normalized Weber functions $(\mathfrak{u}_0^m : \mathfrak{u}_1^m : \mathfrak{u}_2^m : 1)$ determines a Weber modular curve $\mathcal{W}_{3n}$ in $\mathbb{P}^3$

$$\mathcal{W}_{3n} : \begin{cases} X_0^n + X_1^n + X_2^n = 0, \\ X_0 X_1 X_2 = \sqrt{2}^m X_3^3 \end{cases}$$

for m and n such that $mn = 8$

WEBER MODULAR CURVES

The quotient Weber curve $\mathcal{W}_n$ is defined as the image of $(\mathfrak{u}_0^{3m} : \mathfrak{u}_1^{3m} : \mathfrak{u}_2^{3m} : 1)$:

$$\mathcal{W}_n : \begin{cases} X_0^n + X_1^n + X_2^n = 48 X_3^n, \\ X_0 X_1 X_2 = \sqrt{2}^{3m} X_3^3. \end{cases}$$

These curves are equipped with maps $\mathcal{W}_{mn} \to \mathcal{W}_n$ for each product $mn \mid 24$.

To each factorization $mn = 24$, the Weber curve $\mathcal{W}_n$ in $\mathbb{P}^3$, defined by the triple of Weber functions $(\mathfrak{u}_0^m, \mathfrak{u}_1^m, \mathfrak{u}_2^m)$, comes equipped with an action of $\mathsf{PSL}_2(\mathbb{Z})$.

Weber Modular Curves

We denote the kernel of the action by Γ_n, identifying the Weber curves with the modular curve $X(\Gamma_n)$.

WEBER MODULAR GROUPS

The $\mathbf{PSL}_2(\mathbb{Z})$ action on Weber functions induces a representation in $\mathbf{GL}_3(\mathbb{Q}(\zeta_n))$

$$S \longmapsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & \bar{\zeta}_8^m \\ 0 & \zeta_8^m & 0 \end{pmatrix} \qquad T \longmapsto \begin{pmatrix} 0 & \zeta_{24}^m & 0 \\ \bar{\zeta}_{24}^m & 0 & 0 \\ 0 & 0 & \zeta_{24}^m \end{pmatrix}$$

Its kernel Γ_n is a normal congruence subgroup of $\mathbf{PSL}_2(\mathbb{Z})$.

Noting that $\Gamma_1 \subset \Gamma_3 \cap \Gamma_8 = \Gamma_{24}$, we reduce to determining Γ_3 and the chain $\Gamma_1 \subset \Gamma_2 \subset \Gamma_4 \subset \Gamma_8$.

Proposition

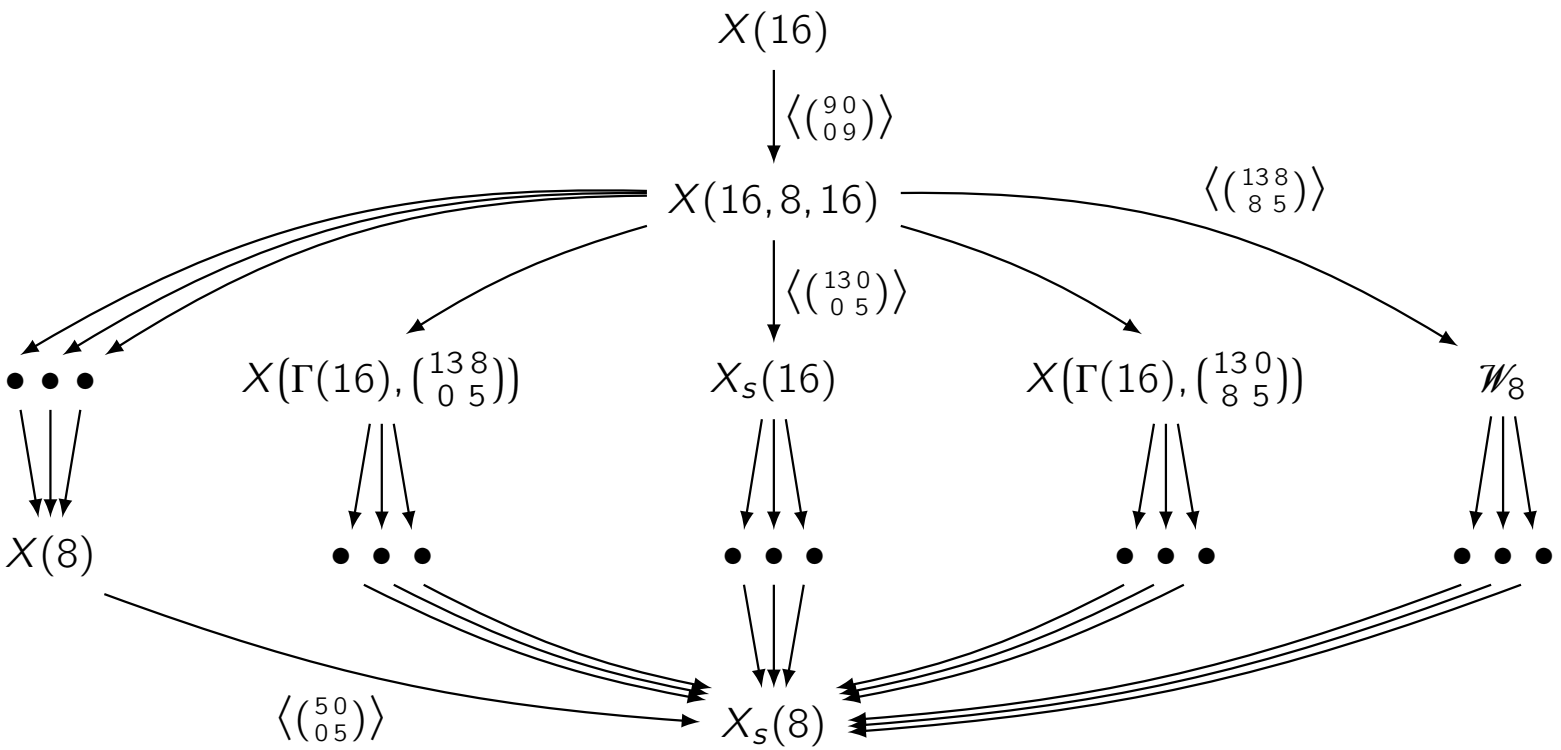
- The Weber kernel group Γ_1 equals $\Gamma(2)$ and $\mathcal{W}_1 \cong X(2)$.
- The Weber kernel group Γ_3 equals $\Gamma(2) \cap \Gamma_{ns}^+(3)$, and for each n dividing 8
$$\Gamma_{3n} = \Gamma_n \cap \Gamma_{ns}^+(3).$$
- The Weber kernel group Γ_2 equals $\Gamma(4)$ and $\mathcal{W}_2 = X(4)$.
- The Weber kernel group Γ_4 equals $\Gamma_s(8)$ and $\mathcal{W}_4 = X_s(8)$.

WEBER MODULAR GROUPS - THE GROUP Γ_8

$$\Gamma(16) \subset \Gamma_8 \subset \Gamma_s(8) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \ : \ b \equiv c \equiv 0 \bmod 8 \right\}$$

Proposition

The Weber kernel subgroup Γ_8 is generated by $\Gamma(16)$ and $\begin{pmatrix} 13 & 8 \\ 8 & 5 \end{pmatrix}$



SUPERSINGULAR FIELDS OF DEFINITIONS

Theorem

The supersingular invariants on the modular curve $X_0(N)$ are defined over $\mathbb{F}_{p^2}$.
If $p \equiv \pm 1 \bmod N$, then the supersingular invariants also split over $\mathbb{F}_{p^2}$ on $X_1(N)$.

- As a consequence, $X_s(N) = X(\Gamma_s(N))$ also splits the supersingular moduli for every N.
- Then the supersingular invariants on the modular curve $X(N)$ are defined over $\mathbb{F}_{p^2}$ for $p \equiv \pm 1 \bmod N$.

Theorem

The supersingular Weber invariants on $\mathcal{W}_{24}$ are defined over $\mathbb{F}_{p^2}$.

$$\begin{array}{ccc} X(16,8,16) & \longrightarrow & X(8) \\ \Big\downarrow{\scriptstyle \left\langle \left(\begin{smallmatrix} 13 & 8 \\ 8 & 5 \end{smallmatrix}\right) \right\rangle} & & \Big\downarrow{\scriptstyle \left\langle \left(\begin{smallmatrix} 5 & 0 \\ 0 & 5 \end{smallmatrix}\right) \right\rangle} \\ \mathcal{W}_8 & \longrightarrow & X_s(8) \end{array}$$

WEBER INITIALIZATION

Let u be a supersingular value of the Weber function,

$$r = f^3 \qquad t = -f^8 \qquad s = t^3$$

along the chain $\mathcal{W}_8 \to Y \to X_0(2) \to X(1)$. we get

$$\Psi_2(x,y) = (x^2 - y)y + 16x \qquad \Psi_3(x,y) = x^4 - x^3y^3 + 8xy + y^4$$

The elliptic curves associated to Weber invariants are the fiber in the family:

$$y^2 + xy = x^3 - \frac{1}{f^{24} - 64}x$$

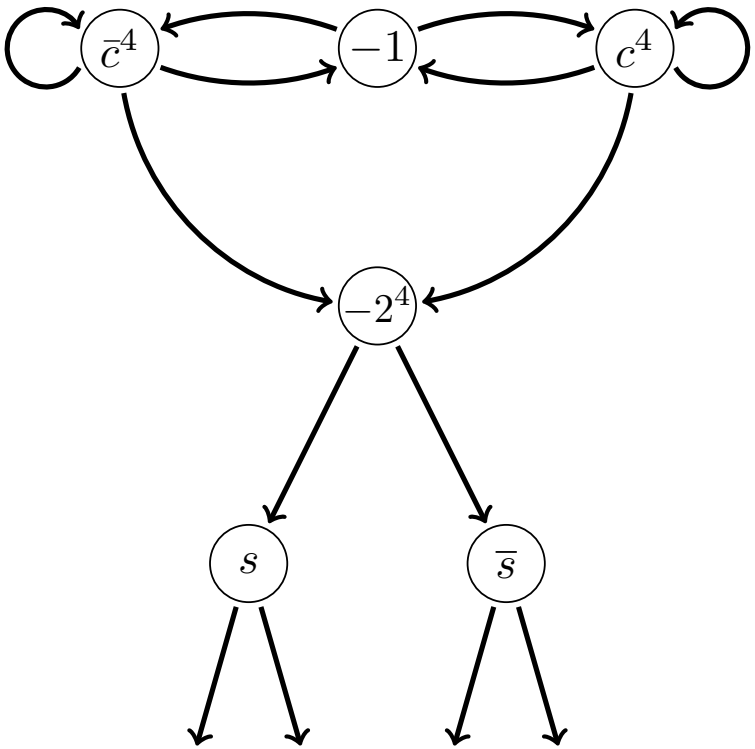
over f on the Weber curve.

The initial values with which to build the public ℓ-isogeny chains are

D	j_0	s_0	t_0	D	j_1	s_1	t_1
-3	0	-2^4	$-(\sqrt[3]{2})^4$	-12	$2^4 15^3$	-2^8	$-(\sqrt[3]{2})^8$
-4	12^3	2^3	2	-16	66^3	2^9	2^3
-7	-15^3	-1	-1	-28	255^3	-2^{12}	-2^4
-8	20^3	2^6	2^2	-32	j_1	t_1^3	$2^3(\sqrt{2}+1)$

WEBER INITIALIZATION - DISCRIMINANT -7

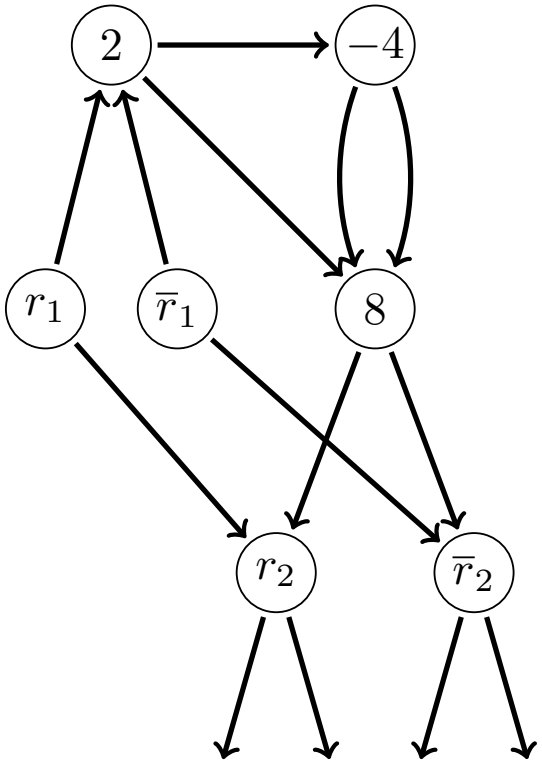
Endomorphism ring is small: generated by an endomorphism of degree 2 we avoid any pathologies associated with the extra automorphisms.



- $t_0 = -1$ and c root of $x^2 - x + 2$.
- c^4 and $\bar{c}^4$ also t-values over $j = -15^3$.
- $\Psi_2(-1, c^4) = \Psi_2(-1, \bar{c}^4) = 0$, the two extensions correspond to the horizontal 2-isogenies.
- $\Psi_2(c^4, c^4) = \Psi_2(c^4, -2^4) = 0$: the former enters a cycle the latter induces a descending isogeny.

Initialization: $(t_0, t_1, t_2, \dots)$ beginning with $(-1, c^4, -2^4, \dots)$.
Successive solutions to $\Psi_2(t_i, t_{i+1}) = 0$ are necessarily descending.
Extension: random choice of root t_{i+1} of $\Psi_2(t_i, x)$.

WEBER INITIALIZATION - DISCRIMINANT -4

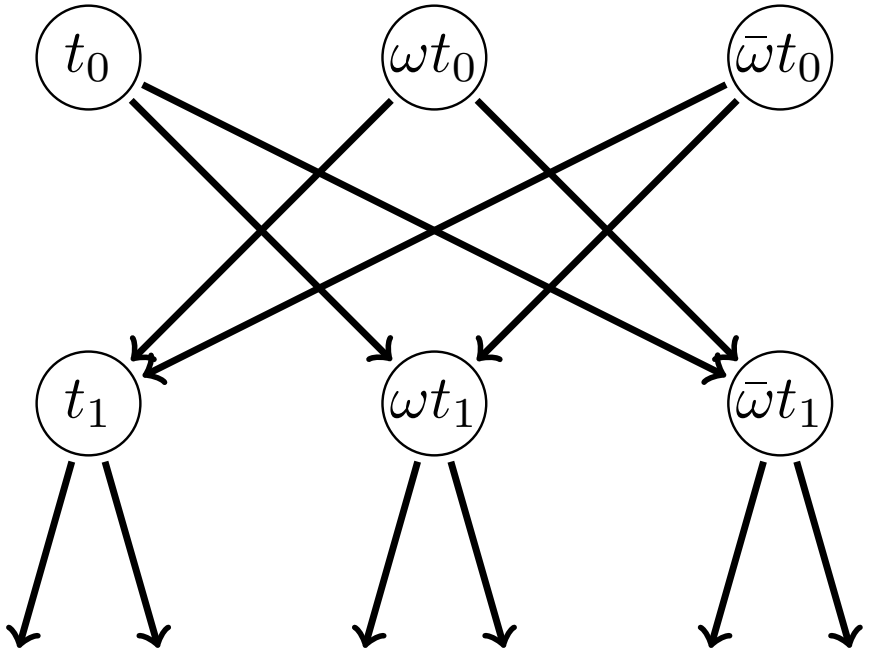


- t-invariants over $j = 12^3$ fall in two orbits of points, $\{2, 2\omega, 2\omega^2\}$ of multiplicity 2, and $\{-4, -4\omega, -4\omega^2\}$ of multiplicity 1.
- These points at the surface are linked by a 2-isogeny and to 2-depth 1, to $t = 8$.
- $\Psi_2(\omega x, \omega^2 y) = \omega \Psi_2(x, y)$: the choice of representative in the orbit gives rise to one of three distinct components of the 2-isogeny graph.

Initialization: $(t_0, t_1, t_2, \dots) = (2, 8, 8c, \dots)$ where c is a root of $x^2 - 8x - 2$.
Extension: random selection of a root t_{i+1} of $\Psi_2(t_i, x)$.

The full 2-isogeny graph has ascending edges from the depth one to $t_0 = 2$
If an isogeny is descending its only extension to a 2-isogeny chain is descending

WEBER INITIALIZATION - DISCRIMINANT -3

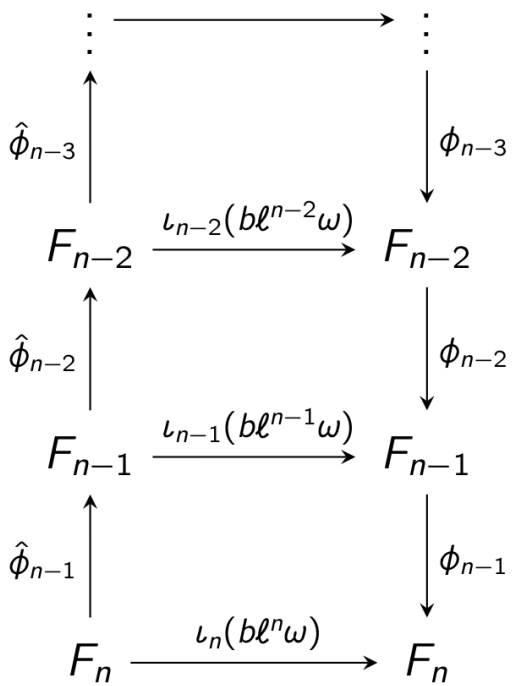


- $t_0 = -(\sqrt[3]{2})^4 = -2\sqrt[3]{2}$.
- $\{t_0, t_0\omega, t_0\omega^2\}$ are t-values over $j = 0$, each of multiplicity 3
- $t_1 = -t_0^2$, and $\Psi_2(t_0, t_1\omega) = \Psi_2(t_0, t_1\omega^2) = 0$,

Since 2 is inert, every path from t_0 is descending, so we may initialize the 2-isogeny chain with $(t_0, t_1\omega)$.

There are additional t-invariants at each depth > 0 which admit ascending and descending isogenies.
Any descending isogenies must rejoin this graph of descending isogenies from the surface.

The knowledge of a K-oriented endomorphism $\iota_n^{(A)}(\beta)$ (respectively $\iota_n^{(B)}(\beta)$) with $\beta \in \mathcal{O}_n \setminus \mathcal{O}_{n+1}$ permits one to recover the curve F_{n-1} (respectively G_{n-1}) and by recursion the whole chain.



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The idea of Dartois and De Feo reduces to finding a short enough non-zero vector of bounded norm in the relation lattice

$$L_n := \left\{ (e_1, \dots, e_t) \in \mathbb{Z}^t \;\middle|\; \prod_{j=1}^{t} \left[\mathfrak{q}_j \cap \mathcal{O}_n\right]^{e_j} = [1] \text{ in } \mathcal{C}\ell(\mathcal{O}_n) \right\}$$

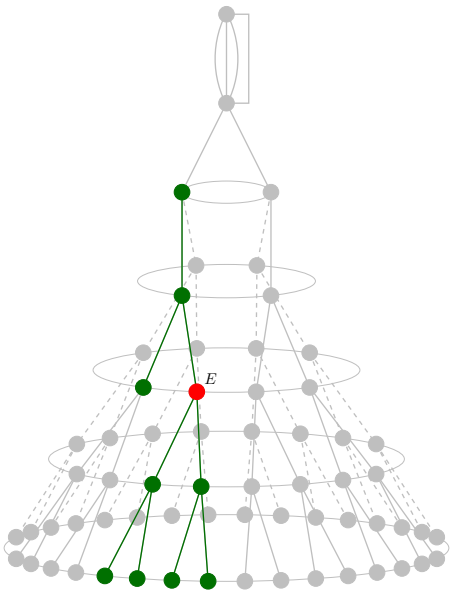
Parameter setup

It consists in fixing a discriminant Δ_K and prime ℓ, and constructing a common public isogeny chain

$$E_0 \longrightarrow E_1 \longrightarrow \ldots \longrightarrow E_n = E$$

Disjoint prime sets: $\mathcal{P}_s$ (split primes), and $\mathcal{P}_A$ and $\mathcal{P}_B$ (non-split primes).

- Split primes: precompute initial directions in $\mathcal{C}\ell(\mathcal{O}_K)$ and push forward to E.
- Non-split primes: precompute labeled clouds and push forward to E.

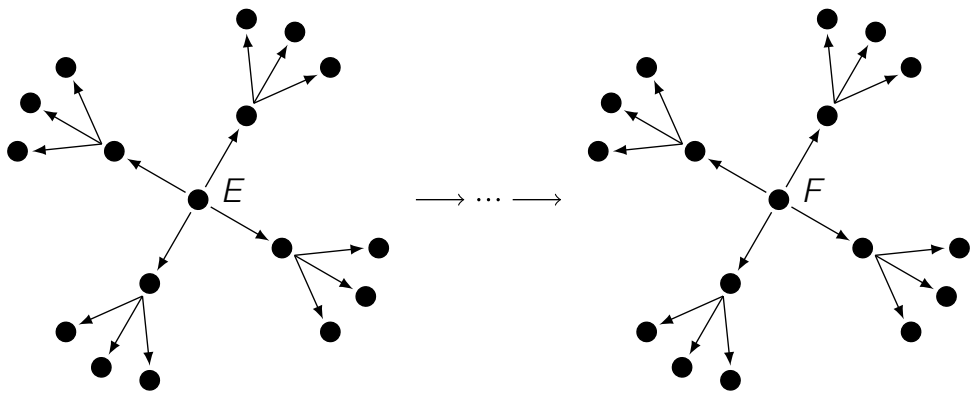


Key generation

On one side, A begins with $F = E$.

- Split primes: for each prime q_i in $\mathcal{P}_S$, choose a random $s_i \in I_i$, constructs the q_i-isogeny walk of length s_i while pushing forward the other direction as well as the q-clouds at each prime q in $\mathcal{P}_A$ and $\mathcal{P}_B$.
- Non-split primes: for each prime choose a random walk in the cloud to a new curve F and push forward the remaining unused q-clouds.

The data F and q-isogeny chains at primes q in $\mathcal{P}_s$ and q-clouds at primes q in $\mathcal{P}_B$ constitute A's public key.



Key generation

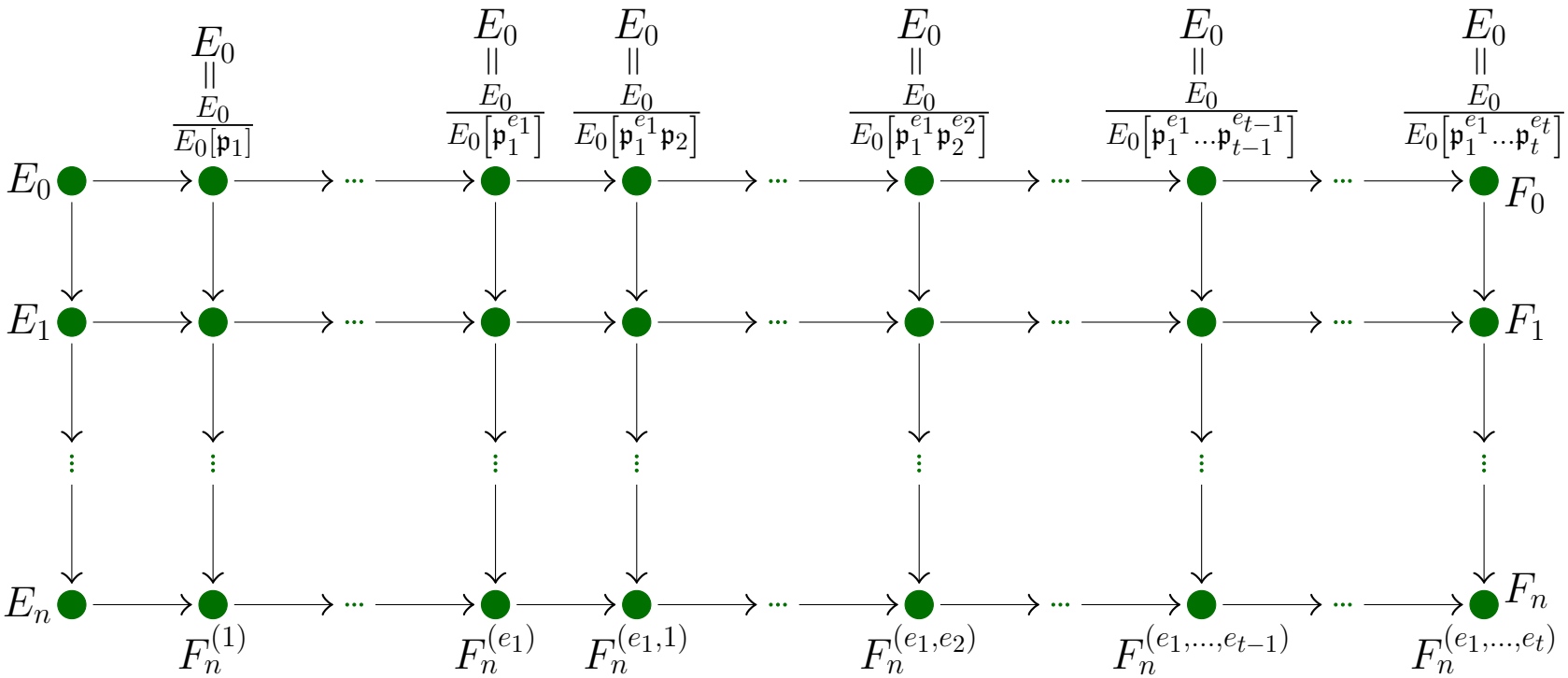
A obtains the public key data from B and reconstructs her isogeny walk from G using B's data for $\mathcal{P}_s$ and $\mathcal{P}_A$.

OSIDH PROTOCOL - GRAPHIC REPRESENTATION I

The first step consists of choosing the secret keys; these are represented by a sequence of integers $(e_1, \dots, e_t)$ such that $|e_i| \leq r$. The bound r is taken so that the number $(2r+1)^t$ of curves that can be reached is sufficiently large. This choice of integers enables Alice to compute a new elliptic curve

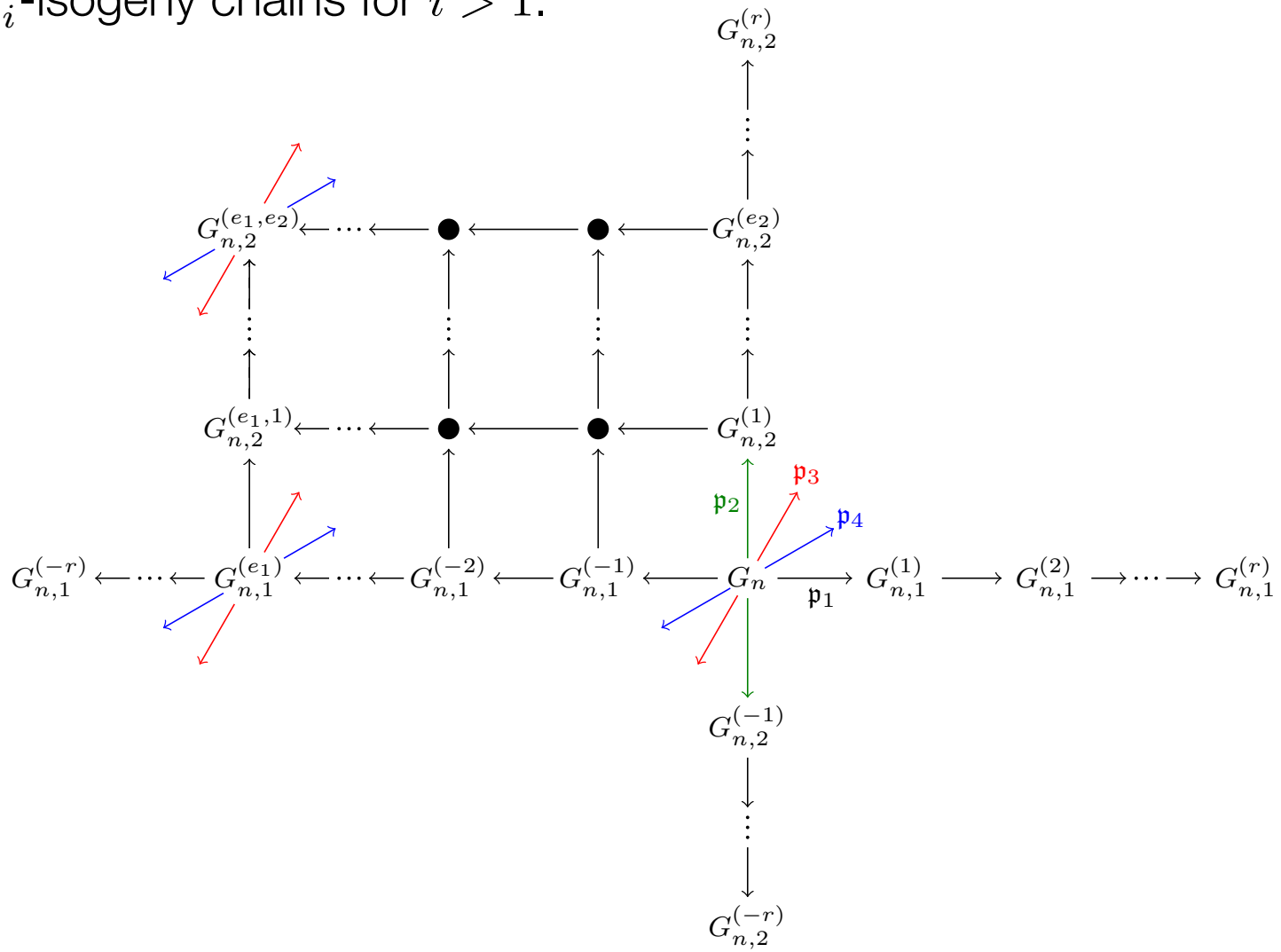
$$F_n = \frac{E_n}{E_n\left[\mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t}\right]}$$

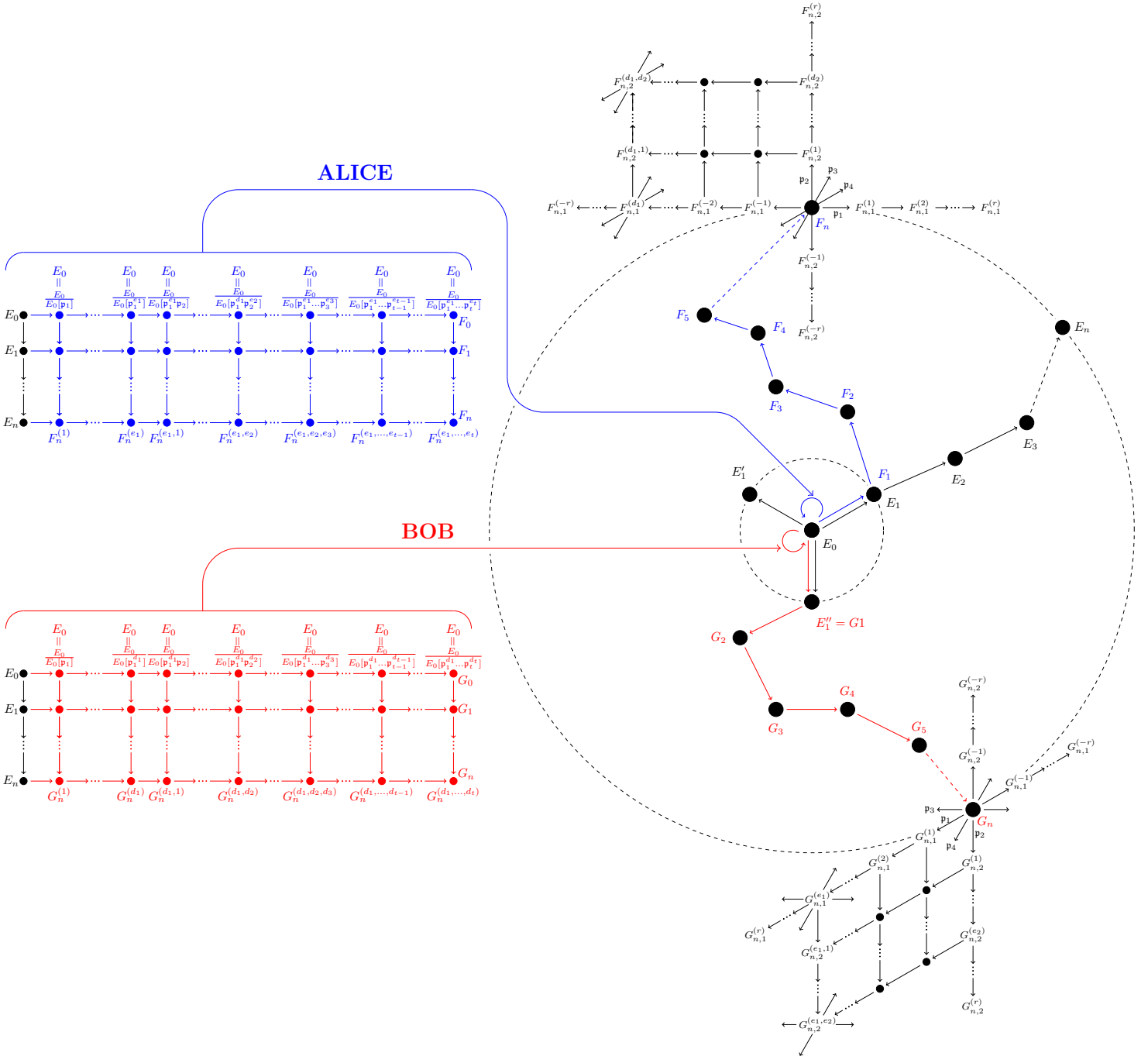
by means of constructing the following commutative diagram



OSIDH PROTOCOL - GRAPHIC REPRESENTATION II

Once that Alice obtain from Bob the curve G_n together with the collection of data encoding the directions, she takes e_1 steps in the $\mathfrak{p}_1$-isogeny chain and push forward all the $\mathfrak{p}_i$-isogeny chains for $i > 1$.




ALICE
BOB
E_0
E_1
E_n
F_0
F_1
F_n
F_2
F_3
F_4
F_5
E_1'
$E_1'' = G1$
G_0
G_1
G_n
G_2
G_3
G_4
G_5